

Dynamics of the Bost–Connes State Space

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Abstract

On the state space obtained by finite p -adic reduction of the Bost–Connes system, we characterize stable states relative to a background and construct a complete cycle of free evolution, formation exchange, and pattern reconstruction, giving a unified description of formation, motion, decay, and thermalization. Exchange between excitations and the background realizes coherent retrieval, exit, reverse formation, and background restoration within the same joint evolution. The state entering the next cycle is reconstructed from these components with their retained mutual coherence, providing a common dynamical basis for excitation motion, formation, and decay, and for relaxation of the ensemble occupation distribution. On this basis, we construct internal observation and recording processes, derive event statistics and convergence to stationarity under the corresponding conditions, and use joint conservation and entropy balances to relate reversible joint evolution to irreversible system relaxation.

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§1 Construction of the State Space

The Bost–Connes system is a quantum statistical mechanical system constructed in 1995 [1]. Starting from its local decomposition, we perform a finite p -adic reduction at a prime p to construct the corresponding state space. Establishing dynamics on this space requires answers to the following questions:

- (1) How does an excitation form from the background? What properties must it have to become a dynamical subject that can be continually identified?
- (2) Given an excitation and its interaction with the background, how does its state change? What determines the form and rate of its evolution?
- (3) Which changes preserve the identity of the subject during evolution, and which lead to its transformation or disappearance? Can these different outcomes be described by the same dynamical law?
- (4) How does the state after one interaction enter subsequent evolution? Under what conditions do repeated interactions maintain an excitation, and under what conditions do they cause it to fade or approach equilibrium?

Addressing these questions first requires specifying the construction of the state space and the arithmetic structure it inherits. We therefore begin with the definition of the Bost–Connes system and the basis for the finite reduction.

§1.1 The Bost–Connes System

The Bost–Connes system consists of an algebra $\mathcal{A}_{\text{BC}}$ and its time evolution σ_t . The two families of algebra generators represent rational phases and integer scaling: phase parameters combine additively, scaling operations compose multiplicatively, and compatibility relations connect the two.

Write $\mathbb{N}^* = \{1, 2, \dots\}$, and let $\mathbb{Q}/\mathbb{Z}$ denote the additive quotient group of rational numbers modulo integers. For each $r \in \mathbb{Q}/\mathbb{Z}$, the phase generator $e(r)$ satisfies

$$e(0) = 1, \quad e(r)^* = e(-r), \quad e(r_1)e(r_2) = e(r_1 + r_2). \quad (1.1)$$

Here 1 is the identity of the algebra, and $*$ denotes the involution. These relations imply $e(r)^*e(r) = 1$, so each phase generator is unitary.

For each $n \in \mathbb{N}^*$, integer scaling is represented by an isometry μ_n , with relations

$$\begin{aligned} \mu_1 &= 1, & \mu_m \mu_n &= \mu_{mn}, & \mu_n^* \mu_n &= 1, \\ \mu_m^* \mu_n &= \mu_n \mu_m^* & (\gcd(m, n) &= 1), & m, n &\in \mathbb{N}^*. \end{aligned} \quad (1.2)$$

These relations represent multiplication of positive integers by composition of operators and specify compatibility between coprime scalings.

The phase and scaling generators further satisfy

$$e(r)\mu_n = \mu_n e(nr), \quad \mu_n e(r)\mu_n^* = \frac{1}{n} \sum_{\substack{s \in \mathbb{Q}/\mathbb{Z} \\ ns=r}} e(s). \quad (1.3)$$

The first identity is the covariance relation, expressing compatibility between integer scaling and multiplication of the phase parameter. The second is the transfer relation, whose sum runs over all n solutions of $ns = r$. These generators and relations define the unital C^* -algebra $\mathcal{A}_{\text{BC}}$ [1, 2].

With $\hbar = 1$, the time evolution of the BC system is defined by

$$\sigma_t(e(r)) = e(r), \quad \sigma_t(\mu_n) = n^{it}\mu_n, \quad t \in \mathbb{R}. \quad (1.4)$$

It fixes the phase generators and assigns a time-dependent phase factor to integer scaling. The identity $(mn)^{it} = m^{it}n^{it}$ shows that this evolution preserves the multiplicative scaling relations; substitution into the covariance and transfer relations shows that the remaining defining relations are also preserved. Addition of parameters corresponds to composition, and σ_{-t} is the inverse of σ_t . Continuity on the generators extends through the dense $*$ -algebra to all of $\mathcal{A}_{\text{BC}}$. Thus σ_t is a one-parameter automorphism group of the algebra.

In the standard representation, the time factor n^{it} corresponds to a logarithmic energy spectrum. Write $\widehat{\mathbb{Z}} = \varprojlim_d \mathbb{Z}/d\mathbb{Z}$, with reduction maps between moduli ordered by divisibility. For $u \in \widehat{\mathbb{Z}}^\times$, define on the standard basis $\{\varepsilon_m\}_{m \geq 1}$ of $\ell^2(\mathbb{N}^*)$

$$\pi_u(e(r))\varepsilon_m = e^{2\pi i m(ur)}\varepsilon_m, \quad \pi_u(\mu_n)\varepsilon_m = \varepsilon_{nm}. \quad (1.5)$$

Here $ur \in \mathbb{Q}/\mathbb{Z}$ is determined by the residue class of u modulo the denominator of r . In this representation, phases act diagonally, whereas scaling changes the integer label m to nm .

Define the Hamiltonian by

$$H\varepsilon_m = (\log m)\varepsilon_m. \quad (1.6)$$

The operator H is self-adjoint on the domain

$$\mathcal{D}(H) = \left\{ \xi = \sum_{m \geq 1} \xi_m \varepsilon_m \in \ell^2(\mathbb{N}^*) : \sum_{m \geq 1} (\log m)^2 |\xi_m|^2 < \infty \right\}.$$

For each basis vector,

$$\begin{aligned} e^{itH}\pi_u(\mu_n)e^{-itH}\varepsilon_m &= e^{it(\log(nm) - \log m)}\varepsilon_{nm} \\ &= n^{it}\pi_u(\mu_n)\varepsilon_m. \end{aligned}$$

The phase generators are diagonal in the same basis as H and are therefore fixed by this evolution. Thus H implements the time evolution defined above.

The logarithm makes integer multiplication correspond to addition of energies:

$$\log(mn) = \log m + \log n.$$

If a real-valued function $f : \mathbb{N}^* \rightarrow \mathbb{R}$ is further required to satisfy

$$f(mn) = f(m) + f(n), \quad f(n) \geq 0, \quad f(n+1) - f(n) \longrightarrow 0, \quad (1.7)$$

then Erdős's theorem on additive functions gives $f(n) = c \log n$, with $c \geq 0$ [3]. When f is not identically zero, $c > 0$, and this positive scale can be absorbed into

the unit of time. The logarithmic spectrum is therefore fixed up to an overall scale by these additivity, nonnegativity, and regularity conditions.

The logarithmic spectrum also determines the equilibrium statistics of the system. Let $\beta > 0$ denote the inverse temperature. For $\beta > 1$, the partition function is

$$Z(\beta) = \text{Tr}(e^{-\beta H}) = \sum_{m \geq 1} m^{-\beta} = \zeta(\beta) = \prod_q (1 - q^{-\beta})^{-1}, \quad (1.8)$$

where the product runs over all primes q . The Euler product follows from unique prime factorization: summing independently over the valuation at each prime produces one geometric series per prime.

The corresponding density operator and state on the BC algebra are

$$D_\beta = \frac{e^{-\beta H}}{\zeta(\beta)}, \quad \phi_{\beta,u}(A) = \text{Tr}(D_\beta \pi_u(A)), \quad A \in \mathcal{A}_{\text{BC}}. \quad (1.9)$$

All representations use the same density operator D_β ; the states differ through the representation π_u . The Bost–Connes theorem states that, for $\beta > 1$, all extremal KMS states are parametrized by $\widehat{\mathbb{Z}}^\times$; for $0 < \beta \leq 1$, the KMS state is unique [1]. In the latter range, the equilibrium state is characterized by the algebraic KMS condition, and the trace-normalized expression above no longer applies.

Unique prime factorization gives both the Euler product for the partition function and the local structure of the spectrum. Substituting $m = \prod_q q^{a_q(m)}$ into the logarithmic spectrum gives

$$\log m = \sum_q a_q(m) \log q.$$

Each positive integer corresponds to a finitely supported sequence of prime valuations, so the correspondence of basis vectors

$$\varepsilon_m \mapsto \bigotimes_q \varepsilon_{a_q(m)} \quad (1.10)$$

is bijective and preserves inner products. Hence

$$\ell^2(\mathbb{N}^*) \cong \bigotimes_q^{\varepsilon_0} \ell^2(\mathbb{Z}_{\geq 0}).$$

The right-hand side is the incomplete infinite tensor product with reference vector ε_0 at each prime. On the dense domain of finite linear combinations,

$$H = \sum_q (\log q) N_q, \quad N_q \varepsilon_m = a_q(m) \varepsilon_m. \quad (1.11)$$

For a fixed prime p , the corresponding valuation direction thus has an equally spaced spectrum with energy spacing $\log p$.

§1.2 Finite p -adic Reduction

The denominators of rational phases decompose prime by prime. Fix a prime p . Phases whose denominators are powers of p form the corresponding local component.

Write

$$\mathbb{Z}[1/q] = \left\{ \frac{a}{q^j} : a \in \mathbb{Z}, j \geq 0 \right\}.$$

The rational phase group has the direct-sum decomposition

$$\mathbb{Q}/\mathbb{Z} = \bigoplus_q \mathbb{Z}[1/q]/\mathbb{Z}, \quad (1.12)$$

where the sum runs over all primes q .

This decomposition follows from unique prime factorization of the phase denominator. Let $n = \prod_{i=1}^s q_i^{a_i} > 1$. The integers $n/q_i^{a_i}$ have greatest common divisor 1, so there are integers c_i such that

$$\sum_{i=1}^s c_i \frac{n}{q_i^{a_i}} = 1.$$

Thus any phase a/n can be written as

$$\frac{a}{n} = \sum_{i=1}^s \frac{ac_i}{q_i^{a_i}}.$$

Each summand on the right has a prime-power denominator involving only one prime. To verify uniqueness, suppose $\sum_i b_i/q_i^{a_i} = 0 \pmod{\mathbb{Z}}$. Multiplying by n and reducing modulo $q_i^{a_i}$ gives

$$b_i \frac{n}{q_i^{a_i}} = 0 \pmod{q_i^{a_i}}.$$

Since $n/q_i^{a_i}$ is invertible modulo this modulus, $b_i = 0 \pmod{q_i^{a_i}}$. Each local component is therefore unique. The zero phase has all local components equal to zero.

The decomposition of the phase group also determines its dual space. A character of the phase group is a group homomorphism into the unit circle. For the discrete group $\mathbb{Q}/\mathbb{Z}$, these characters form its Pontryagin dual [4]. The value of any character χ at $1/n$ is an n th root of unity, so it can be written as

$$\chi(1/n) = e^{2\pi i x_n/n}, \quad x_n \in \mathbb{Z}/n\mathbb{Z}.$$

If $m \mid n$, then $1/m = (n/m)(1/n)$ implies $x_n = x_m \pmod{m}$. Every character thus determines a compatible family of residue classes. Conversely, given such a compatible family, set

$$\chi(a/n) = e^{2\pi i a x_n/n}.$$

Passing to a common denominator and using compatibility verifies that this definition is independent of the fraction representing the phase and respects addition. The two constructions are inverse to one another, yielding

$$\widehat{\mathbb{Q}/\mathbb{Z}} \cong \widehat{\mathbb{Z}} = \varprojlim_n \mathbb{Z}/n\mathbb{Z}. \quad (1.13)$$

The dual topology is determined by evaluation on finite-order phases, so this correspondence is also an isomorphism of topological groups.

In the dual space, the prime decomposition becomes

$$\widehat{\mathbb{Z}} \cong \prod_q \mathbb{Z}_q, \quad \mathbb{Z}_q = \varprojlim_j \mathbb{Z}/q^j\mathbb{Z}. \quad (1.14)$$

Specifically, restriction of a compatible family to all powers of q gives a component in $\mathbb{Z}_q$. Conversely, given components at all primes, the Chinese remainder theorem recovers a unique residue class for each finite modulus. These residue classes are compatible and define an element of $\widehat{\mathbb{Z}}$. The correspondence preserves ring operations and is continuous in every finite coordinate, hence is an isomorphism of topological rings.

The phase algebra can therefore be written as

$$C^*(\mathbb{Q}/\mathbb{Z}) \cong C(\widehat{\mathbb{Z}}).$$

Here $C(X)$ denotes the algebra of continuous complex-valued functions on the compact space X . The prime components of the rational phases correspond to the rings of p -adic integers in the dual space.

Henceforth fix an odd prime p and an integer $K \geq 1$, with K the reduction depth. The phases with denominators dividing p^K form the finite subgroup

$$p^{-K}\mathbb{Z}/\mathbb{Z} = \left\{ \frac{a}{p^K} \pmod{\mathbb{Z}} : a = 0, 1, \dots, p^K - 1 \right\}.$$

This group is generated by $1/p^K$. A character is uniquely determined by its value on this generator, which can be any p^K th root of unity. Consequently,

$$\widehat{p^{-K}\mathbb{Z}/\mathbb{Z}} \cong \mathbb{Z}/p^K\mathbb{Z}, \quad \chi_x(a/p^K) = e^{2\pi i ax/p^K}, \quad x \in \mathbb{Z}/p^K\mathbb{Z}.$$

We take this full residue-class ring as the underlying set of the finite reduction and write

$$X_{p,K} = \mathbb{Z}/p^K\mathbb{Z}.$$

Each residue class corresponds to a distinct character, including the zero element and all nonunits. The corresponding finite phase algebra satisfies

$$C^*(p^{-K}\mathbb{Z}/\mathbb{Z}) \cong C(X_{p,K}), \tag{1.15}$$

where a phase generator corresponds to the function

$$e(a/p^K) \longleftrightarrow \left[x \mapsto e^{2\pi i ax/p^K} \right].$$

The natural reduction map $\pi_K : \mathbb{Z}_p \rightarrow X_{p,K}$ sends a p -adic integer to its residue class modulo p^K [5]. Integer representatives of the residue classes show that π_K is surjective. If $x \in \ker \pi_K$, then for every $r \geq 1$, the component of x modulo p^{K+r} can be divided by p^K . The resulting residue classes modulo p^r are compatible and determine $y \in \mathbb{Z}_p$ with $x = p^K y$. Conversely, every element of $p^K \mathbb{Z}_p$ maps to the zero class. Thus

$$\ker \pi_K = p^K \mathbb{Z}_p, \quad \mathbb{Z}_p/p^K \mathbb{Z}_p \cong X_{p,K}. \tag{1.16}$$

Pullback along π_K gives $C(X_{p,K}) \hookrightarrow C(\mathbb{Z}_p)$. Its image consists of functions constant on each residue class modulo p^K . Selecting phases of finite depth is therefore equivalent to retaining this finite function subalgebra of the local phase algebra.

The transfer action of scaling also involves different reduction depths. The BC transfer relation gives

$$\mu_p e(a/p^K) \mu_p^* = \frac{1}{p} \sum_{\ell=0}^{p-1} e\left(\frac{a + \ell p^K}{p^{K+1}}\right).$$

The right-hand side generally contains phases whose denominators lie one level deeper. Thus a fixed depth determines a finite phase subalgebra, whereas the full transfer action requires relations between adjacent depths.

Reduction at depth K identifies elements differing by $p^K\mathbb{Z}_p$ and gives the full residue-class ring with p^K elements. The zero element, nonunits, and units are all retained, and the phase action on this reduction is determined by rational phases with denominators dividing p^K .

§1.3 Definition of the State Space

The underlying set obtained by finite reduction is $X_{p,K} = \mathbb{Z}/p^K\mathbb{Z}$. On this set, each residue class carries a complex amplitude. The relative amplitudes and phases between points together determine a pattern. Patterns form the complex function space on this set, with linear superposition defined through their complex amplitudes.

Write $d = p^K$. Equip the additive group of $X_{p,K}$ with normalized counting Haar measure, assigning measure $1/d$ to each point. Define the **state space** by

$$\boxed{\mathcal{H} = \mathcal{H}_{p,K} = L^2(X_{p,K}, \mathbb{C}), \quad \dim_{\mathbb{C}} \mathcal{H} = d.} \quad (1.17)$$

For $f, g \in \mathcal{H}$, the inner product and norm are

$$\langle f, g \rangle = \frac{1}{d} \sum_{x \in X_{p,K}} \overline{f(x)} g(x), \quad \|f\|^2 = \langle f, f \rangle. \quad (1.18)$$

This finite-dimensional inner-product space is complete and is therefore a complex Hilbert space. A vector of norm 1 is called normalized, and two vectors with zero inner product are called orthogonal.

For each $x \in X_{p,K}$, define the normalized point vector

$$|x\rangle = \sqrt{d} \mathbf{1}_{\{x\}}, \quad \langle x|y\rangle = \delta_{xy}, \quad \sum_{x \in X_{p,K}} |x\rangle\langle x| = I. \quad (1.19)$$

These vectors form the point basis. The point-basis expansion of any pattern f is

$$f = \sum_{x \in X_{p,K}} \frac{f(x)}{\sqrt{d}} |x\rangle, \quad \langle x|f\rangle = \frac{f(x)}{\sqrt{d}}. \quad (1.20)$$

The function value $f(x)$ thus differs from the coefficient in the normalized point basis by a factor of $\sqrt{d}$.

A pure state is represented by the one-dimensional complex subspace determined by a nonzero vector. Nonzero scalar multiples represent the same pure state, whose rank-one orthogonal projection is

$$P_f = \frac{|f\rangle\langle f|}{\|f\|^2}, \quad f \neq 0.$$

After choosing normalized representatives, different representatives of the same pure state differ only by an overall phase. A general state is represented by a density operator, satisfying $\rho \geq 0$ and $\text{Tr } \rho = 1$. For a normalized vector v , write $P_v = |v\rangle\langle v|$.

In particular, the point projection is $P_x = |x\rangle\langle x|$. The operator norm is denoted by $\|A\|$, and the trace norm by $\|A\|_1 = \text{Tr} \sqrt{A^\dagger A}$.

The finite phase algebra acts on the state space by multiplication. For $a \in C(X_{p,K})$, define $(M_a f)(x) = a(x)f(x)$. In particular, the functions corresponding to finite phases are

$$\chi_\ell^{(\text{add})}(x) = e^{2\pi i \ell x/d}, \quad \ell = 0, 1, \dots, d-1. \quad (1.21)$$

As multiplication functions they define phase actions; as vectors in the state space they form the additive Fourier basis.

The full residue-class ring also has a stratification determined by the p -adic valuation. For $x \in X_{p,K}$, define the truncated valuation

$$j_K(x) = \max\{j \in \{0, \dots, K\} : x \in p^j X_{p,K}\}.$$

Nonzero classes have valuations $0, \dots, K-1$, while the zero class satisfies $j_K(0) = K$. All subsequent uses of $v_p(x)$ for nonzero residue classes refer to this truncated valuation.

Define the j th **valuation shell** by

$$S_j = \{x \in X_{p,K} : j_K(x) = j\}, \quad 0 \leq j \leq K.$$

The shells are pairwise disjoint and cover the full residue-class ring, with terminal shell $S_K = \{0\}$ and

$$|S_j| = \begin{cases} (p-1)p^{K-j-1}, & 0 \leq j < K, \\ 1, & j = K. \end{cases} \quad (1.22)$$

The subspace associated with the j th shell consists of functions supported on S_j , with orthogonal projection

$$\Pi_j = \sum_{x \in S_j} |x\rangle\langle x|, \quad \sum_{j=0}^K \Pi_j = I.$$

Superposing the points in each shell with equal amplitudes and equal phases defines the normalized **shell-average state** and the corresponding Haar shell mass:

$$e_j = \frac{1}{\sqrt{|S_j|}} \sum_{x \in S_j} |x\rangle, \quad h_j = \frac{|S_j|}{d} = \begin{cases} (1-p^{-1})p^{-j}, & 0 \leq j < K, \\ p^{-K}, & j = K. \end{cases} \quad (1.23)$$

Write $z = e_K = |0\rangle$ for the zero-element state. The normalized constant function on the full ring is denoted by $b = \mathbf{1}$, with shell-average expansion

$$b = \sum_{j=0}^K \sqrt{h_j} e_j, \quad \|b\| = 1. \quad (1.24)$$

A single valuation shell also contains distinct direction components. Let $G_{p,r} = (\mathbb{Z}/p^r\mathbb{Z})^\times$ be the unit group modulo p^r . For $x \in S_j$ with $j < K$, the expression $x = p^j u$ uniquely determines $u \in G_{p,K-j}$, denoted by x/p^j . Changing the integer representative of x changes u only by an integer multiple of p^{K-j} .

A **multiplicative character** of the unit group is a homomorphism into the unit circle, satisfying

$$\chi(uv) = \chi(u)\chi(v), \quad |\chi(u)| = 1.$$

For each character of $G_{p,K-j}$, define the corresponding shell direction by

$$f_{j,\chi}(x) = \begin{cases} h_j^{-1/2} \chi(x/p^j), & x \in S_j, \\ 0, & x \notin S_j. \end{cases} \quad (1.25)$$

The trivial character $\chi = 1$ gives the shell-average state e_j ; nontrivial characters give distinct phase distributions within the shell.

The shell-average states span the **shell-average subspace**

$$\mathcal{C} = \text{span}\{e_0, \dots, e_K\}, \quad \Pi = P_{\mathcal{C}} = \sum_{j=0}^K |e_j\rangle\langle e_j|. \quad (1.26)$$

All nontrivial shell directions span the **direction space**

$$\mathcal{M} = \text{span}\{f_{j,\chi} : 0 \leq j < K, \chi \neq 1\}.$$

Write $N_{\text{dir}} = \dim \mathcal{M}$ for the total number of directions. These two types of components give the orthogonal decomposition of the full state space

$$\boxed{\mathcal{H} = \mathcal{C} \oplus \mathcal{M}}, \quad P = P_{\mathcal{M}} = I - \Pi. \quad (1.27)$$

Here Π_j projects onto all components of a single shell, Π onto the average components of all shells, and P onto all nontrivial directions.

Any state vector can be written as

$$\psi = \sum_{j=0}^K a_j e_j + \sum_{j=0}^{K-1} \sum_{\substack{\chi \in \widehat{G_{p,K-j}} \\ \chi \neq 1}} c_{j,\chi} f_{j,\chi},$$

where

$$a_j = \langle e_j, \psi \rangle, \quad c_{j,\chi} = \langle f_{j,\chi}, \psi \rangle.$$

A pattern therefore consists of the average components of the shells and their nontrivial direction components, represented by $\Pi\psi$ and $P\psi$, respectively. A general density operator may contain coherence between these subspaces, so $\Pi\rho P$ need not vanish.

To label the same character consistently across different shells, we define its conductor. For a nontrivial character modulo p^r , the smallest p^c , with $1 \leq c \leq r$, for which the character factors through the natural reduction $G_{p,r} \rightarrow G_{p,c}$ is called its **conductor** [6]. The natural reduction is surjective, so the induced character on $G_{p,c}$ is unique. Denote it by θ and call it a **primitive character**. Composing θ with the natural reduction from a deeper level gives its inflation to the corresponding unit group, again denoted by θ .

A direction of conductor p^c can be defined on the shells $j = 0, \dots, K - c$. Denote the corresponding subspace by

$$\mathcal{F}_{\theta} = \text{span}\{f_{j,\theta} : 0 \leq j \leq K - c\}, \quad \text{cond } \theta = p^c. \quad (1.28)$$

The direction space can thus also be indexed by primitive characters:

$$\mathcal{M} = \bigoplus_{c=1}^K \bigoplus_{\text{cond } \theta=p^c} \mathcal{F}_\theta.$$

Write the direction index as $\nu = (\theta, j_\nu)$, with the convention

$$|\nu\rangle = f_{j_\nu, \theta}, \quad P_\nu = |\nu\rangle\langle\nu|.$$

The finite shell energies use the local spectral spacing $\omega_p = \log p$ and are defined by the truncated valuation as

$$\boxed{H_K = \omega_p \sum_{j=0}^K j \Pi_j.} \tag{1.29}$$

The energy of the j th shell is $E_j = j\omega_p$. Its shell-average state and all its directions have the same energy:

$$H_K e_j = E_j e_j, \quad H_K f_{j, \chi} = E_j f_{j, \chi}.$$

The terminal zero class is assigned energy $K\omega_p$ according to the truncated valuation $j_K(0) = K$. This definition specifies both the shell energies and the terminal energy convention at finite depth.

§2 Structure of the State Space

A pattern in the state space is determined by its complex amplitudes at the points; valuation shells and their directions give its orthogonal decomposition. Arithmetic actions on the underlying ring relate these components and change their amplitudes and phases.

§2.1 Group Structure

Addition in the residue-class ring organizes all points into a group, while its multiplicatively invertible elements form the unit group. The unit group permutes the full ring by multiplication and thereby acts on patterns in the state space. The relation between the full group and the unit group determines how points within valuation shells are related and how direction components transform.

Call the additive group of $X_{p,K}$ the **full group**. Addition is taken modulo p^K , its identity is 0, and the additive inverse of each element x is $-x$. The residue class 1 generates the entire additive group, so

$$(X_{p,K}, +) = \langle 1 \rangle \cong \mathbb{Z}/p^K\mathbb{Z}.$$

The full group contains all p^K residue classes, including the zero element, nonunits, and units.

The invertible elements of the ring form the unit group

$$G_{p,K} = (\mathbb{Z}/p^K\mathbb{Z})^\times = \{u \in X_{p,K} : p \nmid u\}.$$

For each $u \in G_{p,K}$, the multiplication map $\alpha_u(x) = ux$ preserves addition:

$$\alpha_u(x + y) = \alpha_u(x) + \alpha_u(y).$$

Its inverse is $\alpha_{u^{-1}}$, so α_u is an additive automorphism of the full group. Conversely, any additive automorphism α of the full group is determined by $\alpha(1)$. Writing $\alpha(1) = u$ gives $\alpha(x) = xu$. This map is invertible if and only if u is a unit. Together with $\alpha_u \circ \alpha_v = \alpha_{uv}$, this yields

$$\boxed{\text{Aut}(X_{p,K}, +) \cong G_{p,K}.} \quad (2.1)$$

The unit group thus describes precisely all invertible transformations of the full group that preserve its additive structure.

The structure of the unit group itself is determined by reduction modulo p . The natural map

$$G_{p,K} \longrightarrow (\mathbb{Z}/p\mathbb{Z})^\times$$

is a surjective homomorphism, since each nonzero residue class modulo p has an integer representative not divisible by p . Its kernel consists of units reducing to 1 modulo p and is called the **group of principal units**, also called the group of 1-units. It is

$$(1 + p\mathbb{Z}_p)/(1 + p^K\mathbb{Z}_p).$$

Principal units have the form $1 + pa$, where a runs over residue classes modulo p^{K-1} , so the group of principal units has p^{K-1} elements.

The other factor of the unit group is determined by roots of unity. Set $F(t) = t^{p-1} - 1$. By Fermat's little theorem, each nonzero residue class modulo p is a root of F , and

$$F'(t) = (p-1)t^{p-2} \not\equiv 0 \pmod{p}.$$

If t is already a root modulo p^r , all its lifts to the next level have the form $t + ap^r$, where a runs modulo p . Expansion gives

$$F(t + ap^r) \equiv F(t) + ap^r F'(t) \pmod{p^{r+1}}.$$

Since $F'(t)$ is invertible modulo p , exactly one value of a makes the right-hand side vanish. Successive lifting therefore associates each nonzero residue class modulo p with a unique $(p-1)$ st root of unity modulo p^K . These roots form a multiplicative group of order $p-1$, denoted by μ_{p-1} .

For any $u \in G_{p,K}$, choose the unique $\omega \in \mu_{p-1}$ congruent to u modulo p . Then

$$u = \omega v, \quad v = \omega^{-1}u \equiv 1 \pmod{p}.$$

The only element of μ_{p-1} congruent to 1 modulo p is 1, so the root-of-unity group and the group of principal units intersect only at the identity. Elements of the two subgroups commute; hence the decomposition is unique and gives the standard decomposition of the unit group [7]:

$$\boxed{G_{p,K} \cong \mu_{p-1} \times (1 + p\mathbb{Z}_p)/(1 + p^K\mathbb{Z}_p).} \quad (2.2)$$

The group of principal units is generated by $1 + p$. Take an integer

$$a = 1 + p^r b, \quad r \geq 1, \quad p \nmid b.$$

Its binomial expansion is

$$a^p - 1 = p^{r+1}b + \sum_{\ell=2}^{p-1} \binom{p}{\ell} p^{r\ell} b^\ell + p^{rp} b^p.$$

The first term has p -adic valuation $r+1$. The intermediate terms have valuation at least $1+2r$, and the last term has valuation rp . Since p is odd, all these valuations are at least $r+2$. Thus $v_p(a^p - 1) = r+1$. Induction starting from $a = 1+p$ gives

$$v_p((1+p)^{p^n} - 1) = n+1, \quad n \geq 0.$$

For $K \geq 2$,

$$(1+p)^{p^{K-1}} \equiv 1 \pmod{p^K}, \quad (1+p)^{p^{K-2}} \not\equiv 1 \pmod{p^K}.$$

The order of $1+p$ therefore divides p^{K-1} but not p^{K-2} , so it is exactly p^{K-1} . This equals the number of principal units, proving that $1+p$ generates the entire group of principal units. For $K=1$, that group is trivial and $1+p \equiv 1 \pmod{p}$, so the conclusion still holds.

The orbits of the unit group acting on the full group are determined by the valuation shells. For any unit u , the truncated valuation satisfies $j_K(ux) = j_K(x)$, so multiplication by units preserves every valuation shell. Any two points in the same shell can also be related by multiplication by a unit. For $x, y \in S_j$ with $j < K$, write

$$x = p^j a, \quad y = p^j b, \quad a, b \in G_{p, K-j}.$$

Choose the unit ba^{-1} in $G_{p, K-j}$ and lift it to $u \in G_{p, K}$; then $ux = y$. The zero element is fixed by every unit action. Therefore

$$\boxed{G_{p, K} \cdot p^j = S_j \quad (0 \leq j < K), \quad G_{p, K} \cdot 0 = \{0\}.} \quad (2.3)$$

The valuation shells are precisely the orbits of the unit group on the full group.

These multiplicative permutations induce linear actions on the state space. For $u \in G_{p, K}$, define

$$(R_u f)(x) = f(u^{-1}x).$$

These operators satisfy $R_u R_v = R_{uv}$ and $R_1 = I$, and act on the point basis by

$$R_u |x\rangle = |ux\rangle.$$

Since $x \mapsto ux$ is a bijection of the full ring,

$$\langle R_u f, R_u g \rangle = \frac{1}{d} \sum_{x \in X_{p, K}} \overline{f(u^{-1}x)} g(u^{-1}x) = \langle f, g \rangle.$$

Moreover, $R_{u^{-1}} R_u = I$, so $R_u^\dagger = R_{u^{-1}}$. The unit group thus has a unitary representation on the state space.

The unit action preserves each valuation shell, giving

$$[R_u, \Pi_j] = 0, \quad [R_u, H_K] = 0.$$

Since R_u permutes all points in a shell, the shell-average states satisfy $R_u e_j = e_j$. For a shell direction $f_{j,\theta}$, interpret $\theta(u)$ as the character evaluated on the natural reduction of the unit u . Multiplicativity of the character gives, for $x \in S_j$,

$$(R_u f_{j,\theta})(x) = h_j^{-1/2} \theta(u^{-1} x / p^j) = \theta(u)^{-1} f_{j,\theta}(x).$$

Both sides vanish outside the shell, hence

$$\boxed{R_u e_j = e_j, \quad R_u f_{j,\theta} = \theta(u)^{-1} f_{j,\theta}.} \quad (2.4)$$

Multiplication by units therefore preserves shell energies and each character direction, assigning a character-determined phase to each direction.

§2.2 Projection Structure

Positions, valuation shells, and directions describe different components of a pattern. The point projection $P_x = |x\rangle\langle x|$ selects the component at one position, the shell projection Π_j selects all components in the j th shell, and the direction projection $P_\nu = |\nu\rangle\langle \nu|$ selects a particular character direction within that shell. The relations between these projections follow from orthogonality and completeness of the corresponding bases.

The definition $|x\rangle = \sqrt{d} \mathbf{1}_{\{x\}}$ of normalized point vectors immediately gives $\langle x|y\rangle = \delta_{xy}$. For any $f \in \mathcal{H}$, pointwise calculation gives

$$f = \sum_{x \in X_{p,K}} \frac{f(x)}{\sqrt{d}} |x\rangle.$$

Thus the point basis is complete and satisfies $\sum_x P_x = I$.

Position projections can also be represented by the finite phase algebra. The multiplication representation $(M_a f)(x) = a(x)f(x)$ preserves multiplication, the identity, and adjoints. If $M_a = 0$, applying it to each point vector gives $a(x) = 0$, so the representation is faithful. The additive characters satisfy

$$\langle \chi_\ell^{(\text{add})}, \chi_m^{(\text{add})} \rangle = \frac{1}{d} \sum_{x=0}^{d-1} e^{2\pi i(m-\ell)x/d} = \delta_{\ell m}.$$

For $\ell = m$, each summand is one; for $\ell \neq m$, the finite geometric sum vanishes. There are d such orthonormal vectors, so they form the additive Fourier basis. Write the phase multiplier as

$$(M_\ell f)(x) = e^{2\pi i \ell x/d} f(x).$$

The finite Fourier identity [8] gives

$$\boxed{P_x = \frac{1}{d} \sum_{\ell=0}^{d-1} e^{-2\pi i \ell x/d} M_\ell.} \quad (2.5)$$

The finite phase algebra can therefore distinguish every point of the full ring.

For $j < K$, a valuation shell is the difference of two adjacent ideals:

$$S_j = p^j X_{p,K} \setminus p^{j+1} X_{p,K}.$$

Hence $|S_j| = p^{K-j} - p^{K-j-1} = (p-1)p^{K-j-1}$. The terminal shell $S_K = \{0\}$ contains one element. The shells have disjoint supports, so the normalization of their average states gives $\langle e_j, e_l \rangle = \delta_{jl}$. Since the shells cover the ring, $\sum_{j=0}^K h_j = 1$. Pointwise comparison of the constant function with the shell-average states gives

$$b = \sum_{j=0}^K \sqrt{h_j} e_j, \quad \|b\| = 1.$$

Directions within a shell are determined by unit-group characters. For two characters χ, χ' of $G_{p,K-j}$,

$$\langle f_{j,\chi}, f_{j,\chi'} \rangle = \frac{1}{|G_{p,K-j}|} \sum_{u \in G_{p,K-j}} \overline{\chi(u)} \chi'(u).$$

When $\chi = \chi'$, the inner product is one. Otherwise, $\eta = \overline{\chi} \chi'$ is a nontrivial character. Choose a unit u_0 with $\eta(u_0) \neq 1$. The permutation $u \mapsto u_0 u$ in the sum gives

$$\sum_u \eta(u) = \eta(u_0) \sum_u \eta(u).$$

The sum therefore vanishes. Directions on different shells are orthogonal because their supports are disjoint.

A finite abelian group decomposes into a direct product of finite cyclic groups. A cyclic group of order n has n characters, and a character of a direct product is uniquely specified by those of its factors. Thus $G_{p,K-j}$ has exactly $|G_{p,K-j}| = |S_j|$ characters. The orthogonal vectors above therefore form a complete basis of the j th shell. Including e_K from the terminal shell gives a complete orthonormal basis of the entire state space.

The total number of directions can also be counted by conductor. Natural reduction $G_{p,r} \rightarrow G_{p,c}$ for $r \geq c \geq 1$ is surjective, so inflation of characters is injective. For $c \geq 2$, the characters that descend to a shallower modulus are precisely the inflations of characters modulo p^{c-1} . Let a_c be the number of nontrivial primitive characters of conductor exactly p^c . Then

$$a_1 = \varphi(p) - 1 = p - 2, \quad a_c = \varphi(p^c) - \varphi(p^{c-1}) = p^{c-2}(p-1)^2 \quad (c \geq 2).$$

Here $\varphi(p^r) = |G_{p,r}|$. Each character of conductor p^c gives one direction on each of $K - c + 1$ shells. Therefore

$$\begin{aligned} N_{\text{dir}} &= \sum_{c=1}^K (K - c + 1) a_c = \sum_{r=1}^K \sum_{c=1}^r a_c \\ &= \sum_{r=1}^K [\varphi(p^r) - 1] = p^K - K - 1. \end{aligned} \tag{2.6}$$

The shell-average subspace has dimension $K + 1$, and the dimensions of the two types of components sum to d , verifying $\mathcal{H} = \mathcal{C} \oplus \mathcal{M}$.

The full character basis of a shell contains its average state and all nontrivial directions, so

$$\Pi_j = \sum_{x \in S_j} P_x = |e_j\rangle \langle e_j| + \sum_{\nu: j_\nu=j} P_\nu. \tag{2.7}$$

For $j = K$, the direction sum is empty and $e_K = z$. Disjoint shell supports and orthogonality of the direction basis further give

$$\Pi_j \Pi_l = \delta_{jl} \Pi_j, \quad \Pi_j P_\nu = P_\nu \Pi_j = \delta_{j,j_\nu} P_\nu. \quad (2.8)$$

Thus shell and direction projections are compatible. The projections onto all shell-average states and onto all nontrivial directions are, respectively,

$$\Pi = \sum_{j=0}^K |e_j\rangle\langle e_j|, \quad P = \sum_{\nu} P_\nu,$$

and satisfy $\Pi P = 0$ and $\Pi + P = I$.

This decomposition also follows from group averaging in finite-group representation theory [9]. Each e_j is fixed by every R_u . For a nontrivial direction, the character remains nontrivial after inflation to $G_{p,K}$, so its average over the unit group vanishes. Hence

$$\Pi = \frac{1}{|G_{p,K}|} \sum_{u \in G_{p,K}} R_u, \quad P = I - \Pi. \quad (2.9)$$

The shell-average subspace consists of unit-group invariant vectors, and the direction space is its orthogonal complement.

The overlap between position and direction vectors is determined by their normalization factors. For $x \in S_{j_\nu}$,

$$\langle x|\nu\rangle = |S_{j_\nu}|^{-1/2} \theta_\nu(x/p^{j_\nu}), \quad P_x P_\nu = \langle x|\nu\rangle |x\rangle\langle \nu|. \quad (2.10)$$

The overlap vanishes outside the shell, whereas within the shell,

$$0 < |\langle x|\nu\rangle| = \frac{1}{\sqrt{|S_{j_\nu}|}} < 1,$$

since p is odd and each nonzero shell contains at least two points. A position pure state and a direction pure state in the same shell are therefore neither identical nor orthogonal; their rank-one projections do not commute.

By the orbit structure in Section 2.1, for any two points x, y in the same shell, a unit u can be chosen with $uy = x$. Thus

$$P_x R_u P_y = |x\rangle\langle y|. \quad (2.11)$$

These operators give all matrix units within each shell. Conversely, both phase multipliers and unit actions preserve valuation shells. The algebra they generate is therefore precisely the direct sum of the matrix algebras of the shells.

The complete basis gives an operator expansion on the full state space.

Orthogonal decomposition

Denote the complete orthonormal basis by $\{q_\ell\}_{\ell=1}^d = \{e_j\}_{j=0}^K \cup \{|\nu\rangle\}$. The shell-average components and all direction components together resolve the identity. Any operator A is determined by all its matrix elements in this same basis:

$$\begin{aligned} I &= \sum_{\ell=1}^d |q_\ell\rangle\langle q_\ell| = \Pi + P, \\ A &= \sum_{\ell,m=1}^d \langle q_\ell, A q_m \rangle |q_\ell\rangle\langle q_m|. \end{aligned} \quad (2.12)$$

The first identity follows from completeness of the orthonormal basis; the second follows by inserting the first on both sides of A in $A = IAI$. Accordingly,

$$\psi = \sum_{\ell} \langle q_{\ell}, \psi \rangle q_{\ell}, \quad \|\psi\|^2 = \sum_{\ell} |\langle q_{\ell}, \psi \rangle|^2.$$

This decomposition retains the shell-average components, all direction components, and the matrix elements between them, and can therefore represent both shell-preserving and cross-shell actions.

§2.3 Symmetry Structure

Multiplication by units preserves valuation shells and acts by a phase on each character direction. The decomposition of the unit group also defines inversion in each of its two factors. Negation on the full ring and complex conjugation of amplitudes act on positions and amplitudes, respectively. To relate these transformations, we examine their linearity, the structures they preserve, and their action on directions within a shell.

For any nonzero point $x \in S_j$, Eq. (2.2) gives the unique decomposition on that shell

$$x = p^j \omega v, \quad \omega \in \mu_{p-1}, \quad v \in (1 + p\mathbb{Z}_p)/(1 + p^{K-j}\mathbb{Z}_p).$$

Both unit factors are understood modulo p^{K-j} . On the normalized point basis, define the full-ring negation involution ι , inversion of the root-of-unity factor, and inversion of the principal-unit factor, respectively:

$$\begin{aligned} \iota |p^j \omega v\rangle &= |-p^j \omega v\rangle, \\ J_T |p^j \omega v\rangle &= |p^j \omega^{-1} v\rangle, \\ J_P |p^j \omega v\rangle &= |p^j \omega v^{-1}\rangle. \end{aligned} \tag{2.13}$$

All three fix the zero-element state $z = |0\rangle$ and extend linearly from the point basis to the entire state space. The two inversions are performed separately in the corresponding unit factors.

These position maps preserve valuation shells, and applying any of them twice restores the original position. Thus $A^2 = I$ for $A \in \{\iota, J_T, J_P\}$. Each A permutes the orthonormal point basis, so

$$A^\dagger A = I, \quad A^\dagger = A^{-1} = A.$$

All three are therefore self-adjoint unitaries. Since they preserve each shell,

$$[A, \Pi_j] = 0, \quad [A, H_K] = 0.$$

These actions preserve the norm and shell energy, mapping a pattern within a valuation shell to another pattern in that shell.

The two factor inversions are independent, hence $J_T J_P = J_P J_T$. Their product inverts the entire unit part:

$$J_T J_P |p^j \omega v\rangle = |p^j (\omega v)^{-1}\rangle.$$

Full-ring negation is multiplication by the unit -1 , namely $\iota = R_{-1}$. Since p is odd, -1 belongs to the root-of-unity factor; negation changes ω to $-\omega$ and leaves the

principal-unit factor unchanged. Its definition is distinct from inversion of a unit factor.

The action of these position transformations on directions follows from factorization of the character. Write

$$\theta(\omega v) = \theta_T(\omega)\theta_P(v),$$

where $\theta_T(\omega) = \theta(\omega)$ and $\theta_P(v) = \theta(v)$ are the restrictions of θ to the two unit factors, respectively. For a unit-circle-valued character,

$$\theta_T(\omega^{-1}) = \overline{\theta_T(\omega)}, \quad \theta_P(v^{-1}) = \overline{\theta_P(v)}.$$

Each position map is its own inverse, so its action on functions is pullback along the corresponding map. For $x = p^j \omega v \in S_j$, for example,

$$\begin{aligned} (J_T f_{j,\theta})(x) &= f_{j,\theta}(p^j \omega^{-1} v) \\ &= h_j^{-1/2} \overline{\theta_T(\omega)} \theta_P(v). \end{aligned}$$

This gives

$$\begin{aligned} J_T f_{j,\theta_T \theta_P} &= f_{j,\overline{\theta_T} \theta_P}, \\ J_P f_{j,\theta_T \theta_P} &= f_{j,\theta_T \overline{\theta_P}}. \end{aligned}$$

The two inversions conjugate the character of the corresponding factor, leaving the other factor unchanged. Their product satisfies

$$J_T J_P f_{j,\theta} = f_{j,\bar{\theta}}.$$

Negation, in contrast, acts through the unit representation: $\iota f_{j,\theta} = \theta(-1)^{-1} f_{j,\theta}$. Since $(-1)^2 = 1$, we have $\theta(-1)^2 = 1$, so $\theta(-1) \in \{1, -1\}$. The cases $\theta(-1) = 1$ and $\theta(-1) = -1$ correspond to even and odd characters, respectively. Negation preserves each character direction, with its parity determined by the corresponding eigenvalue.

Complex conjugation of point amplitudes is defined by

$$(\mathfrak{c}f)(x) = \overline{f(x)}.$$

It preserves position support, but for complex coefficients satisfies

$$\mathfrak{c}(af + bg) = \bar{a} \mathfrak{c}f + \bar{b} \mathfrak{c}g,$$

and is therefore antilinear. By the definition of the inner product,

$$\langle \mathfrak{c}f, \mathfrak{c}g \rangle = \overline{\langle f, g \rangle}, \quad \mathfrak{c}^2 = I.$$

Thus $\mathfrak{c}$ is an antiunitary involution and preserves the norm. The shell projections and shell Hamiltonian have real matrix elements in the point basis, so

$$\mathfrak{c} \Pi_j \mathfrak{c} = \Pi_j, \quad \mathfrak{c} H_K \mathfrak{c} = H_K.$$

Complex conjugation of amplitudes also preserves each valuation shell and its energy. Conjugating a character direction directly and combining this with the negation action gives

$$\boxed{\iota f_{j,\theta} = \theta(-1) f_{j,\theta}, \quad \mathfrak{c} f_{j,\theta} = f_{j,\bar{\theta}}.} \tag{2.14}$$

Although $J_T J_P$ and $\mathbf{c}$ have the same images on the chosen character basis vectors, their actions on general superpositions differ. For a direction component $\psi = \sum_{\nu} c_{\nu} |\nu\rangle$, write $\bar{\nu} = (\overline{\theta_{\nu}}, j_{\nu})$. Then

$$J_T J_P \psi = \sum_{\nu} c_{\nu} |\bar{\nu}\rangle, \quad \mathbf{c} \psi = \sum_{\nu} \overline{c_{\nu}} |\bar{\nu}\rangle.$$

The former preserves the expansion coefficients, whereas the latter also conjugates them; they are therefore different operators.

Table 1 summarizes these actions.

Table 1: Symmetry actions on the state space. Each character factor is understood on the corresponding unit group.

Action	Linearity	Action on a character direction
ι	Linear	Preserves the direction; multiplies by $\theta(-1)$
J_T	Linear	Conjugates the root-of-unity character factor
J_P	Linear	Conjugates the principal-unit character factor
$\mathbf{c}$	Antilinear	Conjugates the full character and the coefficients

Each shell-average state is fixed by these actions:

$$\iota e_j = J_T e_j = J_P e_j = \mathbf{c} e_j = e_j.$$

The first three equalities follow from permutations within the shell, and the last from the real-valuedness of e_j .

When $K - j = 1$, the principal-unit group of that shell is trivial, and J_P reduces to the identity on the shell. In particular, for $K = 1$, $J_P = I$, while all other relations remain valid.

§2.4 Scaling Structure

Unit actions preserve valuation shells, whereas multiplication by p connects different shells. Scaling in the finite reduction is given by the point map

$$x \mapsto px \pmod{p^K}$$

whose image is $pX_{p,K}$. If $px = py \pmod{p^K}$, then

$$x - y = cp^{K-1} \pmod{p^K}, \quad c = 0, \dots, p-1.$$

Each point in the image therefore has exactly p preimages.

Choose the integer representative $0 \leq x < p^K$ of each residue class and write it uniquely as

$$x = a + cp^{K-1}, \quad 0 \leq a < p^{K-1}, \quad 0 \leq c < p.$$

The scaled point is $px \pmod{p^K} = pa$. Here a determines the output position, while the top digit c is discarded by the point map.

To retain the discarded digit, introduce an auxiliary space $\mathcal{E}_{\text{dig}} = \mathbb{C}^p$ with orthonormal basis $\{|c\rangle\}_{c=0}^{p-1}$, and define

$$\boxed{\mathcal{V}_D |x\rangle = |pa\rangle \otimes |c\rangle.} \tag{2.15}$$

The output position pa uniquely determines a , and the pair (a, c) uniquely determines the input x . Thus distinct point basis vectors have orthogonal unit-norm images, giving $\mathcal{V}_D^\dagger \mathcal{V}_D = I$. This map embeds the original state space isometrically into the joint position–digit space.

Fix an auxiliary input state $|0\rangle_D$. The corresponding input and output families

$$\{|x\rangle \otimes |0\rangle_D\}_{x \in X_{p,K}}, \quad \{\mathcal{V}_D|x\rangle\}_{x \in X_{p,K}}$$

are orthonormal families of the same size in the same joint space. Completing each family to an orthonormal basis constructs a unitary operator U_D such that

$$U_D(|\psi\rangle \otimes |0\rangle_D) = \mathcal{V}_D|\psi\rangle.$$

Finite scaling can therefore be realized, with the digit retained, as a joint unitary action on a fixed auxiliary preparation.

For each digit c , write $c(x) = \lfloor x/p^{K-1} \rfloor$ and define the conditional operator

$$J_c = (I \otimes \langle c|) \mathcal{V}_D = \sum_{a=0}^{p^{K-1}-1} |pa\rangle \langle a + cp^{K-1}|.$$

Distinct inputs with the same digit have distinct output positions, so

$$J_c^\dagger J_c = \sum_{a=0}^{p^{K-1}-1} |a + cp^{K-1}\rangle \langle a + cp^{K-1}|.$$

The input subspaces for different digits are orthogonal and together cover the entire state space. Hence

$$\sum_{c=0}^{p-1} J_c^\dagger J_c = I.$$

If the digit is not read, taking the partial trace over the auxiliary space gives the reduced system action

$$\boxed{\mathcal{D}_K(\rho) = \text{Tr}_{\mathcal{E}_{\text{dig}}}(\mathcal{V}_D \rho \mathcal{V}_D^\dagger) = \sum_{c=0}^{p-1} J_c \rho J_c^\dagger.} \quad (2.16)$$

Each term preserves positivity, and the completeness relation gives $\text{Tr } \mathcal{D}_K(\rho) = \text{Tr } \rho$. For any reference system $\mathcal{R}$ that does not participate in the interaction,

$$(\mathcal{D}_K \otimes \text{id}_{\mathcal{R}})(\rho_{\mathcal{H}\mathcal{R}}) = \sum_c (J_c \otimes I_{\mathcal{R}}) \rho_{\mathcal{H}\mathcal{R}} (J_c^\dagger \otimes I_{\mathcal{R}}),$$

the result is again a positive operator. Thus $\mathcal{D}_K$ is a completely positive, trace-preserving channel, and its system input may be correlated with the reference system. The digit auxiliary used to realize this channel is independently prepared in the fixed state specified above.

The same point map also defines pullback of amplitude functions:

$$(Df)(x) = f(px).$$

In the normalized point basis,

$$D|y\rangle = \sum_{x:px=y} |x\rangle, \quad D^\dagger|x\rangle = |px \bmod p^K\rangle.$$

The point map sends the j th shell to the $(j+1)$ st shell for $j < K$; function pullback sends a pattern supported on the j th shell to its preimage. For $1 \leq j < K$, this preimage is the $(j-1)$ st shell; the zeroth shell has no preimage, and the terminal shell has preimage $S_{K-1} \cup S_K$. Forward motion of points and pullback of amplitude support therefore have different directions.

By the definitions of the digit-conditioned operators,

$$D^\dagger = \sum_{c=0}^{p-1} J_c. \quad (2.17)$$

Hence

$$D^\dagger \rho D = \sum_{c,c'} J_c \rho J_{c'}^\dagger.$$

This expression retains cross terms between different digit components, whereas the reduced channel $\mathcal{D}_K$ retains only terms with $c = c'$ because the digit basis is orthogonal.

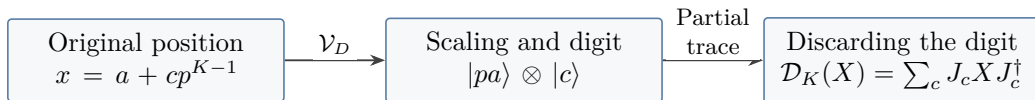
Their normalization properties also differ. Direct calculation gives

$$DD^\dagger|x\rangle = \sum_{r=0}^{p-1} |x + rp^{K-1} \bmod p^K\rangle.$$

Thus $DD^\dagger \neq I$, and

$$\text{Tr}(D^\dagger \rho D) = \text{Tr}(\rho DD^\dagger)$$

which generally differs from $\text{Tr} \rho$. Accordingly, $D^\dagger$ describes coherent combination of amplitudes, whereas $\mathcal{D}_K$ describes the normalized state change after the digit label is discarded (Fig. 1).



Amplitude pullback of the same point map: $Df(x) = f(px)$, with $D^\dagger = \sum_c J_c$.

Figure 1: Finite scaling, amplitude pullback, and digit retention. The pair (pa, c) determines the input position; tracing out the digit and coherently summing amplitudes give different system actions.

§2.5 Spectral Structure

The change of a pattern under a linear action is determined by the matrix elements of that action in a complete basis. The spectral structure further specifies which components share a response and how distinct components are connected by the action.

A linear action on the state space is specified by a finite matrix. Write $A_{xy} = \langle x|A|y\rangle$. For $\psi = \sum_y c_y|y\rangle$, applying the Cauchy–Schwarz inequality row by row gives

$$\|A\psi\|^2 \leq \left(\sum_{x,y} |A_{xy}|^2 \right) \|\psi\|^2. \quad (2.18)$$

Thus every linear operator is bounded. If $A = A^\dagger$, a root of its characteristic polynomial gives a normalized eigenvector u ; its eigenvalue $\lambda = \langle u, Au\rangle$ is real. For $v \perp u$, we have $\langle u, Av\rangle = \langle Au, v\rangle = 0$, so the orthogonal complement of u is invariant under A . Induction on this orthogonal complement gives a complete orthonormal eigenbasis. Let Q_λ be the orthogonal projection onto the eigenspace with eigenvalue λ . Then

$$A = \sum_{\lambda \in \text{spec}(A)} \lambda Q_\lambda, \quad Q_\lambda Q_{\lambda'} = \delta_{\lambda\lambda'} Q_\lambda, \quad \sum_\lambda Q_\lambda = I. \quad (2.19)$$

There are at most d eigenvalues, all real; the spectrum is therefore finite and discrete. The evolution generated by the self-adjoint operator A is $e^{-itA} = \sum_\lambda e^{-it\lambda} Q_\lambda$; Orthogonality of the spectral projections gives $(e^{-itA})^\dagger e^{-itA} = I$, so this evolution preserves inner products and normalization.

The scaling pullback $Df(x) = f(px)$ and multiplication by units together determine pullback under integer multiplication. For a positive integer m , write $T_m f(x) = f(mx)$. Each point has at most d preimages under multiplication, so $\|T_m\| \leq \sqrt{d}$. Writing s for the complex spectral parameter, for $\Re s > 1$ we have

$$\sum_{m \geq 1} \|m^{-s} T_m\| \leq \sqrt{d} \zeta(\Re s) < \infty.$$

The corresponding operator series therefore converges absolutely in norm. Extend a primitive character θ periodically to the integers with period equal to its conductor, and set it to zero on multiples of p . Define

$$L(s, \theta) = \sum_{\substack{u \geq 1 \\ p \nmid u}} \frac{\theta(u)}{u^s}.$$

The full transfer operator and resonance ratio are

$$\mathcal{B}(s)f(x) = \frac{1}{\zeta(s)} \sum_{m \geq 1} m^{-s} f(mx), \quad R_\theta(s) = \frac{L(s, \theta)}{\zeta(s)}. \quad (2.20)$$

The matrix elements of the full transfer operator can be calculated term by term in the common orthogonal decomposition of Section 2.2. Write $m = p^a u$ with $p \nmid u$. Then

$$T_{p^a u} = D^a R_{u^{-1}} = R_{u^{-1}} D^a.$$

Here $R_{u^{-1}} f(x) = f(ux)$, so the response of a direction component to the unit part is $\theta(u)$.

Suppose θ has conductor p^c and $0 \leq j \leq K - c$. For $a \leq j$, the condition $p^a u x \in S_j$ is equivalent to $x \in S_{j-a}$, and $h_j = p^{-a} h_{j-a}$, so the ratio of normalization factors in Eq. (1.25) is $p^{a/2}$. For $a > j$, the product cannot belong to S_j . Thus

$$f_{j,\theta}(p^a u x) = \begin{cases} p^{a/2} \theta(u) f_{j-a,\theta}(x), & a \leq j, \\ 0, & a > j. \end{cases} \quad (2.21)$$

On $\mathcal{F}_\theta$, set $N_\theta f_{j,\theta} = f_{j-1,\theta}$ ($j \geq 1$) and $N_\theta f_{0,\theta} = 0$. Then $N_\theta^{K-c+1} = 0$. Summing over the unit part gives

$$\mathcal{B}(s)|_{\mathcal{F}_\theta} = R_\theta(s) \sum_{a=0}^{K-c} p^{a(1/2-s)} N_\theta^a. \quad (2.22)$$

Before and after inflation of a given character, its Dirichlet coefficients vanish on multiples of p and agree on all other integers, so no new Euler factor appears. Equation (2.22) is a finite-matrix expression and extends coefficientwise to the regular parameters used below.

The shell-average block is retained in the same transfer operator. Using $e_j = h_j^{-1/2} \mathbf{1}_{S_j}$ and the preimages under scaling gives, term by term,

$$De_0 = 0, \quad De_j = \sqrt{p} e_{j-1} \quad (1 \leq j < K), \quad De_K = \sqrt{p-1} e_{K-1} + e_K. \quad (2.23)$$

Write $u_s = p^{-s}$, $\varrho_s = p^{1/2-s}$, and $\mathcal{R}_0 f = f(0)b$. Equation (2.23) shows that $\mathcal{C}$ is invariant under D ; write $D_{\mathcal{C}} = D|_{\mathcal{C}}$. Shell-average functions are invariant under multiplication by units, while $\zeta(s)^{-1} \sum_{p \nmid u} u^{-s} = 1 - p^{-s}$. Thus, for $\Re s > 1$,

$$\mathcal{B}(s)|_{\mathcal{C}} = (1 - u_s) \sum_{a=0}^{\infty} u_s^a D_{\mathcal{C}}^a.$$

For $a \geq K$, $p^a x = 0$, so $D^a = \mathcal{R}_0$. The tail sum is $(1 - u_s) \sum_{a=K}^{\infty} u_s^a \mathcal{R}_0 = u_s^K \mathcal{R}_0$, giving

$$\mathcal{B}(s)|_{\mathcal{C}} = (1 - u_s) \sum_{a=0}^{K-1} u_s^a D_{\mathcal{C}}^a + u_s^K \mathcal{R}_0|_{\mathcal{C}}. \quad (2.24)$$

For $j < K$, Eq. (2.23) gives $D^a e_j = p^{a/2} e_{j-a}$ for $a \leq j$, and zero for $a > j$. For the zero-element state, the recursion instead gives

$$D^a e_K = e_K + \sqrt{p-1} \sum_{r=1}^{\min(a,K)} p^{(r-1)/2} e_{K-r}.$$

Using $(1 - u_s) \sum_{a=r}^{\infty} u_s^a = u_s^r$ gives every column of the shell-average block:

$$\begin{aligned} \mathcal{B}(s)e_j &= (1 - u_s) \sum_{a=0}^j \varrho_s^a e_{j-a}, & j < K, \\ \mathcal{B}(s)e_K &= e_K + \sqrt{1 - p^{-1}} \sum_{i=0}^{K-1} \varrho_s^{K-i} e_i. \end{aligned} \quad (2.25)$$

The matrix elements of this shell-average block are finite polynomials in p^{-s} , and hence give its analytic continuation in the complex parameter s . The constant function satisfies $T_m b = b$, and therefore $\mathcal{B}(s)b = b$. Furthermore, $\langle z|f \rangle = f(0)/\sqrt{d}$ gives $\mathcal{R}_0 = \sqrt{d} |b\rangle \langle z|$, so the zero-element tail term is

$$p^{-Ks} \mathcal{R}_0 = p^{K(1/2-s)} |b\rangle \langle z|.$$

At $s = \frac{1}{2}$, this amplitude component is exactly $|b\rangle \langle z|$.

The directional blocks of the full transfer operator generally do not satisfy the conditions for a self-adjoint spectral decomposition. Write $B_\theta = \mathcal{B}(s)|_{\mathcal{F}_\theta}$. The triangular form of Eq. (2.22) shows that its only eigenvalue is $R_\theta(s)$. When the chain length exceeds one and $R_\theta(s) \neq 0$,

$$\|B_\theta f_{0,\theta}\|^2 = |R_\theta(s)|^2, \quad \|B_\theta^\dagger f_{0,\theta}\|^2 = |R_\theta(s)|^2 \sum_{a=0}^{K-c} |p^{1/2-s}|^{2a} > |R_\theta(s)|^2.$$

A normal operator satisfies $\|Av\| = \|A^\dagger v\|$ for every vector, so B_θ is nonnormal in this case. For any direction with $j > 0$, Eq. (2.22) also gives the nonzero component $R_\theta(s)p^{1/2-s}f_{j-1,\theta}$. It is orthogonal to $f_{j,\theta}$, so that direction is not an eigenstate of the full transfer operator when $R_\theta(s) \neq 0$.

Shell-preserving compression retains only matrix elements within the same shell. Every term with $a > 0$ in Eq. (2.22) changes the shell label, so

$$T_{\text{el}}(s) := P \left(\sum_{j=0}^K \Pi_j \mathcal{B}(s) \Pi_j \right) P = \sum_{\nu} R_{\theta_\nu}(s) P_\nu. \quad (2.26)$$

The same θ has the same resonance ratio on different shells; that ratio alone therefore cannot distinguish shell labels. Applied to the full transfer operator, Eq. (2.12) retains cross-shell matrix elements; when this decomposition is applied to the shell-preserving response, the expansion is diagonal in the direction basis.

Both the full transfer operator and the shell-preserving response are amplitude operators. The former retains cross-shell matrix elements, while the latter is diagonal in the direction basis. These spectral responses do not by themselves define an unconditional trace-preserving evolution.

§2.6 Metric Structure

A pattern in the state space is described by complex amplitudes at individual points; the inner product determines the distance between amplitude vectors. Separations between points of the underlying ring require a choice of coordinates, whereas shell energies determine relative phase changes under free evolution. These structures compare patterns, position labels, and evolution times, respectively.

Using Eq. (1.18), define, for $f, g \in \mathcal{H}$,

$$d_{\mathcal{H}}(f, g) = \|f - g\| = \left(\frac{1}{d} \sum_{x \in X_{p,K}} |f(x) - g(x)|^2 \right)^{1/2}. \quad (2.27)$$

Positive definiteness and symmetry follow from the inner product. The Cauchy–Schwarz inequality gives $\|a+b\|^2 \leq (\|a\| + \|b\|)^2$, which implies the triangle inequality. If (f_n) is a Cauchy sequence in this metric, then $|f_n(x) - f_m(x)| \leq \sqrt{d} \|f_n - f_m\|$ implies that each coordinate converges to some $f(x)$. Since the number of positions is finite,

$$\|f_n - f\|^2 = \frac{1}{d} \sum_{x \in X_{p,K}} |f_n(x) - f(x)|^2 \longrightarrow 0.$$

Thus the metric is complete, and the inner product space defined in Section 1 is a Hilbert space. This metric compares amplitude vectors. A global phase of a

normalized representative vector does not change its pure state; preservation of the excitation pattern is characterized by the corresponding rank-one projection.

Position intervals are defined after coordinates have been chosen on the underlying finite ring. The standard spectral representation in Section 1 gives $E_n^{\text{std}} = \log n$ and $E_{mn}^{\text{std}} = E_m^{\text{std}} + E_n^{\text{std}}$. We adopt the same logarithm as the coordinate convention for position calibration. For each nonzero residue class, take its unique integer representative $m \in \{1, \dots, d-1\}$ and define the oriented interval and the corresponding distance by

$$y(m, n) = \log \frac{m}{n}, \quad d_{\log}(m, n) = |y(m, n)|. \quad (2.28)$$

Since $\log$ is injective on the positive reals, the distance vanishes if and only if $m = n$; symmetry is immediate. The identity $y(m, n) = y(m, \ell) + y(\ell, n)$ and the triangle inequality for the real absolute value imply the triangle inequality for $d_{\log}$. This metric depends on the chosen integer representatives. The zero element is treated separately and no logarithm is assigned to it. Translations and multiplication by units on the finite ring need not preserve this distance; after modular wraparound, it must be recomputed using the new integer representatives.

The distance between adjacent representatives is

$$\Delta_n = d_{\log}(n, n+1) = \log(1 + 1/n), \quad 1 \leq n \leq d-2. \quad (2.29)$$

It decreases strictly with n . For the same expression on the positive integers, a Taylor expansion gives $\log(1 + 1/n) = n^{-1} - (2n^2)^{-1} + O(n^{-3})$ as $n \rightarrow \infty$. For $m < n$, we have $d_{\log}(m, n) = \sum_{r=m}^{n-1} \Delta_r$, so the smallest nonzero distance at finite depth is

$$d_{\log, \min} = \log \frac{d-1}{d-2}. \quad (2.30)$$

The largest adjacent spacing is $\log 2$. Since p is an odd prime, both $d-1$ and $d-2$ are units. Thus Eq. (2.30) still gives the minimum distance after restriction to the unit group. For a positive integer m , an integer $q \geq 2$, and nonnegative integers a, b , if both mq^a and mq^b lie in the chosen range of representatives, then

$$d_{\log}(mq^a, mq^b) = |a - b| \log q. \quad (2.31)$$

Hence a geometric progression without modular wraparound forms an equally spaced sublattice in logarithmic coordinates.

The relation between logarithmic position coordinates and shell energies in the finite model can be decomposed according to the prime valuation. Write the chosen integer representatives uniquely as $m = p^j u$ and $n = p^k v$, with $p \nmid uv$. Equation (1.29) gives $H_K|m\rangle = E_j|m\rangle$, and therefore

$$y(m, n) = (j - k) \log p + \log \frac{u}{v} = (E_j - E_k) + \log \frac{u}{v}. \quad (2.32)$$

Along the sequence without wraparound determined by the same integer unit part u ,

$$d_{\log}(p^j u, p^k u) = |j - k| \log p = |E_j - E_k|. \quad (2.33)$$

A general position interval also contains a contribution from the unit parts. Two positions in the same shell may have a nonzero interval between them while sharing the same shell energy. This distinguishes the logarithmic energy spectrum of the

standard representation, the shell energies of the finite model, and the position coordinates, while specifying their relation along the sequences above.

An energy gap determines the change in relative phase under free evolution. For two nonzero shell components $\psi_j \in \Pi_j \mathcal{H}$ and $\psi_k \in \Pi_k \mathcal{H}$, the spectral decomposition gives

$$e^{-itH_K}(\psi_j + \psi_k) = e^{-itE_j}\psi_j + e^{-itE_k}\psi_k, \quad \Delta\phi_{jk}(t) = -t(E_j - E_k). \quad (2.34)$$

Here $\Delta\phi_{jk}$ is the unwrapped relative phase change from the initial time. For $j \neq k$ and a prescribed positive threshold ϑ_{res} , the time required for the absolute accumulated phase to reach this threshold is

$$t_{jk}^{\text{res}} = \frac{\vartheta_{\text{res}}}{|E_j - E_k|} = \frac{\vartheta_{\text{res}}}{|j - k| \log p}. \quad (2.35)$$

A unit threshold gives the reciprocal energy gap. Interpreting it as an actual resolution time additionally requires a phase readout and its precision to be specified. Components in the same shell acquire the same free phase, so this action produces no relative phase change within a shell. All shell energies are integer multiples of $\log p$, and hence the common free-evolution period satisfies

$$T_p = \frac{2\pi}{\log p}, \quad e^{-iT_p H_K} = I. \quad (2.36)$$

The relative-phase period of a particular superposition can be shorter. The duration of a complete cycle containing exchanges and controls is determined by its actual evolution stages.

§3 Background and Stable States

The first two sections specify the constituents and structure of the state space. Establishing dynamics on this space also requires the subject of evolution to be identified. An excitation is identified relative to a background, and its continued description requires conditions under which its pattern is preserved by the action. The definitions of a background reference and stable states thus provide the basis for describing the dynamical subject.

The background and excitations belong to the same state space. The background specifies the reference pattern against which excitations are identified, whereas an excitation pattern is determined by the relative amplitudes and phases at different positions. When an action preserves these relations, the excitation has a structure that can be identified throughout its evolution.

We use the state space from the preceding two sections, Eq. (1.17),

$$\mathcal{H} = L^2(X_{p,K}, \mathbb{C}), \quad X_{p,K} = \mathbb{Z}/p^K \mathbb{Z}, \quad d = p^K,$$

and the orthogonal decomposition in Eq. (1.27),

$$\mathcal{H} = \mathcal{C} \oplus \mathcal{M}.$$

Here $\mathcal{C}$ is the shell-average subspace and $\mathcal{M}$ the space of nontrivial directions, with respective projections Π and P . The shell-average states, shell masses, and shell energies are denoted by e_j , h_j , and $E_j = j \log p$, respectively.

§3.1 Background

The background reference pattern is invariant under the arithmetic actions employed. The unit group permutes positions within each valuation shell, determining an invariant amplitude distribution within that shell; scaling connects different valuation shells and determines the reference relation between them.

Section 2 shows that the valuation shells are precisely the orbits of the unit group. Thus an amplitude vector satisfying

$$R_u f = f$$

for every unit u is constant on each valuation shell. Conversely, a function constant on each shell is invariant under all unit actions, so

$$\boxed{\{f \in \mathcal{H} : R_u f = f \text{ for all } u\} = \mathcal{C}.}$$

The unit-group average in Eq. (2.9) is the orthogonal projection onto this invariant subspace:

$$\Pi = \frac{1}{|G_{p,K}|} \sum_{u \in G_{p,K}} R_u, \quad P = I - \Pi.$$

The components orthogonal to every shell-average state form the direction space. From

$$\langle e_j, f \rangle = \frac{1}{d\sqrt{h_j}} \sum_{x \in S_j} f(x),$$

we obtain

$$\mathcal{M} = \left\{ f \in \mathcal{H} : \sum_{x \in S_j} f(x) = 0, \quad 0 \leq j \leq K \right\}.$$

This decomposition separates the mean amplitude on each shell from its nontrivial directional structure. A general pattern may contain both components:

$$f = \Pi f + P f.$$

The corresponding density operator may also retain coherence between the two subspaces.

Unit-group invariance here means that the amplitude vector satisfies $R_u f = f$. An individual character direction acquires a global phase under a unit action, while its pure-state projection still satisfies

$$R_u P_\nu R_u^\dagger = P_\nu.$$

Invariance of amplitude vectors and invariance of density operators under the unit-group action are therefore distinct conditions. The distinction between the shell-average and direction subspaces uses the former condition.

Unit-group invariance allows different mean amplitudes on different shells. Requiring the reference pattern to be invariant under scaling further determines the relations between these amplitudes. Denote the normalized constant function by b :

$$b(x) = 1, \quad \|b\| = 1. \quad (3.1)$$

Since the shells cover the entire ring,

$$b = \sum_{j=0}^K \sqrt{h_j} e_j.$$

The pullback of finite scaling on functions is

$$(Df)(x) = f(px).$$

After K successive applications, every point is mapped to the zero element, and hence

$$D^K f = f(0)b.$$

If $Df = f$, then

$$f = D^K f = f(0)b.$$

Conversely, the constant function satisfies $Db = b$, so

$$\boxed{\ker(D - I) = \mathbb{C}b.} \tag{3.2}$$

The same relation determines all eigenvectors of scaling with nonzero eigenvalues. Since

$$D^{K+1} = D^K,$$

if

$$D\psi = \lambda\psi, \quad \psi \neq 0, \quad \lambda \neq 0,$$

then

$$\lambda^{K+1}\psi = \lambda^K\psi,$$

and consequently $\lambda = 1$. Thus the only nonzero eigenvalue of scaling is 1, with eigenspace $\mathbb{C}b$.

Unit-group invariance determines the shell-average subspace, and scaling invariance selects the constant reference direction within it. After normalization, this pattern has equal amplitude and phase at every finite position. Write its pure state as

$$P_b = |b\rangle\langle b|.$$

Substitution of the constant function into each term of the full spectral transfer in Eq. (2.20) also gives

$$\mathcal{B}(s)b = b.$$

Regular continuation of the finite matrix preserves this identity.

Vectors in the direction space are orthogonal to every shell-average state and hence to b . The subspace $\mathcal{C}$ still contains shell-average combinations orthogonal to b . The full orthogonal decomposition therefore remains $\mathcal{C} \oplus \mathcal{M}$; the constant reference is only the direction in $\mathcal{C}$ fixed by scaling.

The constant reference, the zero-element state, and the shell-average state of the unit shell have different supports. The zero-element state is $z = e_K$, the shell-average state of the unit shell is e_0 , and

$$\langle e_j, b \rangle = \sqrt{h_j}, \quad |\langle z, b \rangle|^2 = d^{-1}, \quad |\langle e_0, b \rangle|^2 = 1 - p^{-1}. \tag{3.3}$$

The vector b contains coherent components from every shell, whereas z is supported only at the zero element and e_0 only on the unit shell. Accordingly, the shell energy of z is E_K , while that of e_0 is zero.

Once the reference pattern is fixed, an actual background preparation must specify its shell occupations and coherence. The pure state defined by the constant function b expands as

$$P_b = \sum_{j,\ell=0}^K \sqrt{h_j h_\ell} |e_j\rangle \langle e_\ell|. \quad (3.4)$$

The preparation that retains these shell occupations while removing coherence between different shells is called the **shell-dephased background state**:

$$\sigma_b^{\text{sh}} = \sum_{j=0}^K \Pi_j P_b \Pi_j = \sum_{j=0}^K h_j P_{e_j}. \quad (3.5)$$

The relation between this background preparation and the constant reference can also be expressed by averaging the free phase. Set $\omega_p = \log p$ and use the period in Eq. (2.36),

$$T_p = \frac{2\pi}{\omega_p}.$$

Since $H_K e_j = j\omega_p e_j$,

$$\begin{aligned} & \frac{1}{T_p} \int_0^{T_p} e^{-itH_K} P_b e^{itH_K} dt \\ &= \sum_{j,\ell=0}^K \sqrt{h_j h_\ell} \left(\frac{1}{T_p} \int_0^{T_p} e^{-i(j-\ell)\omega_p t} dt \right) |e_j\rangle \langle e_\ell|. \end{aligned}$$

The coefficient in parentheses is $\delta_{j\ell}$, and therefore

$$\frac{1}{T_p} \int_0^{T_p} e^{-itH_K} P_b e^{itH_K} dt = \sigma_b^{\text{sh}}. \quad (3.6)$$

Here T_p is the common period of the free shell phases and is used to express the average above; it is defined separately from the duration of a complete dynamical cycle.

All h_j are positive and sum to one, so

$$\begin{aligned} \sigma_b^{\text{sh}} &\geq 0, & \text{Tr } \sigma_b^{\text{sh}} &= 1, \\ \text{supp } \sigma_b^{\text{sh}} &= \mathcal{C}, & \text{rank } \sigma_b^{\text{sh}} &= K + 1. \end{aligned}$$

Each e_j is both a shell-energy eigenstate and invariant under the unit group, giving

$$[\sigma_b^{\text{sh}}, H_K] = 0, \quad [\sigma_b^{\text{sh}}, R_u] = 0. \quad (3.7)$$

In the normalized point basis,

$$\langle x | e_j \rangle = |S_j|^{-1/2} \mathbf{1}_{S_j}(x).$$

Combining this with $h_j/|S_j| = 1/d$ gives

$$\langle x|\sigma_b^{\text{sh}}|y\rangle = \frac{1}{d} \sum_{j=0}^K \mathbf{1}_{S_j}(x) \mathbf{1}_{S_j}(y). \quad (3.8)$$

This density operator has diagonal entry $1/d$ at every position and retains coherence between different positions in the same shell. The maximally mixed state I/d on the full space has only position-diagonal entries and represents a different preparation.

The pure state P_b has rank one and contains coherence between different shell energies. Free evolution changes the relative phases of these components, whereas the shell-dephased background state is invariant because it commutes with H_K . We adopt σ_b^{sh} as the baseline preparation for background restoration. The phase-average formula expresses its relation to b . The background can be prepared directly in the state represented by this density operator, without first preparing P_b and then applying the free-evolution phase average.

When the background is used for restoration in the cycle, its occupations must also be examined for thermal equilibrium. A thermal equilibrium preparation is determined by the chosen energies and their degeneracies. For a given inverse temperature β , the Gibbs states on the shell-average subspace and on the full space are, respectively,

$$\begin{aligned} \rho_\beta^{\text{shell}} &= \frac{\sum_{j=0}^K e^{-\beta E_j} P_{e_j}}{\sum_{j=0}^K e^{-\beta E_j}}, \\ \rho_\beta^{\text{full}} &= \frac{\sum_{j=0}^K e^{-\beta E_j} \Pi_j}{\sum_{j=0}^K |S_j| e^{-\beta E_j}}. \end{aligned} \quad (3.9)$$

The former includes one average direction per shell; the latter includes all directions in each shell. The baseline background assigns shares h_j and is defined separately from these thermal preparations.

Different shell-average preparations in the cycle can be written uniformly using fixed positive shares g_j , with

$$g_j > 0, \quad \sum_j g_j = 1.$$

The baseline background corresponds to $g_j = h_j$, and the same shares are used throughout the iteration. Whether the background is also a stationary state of the complete cycle must be checked against the actual exchange action.

§3.2 Stable States

A general excitation is a superposition of several direction components. When these components have different responses, their relative amplitudes and phases change with the number of applications, changing the superposed pattern. If the entire excitation acquires only a common nonzero complex factor, the relative relations between positions are preserved.

An excitation mode whose pattern is preserved under a given propagation action is called a **stable state**; it describes a stable standing-wave pattern under that action. We use the shell-preserving response T_{el} from Section 2 to determine stability.

This response identifies modes that preserve their patterns and supplies the spectral data for the cycle. Its eigenvalues describe amplitude responses; the unitary phases in the actual cycle are implemented through a joint action.

Fix a regular parameter value s_* for the shell-preserving response. A nonzero excitation $\psi \in \mathcal{M}$ preserves its pattern if and only if

$$\boxed{T_{\text{el}}(s_*)\psi = \lambda\psi, \quad \lambda \neq 0.} \quad (3.10)$$

The common complex factor changes only the overall amplitude and phase. For any nonzero complex number c , the vectors ψ and $c\psi$ have the same pure-state projection

$$P_{[\psi]} = \frac{|\psi\rangle\langle\psi|}{\|\psi\|^2}. \quad (3.11)$$

This projection contains both position occupations and coherence between different positions, and therefore determines whether the full pattern is preserved.

Under successive applications of the shell-preserving response, for every integer $n \geq 0$,

$$T_{\text{el}}(s_*)^n\psi = \lambda^n\psi.$$

Since $\lambda \neq 0$, the normalized projection satisfies

$$\frac{T_{\text{el}}(s_*)^n|\psi\rangle\langle\psi|(T_{\text{el}}(s_*)^n)^\dagger}{\|T_{\text{el}}(s_*)^n\psi\|^2} = P_{[\psi]}. \quad (3.12)$$

Conversely, if the vector after one application is nonzero and its normalized projection is still $P_{[\psi]}$, it lies in the same one-dimensional range as ψ and must equal $\lambda\psi$ for some $\lambda \neq 0$. Thus the eigenvalue condition above is equivalent to preservation of a nonzero pattern.

Preservation of squared amplitude moduli at individual positions alone does not ensure preservation of relative phases. Patterns are compared here through their full pure-state projections; normalization does not impart trace preservation to the amplitude operator.

Stable states are classified by the spectral structure of the shell-preserving response. Let θ be a nontrivial primitive character of conductor p^{c_θ} . Its direction on shell j is

$$|\nu\rangle = f_{j_\nu, \theta_\nu}, \quad 0 \leq j_\nu \leq K - c_{\theta_\nu},$$

with corresponding projection

$$P_\nu = |\nu\rangle\langle\nu|.$$

These directions form an orthonormal basis of $\mathcal{M}$.

The spectral response of each direction is given by the resonance ratio

$$R_\theta(s) = \frac{L(s, \theta)}{\zeta(s)}.$$

By the shell-preserving compression in Eq. (2.26) and the operator expansion in Eq. (2.12) of Section 2,

$$T_{\text{el}}(s_*) = P \left(\sum_{j=0}^K \Pi_j \mathcal{B}(s_*) \Pi_j \right) P = \sum_{\nu} R_{\theta_\nu}(s_*) P_\nu. \quad (3.13)$$

Thus, for a general excitation

$$\psi = \sum_{\nu} \psi_{\nu} |\nu\rangle, \quad \psi_{\nu} = \langle \nu | \psi \rangle,$$

we have

$$T_{\text{el}}(s_*)^n \psi = \sum_{\nu} \psi_{\nu} R_{\theta_{\nu}}(s_*)^n |\nu\rangle, \quad n \geq 1. \quad (3.14)$$

The response modulus determines the change in relative amplitude, and its argument determines the change in relative phase.

Substituting this expression into the stability condition and using orthogonality of the direction basis gives, componentwise,

$$(R_{\theta_{\nu}}(s_*) - \lambda) \psi_{\nu} = 0. \quad (3.15)$$

Hence the stable subspace corresponding to a common nonzero response is

$$\boxed{\mathcal{M}_{\lambda}(s_*) = \text{span}\{|\nu\rangle : R_{\theta_{\nu}}(s_*) = \lambda\}, \quad \lambda \neq 0.} \quad (3.16)$$

Every nonzero vector in it is a stable state. A single direction with nonzero response defines a stable pattern; several directions with the same complex response may also form stable patterns through coherent superposition.

The same primitive character has the same response on different shells. Therefore, when $R_{\theta}(s_*) \neq 0$, every nonzero vector in

$$\mathcal{F}_{\theta} = \text{span}\{f_{j,\theta} : 0 \leq j \leq K - c_{\theta}\}$$

satisfies the shell-preserving stability condition, and

$$\dim \mathcal{F}_{\theta} = K - c_{\theta} + 1.$$

This cross-shell degeneracy is retained by the decomposition of the full finite space. Whether different characters share a response is determined by their respective spectral functions.

For real $s > 1$, the Euler products of $L(s, \theta)$ and $\zeta(s)$ converge absolutely and are nonzero [10, 11], so all $R_{\theta}(s)$ are nonzero. The response functions of distinct characters are not identical.

To prove the latter statement, we use the standard separation argument for Dirichlet-series coefficients [12]. Take distinct primitive characters θ, θ' , inflate them to the common unit group $G_{p,K}$, and extend them by zero on integers divisible by p . Primitivity and minimality of the conductors ensure that the inflated characters remain distinct, so there is a first differing Dirichlet coefficient. Write

$$d_m = \theta(m) - \theta'(m), \quad m_* = \min\{m \geq 1 : d_m \neq 0\}.$$

For real $s > 1$, subtraction of the absolutely convergent series gives

$$\begin{aligned} m_*^s (L(s, \theta) - L(s, \theta')) \\ = d_{m_*} + \sum_{m > m_*} d_m \left(\frac{m_*}{m} \right)^s. \end{aligned}$$

Fix $s_1 > 1$. For $s \geq s_1$,

$$\left| d_m \left(\frac{m_*}{m} \right)^s \right| \leq 2 \left(\frac{m_*}{m} \right)^{s_1},$$

and

$$\sum_{m > m_*} 2 \left(\frac{m_*}{m} \right)^{s_1} < \infty.$$

Each term tends to zero as $s \rightarrow +\infty$, so dominated convergence gives

$$m_*^s (L(s, \theta) - L(s, \theta')) \rightarrow d_{m_*} \neq 0. \quad (3.17)$$

Thus the difference of the two L -functions is not identically zero.

This difference is analytic in $\Re s > 1$ and has isolated zeros. Since $\zeta(s) \neq 0$ in this region, coincidence of R_θ and $R_{\theta'}$ has the same zero condition. At finite depth there are only finitely many primitive characters. Consequently, responses of distinct characters coincide only on a locally discrete parameter set in the real interval $s > 1$.

Outside these coincidence parameters,

$$\boxed{\mathcal{M}_{R_\theta(s)}(s) = \mathcal{F}_\theta.} \quad (3.18)$$

If the nonzero responses of several distinct characters coincide at a parameter value, the corresponding stable subspace is the orthogonal direct sum of their shell-chain subspaces.

The spectral reference used in the cycle must also be fixed explicitly. Section 2 writes the zero-element tail of the full transfer as

$$p^{-Ks} \mathcal{R}_0 = p^{K(1/2-s)} |b\rangle \langle z|. \quad (3.19)$$

We impose the normalization that this amplitude component maps the normalized zero-element state z to the normalized constant function b with unit coefficient. For real s , this condition is equivalent to

$$p^{K(1/2-s)} = 1,$$

and hence fixes

$$s_0 = \frac{1}{2}.$$

This normalization is imposed on the separated zero-element tail; the full transfer still contains the remaining matrix elements.

For a nontrivial character θ , the sum over one period vanishes, so its partial sums are bounded. Summation by parts shows that $\sum_{m \geq 1} \theta(m) m^{-s}$ converges uniformly on each compact subset of $\Re s > 0$, and thus $L(s, \theta)$ is analytic near s_0 . The alternating series $\eta(s) = \sum_{n \geq 1} (-1)^{n-1} n^{-s}$ is also analytic in this half-plane. Analytically continuing the identity $\eta(s) = (1 - 2^{1-s}) \zeta(s)$ from $\Re s > 1$ gives

$$\zeta(1/2) = \frac{\sum_{n \geq 1} (-1)^{n-1} n^{-1/2}}{1 - \sqrt{2}} < 0.$$

Pairing adjacent terms in the numerator gives strictly positive terms, so $\zeta(1/2) \neq 0$. Thus s_0 is a regular point of every resonance ratio R_θ . We fix this spectral reference and determine stability of the directions used in the cycle at $s_* = s_0$.

At s_0 , nonvanishing responses and separation between different characters still require separate verification. Nonvanishing and local discreteness for real $s > 1$ do not directly determine the spectrum at this specified parameter value.

On the unit shell $j = 0$, nontrivial characters of conductor p give the **elementary states** modulo p . This name applies to those directions with nonzero response. If all $p - 2$ responses are nonzero, there are $p - 2$ elementary-state directions. Directions of higher conductor or on other valuation shells remain labeled by (θ, j) and are tested by the same stability condition. Zero-response directions still belong to $\mathcal{M}$, but do not satisfy nonzero pattern preservation at this parameter value.

Having specified shell-preserving stability, we examine its relation to full spectral transfer and free evolution. The full transfer retains cross-shell matrix elements, so its eigenvectors must be computed on the corresponding shell chains.

On $\mathcal{F}_\theta$, use the nilpotent operator from Section 2,

$$N_\theta f_{j,\theta} = f_{j-1,\theta} \quad (j \geq 1), \quad N_\theta f_{0,\theta} = 0,$$

and write

$$q_* = p^{1/2-s_*}.$$

The finite-series expression for full transfer in Eq. (2.22) gives

$$\mathcal{B}(s_*)|_{\mathcal{F}_\theta} = R_\theta(s_*)(I - q_* N_\theta)^{-1}.$$

In the basis ordered by shell index, this operator is upper triangular, with every diagonal entry equal to $R_\theta(s_*)$. When $R_\theta(s_*) \neq 0$,

$$\mathcal{B}(s_*) - R_\theta(s_*)I = R_\theta(s_*)q_* N_\theta(I - q_* N_\theta)^{-1} \quad (3.20)$$

holds on the shell chain.

Since N_θ is nilpotent, the inverse exists and commutes with N_θ . Therefore

$$\ker((\mathcal{B}(s_*) - R_\theta(s_*)I)|_{\mathcal{F}_\theta}) = \ker N_\theta = \mathbb{C}f_{0,\theta}.$$

The only eigenvector direction of full transfer on a shell chain with nonzero response is its unit-shell endpoint; the other single-shell directions still satisfy stability under the shell-preserving response.

The pattern of a cross-shell superposition under free evolution also requires a separate calculation. For

$$\psi = \sum_{j=0}^{K-c_\theta} a_j f_{j,\theta},$$

the shell-preserving response multiplies the entire pattern by $R_\theta(s_*)$, whereas free evolution gives

$$e^{-itH_K}\psi = \sum_{j=0}^{K-c_\theta} a_j e^{-itj\omega_p} f_{j,\theta}. \quad (3.21)$$

The relative phases between occupied shells generally vary with time. Preserving the original pattern at a specified time requires all occupied shells to acquire the same free phase, namely,

$$e^{-it(E_j - E_\ell)} = 1 \quad \text{for all } a_j \neq 0, a_\ell \neq 0.$$

Pattern preservation under the shell-preserving response, the eigenvalue condition for full transfer, and phase changes under free evolution are thus determined by their respective actions. The complete cycle must also combine free evolution with actual exchanges. A system density operator that remains unchanged after an entire cycle is called a **stationary state**; its criterion is the fixed-point condition for the complete cycle map.

§3.3 Structure of Stable States

A stable-state pattern is determined jointly by its support and by the relative amplitudes and phases at different positions. A single character mode specifies a basic phase arrangement; coherent superpositions of modes within the same stable subspace give more general patterns. Positions here are the finite positions in the ring underlying the state space.

For a single mode

$$|\nu\rangle = f_{j_\nu, \theta_\nu},$$

the valuation shell determines its support and shell energy, while the character determines the phase relations between positions in that shell:

$$H_K|\nu\rangle = j_\nu\omega_p|\nu\rangle, \quad R_u|\nu\rangle = \theta_\nu(u)^{-1}|\nu\rangle. \quad (3.22)$$

The amplitude vanishes outside the shell and has equal modulus at every position within it. Thus a single mode has a definite shell energy and is distributed over its entire shell. The direction label describes the phase arrangement of this distribution, without selecting an individual position in the shell.

Fix a shell $j = j_\nu$ and denote its corresponding unit group by

$$G_j = G_{p, K-j}.$$

Every $x \in S_j$ has the unique representation

$$x = p^j u, \quad u \in G_j.$$

For $u, v \in G_j$, the ratio of complex amplitudes is

$$\boxed{\frac{f_{j, \theta}(p^j u)}{f_{j, \theta}(p^j v)} = \frac{\theta(u)}{\theta(v)} = \theta(uv^{-1})}. \quad (3.23)$$

The two positions have the same complex amplitude if and only if

$$uv^{-1} \in \ker \theta.$$

Hence the positions with a given phase value form a coset of $\ker \theta$.

The number of phase values and their multiplicities are determined by the order of the character. Write

$$o_\theta = \text{ord } \theta.$$

The image of the character is a finite subgroup of the unit circle, with order o_θ . The isomorphism between the quotient by the kernel and the image,

$$G_j / \ker \theta \simeq \theta(G_j),$$

gives

$$|\theta(G_j)| = o_\theta, \quad |\ker \theta| = \frac{|S_j|}{o_\theta}. \quad (3.24)$$

Every coset has the same cardinality as the kernel. There are therefore o_θ phase values on the shell, each occurring $|S_j|/o_\theta$ times.

The modulus of the single-mode function on its shell is

$$|f_{j,\theta}(p^j u)| = h_j^{-1/2},$$

and the modulus of its coefficient in the normalized point basis is

$$|\langle p^j u | \nu \rangle| = |S_j|^{-1/2}.$$

The character kernel describes the positions at which phases repeat. A single character has no amplitude zeros within its support shell.

The phase structure also has a reduction depth determined by the conductor. Let the conductor of θ be p^{c_θ} , with natural quotient map

$$\pi_{j,c_\theta} : G_j \longrightarrow (\mathbb{Z}/p^{c_\theta}\mathbb{Z})^\times.$$

Since θ is inflated from a primitive character on this quotient, for

$$v \in \ker \pi_{j,c_\theta}$$

we have

$$\theta(uv) = \theta(u), \quad f_{j,\theta}(p^j uv) = f_{j,\theta}(p^j u). \quad (3.25)$$

The exponent c_θ is the smallest one permitting this factorization. Verifying the invariance above alone shows only that the conductor divides the specified modulus; determining the conductor also requires minimality. When $c_\theta = K - j$, the quotient map is the identity and its kernel is $\{1\}$.

The order, kernel, and conductor describe, respectively, the number of phase values, the cosets on which phases repeat, and the minimum depth of natural reduction. The actual phase value assigned to each coset is still determined by the full character.

Position occupation and coherence of a single mode are both contained in its density matrix. For $x, y \in S_j$,

$$\langle x | P_\nu | y \rangle = \frac{1}{|S_j|} \theta_\nu(x/p^j) \overline{\theta_\nu(y/p^j)}. \quad (3.26)$$

The matrix element vanishes if either position lies outside the shell. Different characters on the same shell have identical position-diagonal entries, but may have different off-diagonal entries. Shell energy and position occupation alone therefore do not determine the excitation pattern; relative phases are also part of it.

The symmetries in Section 2 further specify the relations between these phase structures. Negation satisfies

$$\iota f_{j,\theta} = \theta(-1) f_{j,\theta}, \quad \theta(-1) \in \{1, -1\},$$

and hence changes only the overall sign of a single mode. Complex conjugation of amplitudes satisfies

$$\mathfrak{c} f_{j,\theta} = f_{j,\bar{\theta}},$$

and conjugates the phase at every position.

In the region of absolute convergence, the Dirichlet series gives

$$L(s, \bar{\theta}) = \overline{L(\bar{s}, \theta)}.$$

Combining this with

$$\zeta(s) = \overline{\zeta(\bar{s})},$$

and continuing the expressions yields

$$R_{\bar{\theta}}(s) = \overline{R_{\theta}(\bar{s})}. \quad (3.27)$$

Thus, at a real regular parameter value, conjugate directions have conjugate responses of equal modulus, although the responses need not be equal. Self-conjugate characters are fixed by complex conjugation of amplitudes; the remaining characters form pairs of distinct conjugate directions.

These relations determine the support, phases, and symmetries of an individual character mode. When several modes share a nonzero response, the complete stable pattern also depends on their mutual coherence. Its structure follows directly from the single-mode expansion.

Fix $\lambda \neq 0$ and, for each nonzero shell, write

$$\Theta_j(\lambda) = \{\theta : c_{\theta} \leq K - j, R_{\theta}(s_*) = \lambda\}.$$

Only nontrivial primitive characters are included, inflated to the unit group of their shell. Every normalized state in the stable subspace $\mathcal{M}_{\lambda}(s_*)$ can be written as

$$\boxed{\psi = \sum_{j=0}^{K-1} \sum_{\theta \in \Theta_j(\lambda)} a_{j\theta} f_{j,\theta}, \quad \sum_{j,\theta} |a_{j\theta}|^2 = 1.} \quad (3.28)$$

All sums below use these index ranges.

Define the phase superposition on shell j by

$$A_j(u) = \sum_{\theta \in \Theta_j(\lambda)} a_{j\theta} \theta(u).$$

An empty index set gives $A_j = 0$. The amplitude in the normalized point basis is therefore

$$\langle p^j u | \psi \rangle = \frac{A_j(u)}{\sqrt{|S_j|}}. \quad (3.29)$$

Denote the occupation share of shell j by

$$q_j = \langle \psi | \Pi_j | \psi \rangle.$$

Orthogonality of characters on the same shell gives

$$\begin{aligned} q_j &= \frac{1}{|S_j|} \sum_{u \in G_j} |A_j(u)|^2 \\ &= \sum_{\theta, \theta' \in \Theta_j(\lambda)} a_{j\theta} \overline{a_{j\theta'}} \delta_{\theta\theta'} \\ &= \sum_{\theta \in \Theta_j(\lambda)} |a_{j\theta}|^2. \end{aligned}$$

The zero element supports no nontrivial direction, so

$$\langle 0|\psi\rangle = 0, \quad q_K = 0, \quad \sum_{j=0}^{K-1} q_j = 1.$$

For positions in two nonzero shells, the full density-matrix element is

$$\boxed{\langle p^j u | P_{[\psi]} | p^\ell v \rangle = \frac{A_j(u) \overline{A_\ell(v)}}{\sqrt{|S_j| |S_\ell|}}, \quad j, \ell < K.} \quad (3.30)$$

It vanishes if either position is the zero element. The diagonal entries

$$\frac{|A_j(u)|^2}{|S_j|}$$

determine point occupations, while the off-diagonal entries describe both intra-shell and inter-shell coherence. A stable pattern is determined jointly by the participating directions and their complex coefficients. A common phase of all coefficients does not change the pure-state projection.

When only one primitive character participates and $R_\theta(s_*) \neq 0$,

$$\psi = \sum_{j=0}^{K-c_\theta} a_j f_{j,\theta}, \quad \sum_j |a_j|^2 = 1. \quad (3.31)$$

Then

$$A_j(u) = a_j \theta(u), \quad q_j = |a_j|^2,$$

and

$$\langle p^j u | P_{[\psi]} | p^\ell v \rangle = \frac{a_j \overline{a_\ell} \theta(u) \overline{\theta(v)}}{\sqrt{|S_j| |S_\ell|}}.$$

Each occupied shell has equal amplitude moduli and no amplitude zeros. Inter-shell phases are retained in $a_j \overline{a_\ell}$ and cannot be recovered from the occupations $|a_j|^2$ alone. This pattern, stable under the shell-preserving response, may contain components with different shell energies; its free evolution is still determined by the shell-phase formula in Eq. (3.21).

If several distinct characters on the same shell share a nonzero response and enter the superposition, then $A_j(u)$ may cancel exactly at some positions and the position occupations within the shell need not be uniform. Such a stable superposition requires the corresponding spectral coincidence. Equal response moduli with different complex responses are insufficient for stability.

The symmetries of a general stable pattern are likewise determined by the full superposition. Acting term by term gives

$$\begin{aligned} \iota\psi &= \sum_{j,\theta} a_{j\theta} \theta(-1) f_{j,\theta}, \\ \mathfrak{c}\psi &= \sum_{j,\theta} \overline{a_{j\theta}} f_{j,\bar{\theta}}. \end{aligned} \quad (3.32)$$

By orthogonality of the direction basis,

$$\iota\psi = \epsilon\psi, \quad \epsilon = \pm 1,$$

if and only if the character associated with every nonzero coefficient satisfies

$$\theta(-1) = \epsilon.$$

At a real regular parameter value, conjugacy of responses also gives

$$\mathbf{c}\mathcal{M}_\lambda(s_*) = \mathcal{M}_{\bar{\lambda}}(s_*).$$

The conjugate pattern still satisfies the stability condition. It belongs to the same stable subspace as the original pattern only if λ is real. For the two to have the same pure-state pattern, there must additionally be a real α such that

$$\mathbf{c}\psi = e^{i\alpha}\psi.$$

After re-expansion in the same character basis, the coefficients of $\mathbf{c}\psi$ must equal those of ψ multiplied by one common phase.

Position components and direction components are selected by different projections. Section 2 shows that position projections within a shell generally do not commute with direction projections. A position outcome therefore cannot simply be included among simultaneously definite direction eigenvalues. The distribution and coherence formulas here describe excitation patterns; actual position records and their conditional states must be determined by the corresponding readout action.

Figure 2 shows the relation between the background and stable states in the same space.

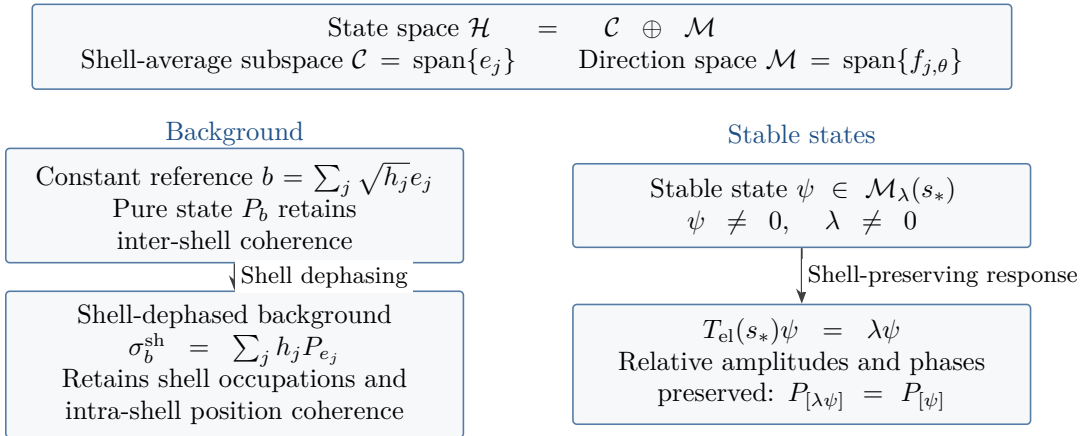


Figure 2: Background and stable states in the state space. The prescribed background density operator may be prepared directly, without first preparing the pure state defined by the constant function b . Stability refers to the specified shell-preserving response. Directions with a common nonzero response may be coherently superposed, including cross-shell components of one character. The complete cycle also includes their free evolution, exchanges, and phase compensation.

§4 Cyclic Evolution

The stable-state criterion describes preservation of an excitation pattern under the shell-preserving response. When an excitation exchanges with the background, the

continuation of its original pattern and the formation of new excitations must be determined by the actual interaction.

These changes involve both the system and the environment participating in the exchange and must be described through their joint evolution. The initial system state specifies the components and their coherence; the environment preparation and interaction jointly determine their subsequent changes.

We use the state space $\mathcal{H} = L^2(\mathbb{Z}/p^K\mathbb{Z})$ and normalized counting inner product of the preceding sections, where p is an odd prime, $K \geq 1$, and $d = p^K$. The shell-average and direction subspaces are $\mathcal{C}$ and $\mathcal{M}$, with projections Π, P . The shell-average states are e_j , the zero-element state is $z = e_K$, and the full set of direction modes is denoted by $|\nu\rangle = f_{j\nu, \theta\nu}$, $P_\nu = |\nu\rangle\langle\nu|$. The shell-energy operator is $H_K = \sum_{j=0}^K E_j \Pi_j$, with $E_j = j \log p$; write $E_\nu = E_{j\nu}$. The unit-group action satisfies $R_u|\nu\rangle = \theta_\nu(u)^{-1}|\nu\rangle$ and $R_u e_j = e_j$. Directions with nonzero response satisfy the stable-state definition in the preceding section; the full space also retains zero-response directions.

Initial state The system enters the cycle in an arbitrary density operator ρ_0 on $\mathcal{H}$, with $\rho_0 \geq 0$ and $\text{Tr} \rho_0 = 1$. Using Eq. (1.27), decompose it as

$$\rho_0 = \Pi \rho_0 \Pi + P \rho_0 P + \Pi \rho_0 P + P \rho_0 \Pi. \quad (4.1)$$

The first two terms give the states within the shell-average and direction subspaces, while the last two retain their mutual coherence. Denote the shares of the two subspaces by

$$q_{\mathcal{C}} = \text{Tr}(\Pi \rho_0), \quad q_{\mathcal{M}} = \text{Tr}(P \rho_0). \quad (4.2)$$

Positivity and $\Pi + P = I$ imply $q_{\mathcal{C}}, q_{\mathcal{M}} \geq 0$ and $q_{\mathcal{C}} + q_{\mathcal{M}} = 1$. When these shares are nonzero, the corresponding normalized density operators are $\rho_{\mathcal{C}} = \Pi \rho_0 \Pi / q_{\mathcal{C}}$ and $\rho_{\mathcal{M}} = P \rho_0 P / q_{\mathcal{M}}$. In general, $\rho_{\mathcal{C}}$ need not equal the chosen restoration background σ_b^{sh} . The latter specifies how the cycle restores the background, whereas ρ_0 specifies the state from which this evolution starts.

For $c \in \mathcal{C}$ and $m \in \mathcal{M}$, the existence of $\rho_0^{1/2}$ and the Cauchy–Schwarz inequality give

$$|\langle c | \rho_0 | m \rangle|^2 \leq \langle c | \rho_0 | c \rangle \langle m | \rho_0 | m \rangle. \quad (4.3)$$

If one subspace has zero share, its positive compression has zero trace and is therefore the zero operator. Equation (4.3) then makes all cross matrix elements vanish. Thus a zero subspace share also implies zero cross coherence. When both shares are nonzero, the initial state may retain cross coherence, and the shares alone do not justify replacing Eq. (4.1) by a classical mixture. Only when the initial state is explicitly chosen without cross coherence and with background preparation σ_b^{sh} do we have

$$\rho_0 = q_{\mathcal{M}} \rho_{\mathcal{M}} + q_{\mathcal{C}} \sigma_b^{\text{sh}}. \quad (4.4)$$

A term with zero share is omitted. For example, a single-direction stable state has $\rho_{\mathcal{M}} = P_\nu$ and $R_{\theta_\nu}(s_0) \neq 0$; general $\rho_{\mathcal{M}}$ also describes multidirection superpositions and mixtures.

Environment preparation Both the background preparation and excitations belong to the system state space; the auxiliary spaces participating in exchanges form the joint space together with the system. The system interacts jointly with the thermal environment $\mathcal{E}_T$ and the auxiliary space $\mathcal{E}_R$ for retrieval and restoration. The thermal environment is prepared in a density operator η_T , with $\eta_T \geq 0$ and $\text{Tr } \eta_T = 1$; the auxiliary system for retrieval and restoration is prepared in the pure state $P_g = |g\rangle\langle g|$.

The shell shares for background restoration are fixed before iteration and written as

$$\sigma_g^{\text{sh}} = \sum_{a=0}^K g_a P_{e_a}, \quad g_a > 0, \quad \sum_a g_a = 1. \quad (4.5)$$

Here g is an auxiliary state, whereas g_a are numerical shares. The shell-dephased background of the preceding section corresponds to $g_a = h_a$, so $\sigma_g^{\text{sh}} = \sigma_b^{\text{sh}}$. Different preparations specify different cycle data; the same g_a are used throughout the entire iterative process.

Tripartite joint states below are always ordered as system, auxiliary system for retrieval and restoration, and thermal environment. The cycle starting from ρ_n uses a fresh preparation independent of the system and all previous environments. The input system–auxiliary state is

$$\Omega_n^{(0)} = \rho_n \otimes P_g \otimes \eta_T. \quad (4.6)$$

If previous environments are retained, the total input state is $\Omega_{S,\text{old}} \otimes P_g \otimes \eta_T$, where $\text{Tr}_{\text{old}} \Omega_{S,\text{old}} = \rho_n$. Existing correlations between the system and previous environments may remain, but those environments no longer participate in this cycle’s interaction. Using an independent fresh environment unit in each cycle gives a repeated-interaction model [13]. Each local joint action is extended by identity factors on the spaces not involved and embedded in the stated tensor order.

The inverse temperature β and within-block shares of the thermal environment, exchange channels, exchange angles, phases, and stage durations are specified in the construction of the respective actions and held fixed in repeated cycles.

§4.1 Free Evolution

Before the system and auxiliary systems exchange, each component evolves according to its own energy. Let the noninteracting total Hamiltonian be $H_0 = H_K + H_R + H_T$, with identity factors inserted in the tensor order above. The auxiliary energies H_R, H_T are assigned according to the exchange blocks below. The energy operators act on different spaces and commute, so joint free evolution factors into three unitary operators. The thermal preparation is diagonal in the energy basis and g is an energy eigenstate; hence, after a duration t_f , the state is

$$\Omega_n^{(1)} = e^{-it_f H_0} \Omega_n^{(0)} e^{it_f H_0}, \quad \rho_n^{(1)} = U_f \rho_n U_f^\dagger, \quad U_f = e^{-it_f H_K}. \quad (4.7)$$

Both auxiliary density operators remain unchanged during this evolution.

For $\psi = \sum_\nu \psi_\nu |\nu\rangle + \sum_j b_j e_j$,

$$U_f \psi = \sum_\nu \psi_\nu e^{-it_f E_\nu} |\nu\rangle + \sum_j b_j e^{-it_f E_j} e_j. \quad (4.8)$$

A single direction acquires only a global phase, components in the same shell acquire a common phase, and relative phases between different shells vary with time. For a general density operator, the inter-shell matrix blocks satisfy

$$\Pi_j \rho_n^{(1)} \Pi_\ell = e^{-it_f(E_j - E_\ell)} \Pi_j \rho_n \Pi_\ell.$$

For the cross-shell stable pattern in Eq. (3.31) of Section 3, free evolution changes the shell coefficients to $a_j e^{-it_f E_j}$. At a specified duration t_f , the pattern retains its original normalized projection if and only if all occupied shells have the same phase factor. Stability under the shell-preserving response and pattern changes under free evolution are thus determined separately.

Free evolution satisfies

$$[U_f, \Pi_j] = [U_f, R_u] = 0, \quad U_f^\dagger U_f = I,$$

and therefore preserves shell occupations, the unit-group structure, and total norm. Since $[U_f, \Pi] = [U_f, P] = 0$, the initial cross block between background and directions evolves as

$$\Pi \rho_n^{(1)} P = U_f (\Pi \rho_n P) U_f^\dagger.$$

Free evolution does not require this coherence to be removed beforehand. The background preparation σ_g^{sh} commutes with H_K and is unchanged during the free stage. Point projections P_x also commute with H_K , giving

$$\text{Tr}(P_x \rho_n^{(1)}) = \text{Tr}(P_x \rho_n).$$

Thus this free action preserves point-basis weights; transfer between excitations and the background must be determined by a joint action.

§4.2 Formation Exchange

Formation exchange determines transfer between excitations and the background, as well as the return of components remaining in the direction space. A specified exchange first couples a direction state to a shell-average state. Direction components are then retrieved coherently through the zero element, and background components are restored according to the prescribed preparation. Each step acts on the same joint state. The requirement that every direction have a positive exchange channel is used only later, in proving strict contraction.

Establishing the exchange The matrix element connecting direction ν and shell-average state e_a is

$$F_{a\nu} = |e_a\rangle\langle\nu|, \quad F_{a\nu}^\dagger = |\nu\rangle\langle e_a|. \quad (4.9)$$

These matrix elements follow from direction selection, character-phase removal, and shell shifts in the finite space. Specifically, for $G = (\mathbb{Z}/p^K\mathbb{Z})^\times$,

$$P_\nu = \frac{\Pi_{j_\nu}}{|G|} \sum_{u \in G} \theta_\nu(u) R_u, \quad M_\nu P_\nu = |e_{j_\nu}\rangle\langle\nu|, \quad (4.10)$$

where M_ν multiplies by $\overline{\theta_\nu(x/p^{j_\nu})}$ on the selected shell and vanishes outside it. Here θ_ν is pulled back to the full unit group G . Since $R_u|\mu\rangle = \theta_\mu(u)^{-1}|\mu\rangle$, the group

average retains only the direction with $\theta_\mu = \theta_\nu$ on the selected shell, proving the first identity. Multiplication by the conjugate character turns the direction function into the normalized constant function on that shell, proving the second. For $Df(x) = f(px)$, finite summation gives $(D^\dagger f)(y) = \sum_{x:px=y} f(x)$, because

$$\langle Df, g \rangle = \frac{1}{d} \sum_y \overline{f(y)} \sum_{x:px=y} g(x).$$

For $0 \leq j \leq K-2$, every point of shell $j+1$ has p preimages in shell j ; the zero element has $p-1$ preimages in shell $K-1$. Thus

$$V_{\text{sh}} = \frac{1}{\sqrt{p}} \sum_{j=0}^{K-2} P_{e_{j+1}} D^\dagger P_{e_j} + \frac{1}{\sqrt{p-1}} P_{e_K} D^\dagger P_{e_{K-1}} = \sum_{j=0}^{K-1} |e_{j+1}\rangle \langle e_j|. \quad (4.11)$$

For $K=1$, the first sum above is empty. Left multiplication of $M_\nu P_\nu$ by the required power of V_{sh} or $V_{\text{sh}}^\dagger$ gives

$$F_{a\nu} = \begin{cases} V_{\text{sh}}^{a-j_\nu} M_\nu P_\nu, & a \geq j_\nu, \\ (V_{\text{sh}}^\dagger)^{j_\nu-a} M_\nu P_\nu, & a < j_\nu. \end{cases}$$

These actions satisfy $[H_K, F_{a\nu}] = (E_a - E_\nu)F_{a\nu}$ and $R_u F_{a\nu} R_u^\dagger = \theta_\nu(u) F_{a\nu}$. These arithmetic factors determine the available matrix elements; actual exchange uses the following joint unitary action.

Choose a channel set $\mathcal{A}$ among all pairs (a, ν) of shell-average states and directions, where $0 \leq a \leq K$ and ν ranges over all direction indices. Assign a positive weight $a_{a\nu}$ to each selected channel and require $a_0 = 1 - \sum_{\mathcal{A}} a_{a\nu} > 0$, with exchange angles $0 < \alpha_{a\nu} < \pi/2$. All channel sums involving exchange weights and angles run only over $\mathcal{A}$. Take the thermal environment space to be the orthogonal direct sum

$$\mathcal{E}_T = \mathbb{C}[*] \oplus \bigoplus_{(a,\nu) \in \mathcal{A}} \text{span}\{|u_{a\nu}\rangle, |v_{a\nu}\rangle\}. \quad (4.12)$$

The auxiliary basis vectors above are orthonormal. An auxiliary basis state $|e\rangle$ has unit-group charge given by the character χ when $R_u^\mathcal{E}|e\rangle = \chi(u)|e\rangle$ for every unit-group element u . Linear extension from the specified orthonormal basis defines the group action on the auxiliary space. Fix a channel, suppress its auxiliary subscripts, and write $\Delta_{a\nu} = E_a - E_\nu$. The states u, v have energies $\max(\Delta_{a\nu}, 0)$ and $\max(-\Delta_{a\nu}, 0)$, and unit-group charges θ_ν and 1, respectively. Thus $|\nu, u\rangle$ and $|e_a, v\rangle$ have equal total energy and total charge 1. Such exchanges act between orthogonal joint subspaces and share the following construction.

Joint Exchange

Let an operator $\mathbf{L}$ on a finite joint space satisfy $\mathbf{L}^\dagger \mathbf{L} = \mathbf{P}_i$, $\mathbf{L} \mathbf{L}^\dagger = \mathbf{P}_f$, where $\mathbf{P}_i, \mathbf{P}_f$ are orthogonal projections and $\mathbf{P}_i \mathbf{P}_f = 0$. For real α, ϑ , set $\mathbf{B}_\vartheta = e^{i\vartheta} \mathbf{L} - e^{-i\vartheta} \mathbf{L}^\dagger$. Then

$$\begin{aligned} \mathbf{U}_{\alpha, \vartheta} &= e^{\alpha \mathbf{B}_\vartheta} \\ &= I + (\cos \alpha - 1)(\mathbf{P}_i + \mathbf{P}_f) + \sin \alpha \mathbf{B}_\vartheta. \end{aligned} \quad (4.13)$$

This action is unitary and is the identity on the orthogonal complement of the two subspaces. If $\mathbf{L}$ commutes with a self-adjoint total quantity or with a joint group action, then so does $\mathbf{U}_{\alpha, \vartheta}$. A concrete cycle requires the exchange matrix element $\mathbf{L}$, angle α , and phase ϑ to be specified.

The partial-isometry relations give $\mathbf{L} = \mathbf{P}_f \mathbf{L} \mathbf{P}_i$, so $\mathbf{L}^2 = (\mathbf{L}^\dagger)^2 = 0$. Consequently, $\mathbf{B}_\vartheta^\dagger = -\mathbf{B}_\vartheta$, $\mathbf{B}_\vartheta^2 = -(\mathbf{P}_i + \mathbf{P}_f)$, and $\mathbf{B}_\vartheta$ vanishes outside the two subspaces. Summing the odd and even terms of the exponential series separately gives Eq. (4.13). Commutation with a self-adjoint total quantity or a unitary group action passes to $\mathbf{L}^\dagger$, $\mathbf{B}_\vartheta$, and its exponential.

For a normalized $\psi \in \text{Ran } \mathbf{P}_i$, this becomes $\mathbf{U}_{\alpha, \vartheta} \psi = \cos \alpha \psi + e^{i\vartheta} \sin \alpha \mathbf{L} \psi$. The terms are orthogonal and have shares $\cos^2 \alpha$ and $\sin^2 \alpha$. Retaining the same auxiliary space and taking another real phase ϑ' also gives

$$\mathbf{U}_{-\pi/2, \vartheta'} \mathbf{U}_{\pi/2, \vartheta} \psi = e^{i(\vartheta - \vartheta')} \psi. \quad (4.14)$$

This first sends ψ to $e^{i\vartheta} \mathbf{L} \psi$ and then retrieves it using $\mathbf{L}^\dagger \mathbf{L} \psi = \psi$.

For thermal exchange, take $\mathbf{L} = F_{a\nu} \otimes |v\rangle\langle u|$ and $\vartheta = 0$, and define

$$B_{a\nu} = F_{a\nu} \otimes |v\rangle\langle u| - F_{a\nu}^\dagger \otimes |u\rangle\langle v|, \quad W_{a\nu} = e^{\alpha_{a\nu} B_{a\nu}}. \quad (4.15)$$

By Eq. (4.13), the complete outputs of the two inputs are

$$\begin{aligned} W_{a\nu} |\nu, u\rangle &= c_{a\nu} |\nu, u\rangle + s_{a\nu} |e_a, v\rangle, \\ W_{a\nu} |e_a, v\rangle &= c_{a\nu} |e_a, v\rangle - s_{a\nu} |\nu, u\rangle, \end{aligned} \quad c_{a\nu} = \cos \alpha_{a\nu}, \quad s_{a\nu} = \sin \alpha_{a\nu}. \quad (4.16)$$

The first identity describes transfer from an excitation to the background; the second describes formation of a directional excitation from the background. The two processes have the same exchange angle and matrix element, with the environment receiving opposite changes in energy and charge.

Each thermal block is prepared in $t_u P_u + t_v P_v$, where

$$t_u = \frac{1}{1 + e^{\beta \Delta_{a\nu}}}, \quad t_v = \frac{e^{\beta \Delta_{a\nu}}}{1 + e^{\beta \Delta_{a\nu}}}, \quad \eta_T = a_0 P_* + \sum_{\mathcal{A}} a_{a\nu} (t_u P_u + t_v P_v). \quad (4.17)$$

The action on $|*\rangle$ is the identity; this state has zero energy and charge 1. Each $W_{a\nu}$ is the identity outside its joint exchange subspace, and different thermal blocks are orthogonal. Hence

$$U_{\text{th}} = \prod_{(a, \nu) \in \mathcal{A}} W_{a\nu} = \exp \left(\sum_{(a, \nu) \in \mathcal{A}} \alpha_{a\nu} B_{a\nu} \right).$$

This joint action preserves norm, total energy, and the total unit-group action. Weights of different thermal blocks belong to the specified preparation; Eq. (4.17) has the same inverse temperature within each block.

Retrieval through the zero element After thermal exchange, components remaining in the direction space undergo a coherent round trip through the zero element with the same auxiliary system. The input state g of the retrieval auxiliary has energy E_K and charge 1. For each direction, introduce an auxiliary state a_ν , forming an orthonormal family together with g . Each a_ν has energy E_ν and charge θ_ν^{-1} . Write $I_\nu = |\nu, g\rangle$ and $Z_\nu = |z, a_\nu\rangle$. Both have total energy $E_\nu + E_K$ and total unit-group charge θ_ν^{-1} . The two vector families are orthonormal internally and mutually orthogonal. Given $|w_\nu| = 1$, set

$$B(w) = \sum_{\nu} (w_\nu |Z_\nu\rangle\langle I_\nu| - \bar{w}_\nu |I_\nu\rangle\langle Z_\nu|), \quad U_{\text{out}} = e^{\pi B(w)/2}, \quad U_{\text{back}} = e^{-\pi B(1)/2}. \quad (4.18)$$

On each orthogonal joint block, taking in Eq. (4.14) $e^{i\vartheta} = w_\nu$, $\vartheta' = 0$, gives $I_\nu \mapsto w_\nu Z_\nu \mapsto w_\nu I_\nu$. Products of shell-average states with g lie in the orthogonal complement and remain unchanged. Thus, for any input,

$$U_{\text{back}} U_{\text{out}}(\psi \otimes g) = \left(\Pi + \sum_\nu w_\nu P_\nu \right) \psi \otimes g. \quad (4.19)$$

For a general input, the first action gives

$$U_{\text{out}}(\psi \otimes g) = \Pi \psi \otimes g + z \otimes \sum_\nu w_\nu \langle \nu | \psi \rangle a_\nu.$$

During transfer to the zero element, direction coefficients are stored in $\sum_\nu \psi_\nu w_\nu a_\nu$. After the return, they again become system direction coefficients, and the auxiliary system returns to the common state g . This round trip preserves the direction coherence present on entry: the matrix element $|\nu\rangle\langle\mu|$ acquires the factor $w_\nu \bar{w}_\mu$. No auxiliary readout or reparation is inserted between the two actions. Tracing out the auxiliary system between the exchanges gives the same reduced system state P_z for every direction input. Retrieval of the original direction still depends on the states a_ν and their coherence, so the same auxiliary system is used for both exchanges. For a system input Ω_{ST} correlated with the thermal environment, the same round trip gives $(U_w \otimes I_T) \Omega_{ST} (U_w^\dagger \otimes I_T)$, with the retrieval auxiliary returning to an independent P_g , where $U_w = \Pi + \sum_\nu w_\nu P_\nu$. The returning component of a single direction preserves its pattern; a general superposition acquires relative phases, so its reconstructed pattern may change. The auxiliary system is retained throughout the two actions. Retrieval cannot be calculated by replacing the full joint state with the intermediate reduced system state.

Background restoration Components transferred to the background continue through the cycle. Introduce orthonormal auxiliary states r_{ab} , orthogonal to g and all a_ν , with energy $E_K + E_b - E_a \geq 0$ and charge 1. Set

$$Y_b = \sum_a \sqrt{g_a} |e_a, r_{ab}\rangle. \quad (4.20)$$

Since $\sum_a g_a = 1$ and the auxiliary states are orthogonal, $\langle Y_b, Y_c \rangle = \delta_{bc}$. The vectors Y_b and $|e_b, g\rangle$ have the same total energy and charge, and the two families are mutually orthogonal. With $\mathbf{L} = \sum_b |Y_b\rangle\langle e_b, g|$, we have

$$\mathbf{L}^\dagger \mathbf{L} = \Pi \otimes P_g, \quad \mathbf{L} \mathbf{L}^\dagger = \sum_b P_{Y_b}, \quad (\Pi \otimes P_g) \sum_b P_{Y_b} = 0.$$

These orthogonality relations meet the conditions of Eq. (4.13). Taking $\alpha = \pi/2$ and $\vartheta = 0$ gives a joint unitary U_{res} that preserves direction components and satisfies

$$U_{\text{res}} |e_b, g\rangle = Y_b, \quad U_{\text{res}} |\nu, g\rangle = |\nu, g\rangle. \quad (4.21)$$

Its anti-self-adjoint generator is $\mathbf{L} - \mathbf{L}^\dagger$. Auxiliary orthogonality further gives

$$\text{Tr}_R |Y_b\rangle\langle Y_c| = \delta_{bc} \sigma_g^{\text{sh}}.$$

Thus every shell-average input is restored, after reduction, to the specified background with its total share; distinctions between input components remain in the joint final state. Composing retrieval and restoration gives the isometry

$$S\psi = \sum_{\nu} w_{\nu} P_{\nu} \psi \otimes g + \sum_{a,b} \sqrt{g_a} |e_a\rangle \langle e_b, \psi\rangle \otimes r_{ab}, \quad S^{\dagger} S = P + \Pi = I. \quad (4.22)$$

Write $U_w = \Pi + \sum_{\nu} w_{\nu} P_{\nu}$. Since g is orthogonal to every r_{ab} , cross terms between direction and shell-average components vanish upon tracing out the restoration auxiliary. Using $\langle r_{ab}, r_{a'b'} \rangle = \delta_{aa'} \delta_{bb'}$ then gives

$$\text{Tr}_R(S\rho S^{\dagger}) = U_w P \rho P U_w^{\dagger} + \text{Tr}(\Pi \rho) \sigma_g^{\text{sh}}. \quad (4.23)$$

This formula holds for any density operator. For an input correlated with the thermal environment, extend restoration by the identity on that environment and trace out all auxiliary systems. The same system marginal is obtained, although the system and thermal environment need not form a product state. Their correlations remain in the full joint state.

Energy and charge matching at each exchange can be listed directly as

Coupled joint vectors	Common total energy	Common total unit-group charge
$ \nu, u\rangle, e_a, v\rangle$	$\max(E_{\nu}, E_a)$	1
$ \nu, g\rangle, z, a_{\nu}\rangle$	$E_{\nu} + E_K$	θ_{ν}^{-1}
$ e_b, g\rangle, Y_b$	$E_b + E_K$	1

The first row uses the assigned thermal-environment energies; every component of Y_b in the third row has the same total quantities. By Eq. (4.13), these equalities extend to the complete rotations.

Destinations of exchanged components Starting with the two inputs of a thermal block, and applying thermal exchange, transfer and retrieval, and background restoration in that order, gives

$$\begin{aligned} |\nu, g, u\rangle &\mapsto c_{a\nu} w_{\nu} |\nu, g, u\rangle + s_{a\nu} \sum_b \sqrt{g_b} |e_b, r_{ba}, v\rangle, \\ |e_a, g, v\rangle &\mapsto c_{a\nu} \sum_b \sqrt{g_b} |e_b, r_{ba}, v\rangle - s_{a\nu} w_{\nu} |\nu, g, u\rangle. \end{aligned} \quad (4.24)$$

The two terms in each expression have squared norms $c_{a\nu}^2, s_{a\nu}^2$, which sum to one. In r_{ba} , the first index denotes the shell after restoration and the second the shell entering restoration. These shares are conditional on the specified joint input. Including the full thermal preparation requires multiplication by $a_{a\nu} t_u$ or $a_{a\nu} t_v$, respectively. For an original excitation input, the component returning to the direction space in the cycle's final state represents continuation of that excitation. The component transferred to the background represents its **decay**; the corresponding transfer is called **exit**. For a background input, the component transferred to the direction space and entering pattern reconstruction represents formation of a new excitation. Table 2 lists the subsequent actions on the four components. They belong to the same joint final state; events have not yet been defined by pointer outcomes.

Table 2: Destinations of components in formation exchange. Retrieval is the identity on shell-average components; restoration is the identity on direction components that have returned.

Exchange input	Component after thermal exchange	Subsequent action	Meaning at cycle end
Directional excitation	Direction component	Retrieval through zero	Continuation of original excitation
Directional excitation	Shell-average component	Background restoration	Decay of original excitation (exit)
Shell-average state	Direction component	Retrieval through zero	Formation of new excitation
Shell-average state	Shell-average component	Background restoration	Continuation of background

After retrieval or background restoration, all components enter pattern reconstruction and then the next cycle. Continuation in the table means that the component remains in the direction space; conditions for preservation of its full pattern are given in Section 4.4. Reverse formation uses the same thermal rotation, while thermalization over multiple cycles is determined by the complete iteration and its stationarity conditions.

Completeness for a general input can be checked directly. Set $F = F_{av}$, $c = c_{av}$, $s = s_{av}$, $D_u = I - (1 - c)P_\nu$, and $D_v = I - (1 - c)P_{e_a}$. Use the tensor order system, restoration auxiliary, thermal environment. For any normalized ψ ,

$$\begin{aligned}
\Xi_* &= S\psi \otimes |*\rangle, \\
\Xi_u &= SD_u\psi \otimes |u\rangle + sSF\psi \otimes |v\rangle, \\
\Xi_v &= -sSF^\dagger\psi \otimes |u\rangle + SD_v\psi \otimes |v\rangle.
\end{aligned} \tag{4.25}$$

The identities $D_u^2 + s^2P_\nu = D_v^2 + s^2P_{e_a} = I$ and $S^\dagger S = I$ show that every output vector is normalized. Amplitudes within each pure input combine coherently; the actual mixed preparation gives the density operator $a_0P_{\Xi_*} + \sum_{\mathcal{A}} a_{av}(t_uP_{\Xi_u} + t_vP_{\Xi_v})$. For a general mixed input $\rho = \sum_k p_k P_{\psi_k}$, sum these joint density operators with weights p_k . This expression includes all directions, background components, and cross coherence without prematurely replacing amplitudes within a pure component by a mixture.

Cycle phase The exchanges above determine transfer and retrieval of the components; the net phase of the coherent round trip must also be specified as part of the cycle data. Given the spectral responses of Section 3, we adopt the following phase assignment. A continuous reading additive under multiplication of positive real amplitudes satisfies $\ell(ab) = \ell(a) + \ell(b)$. Set $\tilde{\ell}(u) = \ell(e^u)$. Then $\tilde{\ell}(u + w) = \tilde{\ell}(u) + \tilde{\ell}(w)$; integer multiples, rational multiples, and continuity successively give $\tilde{\ell}(u) = c_\ell u$, and hence $\ell(a) = c_\ell \log a$. This property fixes the logarithmic form of a continuous multiplicatively additive reading; the unit scale and phase orientation below remain specified by the cycle construction.

The spectral reference $s_0 = 1/2$ is the normalization choice in Eq. (3.19). For $0 < r_\theta = |R_\theta(s_0)| < \infty$, let $P_+ = \sum_{\nu: R_{\theta_\nu} \neq 0} P_\nu$ project onto all nonzero-response

directions. On its range, define the positive operator

$$|T_{\text{el}}| = \sum_{\nu: R_{\theta_\nu} \neq 0} r_{\theta_\nu} P_\nu, \quad F_{\text{el}} = \sum_{\nu: R_{\theta_\nu} \neq 0} \min(r_{\theta_\nu}, r_{\theta_\nu}^{-1}) P_\nu. \quad (4.26)$$

On $P_+ \mathcal{H}$, we have $0 < F_{\text{el}} \leq I$, so its matrix logarithm exists. Calculation on each eigenvalue gives

$$-\log F_{\text{el}} = \sum_{\nu: R_{\theta_\nu} \neq 0} |\log r_{\theta_\nu}| P_\nu. \quad (4.27)$$

We adopt this log modulus as the absolute value of the cycle phase and fix an orientation $\varsigma_\theta \in \{1, -1\}$ for each θ , obtaining

$$\varphi_\nu = \varsigma_{\theta_\nu} |\log r_{\theta_\nu}|, \quad U_\varphi = \Pi + \sum_{\nu} e^{i\varphi_\nu} P_\nu. \quad (4.28)$$

For zero-response directions, set $\varphi_\nu = 0$, so that U_φ is defined on all of $\mathcal{H}$. This zero phase is an extension convention; Eq. (4.27) is used only on $P_+ \mathcal{H}$. The spectral reference, log-modulus phase assignment and its unit scale, phase orientations, and zero-response extension together specify the phase data; none is determined by the definition of the finite state space alone. Equation (4.27) gives the spectral expression for the log modulus, whereas Eq. (4.28) assigns it as a cycle phase. The frequency readout of each direction also depends on the actual cycle period; correspondence between phase and exchange energy must satisfy the compatibility condition in Eq. (8.14). The orthogonal projection decomposition gives $U_\varphi^\dagger U_\varphi = \Pi + P = I$.

By Eq. (3.27), conjugate characters have conjugate resonance ratios at real spectral parameters, and hence equal moduli. For two distinct conjugate directions, opposite choices of ς_θ may be used. The phase orientation of a self-conjugate direction is specified separately, without imposing the opposite-sign condition for conjugate pairs. Every character mode acquires only a global phase under U_φ and preserves its pattern. A superposition of modes preserves its pattern under this action if and only if all occupied modes have the same phase factor.

Here s_0 is the spectral reference; the inverse temperature of actual thermal exchange is denoted separately by β . They enter the spectral response and environment preparation, respectively.

The joint actions above are written in the interaction picture. The four durations are $t_T, t_{\text{out}}, t_{\text{back}}, t_{\text{res}}$. Write $\tau = t_f + t_T + t_{\text{out}} + t_{\text{back}} + t_{\text{res}}$. Each action $e^{\theta B}$ generated by an anti-self-adjoint operator above can be implemented over a prescribed positive duration t by the self-adjoint interaction $H_{\text{int}} = i\theta B/t$. All generators commute with H_0 , so the joint action of the complete cycle without control is

$$\mathbb{U}_{\text{cyc}} = e^{-i\tau H_0} U_{\text{res}} U_{\text{back}} U_{\text{out}} U_{\text{th}}. \quad (4.29)$$

Applying the common free factor to all final states above restores the actual joint evolution and its full phases. To prescribe the net direction phase per cycle as φ_ν in Eq. (4.28), take

$$w_\nu = e^{i(\varphi_\nu + \tau E_\nu)}, \quad U_\varphi = \Pi + \sum_{\nu} e^{i\varphi_\nu} P_\nu. \quad (4.30)$$

Calculation on each direction gives $e^{-i\tau H_K} U_w P = U_\varphi P$. The restored background σ_g^{sh} commutes with H_K , while the free phase of a returning direction combines with w_ν to give $e^{i\varphi_\nu}$. This compensation fixes the net phase of system direction components; all auxiliary free phases remain in the full joint final state.

Orientation and displacement Retrieval specifies the return phase of each direction. Adding shell-preserving input control V at the entrance to formation exchange can produce position changes of these returning components within the same shell. The orientation structure of the control, its coherent implementation, and its reference preparation jointly determine the action. Fix a shell and an orientation class in the same finite state space, and construct the action as follows. Take $p \geq 5$, $j < K$, $q = p - 1$, and $L = p^{K-j-1}$. On this shell, the unit group decomposes using a Teichmüller generator ζ and $h = 1 + p$, and positions are written as

$$x(s, \ell) = p^j \zeta^s h^\ell, \quad s \in \mathbb{Z}_q, \quad \ell \in \mathbb{Z}_L. \quad (4.31)$$

Choose $2a \not\equiv 0 \pmod{q}$ and define

$$|\pm, \ell\rangle = \frac{1}{\sqrt{q}} \sum_s e^{\pm 2\pi i a s / q} |x(s, \ell)\rangle, \quad \mathcal{H}_a = \text{span}\{|\pm, \ell\rangle\} \simeq \mathbb{C}^2 \otimes \mathbb{C}^L \subset \mathcal{M}. \quad (4.32)$$

Point sets with different ℓ are disjoint. At the same ℓ , the inner product of the two orientations is $q^{-1} \sum_s e^{-4\pi i a s / q} = 0$; the amplitudes sum to zero over the torsion coordinate, giving orthogonality to the shell-average state. The space above thus has dimension $2L$. Let $Q_a, Q_\pm$ be the corresponding projections. For $L = 1$, there is no nonzero principal-unit displacement.

Multiplicative translation satisfies $R_h |\pm, \ell\rangle = |\pm, \ell + 1\rangle$. Define an action translating opposite orientations in opposite directions by

$$T = Q_+ R_h + Q_- R_h^\dagger + (I - Q_a). \quad (4.33)$$

Each orthogonal block is unitary, so $T^\dagger T = I$ and $[T, H_K] = 0$. Set $B = 2^{-1/2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \otimes I_L$, and define coherent mixing $C = B U_\varphi B^\dagger$ and the propagation operator $W = CT$ on $\mathcal{H}_a$, extending both by the identity on its orthogonal complement. The principal-unit Fourier modes are

$$|\pm; r\rangle = L^{-1/2} \sum_\ell e^{i\kappa_r \ell} |\pm, \ell\rangle, \quad \kappa_r = 2\pi r / L. \quad (4.34)$$

The two full characters take the values $e^{\pm 2\pi i a s / q + i\kappa_r \ell}$ at $x(s, \ell) / p^j$; full complex conjugation sends (a, r) to $(-a, -r)$. Thus the two orientations at fixed r generally do not form a conjugate pair of full characters. The fixed object here is the conjugate torsion class $\{a, -a\}$, with all principal-unit position modes retained. Lifted states depending only on reduction modulo p correspond to $r = 0$ and have a uniform principal-unit position distribution. The involutions from Section 2 act on this space as

$$\iota|_{\mathcal{H}_a} = (-1)^a I, \quad J_T|_{\mathcal{H}_a} = \sigma_1 \otimes I_L, \quad J_P |\pm; r\rangle = |\pm; -r\rangle. \quad (4.35)$$

The first identity follows by shifting the torsion position by $q/2$; the other two reverse the signs of torsion and principal-unit positions, respectively. Here σ_1 is the Pauli matrix exchanging the two orientations, and ι acts as a scalar on this space. Denote the prescribed phases of the full characters by $\varphi_{\pm, r}$ and set $\alpha_r = (\varphi_{+, r} + \varphi_{-, r})/2$ and $\delta_r = (\varphi_{+, r} - \varphi_{-, r})/2$. Then

$$C_r = e^{i\alpha_r} e^{i\delta_r \sigma_1}, \quad W_r = C_r e^{-i\kappa_r \sigma_3}, \quad \cos \vartheta_r = \cos \delta_r \cos \kappa_r. \quad (4.36)$$

Here σ_1, σ_3 are Pauli matrices on the two orientations. The last relation follows because $e^{-i\alpha_r} W_r$ has determinant one and trace $2 \cos \delta_r \cos \kappa_r$; the eigenvalues of W_r are $e^{i(\alpha_r \pm \vartheta_r)}$. This specifies the cycle-phase relation of position modes within the same orientation class.

To implement this propagation at the entrance to formation exchange, prescribe the shell-preserving input control

$$V|_{\mathcal{H}_a} = U_\varphi^\dagger W|_{\mathcal{H}_a}, \quad V|_{\mathcal{H}_a^\perp} = I. \quad (4.37)$$

It preserves $\mathcal{C}$ and $\mathcal{M}$ and is the identity on $\mathcal{C}$. It also satisfies $[V, H_K] = 0$ and $U_\varphi V Q_a = W Q_a$. This relation is restricted to $\mathcal{H}_a$; other components of the complete cycle are still determined by their respective final states. Control is applied after free evolution and before thermal exchange.

Coherent round trip for motion The control is implemented coherently by a round trip through the zero element. Take the $2L$ orthonormal eigenvectors of V on $\mathcal{H}_a$, satisfying $V w_\lambda = e^{i\theta_\lambda} w_\lambda$. The input auxiliary state $|\text{in}\rangle$ has energy $E_K - E_j$, and the orthogonal states $|\lambda\rangle$ have zero energy. Set

$$A_\lambda = |z, \lambda\rangle \langle w_\lambda, \text{in}|, \quad \mathbf{P}_d = Q_a \otimes P_{\text{in}}, \quad \mathbf{P}_z = P_z \otimes \sum_\lambda P_\lambda. \quad (4.38)$$

The initial and final spaces of each A_λ are orthogonal, and the joint blocks for distinct λ are mutually orthogonal. The joint bare energy $H_{\text{bare}} = H_K \otimes I + I \otimes H_A$ equals $E_K I$ on both spaces, so $[A_\lambda, H_{\text{bare}}] = 0$. Fix $g_V > 0$ and define, with control phases η_λ ,

$$H(\eta) = g_V \sum_\lambda (e^{i\eta_\lambda} A_\lambda + e^{-i\eta_\lambda} A_\lambda^\dagger), \quad \tau_{\text{ex}} = \frac{\pi}{2g_V}. \quad (4.39)$$

On each joint block, substituting $\mathbf{L} = A_\lambda$ and $\vartheta = \eta_\lambda - \pi/2$ into Eq. (4.13) gives the actual pulse $U(\eta) = e^{-i\tau_{\text{ex}} H(\eta)}$. Each joint block satisfies $U(\eta)|w_\lambda, \text{in}\rangle = -ie^{i\eta_\lambda}|z, \lambda\rangle$ and $U(\eta)|z, \lambda\rangle = -ie^{-i\eta_\lambda}|w_\lambda, \text{in}\rangle$. Both actions last a positive duration τ_{ex} , with phases 0 and $\pi - \theta_\lambda$ in that order, yielding

$$|w_\lambda, \text{in}\rangle \mapsto -i|z, \lambda\rangle \mapsto e^{i\theta_\lambda}|w_\lambda, \text{in}\rangle. \quad (4.40)$$

The transfer coefficients multiply to $(-i)^2 e^{-i(\pi - \theta_\lambda)} = e^{i\theta_\lambda}$. For a general superposition, the first action transfers the direction coefficients into a single auxiliary vector and the second retrieves them coherently. Linearity gives

$$U(\eta^{(2)})U(\eta^{(1)})(\psi \otimes |\text{in}\rangle) = V\psi \otimes |\text{in}\rangle, \quad \psi \in \mathcal{H}_a. \quad (4.41)$$

For $\psi \perp \mathcal{H}_a$, both pulses leave $\psi \otimes |\text{in}\rangle$ unchanged. The construction therefore implements V on the full space and also applies to inputs correlated with a reference system that does not participate in the action. The auxiliary system is retained between exchanges and returns to the same input state; its final state therefore does not distinguish system directions. Reading out or refreshing this auxiliary system between the actions changes the coherent round trip. The calculation uses only the orthonormal eigenbasis and phases of V , and remains valid when V is replaced by any unitary operator Y on $\mathcal{H}_a$.

Including free evolution during $t_V = 2\tau_{\text{ex}} = \pi/g_V$, the actual joint output is

$$e^{-it_V H_{\text{bare}}}(V\psi \otimes |\text{in}\rangle) = e^{-it_V H_K} V\psi \otimes e^{-it_V (E_K - E_j)} |\text{in}\rangle.$$

The auxiliary density operator returns to P_{in} . After the system completes the free stage and then the two pulses, its state at the entrance to thermal exchange is

$$e^{-i(t_f+t_V)H_K} V\rho_n V^\dagger e^{i(t_f+t_V)H_K}.$$

Here $[V, H_K] = 0$ combines the free phases. The time spent on control must be included in the total cycle period.

Reference preparation Orientation mixing also requires the corresponding unit-group reference. Suppose the auxiliary preparation η_A is unit-group invariant and the joint U preserves the total system–auxiliary group action. Write $\mathcal{G}_U(\rho) = \text{Tr}_A[U(\rho \otimes \eta_A)U^\dagger]$. Conjugating the joint input by $R_u \otimes R_u^A$, using invariance of the auxiliary preparation and joint commutation, and tracing out the auxiliary system gives covariance:

$$\mathcal{G}_U(R_u \rho R_u^\dagger) = R_u \mathcal{G}_U(\rho) R_u^\dagger. \quad (4.42)$$

In a fixed r block, $R_\zeta = e^{-2\pi i a \sigma_3/q}$. When $\sin \delta_r \cos \delta_r \neq 0$, C_r maps the group-invariant input density operator $|+; r\rangle\langle +; r|$ to a state with coherence between two distinct torsion charges, since $C_r|+; r\rangle = e^{i\alpha_r}(\cos \delta_r|+; r\rangle + i \sin \delta_r| -; r\rangle)$, whose cross matrix element is $-i \cos \delta_r \sin \delta_r|+; r\rangle\langle -; r|$. For $2a \not\equiv 0 \pmod{q}$, this term changes under conjugation by R_ζ , so the output is no longer invariant under that group action. Both T and U_φ commute with the unit group. Thus the actual control $V = U_\varphi^\dagger C T$ also requires an asymmetric reference preparation at these angles. Covariance at special angles must be checked from the actual matrices.

Take the finite group reference $\mathcal{H}_{\text{ref}} = \ell^2(G)$, where $G = (\mathbb{Z}/d\mathbb{Z})^\times$, with zero energy, input state $|1_G\rangle$, and group action $L_v|u\rangle = |vu\rangle$. Use the trivial group action on the control auxiliary and write $D_u = R_u \otimes I_A$. For any operator X on the system and control auxiliary, define

$$\mathfrak{L}(X) = \sum_{u \in G} D_u X D_u^\dagger \otimes |u\rangle\langle u|, \quad \tilde{H} = \mathfrak{L}(H), \quad \tilde{U} = \mathfrak{L}(U). \quad (4.43)$$

Multiplication of orthogonal blocks gives $\mathfrak{L}(XY) = \mathfrak{L}(X)\mathfrak{L}(Y)$, $\mathfrak{L}(X^\dagger) = \mathfrak{L}(X)^\dagger$, and $\mathfrak{L}(I) = I$. Unitary operators and complete orthogonal projection families therefore map to objects of the same type, and $\mathfrak{L}(e^{-itH}) = e^{-it\mathfrak{L}(H)}$. Conjugation by $D_v \otimes L_v$ replaces the index u by vu , so every $\mathfrak{L}(X)$ commutes with the total group action. If X commutes with the joint bare energy, energy preservation by R_u also makes $\mathfrak{L}(X)$ energy preserving. For any joint vector ξ of the system and control auxiliary,

$$\tilde{U}(\xi \otimes |1_G\rangle) = U\xi \otimes |1_G\rangle. \quad (4.44)$$

The same reference returns after each stage and each cycle. Its initial state P_{1_G} is a unit-group-asymmetric pure state, whereas the Gibbs state of the zero-energy space is $I_{\text{ref}}/|G|$. Using the latter and discarding the reference gives

$$\frac{1}{|G|} \sum_{u \in G} D_u U D_u^\dagger \rho D_u U^\dagger D_u^\dagger, \quad (4.45)$$

where ρ is the joint input of the system and control auxiliary. This action generally differs from $U\rho U^\dagger$. Thus this implementation explicitly uses the asymmetric preparation $|1_G\rangle$; pulse couplings, phases, and start and end times remain prescribed data.

Complete cycle Free evolution, formation exchange, and background restoration have been specified on the system and auxiliary spaces. Composing them in chronological order gives the complete evolution from one cycle's initial state to the next. For a cycle with input control, the two control pulses act after free evolution and before thermal exchange, with total duration $t_V = \pi/g_V$.

Using the tensor order system, restoration auxiliary, thermal environment, control auxiliary, and group reference, denote the full auxiliary space by

$$\mathcal{E} = \mathcal{E}_R \otimes \mathcal{E}_T \otimes \mathcal{E}_A \otimes \mathcal{H}_{\text{ref}}, \quad \eta_{\text{full}} = P_g \otimes \eta_T \otimes P_{\text{in}} \otimes P_{1_G}.$$

The thermal environment and restoration auxiliary are prepared as above; the control auxiliary and group reference return to their respective preparations after the coherent round trip. Insert identity factors into the energy operators in this tensor order. The group reference has zero energy, and the noninteracting total energy on the enlarged space is $H_{\text{tot}} = H_K + H_R + H_T + H_A$. Write $t_X = t_T + t_{\text{out}} + t_{\text{back}} + t_{\text{res}}$ and define on the same joint space

$$\begin{aligned} \mathbb{F}_f &= e^{-it_f H_{\text{tot}}}, \\ \mathbb{C}_V &= e^{-it_V H_{\text{tot}}} \tilde{U}_{V,2} \tilde{U}_{V,1}, \\ \mathbb{X} &= e^{-it_X H_{\text{tot}}} U_{\text{res}} U_{\text{back}} U_{\text{out}} U_{\text{th}}. \end{aligned}$$

Here $\tilde{U}_{V,1}, \tilde{U}_{V,2}$ are the group-reference extensions of the two control pulses above. The actual time order is $\mathbb{F}_f \rightarrow \mathbb{C}_V \rightarrow \mathbb{X}$, with joint operator $\mathbb{U}_{\text{tot}} = \mathbb{X} \mathbb{C}_V \mathbb{F}_f$. Each interaction preserves H_{tot} . Combining the free factors gives

$$\begin{aligned} \mathbb{U}_{\text{tot}} &= e^{-i\tau_{\text{tot}} H_{\text{tot}}} U_{\text{res}} U_{\text{back}} U_{\text{out}} U_{\text{th}} \tilde{U}_{V,2} \tilde{U}_{V,1}, \\ \tau_{\text{tot}} &= t_f + t_V + t_T + t_{\text{out}} + t_{\text{back}} + t_{\text{res}} = \tau + t_V. \end{aligned} \tag{4.46}$$

This combination retains the order of the interactions. In particular, $[\mathbb{C}_V, \mathbb{F}_f] = 0$, so $\mathbb{X} \mathbb{C}_V \mathbb{F}_f = \mathbb{X} \mathbb{F}_f \mathbb{C}_V$. The equality gives the same cycle-end state for the fixed preparation; the intra-stage evolution under each arrangement is still determined by its actual time order.

The control duration also contributes to the free phase, so Eq. (4.30) becomes

$$w_\nu = e^{i(\varphi_\nu + \tau_{\text{tot}} E_\nu)}, \quad e^{-i\tau_{\text{tot}} E_\nu} w_\nu = e^{i\varphi_\nu}.$$

This compensation fixes the net phase of system direction components, while auxiliary phases remain in the full free factor. Equation (4.25) gives the joint outputs $\Xi_\kappa(\psi)$ associated with the thermal-environment input labels κ . After fixing their retrieval phases using the total duration above, the complete cycle has pure-state output

$$\hat{\Xi}_\kappa(\psi) = e^{-i\tau_{\text{tot}} H_{\text{tot}}} [\Xi_\kappa(V\psi) \otimes |\text{in}\rangle \otimes |1_G\rangle].$$

This expression includes continuation and decay of the original excitation, formation of new excitations, and restoration of background components. All processes remain in the same joint final state. For an arbitrary initial system state ρ_n , write uniformly

$$\Omega_n^{\text{out}} = \mathbb{U}_{\text{tot}}(\rho_n \otimes \eta_{\text{full}})\mathbb{U}_{\text{tot}}^\dagger, \quad \rho_{n+1} = \text{Tr}_{\mathcal{E}} \Omega_n^{\text{out}}. \quad (4.47)$$

This also applies when the system is already correlated with previous environments: the fresh auxiliary preparation is independent of the system and all previous environments jointly, and previous environments no longer participate, so their partial trace commutes with this cycle's evolution. Without input control, omit the control auxiliary and group reference and set $V = I$, $t_V = 0$, recovering Eq. (4.29).

§4.3 Pattern Reconstruction

Returning excitations, newly formed excitations, and restored background components jointly constitute the pattern at the end of the cycle. Expanding the joint final state in the point basis gives their amplitudes and correlations with auxiliary states. Pattern reconstruction is the point-basis representation of the same joint final state; it introduces no additional interaction or duration.

Abbreviate the direction functions as $f_\nu = f_{j_\nu, \theta_\nu}$. The shell-average states and all directions form the orthonormal basis $\{q_\ell\} = \{e_j\} \cup \{f_\nu\}$. The normalized point basis is $|x\rangle = \sqrt{d} \mathbf{1}_{\{x\}}$, where $x \in X_{p,K}$. The change-of-basis matrix is

$$A_{x\ell} = \langle x | q_\ell \rangle = \frac{q_\ell(x)}{\sqrt{d}}, \quad (A^\dagger A)_{\ell m} = \frac{1}{d} \sum_x \overline{q_\ell(x)} q_m(x) = \delta_{\ell m}.$$

Both bases have dimension d , so A is unitary. This transformation preserves inner products and normalization, expressing the shell and direction expansion as point amplitudes.

First consider a normalized pure joint final state, written as

$$\begin{aligned} |\Psi_{\text{out}}\rangle &= \sum_j |e_j\rangle \otimes \mathbf{b}_j + \sum_\nu |\nu\rangle \otimes \mathbf{c}_\nu, \\ \Psi_{\text{out}}(x) &= \sum_j e_j(x) \mathbf{b}_j + \sum_\nu f_\nu(x) \mathbf{c}_\nu. \end{aligned} \quad (4.48)$$

The auxiliary vectors $\mathbf{b}_j, \mathbf{c}_\nu \in \mathcal{E}$ are determined componentwise by the complete cycle and already include the amplitudes; they need not be individually normalized or mutually orthogonal. The vector-valued amplitude $\Psi_{\text{out}}(x)$ at point x satisfies

$$|\Psi_{\text{out}}\rangle = \frac{1}{\sqrt{d}} \sum_x |x\rangle \otimes \Psi_{\text{out}}(x).$$

Orthogonality of shell-average states and direction functions gives

$$\frac{1}{d} \sum_x \|\Psi_{\text{out}}(x)\|^2 = \sum_j \|\mathbf{b}_j\|^2 + \sum_\nu \|\mathbf{c}_\nu\|^2 = 1. \quad (4.49)$$

Cross terms here cancel after summation over all points; at an individual point, the pattern still combines the actual amplitudes and overlaps of auxiliary states. When

the system and auxiliary space are entangled, the system is described by a reduced density operator, with coherence determined by reduction of the joint final state.

Environment Overlap and Final-State Reduction

Let the full joint final state be Ω , with system state $\rho_S = \text{Tr}_{\mathcal{E}} \Omega$. For a pure state $\Omega = P_{\Psi}$, write in any orthonormal system basis $\Psi = \sum_i |i\rangle \otimes \eta_i$, with the expansion amplitudes included in η_i . Then

$$\langle i | \rho_S | j \rangle = \langle \eta_j, \eta_i \rangle. \quad (4.50)$$

The reduced state of a mixed joint state is the weighted sum of the reduced density operators of its pure-state components.

Expanding the projection operator gives

$$P_{\Psi} = \sum_{i,j} |i\rangle \langle j| \otimes |\eta_i\rangle \langle \eta_j|.$$

Since $\text{Tr}_{\mathcal{E}}(|\eta_i\rangle \langle \eta_j|) = \langle \eta_j, \eta_i \rangle$, partial trace gives Eq. (4.50); the mixed-state statement follows by linearity. Applied to the point basis, this yields

$$\langle x | \rho_{\text{out}} | y \rangle = \frac{1}{d} \langle \Psi_{\text{out}}(y), \Psi_{\text{out}}(x) \rangle, \quad \rho_{\text{out}} = \text{Tr}_{\mathcal{E}} P_{\Psi_{\text{out}}}. \quad (4.51)$$

In particular, the point weights are

$$p_{\text{out}}(x) = \langle x | \rho_{\text{out}} | x \rangle = \frac{1}{d} \|\Psi_{\text{out}}(x)\|^2, \quad \sum_x p_{\text{out}}(x) = 1.$$

Off-diagonal entries retain coherence between points; point weights alone do not determine the full system state.

Write each nonzero auxiliary component as $\eta_i = A_i |\varepsilon_i\rangle$, with $\|\varepsilon_i\| = 1$. Then

$$\langle i | \rho_S | j \rangle = A_i \overline{A_j} \langle \varepsilon_j, \varepsilon_i \rangle.$$

Identical auxiliary states retain full coherence between components; orthogonal auxiliary states remove the corresponding cross terms from the reduced system state. For partial overlap, the actual inner product remains as the cross-term coefficient. Each branch thus contributes to the same output pattern according to correlations in the full joint final state. For a general mixed joint final state $\Omega_{\text{out}} = \sum_r t_r P_{\Psi_r}$, with $t_r \geq 0$ and $\sum_r t_r = 1$,

$$\langle x | \rho_{\text{out}} | y \rangle = \frac{1}{d} \sum_r t_r \langle \Psi_r(y), \Psi_r(x) \rangle.$$

The result depends only on Ω_{out} , not on the chosen pure-state decomposition. The reduced state becomes the initial system state of the next cycle, connecting successive cycles.

The next cycle Pattern reconstruction determines the system state at cycle end. To use it as the next input, take auxiliary-space matrix elements of the full joint evolution to obtain the evolution of the system density operator. The joint dilation gives a completely positive map [14], expressed in finite dimension as an operator sum [15, 16]. For each thermal channel (a, ν) , use $D_u = I - (1 - c_{a\nu})P_\nu$ and $D_v = I - (1 - c_{a\nu})P_{e_a}$. Absorbing the thermal preparation weights into the matrix elements gives $A_* = \sqrt{a_0}I$ and

$$\begin{aligned} A_{uu}^{(a\nu)} &= \sqrt{a_{a\nu}t_u}D_u, & A_{vu}^{(a\nu)} &= \sqrt{a_{a\nu}t_u}s_{a\nu}F_{a\nu}, \\ A_{uv}^{(a\nu)} &= -\sqrt{a_{a\nu}t_v}s_{a\nu}F_{a\nu}^\dagger, & A_{vv}^{(a\nu)} &= \sqrt{a_{a\nu}t_v}D_v. \end{aligned} \quad (4.52)$$

The first subscript labels the environment output, the second its input; t_u, t_v are the preparation shares of the corresponding channel. Denote the full finite operator family by $\{A_\kappa\}$. Different environment labels retain their own weights and are summed separately even when their system matrices coincide. Since $F_{a\nu}^\dagger F_{a\nu} = P_\nu$ and $F_{a\nu} F_{a\nu}^\dagger = P_{e_a}$,

$$D_u^2 + s_{a\nu}^2 P_\nu = I, \quad D_v^2 + s_{a\nu}^2 P_{e_a} = I.$$

Each thermal channel therefore satisfies

$$\begin{aligned} \sum_{\xi, \zeta \in \{u, v\}} A_{\xi\zeta}^{(a\nu)\dagger} A_{\xi\zeta}^{(a\nu)} &= a_{a\nu} [t_u(D_u^2 + s_{a\nu}^2 P_\nu) + t_v(D_v^2 + s_{a\nu}^2 P_{e_a})] \\ &= a_{a\nu} I. \end{aligned}$$

Including the star component and using $a_0 + \sum_A a_{a\nu} = 1$ gives $\sum_\kappa A_\kappa^\dagger A_\kappa = I$.

The matrix elements of background restoration for auxiliary preparation g are $B_0 = P$, $B_{ab} = \sqrt{g_a}|e_a\rangle\langle e_b|$. Here B_0 retains direction components, while B_{ab} sends background input e_b to the restored shell-average component e_a . Since $\sum_a g_a = 1$,

$$\sum_\lambda B_\lambda^\dagger B_\lambda = P + \sum_{a,b} g_a P_{e_b} = P + \Pi = I.$$

When free evolution over the complete cycle is included, the retrieval phases give, direction by direction,

$$e^{-i\tau_{\text{tot}}H_K} U_w P = U_\varphi P.$$

The background restoration components satisfy

$$e^{-i\tau_{\text{tot}}H_K} B_{ab} = e^{-i\tau_{\text{tot}}E_a} B_{ab}, \quad U_\varphi B_{ab} = B_{ab}.$$

Every auxiliary output label has a definite energy. For fixed input and output labels, the remaining free phase is a global phase of that matrix element and cancels in its product with the adjoint. The system evolution of the complete cycle can therefore be written as

$$\begin{aligned} K_{\lambda\kappa}^{(V)} &= U_\varphi B_\lambda A_\kappa V, \\ \mathcal{F}_{V,g}(\rho) &= \sum_{\lambda,\kappa} K_{\lambda\kappa}^{(V)} \rho K_{\lambda\kappa}^{(V)\dagger}. \end{aligned} \quad (4.53)$$

This sum is the auxiliary partial trace of the full joint final state and includes every component. The joint final state itself remains that of Eq. (4.46). Record criteria and

readouts are determined separately by actual auxiliary actions; the matrix-element labels here expand the reduced state.

Using completeness of the two matrix-element families and unitarity of U_φ, V ,

$$\begin{aligned}\sum_{\lambda, \kappa} K_{\lambda\kappa}^{(V)\dagger} K_{\lambda\kappa}^{(V)} &= V^\dagger \sum_{\kappa} A_\kappa^\dagger \left(\sum_{\lambda} B_\lambda^\dagger B_\lambda \right) A_\kappa V \\ &= V^\dagger \left(\sum_{\kappa} A_\kappa^\dagger A_\kappa \right) V = I.\end{aligned}$$

Hence $\text{Tr } \mathcal{F}_{V,g}(\rho) = \text{Tr } \rho$. For any auxiliary space $\mathcal{R}$ and positive operator $X \geq 0$,

$$(\mathcal{F}_{V,g} \otimes \text{id}_{\mathcal{R}})(X) = \sum_{\lambda, \kappa} (K_{\lambda\kappa}^{(V)} \otimes I_{\mathcal{R}}) X (K_{\lambda\kappa}^{(V)\dagger} \otimes I_{\mathcal{R}}) \geq 0.$$

Thus $\mathcal{F}_{V,g}$ is completely positive and trace preserving, mapping every system density operator to a valid state for the next cycle.

In repeated cycles, denote all previous environments by $\mathcal{E}_{\text{old}}$ and allow correlations in the joint state $\Omega_{S,\text{old}}$, with $\text{Tr}_{\text{old}} \Omega_{S,\text{old}} = \rho_n$. The fresh preparation is independent of the system and all previous environments jointly; the latter no longer participate. Let $\tilde{\mathbb{U}}_{\text{tot}}$ act on the system and current auxiliary systems and as the identity on previous environments. Then

$$\begin{aligned}\text{Tr}_{\text{old}, \mathcal{E}} [\tilde{\mathbb{U}}_{\text{tot}}(\Omega_{S,\text{old}} \otimes \eta_{\text{full}}) \tilde{\mathbb{U}}_{\text{tot}}^\dagger] \\ = \text{Tr}_{\mathcal{E}} [\mathbb{U}_{\text{tot}}(\rho_n \otimes \eta_{\text{full}}) \mathbb{U}_{\text{tot}}^\dagger] = \mathcal{F}_{V,g}(\rho_n).\end{aligned}$$

This equality follows because the previous-environment partial trace commutes with the current action; it does not require removing their correlations with the system. Keeping preparation, control, and interaction data fixed throughout gives

$$\rho_{n+1} = \mathcal{F}_{V,g}(\rho_n), \quad \rho_n = \mathcal{F}_{V,g}^n(\rho_0). \quad (4.54)$$

Figure 3 relates the destinations of joint components to the complete cycle.

§4.4 Continuation of the Cycle

With fixed environment preparation and interactions, each final system state becomes the next cycle's input. Returning excitations continue to evolve, while components transferred to the background participate in subsequent formation.

Coherent propagation The star component of the thermal preparation undergoes no thermal exchange. In this component, an excitation entering formation exchange returns to the direction space after coherent retrieval, with the auxiliary state remaining independent and the system pattern updated by the corresponding unitary action. Use the intra-shell propagation space $\mathcal{H}_a$ and projection Q_a defined in Eq. (4.32). Since $\mathcal{H}_a \subset \mathcal{M}$ and $U_\varphi V Q_a = W Q_a$, the cycle matrix element satisfies

$$K_{0*}^{(V)} Q_a = \sqrt{a_0} U_\varphi V Q_a = \sqrt{a_0} W Q_a. \quad (4.55)$$

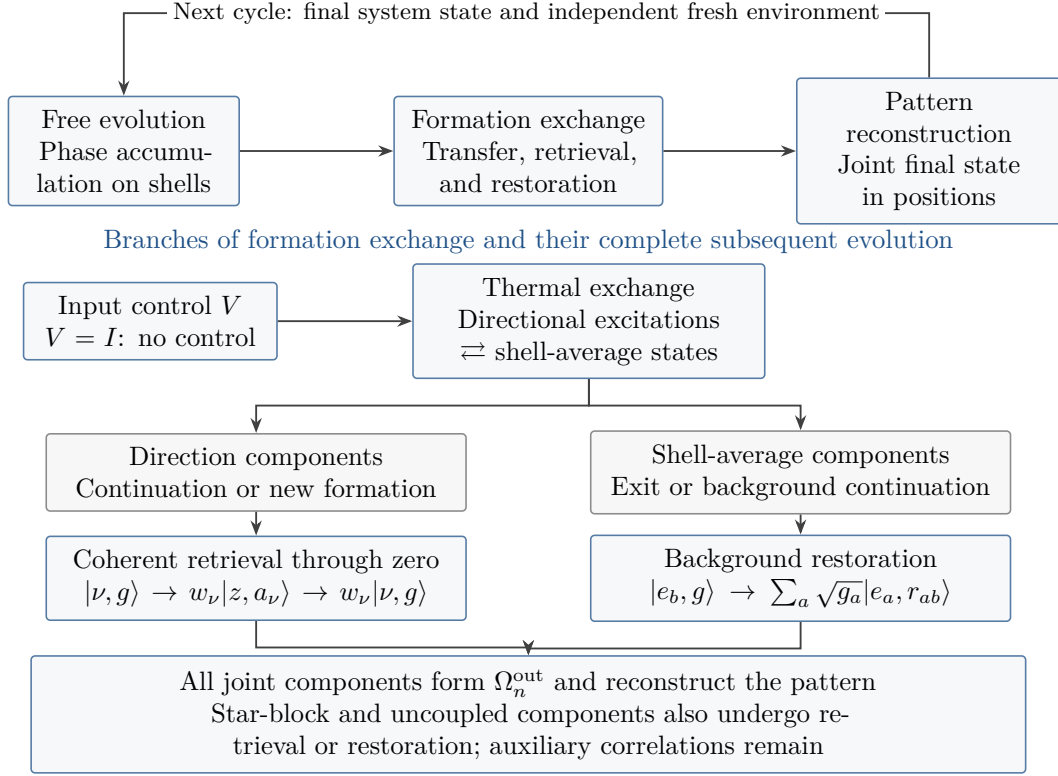


Figure 3: The complete cycle of background and excitations. Control V acts after free evolution and before thermal exchange. The two subsequent paths are actions on different components of the same joint evolution: retrieval is the identity on shell-average components, and restoration on returning direction components. Equations (4.24) and (4.25) retain all joint amplitudes for every input. These branches have not yet been classified as events by actual pointers. Pattern reconstruction introduces no additional interaction or duration. Tracing out the environment gives the final system state that enters the next cycle.

For a normalized initial state $\psi_0 \in \mathcal{H}_a$, this component has share a_0 , and its normalized pattern evolves under W . Selecting this matrix element for m successive cycles, in the phase convention of Eq. (4.53), gives

$$(K_{0*}^{(V)})^m \psi_0 = a_0^{m/2} W^m \psi_0, \quad \|(K_{0*}^{(V)})^m \psi_0\|^2 = a_0^m.$$

The normalized pattern is $\psi_m = W^m \psi_0$; auxiliary labels and actual free phases of every cycle remain in the full joint component.

In the orthonormal basis $|\pm, \ell\rangle$, write

$$\psi_m(\ell) = \begin{pmatrix} \langle +, \ell | \psi_m \rangle \\ \langle -, \ell | \psi_m \rangle \end{pmatrix}, \quad \sum_{\ell \in \mathbb{Z}_L} \|\psi_m(\ell)\|^2 = 1.$$

Here ℓ is the principal-unit coordinate within the fixed shell, and the two components correspond to the two orientations above. Set $\kappa_r = 2\pi r/L$ and define

$$\tilde{\psi}_m(r) = \frac{1}{\sqrt{L}} \sum_{\ell} e^{-i\kappa_r \ell} \psi_m(\ell).$$

Propagation satisfies $\tilde{\psi}_{m+1}(r) = W_r \tilde{\psi}_m(r)$, where $W_r = C_r e^{-i\kappa_r \sigma_3}$. Iterating and applying Fourier inversion gives

$$\psi_m(\ell) = \sum_{u \in \mathbb{Z}_L} G_m(\ell - u) \psi_0(u), \quad G_m(v) = \frac{1}{L} \sum_{r \in \mathbb{Z}_L} e^{i\kappa_r v} W_r^m. \quad (4.56)$$

This kernel retains coherent superposition between position modes and determines the intra-shell distribution reconstructed after retrieval in each cycle. When the orientation action is identical for all position modes, $C_r = C_0$, the two orientations acquire opposite unit displacements, giving

$$\psi_{m+1}(\ell) = C_0 \begin{pmatrix} \psi_{m,+}(\ell - 1) \\ \psi_{m,-}(\ell + 1) \end{pmatrix}. \quad (4.57)$$

All position indices are modulo L . The two orientations arrive from neighboring positions and are mixed by the same C_0 to form the new pattern at that position. This combination of an internal two-state system and opposite displacements occurs in the original quantum-walk construction [17].

The finite Fourier kernel determines the condition for this local recurrence. With the fixed orientation transformation B , write

$$C_r = B \begin{pmatrix} z_{+,r} & 0 \\ 0 & z_{-,r} \end{pmatrix} B^\dagger, \quad z_{\pm,r} = e^{i\varphi_{\pm,r}}, \quad \hat{C}(v) = \frac{1}{L} \sum_r e^{i\kappa_r v} C_r.$$

Fourier inversion and invertibility of B give

$$\boxed{\begin{aligned} \text{supp } \hat{C} \subset \{0\} &\iff C_r = C_0 \quad (\text{all } r) \\ &\iff z_{\pm,r} = z_{\pm,0} \quad (\text{all } r). \end{aligned}} \quad (4.58)$$

Equality of phase factors is required; real phase values may differ by integer multiples of 2π . In the fixed orientation construction of this section, all C_r belong to $\text{span}\{I, \sigma_1\}$. Writing $\hat{C}(v) = f(v)I + g(v)\sigma_1$, the one-step propagation kernel is

$$\widehat{W}(v) = \begin{pmatrix} f(v-1) & g(v+1) \\ g(v-1) & f(v+1) \end{pmatrix}. \quad (4.59)$$

If the one-step support of every oriented point state is confined to the original position and its two neighbors, then $\text{supp } \widehat{W} \subset \{-1, 0, 1\}$. The first column requires the supports of both f, g to lie in $\{-2, -1, 0\}$, and the second also requires them to lie in $\{0, 1, 2\}$. For $L \geq 5$, these sets intersect only at zero in $\mathbb{Z}_L$, so $\text{supp } \widehat{C} \subset \{0\}$. Conversely, if $\widehat{C}$ is nonzero only at zero, then $\widehat{W}$ can be nonzero only at $v = \pm 1$, giving Eq. (4.57).

Thus, in this fixed orientation construction, one-step propagation has the stated locality for every oriented point state if and only if C_r is independent of r . Since $p \geq 5$, either $L = 1$ or $L \geq 5$. For $L = 1$, there is only one principal-unit position, and only orientation changes remain. For general spectral phases varying with r , Eq. (4.56) still applies and cannot be reduced to the neighboring-position recurrence above. Spectral requirements for position preparation and approximation errors for neighboring-position propagation are discussed further in Appendix A.

The factor a_0^m counts only successive star-return components. Components that return to the direction space for other thermal preparations, as well as exit, formation, and background-continuation components, all remain in the complete cycle map. Full continuation of the original excitation and changes in the full system state require all these components to be included.

Decay and formation Decay and formation correspond to the two transfer directions between the direction and shell-average subspaces. Their shares are squared norms of the corresponding joint components.

For a selected channel (a, ν) , write $\Delta_{a\nu} = E_a - E_\nu$. The thermal-exchange matrix elements give

$$\begin{aligned} A_{vu}^{(a\nu)}|\nu\rangle &= \sqrt{a_{a\nu}t_us_{a\nu}}|e_a\rangle, \\ A_{uv}^{(a\nu)}|e_a\rangle &= -\sqrt{a_{a\nu}t_vs_{a\nu}}|\nu\rangle. \end{aligned}$$

Taking squared norms gives the exit and formation shares per cycle,

$$\epsilon_{a\nu} = a_{a\nu}t_us_{a\nu}^2, \quad f_{\nu a} = a_{a\nu}t_vs_{a\nu}^2, \quad \frac{\epsilon_{a\nu}}{f_{\nu a}} = e^{-\beta(E_a - E_\nu)}. \quad (4.60)$$

Here $\epsilon_{a\nu} + f_{\nu a} = a_{a\nu}s_{a\nu}^2$. The ratio applies to selected positive exchange channels at finite β . The thermal preparation fixes the ratio; channel weights and exchange angles determine the absolute shares. For an unselected channel, define $\epsilon_{a\nu} = f_{\nu a} = 0$. Summing channels for the same input, write

$$\epsilon_\nu = \sum_a \epsilon_{a\nu}, \quad k_a = 1 - \sum_\nu f_{\nu a}.$$

Here ϵ_ν is the total exit share, and k_a is the share of shell-average input e_a that does not form a directional excitation. Since $0 \leq t_u, t_v, s_{a\nu}^2 \leq 1$ and $\sum_A a_{a\nu} = 1 - a_0$, $0 \leq \epsilon_\nu \leq 1 - a_0$, $a_0 \leq k_a \leq 1$.

First consider a cycle without input control, $V = I$. For a single direction input, the returning component remains in that direction and its net phase cancels in the projection operator. All exit components give σ_g^{sh} after restoration. For a shell-average input, each formation channel produces its corresponding direction, and the remaining components return to the background through restoration. Cross terms

between distinct auxiliary labels vanish under partial trace, so

$$\begin{aligned}\mathcal{F}_{I,g}(P_\nu) &= (1 - \epsilon_\nu)P_\nu + \epsilon_\nu\sigma_g^{\text{sh}}, \\ \mathcal{F}_{I,g}(P_{e_a}) &= \sum_\nu f_{\nu a}P_\nu + k_a\sigma_g^{\text{sh}}.\end{aligned}\tag{4.61}$$

Both expressions are system outputs after free evolution, formation exchange, and pattern reconstruction, with total trace one. Every specified formation component has a definite directional structure; their combined state is generally mixed. With input control, $\mathcal{F}_{V,g}(\rho) = \mathcal{F}_{I,g}(V\rho V^\dagger)$, so the same exchange acts on the controlled input.

The energies $E_u = \max(\Delta_{a\nu}, 0)$ and $E_v = \max(-\Delta_{a\nu}, 0)$ of the thermal environment satisfy $E_\nu + E_u = E_a + E_v$. The exit process $|\nu, u\rangle \rightarrow |e_a, v\rangle$ changes the system energy by $\Delta_{a\nu}$ and the thermal-environment energy by $-\Delta_{a\nu}$. Formation proceeds in the opposite direction with opposite energy changes. Holding all other cycle data fixed and taking the thermal environment alone to the zero-temperature limit $\beta \rightarrow +\infty$ gives three cases.

If $E_a < E_\nu$, exit lowers system energy, with $t_u \rightarrow 1$ and $t_v \rightarrow 0$, so

$$\epsilon_{a\nu} \longrightarrow a_{a\nu}s_{a\nu}^2 > 0, \quad f_{\nu a} \longrightarrow 0.\tag{4.62}$$

The original directional excitation can transfer to a lower-energy shell-average component, with the thermal environment receiving $E_\nu - E_a$. Reverse formation requires the environment to supply this energy, and its share vanishes in the zero-temperature limit. If $E_a > E_\nu$, then $t_u \rightarrow 0$ and $t_v \rightarrow 1$, giving

$$\epsilon_{a\nu} \longrightarrow 0, \quad f_{\nu a} \longrightarrow a_{a\nu}s_{a\nu}^2.$$

A shell-average input can then form a directional excitation by lowering its energy and transferring $E_a - E_\nu$ to the thermal environment. If $E_a = E_\nu$, the two auxiliary states are degenerate and the prescribed within-block thermal preparation satisfies

$$t_u = t_v = \frac{1}{2}, \quad \epsilon_{a\nu} = f_{\nu a} = \frac{1}{2}a_{a\nu}s_{a\nu}^2.$$

Exchange changes the composition of direction and background components without changing system energy during this stage.

These energy changes concern thermal exchange. After exit, the shell-average component must still be restored to σ_g^{sh} ; the restoration auxiliary shares the associated energy change, and the energy balance of the complete cycle includes all joint steps. The same formation exchange therefore determines both decay of an original excitation and formation of a new one. Decay of a specific physical subject and its products can be identified only after the system and auxiliary final states receive the corresponding physical interpretation.

Continuation of an excitation After formation exchange, part of an original excitation returns to the direction space and another part enters the background. Tracking its continued evolution requires all returning components to be collected and their combined system state to be determined. The control V here preserves

$\mathcal{C} \oplus \mathcal{M}$ and is the identity on $\mathcal{C}$. For a direction input $X = PXP$, select from the complete cycle the matrix elements returning to the direction space and define

$$\mathbf{L}_\kappa = U_\varphi P A_\kappa V P, \quad \mathcal{Q}_V(X) = \sum_{\kappa} \mathbf{L}_\kappa X \mathbf{L}_\kappa^\dagger, \quad X = PXP. \quad (4.63)$$

The projection P excludes components transferred to the background; zero matrix elements may be retained. This action is completely positive. Since $P \leq I$, completeness of the thermal-exchange matrix elements gives

$$\begin{aligned} \sum_{\kappa} \mathbf{L}_\kappa^\dagger \mathbf{L}_\kappa &= P V^\dagger \left(\sum_{\kappa} A_\kappa^\dagger P A_\kappa \right) V P \\ &\leq P V^\dagger \left(\sum_{\kappa} A_\kappa^\dagger A_\kappa \right) V P = P. \end{aligned}$$

Thus $\text{Tr } \mathcal{Q}_V(X) \leq \text{Tr } X$ for $X \geq 0$, allowing the share of the original excitation to decrease through exit.

Write $Y = V X V^\dagger$. In channel (a, λ) , the returning part is given by D_u for thermal input u ; for input v , the original direction component remains unchanged. Both transfer matrix elements vanish when input and output are restricted to the direction space, so

$$U_\varphi^\dagger \mathcal{Q}_V(X) U_\varphi = a_0 Y + \sum_{(a, \lambda) \in \mathcal{A}} a_{a\lambda} (t_u D_u Y D_u + t_v Y).$$

The total coefficient of a diagonal entry $Y_{\nu\nu}$ is $1 - \sum_a a_{a\nu} t_u (1 - c_{a\nu}^2) = 1 - \epsilon_\nu$. For $\nu \neq \mu$, each thermal block changes the return amplitude of only one of these directions. Set

$$\begin{aligned} d_\nu &= \sum_a a_{a\nu} t_u (1 - c_{a\nu}), \quad \gamma_{\nu\mu} = 1 - d_\nu - d_\mu, \\ \Gamma_{\nu\nu} &= 1 - \epsilon_\nu, \quad \Gamma_{\nu\mu} = \gamma_{\nu\mu} \quad (\nu \neq \mu). \end{aligned} \quad (4.64)$$

Then

$$\boxed{\mathcal{Q}_V(X) = U_\varphi [\Gamma \circ (V X V^\dagger)] U_\varphi^\dagger, \quad X = PXP.} \quad (4.65)$$

Here $\circ$ denotes entrywise multiplication in the direction basis, $(\Gamma \circ Y)_{\nu\mu} = \Gamma_{\nu\mu} Y_{\nu\mu}$. Since $0 < \alpha_{a\nu} < \pi/2$, we have $0 < 1 - c_{a\nu} < 1$. Channels for distinct directions have distinct labels, so

$$0 \leq d_\nu + d_\mu \leq \sum_{\substack{(a, \lambda) \in \mathcal{A} \\ \lambda \in \{\nu, \mu\}}} a_{a\lambda} \leq 1 - a_0, \quad a_0 \leq \gamma_{\nu\mu} \leq 1.$$

In the unnormalized return matrix, this positive real factor changes the modulus of coherence entries, while U_φ gives the net phase change. The control V may also mix directions, so returning components generally form a new pattern.

For a normalized excitation $\psi \in \mathcal{M}$, let $P_{[\psi]} = |\psi\rangle\langle\psi|$ describe its original pattern, and write $s_\psi = \text{Tr } \mathcal{Q}_V(P_{[\psi]})$ for the total return share. The star term $\mathbf{L}_* = \sqrt{a_0} U_\varphi V P$ gives $a_0 \leq s_\psi \leq 1$, so the normalized state of the returning component is always

defined. Using equality of normalized projections as the criterion for preserving the original pattern,

$$\frac{\mathcal{Q}_V(P_{[\psi]})}{s_\psi} = P_{[\psi]} \iff \mathsf{L}_\kappa \psi = c_\kappa \psi \quad (\text{all } \kappa), \quad (4.66)$$

$$\sum_\kappa |c_\kappa|^2 = s_\psi.$$

Zero coefficients c_κ are allowed. The return share in the orthogonal complement of the original pattern is

$$\text{Tr}[(I - P_{[\psi]})\mathcal{Q}_V(P_{[\psi]})] = \sum_\kappa \|(I - P_{[\psi]})\mathsf{L}_\kappa \psi\|^2.$$

If the normalized return state is $P_{[\psi]}$, the left side vanishes, so every nonnegative term vanishes and every return vector lies in $\text{span}\{\psi\}$. Conversely, substituting $\mathsf{L}_\kappa \psi = c_\kappa \psi$ gives $\mathcal{Q}_V(P_{[\psi]}) = (\sum_\kappa |c_\kappa|^2)P_{[\psi]}$, with its coefficient fixed by the trace. The stable-state condition of Section 3 concerns a specified shell-preserving response. In a complete cycle, thermal exchange, retrieval, and control must also be included; Equation (4.66) determines whether the returning component preserves the original pattern.

For a normalized initial state $\rho_0 = P\rho_0P$, the state and share continuing through direction returns for m consecutive cycles are $\mathcal{Q}_V^m(\rho_0)$ and $s_m = \text{Tr } \mathcal{Q}_V^m(\rho_0)$. Trace nonincrease and the star term give $a_0^m \leq s_m \leq 1$ and $s_{m+1} \leq s_m$. This share sums all successive return components, including the star contribution above. In particular, for $V = I$ and $\rho_0 = P_\nu$, iteration gives

$$\mathcal{Q}_I^m(P_\nu) = (1 - \epsilon_\nu)^m P_\nu, \quad s_m = (1 - \epsilon_\nu)^m. \quad (4.67)$$

The normalized pattern of the original direction is unchanged; when $\epsilon_\nu > 0$, its uninterrupted continuation share decreases geometrically. Direction occupations formed after exit are supplied by formation terms; both contributions determine the total occupation over repeated cycles.

Occupation dynamics Uninterrupted continuation of original excitations accounts for part of the direction occupation. The background can also form new excitations in every exchange, while exit components replenish the background through restoration. Total occupation must include all these processes. To expand the complete cycle, write

$$\tilde{\rho}_n = \sum_\kappa A_\kappa V \rho_n V^\dagger A_\kappa^\dagger.$$

Completeness of the matrix elements ensures that this is a normalized positive operator. The reduced expression for restoration gives

$$\rho_{n+1} = U_\varphi P \tilde{\rho}_n P U_\varphi^\dagger + \text{Tr}(\Pi \tilde{\rho}_n) \sigma_g^{\text{sh}}.$$

The direction output is supported on $\mathcal{M}$, while the background output always has the fixed shell shares g_a . Thus, for any initial state, the first-cycle output satisfies

$$P\rho_1\Pi = \Pi\rho_1P = 0, \quad \rho_1 = X_1 + y_1\sigma_g^{\text{sh}}, \quad X_1 = PX_1P \geq 0, \quad y_1 \geq 0.$$

This follows from the complete cycle and the auxiliary partial trace, allowing the initial cross coherence in Eq. (4.1). The output cross blocks vanish because return and restoration components carry orthogonal auxiliary labels; their correlations remain in the full joint final state. Every subsequent state has the form

$$\rho_n = X_n + y_n \sigma_g^{\text{sh}}, \quad X_n = P \rho_n P, \quad y_n = \text{Tr}(\Pi \rho_n), \quad \text{Tr } X_n + y_n = 1, \quad n \geq 1.$$

Here X_n is the unnormalized direction state, including occupations and coherence, and y_n is the total background share.

Using $\sigma_g^{\text{sh}} = \sum_a g_a P_{e_a}$, define the formation component produced by a unit background share as

$$f_\nu(g) = \sum_a g_a f_{\nu a}, \quad B_g = \sum_\nu f_\nu(g) P_\nu, \quad f_\Sigma = \text{Tr } B_g.$$

The operator B_g is positive and unnormalized; its trace is the total formation share. Since $\sum_\nu f_{\nu a} = 1 - k_a$ and $k_a \geq a_0$, $0 \leq f_\Sigma = \sum_a g_a (1 - k_a) \leq 1 - a_0$. The control V is the identity on $\mathcal{C}$, so

$$\begin{aligned} \mathcal{F}_{V,g}(\sigma_g^{\text{sh}}) &= \sum_a g_a \left(\sum_\nu f_{\nu a} P_\nu + k_a \sigma_g^{\text{sh}} \right) \\ &= B_g + (1 - f_\Sigma) \sigma_g^{\text{sh}}. \end{aligned}$$

For any direction input $X = PXP \geq 0$, the returning part is $\mathcal{Q}_V(X)$ and trace preservation fixes the remaining share. Hence

$$\mathcal{F}_{V,g}(X) = \mathcal{Q}_V(X) + [\text{Tr } X - \text{Tr } \mathcal{Q}_V(X)] \sigma_g^{\text{sh}}.$$

Applying these to X_n and $y_n \sigma_g^{\text{sh}}$ gives

$$\boxed{\begin{aligned} X_{n+1} &= \mathcal{Q}_V(X_n) + y_n B_g, \\ y_{n+1} &= \text{Tr}[X_n - \mathcal{Q}_V(X_n)] + (1 - f_\Sigma) y_n, \quad n \geq 1. \end{aligned}} \quad (4.68)$$

The first identity combines continuation of original directions and new formation; the second combines exit components and background shares that do not form excitations. If the initial state already has this closed form, the recurrence may start at $n = 0$. Every term is nonnegative, and

$$\begin{aligned} \text{Tr } X_{n+1} + y_{n+1} &= \text{Tr } \mathcal{Q}_V(X_n) + y_n f_\Sigma + \text{Tr } X_n - \text{Tr } \mathcal{Q}_V(X_n) + (1 - f_\Sigma) y_n \\ &= \text{Tr } X_n + y_n = 1. \end{aligned}$$

Thus each cycle again gives a normalized positive state. The operator X_n retains all direction matrix elements; control changes its distribution and coherence, formation adds direction occupation, and exit transfers shares to the background. This closed recurrence describes both occupation and coherence.

Iterating the first identity gives

$$X_n = \mathcal{Q}_V^{n-1}(X_1) + \sum_{j=1}^{n-1} y_j \mathcal{Q}_V^{n-1-j}(B_g), \quad n \geq 1.$$

For $n = 1$, the sum is empty. The first term is uninterrupted continuation of the first-cycle direction component; the term indexed by j is the excitation formed during the cycle with input ρ_j , continued through the subsequent $n - 1 - j$ cycles. The operator X_1 already includes first-cycle formation. Total direction occupation combines components formed in different cycles; its long-time behavior requires both the return action and formation input.

Stationary state Over repeated cycles, direction components return or exit, and the background continually forms new excitations. When the full system state produced by these processes is unchanged after one cycle, it is a stationary state of the cycle:

$$\mathcal{F}_{V,g}(\rho_*) = \rho_*, \quad \rho_* \geq 0, \quad \text{Tr } \rho_* = 1.$$

Stationarity here concerns the full density operator, whereas stability in Section 3 concerns an excitation pattern under a specified propagation action. Fix all cycle data, assume finite β , $g_a > 0$, and at least one positive exchange channel for every direction. Then $\epsilon_\nu > 0$ and $f_\nu(g) > 0$: every direction can exit and can be formed again from the background.

Each thermal channel satisfies

$$\begin{aligned} \sum_{\xi, \zeta \in \{u, v\}} A_{\xi\zeta}^{(a\nu)\dagger} \Pi A_{\xi\zeta}^{(a\nu)} \\ = a_{a\nu} [t_u(\Pi + s_{a\nu}^2 P_\nu) + t_v(\Pi - s_{a\nu}^2 P_{e_a})]. \end{aligned}$$

Adding the star term and summing gives the total effect for background output,

$$\sum_{\kappa} A_{\kappa}^\dagger \Pi A_{\kappa} = D_g, \quad D_g = \sum_a k_a P_{e_a} + \sum_\nu \epsilon_\nu P_\nu.$$

The background output of an arbitrary input X is $\text{Tr}(V^\dagger D_g V X) \sigma_g^{\text{sh}}$. Since $k_a \geq a_0 > 0$ and there are finitely many directions, take

$$c = \min\{a_0, \min_\nu \epsilon_\nu\} > 0, \quad D_g \geq cI, \quad 0 < c < 1.$$

Each cycle therefore sends a positive input-independent share to the same background preparation, yielding the decomposition

$$\mathcal{F}_{V,g}(X) = c \text{Tr}(X) \sigma_g^{\text{sh}} + (1 - c) \mathcal{G}_{V,g}(X), \quad (4.69)$$

where $\mathcal{G}_{V,g}$ is completely positive and trace preserving. To verify this, write $D_g = \sum_i \delta_i P_{q_i}$ with $\{q_i\} = \{e_a\} \cup \{|\nu\rangle\}$. After subtracting $c \text{Tr}(X) \sigma_g^{\text{sh}}$ from the background output, the remainder is

$$\sum_{a,i} L_{ai} X L_{ai}^\dagger, \quad L_{ai} = \sqrt{g_a(\delta_i - c)} |e_a\rangle \langle q_i| V.$$

Since $\delta_i \geq c$, these matrix elements are well defined. Retaining all direction-output matrix elements of the original channel leaves a completely positive remainder with total trace $(1 - c) \text{Tr } X$. Division by $1 - c$ gives the trace-preserving map in Eq. (4.69).

Compare states using the trace norm $\|X\|_1 = \text{Tr} \sqrt{X^\dagger X}$. For the positive and negative decomposition $Z = Z_+ - Z_-$ of a self-adjoint operator, positivity and trace preservation give

$$\|\mathcal{G}_{V,g}(Z)\|_1 \leq \text{Tr} \mathcal{G}_{V,g}(Z_+) + \text{Tr} \mathcal{G}_{V,g}(Z_-) = \text{Tr} Z_+ + \text{Tr} Z_- = \|Z\|_1.$$

The difference of two normalized states has zero trace, so the common background term cancels. Iterating the contraction gives

$$\|\mathcal{F}_{V,g}^n(\rho) - \mathcal{F}_{V,g}^n(\rho')\|_1 \leq (1-c)^n \|\rho - \rho'\|_1. \quad (4.70)$$

Set $\rho_n = \mathcal{F}_{V,g}^n(\rho_0)$. The triangle inequality and geometric series give

$$\|\rho_{n+k} - \rho_n\|_1 \leq \sum_{j=n}^{n+k-1} (1-c)^j \|\rho_1 - \rho_0\|_1 \leq \frac{(1-c)^n}{c} \|\rho_1 - \rho_0\|_1.$$

The right side tends to zero, so the iterates form a Cauchy sequence. The set of density operators is closed and complete in trace norm, and its limit is normalized and positive. Continuity of the map makes the limit stationary. If two stationary states existed, Eq. (4.70) would force their trace-norm difference to vanish, proving uniqueness. For every initial state,

$$\|\rho_n - \rho_*\|_1 \leq (1-c)^n \|\rho_0 - \rho_*\|_1.$$

This proves geometric convergence for a strictly contractive quantum channel [18]. Its stationary state can be computed from the occupation recurrence. For a positive operator X on the direction space, at least a share c exits in each cycle, so $\text{Tr} \mathcal{Q}_V^n(X) \leq (1-c)^n \text{Tr} X$. Positive and negative decomposition gives the same trace-norm bound for self-adjoint inputs. Splitting a general complex operator into real and imaginary parts then gives

$$\|\mathcal{Q}_V^n(X)\|_1 \leq 2(1-c)^n \|X\|_1.$$

The trace norms of both parts are at most $\|X\|_1$; the factor 2 is unnecessary for self-adjoint inputs. The following series therefore converges in the induced operator norm:

$$\mathcal{R}_V := \sum_{n=0}^{\infty} \mathcal{Q}_V^n = (I - \mathcal{Q}_V)^{-1} \quad \text{on } \mathcal{B}(\mathcal{M}). \quad (4.71)$$

Here I is the identity map on the operator space. Multiplying the finite partial sums by $I - \mathcal{Q}_V$ on either side gives $I - \mathcal{Q}_V^{N+1}$; taking the limit proves the inverse relation. Every summand preserves positivity, so $\mathcal{R}_V$ does as well. The first-cycle closure gives $\rho_* = X_* + y_* \sigma_g^{\text{sh}}$, with direction stationarity equation $X_* = \mathcal{Q}_V(X_*) + y_* B_g$. Combining it with normalization yields

$$\boxed{X_* = y_* \mathcal{R}_V(B_g), \quad y_* = [1 + \text{Tr} \mathcal{R}_V(B_g)]^{-1}.} \quad (4.72)$$

Taking the trace of the direction stationarity equation gives $\text{Tr} X_* - \text{Tr} \mathcal{Q}_V(X_*) = y_* f_\Sigma$, which also satisfies background stationarity. The direction share exiting each cycle exactly equals the share formed again from the background.

The operator B_g is positive definite on $\mathcal{M}$; the first series term gives $X_* \geq y_* B_g > 0$ on $\mathcal{M}$. Also $y_* > 0$ and σ_g^{sh} is positive definite on $\mathcal{C}$, so ρ_* has full rank on the entire state space. At stationarity, uninterrupted continuation of original excitations may still decay, while formation from the background maintains positive total direction occupation. Stationarity expresses the balance of these shares in the complete cycle.

Thermalization A stationary state is a system state that is unchanged by one complete cycle. Thermalization further requires convergence of the complete cycle to the Gibbs state associated with H_K and the thermal-environment inverse temperature β ; its conditions involve both thermal exchange and background restoration. For finite β , the Gibbs state on the full state space is

$$\rho_\beta = \frac{e^{-\beta H_K}}{Z_{\text{full}}}, \quad Z_{\text{full}} = \text{Tr } e^{-\beta H_K} = \sum_{j=0}^K |S_j| e^{-\beta E_j}.$$

Here $|S_j|$ counts every independent state at that energy. On the shell-average subspace $\mathcal{C}$, the corresponding conditional Gibbs state is

$$\sigma_\pi^{\text{sh}} = \sum_a \pi_a P_{e_a}, \quad \pi_a = \frac{e^{-\beta E_a}}{Z_{\mathcal{C}}}, \quad Z_{\mathcal{C}} = \sum_a e^{-\beta E_a}. \quad (4.73)$$

Each e_a occupies one dimension in $\mathcal{C}$, so $Z_{\mathcal{C}}$ contains no full-shell degeneracy factor. The full Gibbs state decomposes as

$$\rho_\beta = X_\beta + y_\beta \sigma_\pi^{\text{sh}}, \quad X_\beta = \frac{1}{Z_{\text{full}}} \sum_\nu e^{-\beta E_\nu} P_\nu, \quad y_\beta = \frac{Z_{\mathcal{C}}}{Z_{\text{full}}} > 0.$$

Each thermal block satisfies $e^{-\beta E_\nu} t_u = e^{-\beta E_a} t_v$. The coupled joint states $|\nu, u\rangle$ and $|e_a, v\rangle$ consequently have equal weights in $\rho_\beta \otimes \eta_T$:

$$\frac{a_{a\nu}}{Z_{\text{full}}} e^{-\beta E_\nu} t_u = \frac{a_{a\nu}}{Z_{\text{full}}} e^{-\beta E_a} t_v.$$

The restriction of the joint density operator to this two-dimensional exchange subspace is proportional to the identity and invariant under the exchange rotation. There is no initial coherence between thermal blocks, and the action on the star block and on each exchange block's orthogonal complement is the identity. Hence

$$U_{\text{th}}(\rho_\beta \otimes \eta_T) U_{\text{th}}^\dagger = \rho_\beta \otimes \eta_T.$$

This invariance uses the Gibbs ratio within each thermal block; the total weights $a_{a\nu}$ across blocks remain those of the prescribed preparation. Channel-by-channel detailed balance in this thermal-exchange stage can also be written as

$$\pi_a f_{\nu a} = \frac{e^{-\beta E_\nu}}{Z_{\mathcal{C}}} \epsilon_{a\nu}.$$

The shell-preserving input control and net phase satisfy $[V, H_K] = [U_\varphi, H_K] = 0$; both preserve ρ_β , as does free evolution. Preservation of the Gibbs state by the

complete cycle therefore depends on background restoration. For any prescribed restoration shares g ,

$$\mathcal{F}_{V,g}(\rho_\beta) = X_\beta + y_\beta \sigma_g^{\text{sh}}.$$

Since $y_\beta > 0$ and the projections P_{e_a} are linearly independent,

$$\boxed{\mathcal{F}_{V,g}(\rho_\beta) = \rho_\beta \iff g_a = \pi_a \quad (\text{all } a).}$$

This is the necessary and sufficient condition for the complete cycle to preserve the Gibbs state at the same inverse temperature. If the convergence condition of a positive exchange channel for every direction also holds, fixing $g = \pi$ before iteration gives

$$\rho_* = \frac{e^{-\beta H_K}}{Z_{\text{full}}}, \quad Z_{\text{full}} = \sum_{j=0}^K |S_j| e^{-\beta E_j}. \quad (4.74)$$

Moreover, for every initial state,

$$\|\mathcal{F}_{V,\pi}^n(\rho_0) - \rho_\beta\|_1 \leq (1 - c)^n \|\rho_0 - \rho_\beta\|_1 \longrightarrow 0.$$

Thermal exchange preserves the Gibbs weights, restoration preserves their background part, and positive exchange in every direction ensures convergence to the unique state from any initial state.

The Haar shell-mass preparation of the preceding section requires a separate comparison. Its shares are

$$h_j = (p - 1)p^{-j-1} \quad (0 \leq j < K), \quad h_K = p^{-K}.$$

For $K \geq 2$, $h_{j+1}/h_j = p^{-1}$ for $0 \leq j \leq K - 2$, while the terminal ratio is $h_K/h_{K-1} = 1/(p-1)$. The Gibbs shares, by contrast, satisfy $\pi_{j+1}/\pi_j = e^{-\beta \log p}$ for every adjacent pair of shells. The two distinct Haar ratios cannot both satisfy this relation, so no inverse temperature gives $h = \pi$ when $K \geq 2$. A cycle with $g = h$ still has a unique stationary state under the convergence conditions, with occupations determined by Eq. (4.72). For $K = 1$, there is only one adjacent-shell ratio, and $h = \pi$ at $\beta = \log(p-1)/\log p$.

§5 Observational Spacetime

The final cycle state contains system occupations and coherence, as well as correlations between the system and the auxiliary spaces participating in exchanges. Further description of the dynamics requires these changes to be distinguished and their observational representation to be determined.

Observation rests on a reference and the corresponding distinguishing action. The reference specifies the object against which states are compared, and the observational action determines which state differences can be distinguished. The reference, observed excitations, and auxiliary systems involved in observation are represented within the same joint description; their internal states and actions establish the observational relation.

§5.1 Reference for Observation

Position resolution distinguishes different position components in the state space, while negation relates opposite positions. Use the state space, inner product, and normalized point basis of Eqs. (1.17)–(1.19), and write $d = p^K$. Equation (2.5) expresses the position projections $P_x = |x\rangle\langle x|$ as finite Fourier sums of phase multipliers. Negation is the linear involution of Section 2.3, satisfying

$$\iota|x\rangle = |-x\rangle, \quad \iota^\dagger = \iota, \quad \iota^2 = I, \quad \iota P_x \iota = P_{-x}.$$

Since p is odd, 2 is invertible in $\mathbb{Z}/d\mathbb{Z}$, so $a = -a$ holds only for $a = 0$. For any nonzero a , the smallest subspace containing $|a\rangle$ and invariant under position projections and negation is

$$\mathcal{H}_{D_a} = \text{span}\{|a\rangle, |-a\rangle\}, \quad D_a = \{a, -a\}, \quad Q_{D_a} = P_a + P_{-a}.$$

Indeed, a negation-invariant subspace containing $|a\rangle$ must contain $|-a\rangle$, and the two-dimensional space above is invariant under all position projections and ι . The two-point negation orbit D_a is called a doublet, and its two-dimensional space $\mathcal{H}_{D_a}$ is the corresponding **observation unit**. The zero element forms a separate one-dimensional invariant space.

On the full state space, position projections and negation generate the complex operator algebra

$$\mathfrak{A}_\iota = \text{alg}_{\mathbb{C}}^*\{P_x : x \in \mathbb{Z}/d\mathbb{Z}; \iota\}. \quad (5.1)$$

Since $\sum_x P_x = I$, the algebra is unital. Let $\mathfrak{D}$ be the set of distinct nonzero negation orbits and abbreviate $\mathcal{H}_D = Q_D \mathcal{H}$. All generators preserve each orbit. On any $\mathcal{H}_D$, the algebra contains

$$\begin{aligned} E_{++} &= P_a, & E_{--} &= P_{-a}, \\ E_{+-} &= P_a \iota P_{-a} = |a\rangle\langle -a|, & E_{-+} &= P_{-a} \iota P_a = |-a\rangle\langle a|. \end{aligned} \quad (5.2)$$

These satisfy $E_{ij}E_{k\ell} = \delta_{jk}E_{i\ell}$ and $E_{ij}^\dagger = E_{ji}$, where $i, j, k, \ell \in \{+, -\}$, and span all complex 2×2 matrices on the orbit. Together with orbit preservation by the generators, this gives

$$\boxed{\mathfrak{A}_\iota = \mathbb{C}P_0 \oplus \bigoplus_{D \in \mathfrak{D}} \mathcal{B}(\mathcal{H}_D)}. \quad (5.3)$$

Here $\mathcal{B}(\mathcal{H}_D)$ denotes all linear operators on $\mathcal{H}_D$. An operator commuting with every P_x is diagonal in the point basis and can be written as $\sum_x c_x P_x$. Commutation with ι further requires $c_a = c_{-a}$. Conversely, diagonal operators satisfying this condition commute with every generator. Hence

$$Z(\mathfrak{A}_\iota) = \mathbb{C}P_0 \oplus \bigoplus_{D \in \mathfrak{D}} \mathbb{C}Q_D. \quad (5.4)$$

Thus P_0 and the projections Q_D are precisely the minimal nonzero central projections. Equation (5.3) is the simple-block decomposition of the algebra; every nonzero observation unit corresponds to a 2×2 simple block.

The multiplicative identity of the finite ring satisfies $1 \cdot x = x$. Fix 1 as the reference point for observation and take

$$D_1 = \{1, -1\}, \quad \mathcal{H}_{D_1} = \text{span}\{|1\rangle, |-1\rangle\}. \quad (5.5)$$

Within this reference unit, abbreviate $D = D_1$, so $Q_D = P_1 + P_{-1}$. In the ordered basis $(|1\rangle, |-1\rangle)$, using Eq. (5.2) for identification, its observable algebra is

$$\mathcal{A}_{\text{obs}} = (Q_D \mathfrak{A}_i Q_D)|_{\mathcal{H}_{D_1}} = M_2(\mathbb{C}). \quad (5.6)$$

The unit of this block is Q_D on the full state space and is denoted by I_2 in the two-dimensional representation.

The reference point for observation and the background in Section 3 have distinct roles. The background reference b specifies the equal-amplitude, equal-phase pattern, and σ_g^{sh} specifies the background restoration preparation used in the cycle. Position 1 fixes the current reference point for comparing position components; I_2 is the unit of the observable algebra. The background state in an observation unit is determined by compressing b or σ_g^{sh} to that unit.

The reference point 1 lies in the unit shell S_0 . For the shell Hamiltonian $H_K = (\log p) \sum_{j=0}^K j \Pi_j$ used above,

$$\ker H_K = L^2(S_0), \quad \dim \ker H_K = p^{K-1}(p-1), \quad H_K Q_D = 0. \quad (5.7)$$

The entire unit shell is a degenerate zero-energy subspace, and $|1\rangle$ is one normalized point-basis vector in it. Zero energy alone does not select this point. The unit-group action from Section 2, $R_u|x\rangle = |ux\rangle$, sends D_a to D_{ua} , unitarily relating observation units in the same shell. Its transitivity on every nonzero shell connects any two units within that shell. Every nonzero unit has the algebraic structure of Eq. (5.6); the zero element remains a separate one-dimensional block in Eq. (5.3). This decomposition concerns the algebra generated by position resolution and negation. The complete cycle continues to act on $\mathcal{H}$ and its auxiliary spaces, and the state on an observation unit is computed from the joint final state. Self-adjoint actions within the unit represent real-valued observables; their real-linear relations and positivity determine the corresponding geometry.

§5.2 Structure of Observational Spacetime

The algebra of an observation unit determines the available actions. Relations between real-valued observables are described by its self-adjoint part. This part forms the **observable space** $\mathcal{V}_{\text{obs}} = \text{Herm}_2(\mathbb{C})$. Take the Pauli matrices

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Every $A \in \mathcal{V}_{\text{obs}}$ has a unique expression

$$A = tI_2 + \sum_{a=1}^3 x_a \sigma_a = \begin{pmatrix} t + x_3 & x_1 - ix_2 \\ x_1 + ix_2 & t - x_3 \end{pmatrix}, \quad t, x_1, x_2, x_3 \in \mathbb{R}. \quad (5.8)$$

The diagonal entries determine t, x_3 , and the real and imaginary parts of the off-diagonal entries determine x_1, x_2 ; equivalently, $t = \text{Tr } A/2$ and $x_a = \text{Tr}(A\sigma_a)/2$. Thus $\dim_{\mathbb{R}} \mathcal{V}_{\text{obs}} = 4$.

The positive semidefinite matrices form a positive cone [19], denoted by $\mathcal{P}_{\text{obs}}$. The characteristic polynomial of A gives eigenvalues $t \pm |\mathbf{x}|$, where $|\mathbf{x}| = (x_1^2 + x_2^2 + x_3^2)^{1/2}$. Hence

$$\boxed{A \geq 0 \iff t \geq |\mathbf{x}|.} \quad (5.9)$$

The nonzero boundary consists of rank-one positive operators, with $t = |\mathbf{x}| > 0$; positive definite matrices satisfy $t > |\mathbf{x}|$ and lie in the interior. Every self-adjoint matrix is the difference of two positive matrices, obtained from its positive and negative spectral parts, so the real span of the cone is the entire $\mathcal{V}_{\text{obs}}$.

A real homogeneous quadratic form compatible with this cone boundary vanishes on every nonzero rank-one positive operator, and $q(I_2) = 1$ fixes its unit normalization. These two requirements specify the geometric construction used here. Write a general real quadratic form as

$$q(A) = ct^2 + 2t\mathbf{b} \cdot \mathbf{x} + \mathbf{x}^\top C \mathbf{x}, \quad C = C^\top \in M_3(\mathbb{R}).$$

For every real unit vector $\mathbf{n}$, both $I_2 \pm \mathbf{n} \cdot \boldsymbol{\sigma}$ are rank-one positive operators. Substituting both signs and subtracting gives $\mathbf{b} \cdot \mathbf{n} = 0$, hence $\mathbf{b} = 0$; adding gives $\mathbf{n}^\top C \mathbf{n} = -c$. Using coordinate unit vectors and normalized sums of each pair gives diagonal entries $-c$ and zero off-diagonal entries for C . Normalization then fixes $c = 1$, so the quadratic form is unique and equals

$$\boxed{q(A) = \det A = t^2 - x_1^2 - x_2^2 - x_3^2.} \quad (5.10)$$

Polarization gives

$$B(A, A') = \frac{1}{2}[q(A + A') - q(A) - q(A')] = \frac{1}{2}[\text{Tr } A \text{Tr } A' - \text{Tr}(AA')]. \quad (5.11)$$

In this basis, $B(A, A') = tt' - \sum_a x_a x'_a$, which is nondegenerate with signature $(1, 3)$. Construct an affine space modeled on the difference space $\mathcal{V}_{\text{obs}}$. After choosing an origin, label its points by (t, x_1, x_2, x_3) ; their quadratic interval is $q(A - A')$. We define this affine space with this metric to be **observational spacetime**, with difference space $\mathcal{V}_{\text{obs}}$ and metric signature $(1, 3)$. The positive direction of I_2 selects the future orientation, and the positive cone in Eq. (5.9) is the **closed future light cone**.

A complete representation of the same observational content must preserve real-linear relations of self-adjoint actions, positivity, and the normalized unit. Let another representation have real vector space $\mathcal{V}'$, positive cone $\mathcal{P}'$, and unit e' , with a real-linear bijection $F : \mathcal{V}_{\text{obs}} \rightarrow \mathcal{V}'$ satisfying $F(\mathcal{P}_{\text{obs}}) = \mathcal{P}'$ and $F(I_2) = e'$. The unit here is the algebra unit, defined separately from the background state of Section 3. Writing $A \leq B$ for $B - A \geq 0$, the positive order and unit determine two spectral endpoints:

$$\begin{aligned} \lambda_{\max}(A) &= \inf\{s \in \mathbb{R} : A \leq sI_2\}, \\ \lambda_{\min}(A) &= \sup\{s \in \mathbb{R} : sI_2 \leq A\}. \end{aligned} \quad (5.12)$$

Spectral decomposition identifies them as the largest and smallest eigenvalues of A . Since F and its inverse preserve positivity, $A \leq sI_2$ if and only if $F(A) \leq se'$, and likewise for the other inequality. Both endpoints, defined in the other representation by the same order relations, are therefore preserved:

$$q'(F(A)) = \lambda_{\min}(F(A))\lambda_{\max}(F(A)) = \lambda_{\min}(A)\lambda_{\max}(A) = \det A. \quad (5.13)$$

Since F is a real-linear bijection, q' remains a real quadratic form with the same dimension and signature. For any two such representations F_1, F_2 , the map $F_2 F_1^{-1}$ preserves the quadratic form, positive cone, and unit. All nonzero observation units

consequently have the same real four-dimensional geometric type. In these coordinates, a unit-preserving change of representation preserves I_2 and the polar form B , and hence preserves the B -orthogonal complement $\{t = 0\}$ of I_2 . The quadratic form there is $-|\mathbf{x}|^2$, so the restriction belongs to $O(3)$ and is a spatial orthogonal transformation.

Geometric transformations are also expressed through the same matrices. For $g \in \text{SL}(2, \mathbb{C})$,

$$A \mapsto gAg^\dagger, \quad \det(gAg^\dagger) = |\det g|^2 \det A = \det A. \quad (5.14)$$

This action preserves the positive cone and gives a future-preserving Lorentz transformation [20]. A general g sends I_2 to $gg^\dagger$, allowing the timelike unit axis to change with the frame; unitary g preserves this axis.

In a chosen frame, I_2 gives the timelike axis and the three Pauli directions give spatial axes. Define the geometric cone order by $A \preceq A'$ if and only if $A' - A \geq 0$, and distinguish timelike, null, and spacelike intervals by the sign of $q(A' - A)$. Whether a particular record history obeys a propagation bound must be determined by the cycle action and actual position readout. These congruence transformations describe symmetries of the observational geometry. Nonunitary g generally changes matrix traces, so its implementation as a physical operation must be distinguished from trace-preserving system evolution. This determines the observational geometry; the final state of a concrete cycle defines a state and its pairing with observables on the same algebra.

§5.3 Observables and States

The same observational geometry accommodates different cycle states. Position amplitudes produced by the cycle and their environment correlations jointly determine occupation and coherence on an observation unit. Use the joint final state Ω_n^{out} of Section 4 for a cycle with input ρ_n . The system state entering the next cycle is

$$\rho_{n+1} = \text{Tr}_{\mathcal{E}} \Omega_n^{\text{out}}. \quad (5.15)$$

To represent the two position components of the observation unit, take the isometric embedding $j_D : \mathbb{C}^2 \rightarrow \mathcal{H}$, $j_D(c_+, c_-)^\top = c_+|1\rangle + c_-|-1\rangle$. The two point-basis vectors are orthonormal, so

$$j_D^\dagger j_D = I_2, \quad j_D j_D^\dagger = Q_D.$$

The state matrix on the observation unit is

$$R_{n+1} = j_D^\dagger \rho_{n+1} j_D, \quad R_{n+1} \geq 0, \quad \text{Tr } R_{n+1} = \text{Tr}(Q_D \rho_{n+1}) \leq 1. \quad (5.16)$$

Positivity follows from $\mathbf{c}^\dagger R_{n+1} \mathbf{c} = \langle j_D \mathbf{c}, \rho_{n+1} j_D \mathbf{c} \rangle \geq 0$, and the trace bound follows from $0 \leq Q_D \leq I$ and $\text{Tr } \rho_{n+1} = 1$. If $\text{Tr } R_{n+1} = 0$, positivity forces both eigenvalues to vanish and hence $R_{n+1} = 0$. For positive trace, $R_{n+1} / \text{Tr } R_{n+1}$ is the normalized state conditional on occupation of this unit. By Eqs. (5.8) and (5.9), write the state matrix as

$$R = t_R I_2 + \mathbf{x}_R \cdot \boldsymbol{\sigma}, \quad R \geq 0, \quad \text{Tr } R \leq 1 \quad \Longleftrightarrow \quad |\mathbf{x}_R| \leq t_R \leq \frac{1}{2}. \quad (5.17)$$

State matrices occupy only the portion of the positive cone satisfying this trace bound. The observable space has real dimension four; after normalization to trace

one, a nonzero state matrix generally has only three independent real parameters, giving the Bloch-ball representation of a two-level state [21]. For another nonzero orbit D_a , replace j_D by $(c_+, c_-)^T \mapsto c_+|a\rangle + c_-|-a\rangle$ to obtain the same form. Denote this embedding by j_a and write $R_a[\rho] = j_a^\dagger \rho j_a$. The state compression and linear pairing below apply to every nonzero observation unit.

The joint final state has spectral decomposition $\Omega_n^{\text{out}} = \sum_\alpha w_\alpha |\Psi_\alpha\rangle\langle\Psi_\alpha|$, where $w_\alpha \geq 0$, $\sum_\alpha w_\alpha = 1$, and each Ψ_α is normalized. Let $\eta_{\epsilon,\alpha} = (\langle\epsilon 1| \otimes I_{\mathcal{E}})\Psi_\alpha$ be its environment vectors, with $\epsilon \in \{+, -\}$. These vectors include the amplitudes of their position components and need not be individually normalized. Expanding the joint state in the normalized point basis and tracing out the environment gives the following two-position form of Eq. (4.50):

$$(R_{n+1})_{\epsilon\epsilon'} = \sum_\alpha w_\alpha \langle \eta_{\epsilon',\alpha}, \eta_{\epsilon,\alpha} \rangle. \quad (5.18)$$

Diagonal entries are squared position-amplitude moduli; off-diagonal entries include overlaps of the associated environment states. Since $|x\rangle$ is normalized, the environment vectors give the entries directly, with no additional position-normalization factor. If the system and environment form a product state and the system is pure, write its amplitude in the unit as $\psi_D = (c_+, c_-)^T$. The expression becomes

$$R = \psi_D \psi_D^\dagger = \begin{pmatrix} |c_+|^2 & c_+ \overline{c_-} \\ c_- \overline{c_+} & |c_-|^2 \end{pmatrix}. \quad (5.19)$$

A global phase $\psi_D \mapsto e^{i\gamma} \psi_D$ leaves R invariant, while the relative phase remains in its off-diagonal entries. In Eq. (5.18), orthogonal environment components contribute zero to the corresponding off-diagonal entry.

Real-valued observables are represented by self-adjoint operators. For any $A = A^\dagger \in \mathcal{A}_{\text{obs}}$, the state defines the linear functional

$$\ell_{n+1}(A) = \text{Tr}(R_{n+1}A) = \text{Tr}[\rho_{n+1} j_D A j_D^\dagger] \in \mathbb{R}. \quad (5.20)$$

Complex conjugation and cyclicity of the trace verify that it is real; for positive observables it is positive:

$$\begin{aligned} \overline{\ell_{n+1}(A)} &= \text{Tr}(A R_{n+1}) = \ell_{n+1}(A), \\ \ell_{n+1}(A) &= \text{Tr}(A^{1/2} R_{n+1} A^{1/2}) \geq 0 \quad (A \geq 0). \end{aligned}$$

The final cycle state thus determines a positive linear functional on the observable algebra, with $\ell_{n+1}(I_2) = \text{Tr} R_{n+1}$. When this value is positive, the normalized pairing within the unit is

$$\ell_{D,n+1}(A) = \frac{\text{Tr}(R_{n+1}A)}{\text{Tr} R_{n+1}}, \quad \ell_{D,n+1}(I_2) = 1. \quad (5.21)$$

This is the linear pairing of states and observables. An actual pointer determines an individual reading; Section 6 establishes the probabilistic interpretation of this pairing under the stated statistical consistency conditions.

Background and stable states define their respective states on the same observable algebra. Substitute the background reference b and shell-dephased background

of Eq. (3.5) into the compression formula. Using $\langle x|b\rangle = d^{-1/2}$ and $h_0/|S_0| = 1/d$ gives

$$\begin{aligned} Q_D b &= d^{-1/2}(|1\rangle + |-1\rangle), \\ Q_D P_b Q_D|_{\mathcal{H}_{D_1}} &= Q_D \sigma_b^{\text{sh}} Q_D|_{\mathcal{H}_{D_1}} = \frac{1}{d} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}. \end{aligned} \quad (5.22)$$

Both retain intra-shell coherence in the unit and have trace $2/d$, while their inter-shell coherence on the full space differs. For general restoration shares g and $a \in S_j$, only $|e_j\rangle\langle e_j|$ contributes to the unit, so

$$\mathbf{R}_a[\sigma_g^{\text{sh}}] = \frac{g_j}{|S_j|} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \quad \text{Tr } \mathbf{R}_a[\sigma_g^{\text{sh}}] = \frac{2g_j}{|S_j|}. \quad (5.23)$$

The reference unit has $j = 0$. For $g = h$, every nonzero shell satisfies $h_j/|S_j| = 1/d$, and every nonzero unit has the background compression in Eq. (5.22). Whenever $g_j > 0$, the normalized background state in that unit is $(I_2 + \sigma_1)/2$. Background occupation coefficients do not change the observable algebra or its geometry.

The zero-element state $z = e_K$ satisfies $Q_D z = 0$, so $\mathbf{R}_1[P_z] = 0$. This zero compression means that the current state has no occupation in the unit; subsequent occupation is still determined by the full-space cycle. When V is the identity on $\mathcal{C}$, Eq. (4.61) gives

$$\mathcal{F}_{V,g}(P_z) = \sum_{\nu} f_{\nu K} P_{\nu} + k_K \sigma_g^{\text{sh}}.$$

Every term is positive. If the restoration preparation has $g_0 > 0$, then $k_K \geq a_0 > 0$ gives

$$\text{Tr } \mathbf{R}_1[\mathcal{F}_{V,g}(P_z)] \geq \frac{2k_K g_0}{|S_0|} > 0.$$

Thus an initially unoccupied unit still participates in later cycles; the complete action can supply occupation from background and other components.

The same compression gives local occupation and coherence for the stable states of Section 3. First take a normalized stable mode $|\nu\rangle = f_{j,\theta_{\nu}}$ on shell S_j , with nonzero response at the selected parameter. For $a = p^j u \in S_j$, Equation (3.26) gives

$$\mathbf{R}_a[P_{\nu}] = \frac{1}{|S_j|} \begin{pmatrix} 1 & \theta_{\nu}(-1) \\ \theta_{\nu}(-1) & 1 \end{pmatrix}, \quad \text{Tr } \mathbf{R}_a[P_{\nu}] = \frac{2}{|S_j|}. \quad (5.24)$$

Since $\theta_{\nu}(-1)^2 = \theta_{\nu}(1) = 1$, we have $\theta_{\nu}(-1) = \pm 1$. Also $\theta_{\nu}(-u) = \theta_{\nu}(-1)\theta_{\nu}(u)$, so the common phase at position u cancels in the compressed matrix. For a nonzero position a outside S_j , both points lie outside the mode's support and the compressed matrix is zero. In an occupied unit, the normalized state is $(I_2 + \theta_{\nu}(-1)\sigma_1)/2$. This local representation of a single mode therefore gives its occupation share and negation parity. Different modes with the same parity may have identical representations in units of the same shell.

A general stable pattern allows superposition of modes sharing the same nonzero response λ . Using the direction sets $\Theta_j(\lambda)$ of Eq. (3.28), write

$$\psi = \sum_j \sum_{\theta \in \Theta_j(\lambda)} a_{j\theta} f_{j,\theta}, \quad \sum_{j,\theta} |a_{j\theta}|^2 = 1.$$

On S_j , set

$$A_j(u) = \sum_{\theta \in \Theta_j(\lambda)} a_{j\theta} \theta(u).$$

By Eq. (3.29), the amplitudes at the two positions of D_a are $A_j(u)/\sqrt{|S_j|}$ and $A_j(-u)/\sqrt{|S_j|}$. Substitution into Eq. (5.19) gives

$$\begin{aligned} \mathbf{R}_a[P_\psi] &= \frac{1}{|S_j|} \begin{pmatrix} |A_j(u)|^2 & A_j(u)\overline{A_j(-u)} \\ A_j(-u)\overline{A_j(u)} & |A_j(-u)|^2 \end{pmatrix}, \\ \text{Tr } \mathbf{R}_a[P_\psi] &= \frac{|A_j(u)|^2 + |A_j(-u)|^2}{|S_j|} \leq 1. \end{aligned} \tag{5.25}$$

If more than one direction in the shell is occupied, their relative coefficients and phases enter the local representation through A_j and can change both position occupation and coherence. If all occupied directions in the shell have the same negation parity ϵ , then $A_j(-u) = \epsilon A_j(u)$. For nonzero occupation in the unit, its normalized state remains $(I_2 + \epsilon \sigma_1)/2$, so relative phases between these directions can appear in this unit only through its occupation share. Since both points of D_a lie in the same shell, relative phases between overall amplitudes on different shells do not enter this compression. In particular, for the same-character inter-shell superposition in Eq. (3.31), the compression on shell j is Eq. (5.24) multiplied by $|a_j|^2$. These compressions relate stable patterns to observational states and show that a single observation unit generally does not suffice to reconstruct the full-space pattern. Mixed states and environment-correlated states produced by the cycle are still calculated using Eqs. (5.15)–(5.18).

Figure 4 shows the relation between observational geometry and cycle states on this unit.

Reference for observation and geometric structure

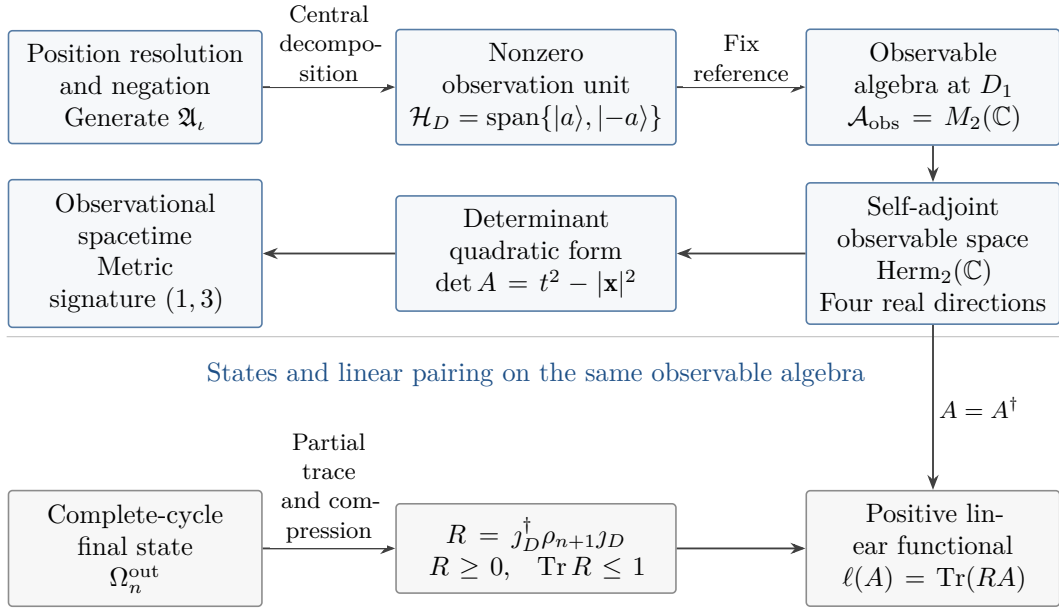


Figure 4: From observational actions in the state space to observational geometry. Central decomposition of the algebra generated by position projections and negation gives the two-dimensional observation units associated with all nonzero doublets, with the zero element forming a separate one-dimensional block. After fixing a reference, self-adjoint actions give four real directions. The normalized quadratic form compatible with the positive-cone boundary is the determinant, whose polarization gives the metric. Below are the state matrix and linear pairing determined by the complete cycle; individual outcomes are realized by actual pointers. A nonzero state matrix normalized to trace one generally has three independent real parameters.

Each cycle produces a corresponding observational state. Comparing these states at different times requires the cycle's time calibration and phase readings.

§5.4 Time and Frequency

Cycle time is determined by actual evolution durations, while phase changes provide readings associated with cycle counts. The position set $X_{p,K}$ is finite, while the amplitudes vary continuously during each actual unitary stage of Section 4. Let $U(s)$ be the actual joint propagator from a common initial time to s , with s and $s + h$ in the same stage. For its constant self-adjoint generator H_{seg} ,

$$U(s + h) = e^{-ihH_{\text{seg}}}U(s), \quad e^{-ihH_{\text{seg}}} - I = -i \int_0^h H_{\text{seg}} e^{-iuH_{\text{seg}}} du.$$

Each exponential factor is unitary, so

$$\|U(s + h) - U(s)\| \leq |h| \|H_{\text{seg}}\|.$$

Set $U' = U(s + h)$ and $U = U(s)$. For any normalized joint density operator Ω , write the difference as

$$U'\Omega U'^\dagger - U\Omega U^\dagger = (U' - U)\Omega U'^\dagger + U\Omega(U'^\dagger - U^\dagger).$$

Using $\|AXB\|_1 \leq \|A\|\|X\|_1\|B\|$ and $\|\Omega\|_1 = 1$ gives

$$\begin{aligned} & \|U(s+h)\Omega U(s+h)^\dagger - U(s)\Omega U(s)^\dagger\|_1 \\ & \leq 2\|U(s+h) - U(s)\| \leq 2|h| \|H_{\text{seg}}\|. \end{aligned} \quad (5.26)$$

The final state of each stage is the initial state of the next, so their finite ordered composition remains continuous. For a self-adjoint difference X , trace-norm duality gives

$$\|\text{Tr}_{\mathcal{E}} X\|_1 = \sup_{L=L^\dagger, \|L\| \leq 1} |\text{Tr}[(L \otimes I_{\mathcal{E}})X]| \leq \|X\|_1.$$

The reduced state is therefore continuous in actual time. Since $\|J_D^\dagger X J_D\|_1 \leq \|X\|_1$, the state matrix on the observation unit is continuous as well. Environment refreshing at cycle end preserves the current reduced system state; a fresh environment is independently prepared in the same state for the next cycle.

The cycle period is the sum of all actual stage durations, and the cycle rate is $v = 1/\tau_{\text{tot}}$. Using the durations in Eq. (4.46) and fixing $t_0 = 0$ gives

$$\begin{aligned} \tau_{\text{tot}} &= t_f + t_V + t_T + t_{\text{out}} + t_{\text{back}} + t_{\text{res}}, \\ t_n &= n\tau_{\text{tot}}, \quad v = \tau_{\text{tot}}^{-1}. \end{aligned} \quad (5.27)$$

Each completed cycle advances time by τ_{tot} , yielding $t_n = n\tau_{\text{tot}}$ by successive addition. Two endpoints separated by m cycles have interval $m\tau_{\text{tot}}$. Without input control, $t_V = 0$ and $\tau_{\text{tot}} = \tau$.

In the geometric frame of Eq. (5.10), choose a time unit and calibrate the timelike coordinate by the cycle period. After fixing the time origin, label cycle-end times by $t_n = n\tau_{\text{tot}}$. A pure time interval is represented by $\Delta t I_2$, satisfying $q(\Delta t I_2) = (\Delta t)^2$. In Eq. (5.8), $t = \text{Tr } A/2$ is the timelike component of a geometric vector; the trace of the state matrix R_n , however, is the occupation share of the observation unit. The geometric timelike component, cycle-end time, and state occupation share have these distinct definitions. Numerical calibration of the period in the geometry requires a time unit; $\text{Tr } R_n/2$ cannot replace the cycle-end time. The reference observation unit satisfies $H_K Q_D = 0$, and the shell-dephased background is invariant under free evolution generated by H_K . Relative phase changes in the complete cycle can be used for cycle timing. Consider a cycle without input control, $V = I$, with the same cycle data and restoration shares g_a in every cycle. Using Eq. (4.64), write

$$d_\nu = \sum_a a_{a\nu} t_u (1 - c_{a\nu}), \quad \gamma_{\nu\mu} = 1 - d_\nu - d_\mu \quad (\nu \neq \mu).$$

Here $c_{a\nu} = \cos \alpha_{a\nu}$, and t_u is taken from the corresponding thermal channel. The preceding section proves $a_0 \leq \gamma_{\nu\mu} \leq 1$, with $a_0 > 0$. Transfer terms in thermal exchange contribute only to occupations, and background restoration produces no coherence between distinct directions. Write $(\rho_n)_{\nu\mu} = \langle \nu | \rho_n | \mu \rangle$. For $\nu \neq \mu$, the diagonal multipliers and return phases computed in Section 4 give

$$(\rho_{n+1})_{\nu\mu} = (1 - d_\nu - d_\mu) e^{i(\varphi_\nu - \varphi_\mu)} (\rho_n)_{\nu\mu}. \quad (5.28)$$

Here φ_ν is the direction's phase in the complete cycle. The positive attenuation factor $\gamma_{\nu\mu}$ does not change the argument of the matrix element. With the same independent environment preparation in every cycle, iteration of Eq. (5.28) gives

$$(\rho_m)_{\nu\mu} = (1 - d_\nu - d_\mu)^m e^{im(\varphi_\nu - \varphi_\mu)} (\rho_0)_{\nu\mu}. \quad (5.29)$$

If the chosen model contains distinct directions ν, μ in the same shell such that

$$(\rho_0)_{\nu\mu} \neq 0, \quad e^{i\Delta\varphi_{\nu\mu}} \neq 1, \quad \Delta\varphi_{\nu\mu} = \varphi_\nu - \varphi_\mu, \quad (5.30)$$

their coherence can be used for phase timing at cycle endpoints. The second condition is equivalent to $\Delta\varphi_{\nu\mu} \notin 2\pi\mathbb{Z}$, ensuring different phase factors at adjacent endpoints. Existence of such a pair and preparation of its initial coherence must both satisfy this criterion. Equation (5.29) keeps the coherence entry nonzero at every finite cycle count, with argument satisfying

$$\arg(\rho_m)_{\nu\mu} = \arg(\rho_0)_{\nu\mu} + m\Delta\varphi_{\nu\mu} \pmod{2\pi}.$$

After choosing an unwrapped phase branch, define the relative angular frequency

$$\Delta\omega_{\nu\mu} = \frac{\Delta\varphi_{\nu\mu}}{\tau_{\text{tot}}} \neq 0.$$

Using the known cycle count m , we write the unwrapped accumulated phase as $\Delta\Phi_{\nu\mu} = m\Delta\varphi_{\nu\mu}$. This gives

$$\Delta t = m\tau_{\text{tot}} = \frac{\Delta\Phi_{\nu\mu}}{\Delta\omega_{\nu\mu}}. \quad (5.31)$$

Equation (5.31) relates unwrapped accumulated phase to cycle count. A single phase reading determines only a value modulo 2π ; accumulated timing uses the cycle count above. In particular, if $\Delta\varphi_{\nu\mu}/(2\pi)$ is rational, endpoint phases repeat after finitely many cycles, and this phase reading alone does not determine the accumulated count. Continuous phase tracking, if used, requires its own readout conditions; nonzero endpoint coherence alone does not establish a readout process within each stage. For $\gamma_{\nu\mu} < 1$, the coherence amplitude decays with the cycle count; for $\gamma_{\nu\mu} = 1$, it is preserved. Actual phase resolution also depends on readout precision. When V mixes directions, the complete state is computed from Eq. (4.53) and generally does not satisfy this directionwise phase recurrence.

Endpoint phase factors admit unwrapped values differing by integer multiples of 2π . Endpoint data alone therefore determine relative angular frequency only modulo $2\pi/\tau_{\text{tot}}$. A continuous interpolation of these endpoints can be fixed within a restricted analytic function class. Suppose two candidates f_1, f_2 are analytic in a neighborhood of the closed right half-plane, and each admits finite constants $A_i, C_i > 0$ and $0 \leq b_i < \pi v$ such that

$$|f_i(z)| \leq C_i e^{A_i|z|} \quad (\Re z \geq 0), \quad |f_i(iy)| \leq C_i e^{b_i|y|} \quad (y \in \mathbb{R}). \quad (5.32)$$

If $f_1(n\tau_{\text{tot}}) = f_2(n\tau_{\text{tot}})$ for every $n \geq 0$, then $g(z) = f_1(\tau_{\text{tot}}z) - f_2(\tau_{\text{tot}}z)$ vanishes at nonnegative integers, has finite exponential type in the right half-plane, and has type strictly less than π on the imaginary axis. Carlson's uniqueness theorem gives $g = 0$ [22]. For a chosen branch $|\varphi_\nu| < \pi$, the function $e^{iv\varphi_\nu z}$ belongs to this class and is therefore the unique interpolation of the sequence $e^{in\varphi_\nu}$ within this class, with angular frequency $\omega_\nu = v\varphi_\nu$. Applying this uniqueness result to a relative-phase sequence additionally requires $|\Delta\varphi_{\nu\mu}| < \pi$; bounds on the two individual phases do not imply the bound on their difference. Without these analytic and growth conditions, endpoint phases still determine only a branch modulo 2π .

This uniqueness concerns the phase factor; it excludes the attenuation factor $\gamma_{\nu\mu}^m$ in Eq. (5.29) and does not specify the timing of controls within a stage. Actual continuous evolution remains determined by the stage generators and durations of Section 4. The instrument in Section 6.3 determines how the real and imaginary parts of same-shell coherence become separate record readings.

§5.5 Position Intervals

Position readings are determined jointly by the chosen reference and coordinate calibration. Section 2 defines logarithmic coordinates and intervals on nonzero position labels. We now express this calibration as an observational action on the state space and examine the structure of its readings.

For each nonzero residue class, take its unique integer representative $m \in \{1, \dots, d-1\}$. Fix the reference representative $u_* \in \{1, \dots, d-1\}$ and define the coordinate of position m relative to it by

$$y_{u_*}(m) = \log \frac{m}{u_*}.$$

The reference has coordinate zero. The reference point 1 used here has $u_* = 1$, giving $y_1(m) = \log m$. The zero element is excluded from this logarithmic calibration and retains its separate observation component P_0 .

On the nonzero-position subspace

$$\mathcal{H}_{\neq 0} = \bigoplus_{m=1}^{d-1} \text{Ran } P_m$$

define the position-readout operator

$$Y_{u_*} = \sum_{m=1}^{d-1} \log \frac{m}{u_*} P_m. \quad (5.33)$$

Its coefficients are real and the position projections are mutually orthogonal, so Y_{u_*} is self-adjoint, with spectrum

$$\text{spec}(Y_{u_*}) = \left\{ \log \frac{m}{u_*} : 1 \leq m < d \right\}.$$

Strict monotonicity of the logarithm assigns distinct readings to distinct integer representatives; the spectral projections are exactly P_m .

Changing the reference gives

$$Y_{u'_*} = Y_{u_*} + \log \frac{u_*}{u'_*} I_{\neq 0}, \quad I_{\neq 0} = \sum_{m=1}^{d-1} P_m. \quad (5.34)$$

Thus every coordinate changes by a common constant, while the oriented interval between positions satisfies

$$y_{u_*}(m) - y_{u_*}(n) = \log \frac{m}{n} = y(m, n),$$

and is reference independent. Its absolute value is the distance defined in Eq. (2.28),

$$d_{\log}(m, n) = \left| \log \frac{m}{n} \right|.$$

These relations use the chosen integer representatives. After modular wraparound under an action on the finite ring, position coordinates must be recomputed from the representatives of the resulting residue classes.

Position readout is directly related to the observable algebra. Every P_m belongs to the full-space algebra $\mathfrak{A}_i$, so Y_{u_*} belongs to the self-adjoint part of $I_{\neq 0} \mathfrak{A}_i I_{\neq 0}$ on $\mathcal{H}_{\neq 0}$. For a nonzero two-point orbit $D_a = \{a, -a\}$, if a has representative m , then $-a$ has representative $d - m$. Using its isometric embedding j_a gives

$$\begin{aligned} j_a^\dagger Y_{u_*} j_a &= \begin{pmatrix} \log(m/u_*) & 0 \\ 0 & \log((d-m)/u_*) \end{pmatrix} \\ &= \frac{1}{2} \log \frac{m(d-m)}{u_*^2} I_2 + \frac{1}{2} \log \frac{m}{d-m} \sigma_3. \end{aligned} \quad (5.35)$$

This is the self-adjoint matrix representation of position readout in the observation unit. Changing the reference affects only the identity-matrix component and leaves the difference of the two position readings unchanged.

Here $d_{\log}$ compares scalar readings in the chosen calibration; the determinant quadratic form of Section 5.2 determines the geometry of observational spacetime. Using position readings as specific spacetime coordinates additionally requires calibration of the respective coordinate components. The scalar distance above cannot be directly identified with a spacetime interval.

The valuation decomposition of integer representatives determines the relation between position coordinates and shell energies. Write

$$m = p^j u, \quad n = p^k v, \quad p \nmid uv,$$

Equation (2.32) then gives

$$y(m, n) = (j - k) \log p + \log \frac{u}{v} = (E_j - E_k) + \log \frac{u}{v}.$$

For positions with the same integer unit part, the oriented interval equals the shell-energy difference; in general, the unit parts also contribute. Positions in the same shell can therefore have different readings but equal shell energy.

At fixed depth K , all oriented intervals form the finite set

$$\mathcal{L}_{p,K} = \left\{ \log \frac{m}{n} : 1 \leq m, n < p^K \right\}. \quad (5.36)$$

With the usual real distance, this set is already complete: distinct elements have a minimum positive separation, so every Cauchy sequence is eventually constant. Position readings and intervals remain finite in number at fixed depth.

An ambient space for continuous scalar intervals can be constructed by comparison across depths. Keeping p fixed, the same pair of positive integer representatives can still be used at greater depths, so

$$\mathcal{L}_{p,K} \subseteq \mathcal{L}_{p,K+1}.$$

Every interval is the logarithm of a positive rational; conversely, both numerator and denominator of any positive rational m/n are below some p^K . Thus

$$\bigcup_{K \geq 1} \mathcal{L}_{p,K} = \log \mathbb{Q}_{>0}. \quad (5.37)$$

For any nonempty real open interval (a, b) , density of the positive rationals gives a positive rational $q \in (e^a, e^b)$ and hence $a < \log q < b$. Therefore $\log \mathbb{Q}_{>0}$ is dense in the real line, and its completion in the usual real metric is

$$\overline{\bigcup_{K \geq 1} \mathcal{L}_{p,K}} = \mathbb{R}. \quad (5.38)$$

This completion concerns the set of real intervals and provides a continuous ambient space for scalar intervals. Each individual position reading at fixed depth is still determined by the finite spectrum in Eq. (5.33). Whether structure-preserving maps exist between state spaces and cycles at different depths must be established separately. A concrete comparison of shell scaling and position-resolving lifts appears in Appendix A.3.

§6 Records and Events

§6.1 Formation of Records

Formation exchange correlates the system with auxiliary spaces. Distinctions left by different processes constitute a **record** if they remain in the actual output after retrieval and pattern reconstruction and are preserved during the subsequent evolution under consideration.

Use the state space $\mathcal{H}$ and normalized point basis $|x\rangle$ of Section 1, with $d = p^K$ and $P_x = |x\rangle\langle x|$. First take the original cycle of Section 4 with $V = I$ and $g = h$, write $\Psi_\beta = \mathcal{F}_{I,h}$, and use the complete auxiliary preparation $\eta = P_g \otimes \eta_T$. Denote its joint cycle unitary, environment preparation, and reduced action by $\mathbb{U}_{\text{cyc}}$, η , and Ψ_β , respectively:

$$\Omega^{\text{out}}(\rho) = \mathbb{U}_{\text{cyc}}(\rho \otimes \eta) \mathbb{U}_{\text{cyc}}^\dagger, \quad \Psi_\beta(\rho) = \text{Tr}_{\mathcal{E}} \Omega^{\text{out}}(\rho). \quad (6.1)$$

Here ρ is the initial system density operator, $\mathcal{E}$ is the full auxiliary space of Section 4, and β is the inverse temperature of the thermal-exchange environment. Equation (6.1) gives system evolution before any end-of-cycle probing is added. The joint definition of records also applies to fixed V, g . The original cycle's matrix elements are first used to explain readout; Section 6.6 then uses the actual periods of the same controlled family throughout.

Unambiguous identification of a process distinction requires specifying the process components in the actual preparation and joint evolution and testing whether their outputs form preservable pointer labels. For a process component α with nonzero share, let ω_α be its normalized conditional environment state and $r = f(\alpha)$ its recorded process class. A class may combine several processes but must distinguish the destinations to be recorded. Process components are determined by the actual preparation and joint evolution, not by an arbitrary Kraus decomposition of the reduced channel. The following three conditions jointly define a **complete record** of the selected process classes.

-
- (1) Distinctions are represented by orthogonal subspaces of an actual pointer. Let Π_r^{rec} project onto these pointer subspaces, and include remaining outputs as a no-response outcome, so all outcomes satisfy

$$\Pi_r^{\text{rec}} \Pi_s^{\text{rec}} = \delta_{rs} \Pi_r^{\text{rec}}, \quad \sum_r \Pi_r^{\text{rec}} = I_{\mathcal{E}}. \quad (6.2)$$

Correspondence between process classes and pointer labels requires

$$\boxed{\Pi_r^{\text{rec}} \omega_\alpha = \delta_{r,f(\alpha)} \omega_\alpha.} \quad (6.3)$$

Thus $\text{Tr}(\Pi_r^{\text{rec}} \omega_\alpha) = \delta_{r,f(\alpha)}$. Conditional states of different classes are supported on orthogonal pointer subspaces; coherence may remain within a class. These projections generate a commutative record algebra, relative to which the labels are classical.

- (2) Distinctions remain in the actual output. Conditional environment states are taken from the joint final state after free evolution, formation exchange, and pattern reconstruction. Record-bearing components must have nonzero share and, under the preservation scheme used, be excluded from subsequent retrieval. Distinctions between intermediate auxiliary states must pass through the remaining joint evolution before Eq. (6.3) can be applied. We use the environment refreshing of Section 4: after the cycle, its environment no longer interacts with the later system; the next cycle receives the same fresh preparation, independent of the system and all existing environments and records jointly.
- (3) Labels remain unchanged during storage. Suppose a record forms at t_{form} and is read at t_{read} . Let $U_{\text{post}}(t, t_{\text{form}})$ be the actual subsequent joint unitary, including the old record and all other degrees of freedom. On this joint space set $\hat{\Pi}_r^{\text{rec}} = I_{\text{rest}} \otimes \Pi_r^{\text{rec}}$. Preservation requires

$$U_{\text{post}}(t, t_{\text{form}})^\dagger \hat{\Pi}_r^{\text{rec}} U_{\text{post}}(t, t_{\text{form}}) = \hat{\Pi}_r^{\text{rec}}, \quad t_{\text{form}} \leq t \leq t_{\text{read}}. \quad (6.4)$$

For any joint state Ω_{form} at record formation, cyclicity of the trace gives

$$\text{Tr} \left[\hat{\Pi}_r^{\text{rec}} U_{\text{post}} \Omega_{\text{form}} U_{\text{post}}^\dagger \right] = \text{Tr} \left[U_{\text{post}}^\dagger \hat{\Pi}_r^{\text{rec}} U_{\text{post}} \Omega_{\text{form}} \right] = \text{Tr} \left[\hat{\Pi}_r^{\text{rec}} \Omega_{\text{form}} \right].$$

The share of each label is therefore constant throughout the storage interval.

Storage and subsequent cycles implement the preservation condition separately. If the environment storage Hamiltonian H_{store} commutes with the record projection Π_r^{rec} , then

$$e^{itH_{\text{store}}} \Pi_r^{\text{rec}} e^{-itH_{\text{store}}} = \Pi_r^{\text{rec}}. \quad (6.5)$$

The stored label share is unchanged with storage time. Subsequent cycles act only on the system and fresh environments. Denote their joint unitary by U_{next} and the old-record projection by $\Pi_{\text{old}}^{\text{rec}}$. Displaying the old-record space explicitly gives

$$[U_{\text{next}} \otimes I_{\text{old}}, I_{S,\text{new}} \otimes \Pi_{\text{old}}^{\text{rec}}] = 0. \quad (6.6)$$

For any joint state, the trace against this projection is thus equal before and after the unitary action. Together with Eq. (6.5), subsequent exchanges and background

restoration preserve marginal shares of old record histories. An auxiliary system still participating in retrieval within the current cycle has not departed and continues to follow the full joint evolution.

Coherence between processes is determined by their associated final environment states. Suppose several processes lead to the same normalized system state $|\phi\rangle$, with amplitudes A_α and normalized environment states $|e_\alpha\rangle$. They combine as $|\phi\rangle \otimes \sum_\alpha A_\alpha |e_\alpha\rangle$. Its squared norm, using the same input normalization, gives the **share**

$$w_\phi = \sum_{\alpha, \alpha'} A_\alpha \overline{A_{\alpha'}} \langle e_{\alpha'} | e_\alpha \rangle. \quad (6.7)$$

For identical environment final states, absorbing their global phases into A_α gives $w_\phi = |\sum_\alpha A_\alpha|^2$. For pairwise orthogonal environment final states, $w_\phi = \sum_\alpha |A_\alpha|^2$. For partial overlap, cross terms remain as in Eq. (6.7). Environment distinguishability and interference are quantitatively related through which-path information and fringe visibility [23]. Environment subspaces of distinct conserved charges are orthogonal; states within the same charge subspace may still differ, and their overlaps must be calculated from their actual final states.

Environment refreshing prevents departed components from being retrieved in later cycles but does not change inner products already present at the same cycle end. Mapping the full supports of two states into separate orthogonal pointer subspaces by an isometry is called perfect discrimination. For two normalized pure environment states, perfect discrimination requires orthogonality: orthogonal output labels make the output inner product zero, and isometry forces the input inner product to be zero as well. For mixed environment states ω_1, ω_2 of different record classes, perfect discrimination requires orthogonal supports. Let $\mathbf{V}$ be the discrimination isometry and Π_1^{rec} the first-label projection. Then $E = \mathbf{V}^\dagger (I \otimes \Pi_1^{\text{rec}}) \mathbf{V}$ satisfies $0 \leq E \leq I$. Each input is mapped entirely into its respective output subspace, giving readout shares $\text{Tr}(E\omega_1) = 1$ and $\text{Tr}(E\omega_2) = 0$. Positivity implies $(I - E)\omega_1^{1/2} = 0$ and $E\omega_2^{1/2} = 0$. The two supports lie in the eigenspaces of E for eigenvalues 1 and 0 and are orthogonal. Conversely, let their support projections T_1, T_2 be orthogonal and set $T_\perp = I - T_1 - T_2$. Using three orthogonal labels, construct

$$\mathbf{V}\psi = T_1\psi \otimes |1\rangle + T_2\psi \otimes |2\rangle + T_\perp\psi \otimes |0\rangle, \quad \mathbf{V}^\dagger\mathbf{V} = T_1 + T_2 + T_\perp = I. \quad (6.8)$$

This sends the two input supports to subspaces with labels 1 and 2, providing perfect discrimination; completion of orthonormal bases extends it to a unitary on the joint space. For finitely many classes, take the projection onto the span of the conditional-state supports in each class and add the projection onto their orthogonal complement; the same construction applies. Thus orthogonality of supports between classes is equivalent to existence of a perfectly discriminating pointer. Equation (6.3) additionally requires the actual pointer to implement this correspondence.

For partially overlapping conditional states in different record classes, Eq. (6.3) fails, but the selected readout shares $w_{r|\alpha} = \text{Tr}(\Pi_r^{\text{rec}}\omega_\alpha)$ may still depend on the process, retaining partial readable distinctions. For such readouts, the actual environment overlaps must be retained; not every outcome can be interpreted as an unambiguous identification of a process class. The complete-record criterion does not require the joint system–environment state to be diagonal in record labels; joint coherence between orthogonal labels may disappear from the system state only after tracing out the environment.

The joint final states of Section 4 determine the individual branches. In Eq. (4.19), the retrieval auxiliary returns to the common state g after its coherent round trip, so its final state no longer retains the direction distinctions carried by intermediate a_ν . Other auxiliary systems must be checked by their own final states. For the two coupled inputs in Eq. (4.24), their respective outputs lie in the orthogonal environment subspaces $\text{span}\{|g, u\rangle\}$ and $\text{span}\{|r_{ba}, v\rangle : b\}$. For direction input, these represent continuation and exit of the original excitation; for shell-average input, formation and background continuation. The output subspaces distinguish the two destinations for a fixed input, but without input discrimination they do not distinguish continuation from formation or exit from background continuation. The actual readout retaining incident labels in Section 6.2 establishes correspondence of these four process classes with pointer labels.

The same environment overlaps determine state matrices on the observation units of Section 5; see Eq. (5.18). Mixed components are weighted at the density-operator level, while amplitudes within each pure component still combine coherently according to Eq. (6.7).

The selected process classes and actual pointer jointly determine record resolution. A system action may have different Kraus representations; changing representation does not change distinctions already formed in the environment.

§6.2 Readout of Events

Records preserve distinctions between processes. A readout identifies them and assigns labels and readings. Use the actual pointer projections Π_r^{rec} of Eq. (6.2); failure to detect a selected record is called the **no-response outcome**. An outcome identified in an actual readout is called an **event**. Its label specifies the distinction read out, and its physical reading is determined by pointer calibration. No-response outcomes remain in the complete statistics, with environment states determined by the actual readout.

Outcome r corresponds to the joint component at cycle end, $(I \otimes \Pi_r^{\text{rec}}) \Omega^{\text{out}}(\rho) (I \otimes \Pi_r^{\text{rec}})$. Tracing out the auxiliary space gives its system component. The completely positive actions for the outcomes form a quantum **instrument** $\{\mathcal{I}_r\}_r$ [24]. Actual readout first selects a record component and then determines its system component by Eq. (4.50) and its linear extension to mixed states.

Actual Records and Observational Readout

Given an actual pointer satisfying Eq. (6.2) and preserved during storage, the conditional action and share of outcome r are

$$\begin{aligned} \mathcal{I}_r(\rho) &= \text{Tr}_{\mathcal{E}} [(I \otimes \Pi_r^{\text{rec}}) \Omega^{\text{out}}(\rho) (I \otimes \Pi_r^{\text{rec}})] , \\ w_r(\rho) &= \text{Tr} \mathcal{I}_r(\rho). \end{aligned} \tag{6.9}$$

Its state on the observation unit of Section 5 is $R_a[\mathcal{I}_r(\rho)] = j_a^\dagger \mathcal{I}_r(\rho) j_a$. The actual pointer and its calibration are readout data. The probabilistic interpretation of shares uses conditions (F0)–(F4) of Section 6.4, with the conclusion in Eq. (6.37).

Joint measurement processes realize completely positive instruments [25]. To expand this action in the present cycle, take a spectral decomposition of the input environment state, $\eta = \sum_j \pi_j |j\rangle\langle j|$, and an orthonormal basis $\{|r, \mu\rangle\}_\mu$ in each

record subspace. Substitution into Eq. (6.9) and expansion of the partial trace give

$$J_{r\mu j} = \sqrt{\pi_j} \langle r, \mu | \mathbb{U}_{\text{cyc}} | j \rangle, \quad \mathcal{I}_r(\rho) = \sum_{\mu, j} J_{r\mu j} \rho J_{r\mu j}^\dagger. \quad (6.10)$$

Each map $X \mapsto J_{r\mu j} X J_{r\mu j}^\dagger$ preserves positivity even after tensoring with the identity map on an arbitrary reference space, so $\mathcal{I}_r$ is completely positive. Completeness of the output basis and joint unitarity give

$$\begin{aligned} \sum_{r, \mu, j} J_{r\mu j}^\dagger J_{r\mu j} &= \sum_j \pi_j \langle j | \mathbb{U}_{\text{cyc}}^\dagger \mathbb{U}_{\text{cyc}} | j \rangle = I, \\ \sum_r \mathcal{I}_r(\rho) &= \Psi_\beta(\rho), \quad \sum_r w_r(\rho) = 1. \end{aligned} \quad (6.11)$$

Each conditional action is therefore trace nonincreasing, and summing all outcomes recovers the trace-preserving cycle of Section 4. Sum over unresolved indices μ within record subspaces and unmonitored incident indices j . The index j in the input spectral decomposition only expands the mixed preparation; unless actually retained, it is not a readout label. The observation compression $R_a[X] = J_a^\dagger X J_a$ of Section 5 is linear, so the same joint final state also satisfies

$$R_a[\Psi_\beta(\rho)] = \sum_r R_a[\mathcal{I}_r(\rho)]. \quad (6.12)$$

Record components retain their unnormalized shares and define positive semidefinite state matrices in the same observation unit. Equation (6.12) shows that summing all record components recovers the full observational state in that unit. Record classification and observational states are determined by the same cycle final state. When retaining outcome r , normalize by w_r only if $w_r > 0$. If one further conditions on occupation of the observation unit, then, provided $\text{Tr } R_a[\mathcal{I}_r(\rho)] > 0$, divide this matrix by its own trace. These conditions refer respectively to event outcome and occupation within the unit. For fixed V, g , replace these objects by the actual instrument and $\mathcal{F}_{V, g}$; the same relations hold.

If the actual readout retains only a union $\mathcal{A}$ of outcomes, then

$$\mathcal{I}_{\mathcal{A}} = \sum_{r \in \mathcal{A}} \mathcal{I}_r, \quad w_{\mathcal{A}} = \sum_{r \in \mathcal{A}} w_r. \quad (6.13)$$

This is coarse graining of already distinguished outcomes. For processes reaching the same system final state, conditioning on record r still gives

$$w_{\phi, r} = \sum_{\alpha, \alpha'} A_\alpha \overline{A_{\alpha'}} \langle e_{\alpha'} | \Pi_r^{\text{rec}} | e_\alpha \rangle. \quad (6.14)$$

A common label specifies only membership in the same record subspace; coherence within that subspace is determined by projected overlaps.

Thermal exchange in Section 4 supplies explicit records. Fix a channel (a, ν) , where e_a is a shell-average state and $|\nu\rangle$ a direction state. Use $F_{a\nu} = |e_a\rangle\langle\nu|$ and abbreviate $c_{a\nu} = \cos \alpha_{a\nu}$, $s_{a\nu} = \sin \alpha_{a\nu}$. The incident levels of the thermal environment unit are u, v , with shares t_u, t_v satisfying $t_u + t_v = 1$. Their joint rotation and the action on its orthogonal complement are given by Eq. (4.16). This readout

stores the channel and incident energy-level labels during preparation and reads the outgoing level at the end. Let $|c_{av,i}\rangle$ be orthogonal preparation labels and use the input

$$\sum_{i=u,v} t_i |i\rangle\langle i| \otimes |c_{av,i}\rangle\langle c_{av,i}|. \quad (6.15)$$

Labels do not participate in exchange; tracing them out recovers the original thermal input. When retaining labels, include their storage unit in the record space of Eq. (6.9) and extend $\mathbb{U}_{\text{cyc}}$ by the identity on it. The original-environment marginal of the enlarged initial state remains η . Denote the extended preparation of all channels and the star component by η_{tag} , with label space C . Then $\text{Tr}_C \eta_{\text{tag}} = \eta$, and

$$\text{Tr}_C [(\mathbb{U}_{\text{cyc}} \otimes I_C)(\rho \otimes \eta_{\text{tag}})(\mathbb{U}_{\text{cyc}}^\dagger \otimes I_C)] = \mathbb{U}_{\text{cyc}}(\rho \otimes \eta)\mathbb{U}_{\text{cyc}}^\dagger. \quad (6.16)$$

Retaining labels therefore leaves nonselective system evolution unchanged while allowing incident and outgoing levels to be compared in each exchange. The preparation labels and their storage are actual readout data.

For the coupled inputs of Eq. (4.24), abbreviate their preparation labels by $c_i = c_{av,i}$. Ordering environment factors as preparation label, restoration output, and thermal output, the four process classes have supports

Process class	Environment subspace
Continuation of original excitation	$\text{span}\{ c_u, g, u\rangle\}$
Exit of original direction	$\text{span}\{ c_u, r_{ba}, v\rangle : b\}$
Formation of new excitation	$\text{span}\{ c_v, g, u\rangle\}$
Background continuation	$\text{span}\{ c_v, r_{ba}, v\rangle : b\}$

(6.17)

The preparation labels c_u, c_v are orthogonal, as are the two outputs for each preparation, so all four subspaces are mutually orthogonal. Including their projections in the actual pointer satisfies Eq. (6.3); leaving b unresolved only combines restoration outcomes in the same class. Preservation must apply to preparation labels, thermal outputs, and restoration outputs until readout is complete. General inputs also contain nontransfer components on exchange-block complements, star components, and other channels; the complete conditional-operator family below includes all outcomes.

Use the complete thermal conditional family $\{A_\ell\}$ from Eq. (4.52), including channel weights $\sqrt{a_{av}}$ and the star term. The thermal label ℓ retains the channel and incident and outgoing levels, and includes a star label. For the coupled inputs of Eq. (4.24), retaining incident labels gives the four process supports in Eq. (6.17); the exchange-block complements remain included in the same complete family. The original cycle has $g = h$ and uses restoration operators B_0, B_{ab} preceding Eq. (4.53): B_0 retains direction coherence, and B_{ab} resolves shell-average components before and after restoration. Write $K_{\lambda\ell} = K_{\lambda\ell}^{(I)}$ here for that complete conditional family at $V = I, g = h$; its completeness follows from the joint implementation and completeness of the two conditional-operator families. For a fixed outcome, system and auxiliary free phases combine into a global phase. Use the auxiliary energies $E(g) = E_K$ and $E(r_{ab}) = E_K + E_b - E_a$ of Section 4, and let E_ℓ^{out} be the outgoing thermal-environment energy. For $\lambda = 0$, the system free phase is already included in U_φ ; the actual matrix element differs by the factor $e^{-i\tau(E_\ell^{\text{out}} + E_K)}$. For $\lambda = (a, b)$, adding the system output energy E_a to the auxiliary energies gives the factor $e^{-i\tau(E_\ell^{\text{out}} + E_K + E_b)}$. These global

phases cancel in $K_{\lambda\ell}\rho K_{\lambda\ell}^\dagger$. The fine outcome (λ, ℓ) has conditional action $K_{\lambda\ell}\rho K_{\lambda\ell}^\dagger$. If only outgoing labels are read, sum over the corresponding incident labels as in Eq. (6.13). Adding the prescribed control V of Section 4 gives conditional operators $K_{\lambda\ell}V$.

Energy readings of records follow componentwise from the same rotation. Energy matching in Section 4 gives $E_\nu + E_u = E_a + E_v$, so for outcome $\nu \rightarrow e_a$,

$$E_v - E_u = E_\nu - E_a. \quad (6.18)$$

Reverse formation reverses this difference; unit-group charges are likewise balanced by equality of joint charges. The restoration auxiliary carries the balance in background restoration. Without retained incident labels, the outgoing energy alone generally does not determine the exchanged energy.

This readout specifies destinations of the selected cycle processes and their exchange readings. Reading out the position and coherence quantities of Section 5 additionally requires pointers corresponding to those observable actions.

§6.3 Position and Coherence Readout

Position readout distinguishes components at different points of the state space; coherence readout determines the complex coherence between selected components. Section 5 gives these observables and their state pairings. To turn their distinctions into records requires joint actions coupling observables to pointers. Detection pointers and their reference preparations belong to the joint system, and their final states are stored under the record conditions above.

Position pointer The exchange labels record channels, incident and outgoing levels, and restoration outputs. Classical relabeling of these generally cannot yield position outcomes. In the original cycle with $V = I$, every conditional operator is diagonal, a single matrix unit, or zero in the $\{e_a, \nu\}$ basis. These forms are closed under multiplication, so every finite label history has a conditional operator of the same form and an effect diagonal in this basis. Classical coarse graining only adds effects and thus retains diagonality. A general point projection P_x has off-diagonal components in this basis; position readout therefore requires its own detection action. With a direction-mixing control V , history effects must be recomputed from $K_{\lambda\ell}V$. The following position-readout construction does not depend on this diagonality.

Use $P_x = |x\rangle\langle x|$, with $x \in \mathbb{Z}/d\mathbb{Z}$. Take the detector space

$$\mathcal{E}_D = \text{span}\{|g_D\rangle, |r_x\rangle : x \in \mathbb{Z}/d\mathbb{Z}\},$$

with the listed vectors orthonormal, $|g_D\rangle$ the preparation state, and $|r_x\rangle$ the position pointer states. Set the free detector Hamiltonian to zero and define

$$\mathcal{B}_D = \sum_x P_x \otimes (|r_x\rangle\langle g_D| - |g_D\rangle\langle r_x|), \quad W_D = e^{\pi \mathcal{B}_D/2}. \quad (6.19)$$

Since $\mathcal{B}_D^\dagger = -\mathcal{B}_D$, the operator W_D is unitary. With $\mathbf{L} = \sum_x P_x \otimes |r_x\rangle\langle g_D|$,

$$\mathbf{L}^\dagger \mathbf{L} = I \otimes P_{g_D}, \quad \mathbf{L} \mathbf{L}^\dagger = \sum_x P_x \otimes P_{r_x}.$$

The projections are orthogonal. Equation (4.13), with angle $\pi/2$ and phase 0, gives

$$\begin{aligned} W_D(|\psi\rangle \otimes |g_D\rangle) &= \sum_x P_x |\psi\rangle \otimes |r_x\rangle, \\ \left\| \sum_x P_x \psi \otimes r_x \right\|^2 &= \sum_x \|P_x \psi\|^2 = \|\psi\|^2. \end{aligned} \quad (6.20)$$

Each position component is thereby correlated with an orthogonal pointer state, preserving the norm of the full input. For a recording duration $t_D > 0$, take the self-adjoint interaction $H_{\text{int},D} = i\pi\mathcal{B}_D/(2t_D)$. Since $[P_x, H_K] = 0$, this interaction preserves total bare energy. For the unit group, set $R_u^D |g_D\rangle = |g_D\rangle$ and $R_u^D |r_x\rangle = |r_{ux}\rangle$. The joint unit-group action on the system and detector simultaneously permutes the summands, so W_D preserves the joint unit-group action and pointer labels transform covariantly with the reference.

In the interaction picture of the recording stage, pointer outcome r_x gives system component and share

$$\mathcal{L}_x(\rho) = P_x \rho P_x, \quad w_x(\rho) = \text{Tr}(P_x \rho), \quad \sum_x \mathcal{L}_x(\rho) = \Delta_{\text{pos}}(\rho). \quad (6.21)$$

Here $\Delta_{\text{pos}}(\rho) = \sum_x P_x \rho P_x$ is position dephasing. Bare free evolution leaves position shares unchanged; for $w_x(\rho) > 0$, the normalized conditional final system state is P_x . After formation, decoupling the detector from the system and retaining its zero free Hamiltonian preserves every pointer projection during storage, as required in Section 6.1.

Joint readout of exchange and position records Let the original cycle have joint final state $\Omega^{\text{out}}(\rho)$ and record instrument $\{\mathcal{I}_r\}$. Independently prepare the position detector in P_{g_D} and apply W_D after the cycle. In the interaction picture, the resulting joint state is

$$\sum_{x,y} (P_x \otimes I_{\mathcal{E}}) \Omega^{\text{out}}(\rho) (P_y \otimes I_{\mathcal{E}}) \otimes |r_x\rangle \langle r_y|.$$

The original records are stored during detection under the conditions of Section 6.1. Reading the original environment label r and position label x , then tracing out the auxiliary spaces, gives

$$\begin{aligned} \mathcal{I}_{r,x}^{\text{pos}}(\rho) &= P_x \mathcal{I}_r(\rho) P_x, \\ \sum_r \mathcal{I}_{r,x}^{\text{pos}}(\rho) &= P_x \Psi_\beta(\rho) P_x, \\ \sum_{r,x} \mathcal{I}_{r,x}^{\text{pos}}(\rho) &= \Delta_{\text{pos}}(\Psi_\beta(\rho)). \end{aligned} \quad (6.22)$$

The position-branch free phase cancels in the conditional action, so Eq. (6.22) also gives the system component after the actual recording stage. Marginal shares of exchange labels remain unchanged:

$$\sum_x \text{Tr} \mathcal{I}_{r,x}^{\text{pos}}(\rho) = \text{Tr} \mathcal{I}_r(\rho).$$

Position readout preserves existing record shares and causes position dephasing of the final system state. If the system completes n original cycles before one position readout, its position share is $w_x^{(n)} = \text{Tr}[P_x \Psi_\beta^n(\rho_0)]$. If a position record is added at every cycle end, the nonselective one-cycle action becomes $\mathcal{F}_{\text{pos}} = \Delta_{\text{pos}} \circ \Psi_\beta$. Successive records and subsequent states must then be computed using this action.

For nonzero position outcomes, use the calibration of Eq. (5.33), $y_x = \log(x/u_*)$, with fixed integer representative $1 \leq x < d$; retain the zero element as a separate outcome. Time calibration includes the actual detection duration. For one probe after n original cycles, readout completes at $t_{\text{read}} = n\tau + t_D$. With a probe in every cycle, the complete repeated period is $\tau + t_D$, and the n th readout completes at $t_n = n(\tau + t_D)$. For a controlled cycle, replace τ by its actual period τ_{tot} .

Spectral readout of an observation unit A nonzero observation unit of Section 5 is determined by a negation orbit $D_a = \{a, -a\}$. Let its embedding be $j_a : \mathbb{C}^2 \rightarrow \mathcal{H}$ and write $\mathbf{R}_a[\rho] = j_a^\dagger \rho j_a$. A self-adjoint observable $A = A^\dagger \in M_2(\mathbb{C})$ on the unit has spectral decomposition $A = \sum_\kappa a_\kappa E_\kappa$, with $\sum_\kappa E_\kappa = I_2$. Define on the full state space

$$Q_\kappa = j_a E_\kappa j_a^\dagger, \quad Q_\perp = I - j_a j_a^\dagger. \quad (6.23)$$

Since $j_a^\dagger j_a = I_2$,

$$Q_\kappa Q_\lambda = \delta_{\kappa\lambda} Q_\kappa, \quad Q_\kappa Q_\perp = 0, \quad \sum_\kappa Q_\kappa + Q_\perp = I.$$

These form a complete orthogonal readout on the full space, where $\perp$ denotes components outside the selected unit. Replacing $\{P_x\}$ in position detection by this family gives a spectral instrument through the same block rotations. For outcomes inside the unit, the conditional actions, shares, and weighted readings are

$$\begin{aligned} \mathcal{L}_\kappa^A(\rho) &= Q_\kappa \rho Q_\kappa, \\ w_\kappa^A(\rho) &= \text{Tr}(\mathbf{R}_a[\rho] E_\kappa), \\ \sum_\kappa a_\kappa w_\kappa^A(\rho) &= \text{Tr}(\mathbf{R}_a[\rho] A). \end{aligned} \quad (6.24)$$

The outside outcome has action $\mathcal{L}_\perp^A(\rho) = Q_\perp \rho Q_\perp$ and share $w_\perp^A(\rho) = 1 - \text{Tr} \mathbf{R}_a[\rho]$. Conditional on a response within the unit, if $\text{Tr} \mathbf{R}_a[\rho] > 0$, the normalized weighted reading is

$$\frac{\text{Tr}(\mathbf{R}_a[\rho] A)}{\text{Tr} \mathbf{R}_a[\rho]}.$$

This is exactly the normalized state–observable pairing of Eq. (5.21).

Both points of a negation orbit belong to the same shell, so every Q_κ and $Q_\perp$ commutes with H_K . Including bare free evolution during recording and appending spectral readout to the original cycle gives the actual conditional action for (r, κ) :

$$\mathcal{I}_{r,\kappa}^A(\rho) = e^{-it_D H_K} Q_\kappa \mathcal{I}_r(\rho) Q_\kappa e^{it_D H_K}. \quad (6.25)$$

Here κ includes the outside outcome $\perp$. Components inside the unit lie in one shell and their global free phase cancels; outside components may contain inter-shell coherence and require the free evolution above.

A general spectral projection family need not be closed under the unit group by permutation of outcomes. To preserve the joint unit-group action during readout, introduce a finite reference $\mathcal{E}_{\text{ref}} = \ell^2(G)$, with $G = (\mathbb{Z}/d\mathbb{Z})^\times$. Give it zero energy, group action $L_u|g\rangle = |ug\rangle$, and preparation $|1_G\rangle$. For the complete projection family including the complementary outcome, define

$$\tilde{Q}_\kappa = \sum_{g \in G} (R_g Q_\kappa R_g^\dagger) \otimes |g\rangle\langle g|. \quad (6.26)$$

This applies Eq. (4.43) to a projection family. Orthogonality of the reference basis gives

$$\tilde{Q}_\kappa \tilde{Q}_\lambda = \delta_{\kappa\lambda} \tilde{Q}_\kappa, \quad \sum_\kappa \tilde{Q}_\kappa = I.$$

Under the joint group action, the change of index $g \mapsto ug$ gives

$$\begin{aligned} (R_u \otimes L_u) \tilde{Q}_\kappa (R_u^\dagger \otimes L_u^\dagger) \\ = \sum_{g \in G} (R_{ug} Q_\kappa R_{ug}^\dagger) \otimes |ug\rangle\langle ug| = \tilde{Q}_\kappa. \end{aligned}$$

Both R_g and Q_κ also preserve shell energy, so $[\tilde{Q}_\kappa, H_K \otimes I] = 0$. Using these joint projections to control a zero-energy pointer with trivial group action, the block rotations give a recording unitary preserving total energy and total group action. For the specified reference preparation,

$$\tilde{Q}_\kappa(\psi \otimes |1_G\rangle) = Q_\kappa \psi \otimes |1_G\rangle.$$

The resulting system instrument is therefore still $Q_\kappa \rho Q_\kappa$, with the reference returning to its initial state. This reference preparation is readout data and must be specified separately from the thermal preparation.

For the Pauli observable σ_i , the spectral projections are $E_{i,\pm} = (I_2 \pm \sigma_i)/2$, and their shares satisfy

$$\begin{aligned} w_{i,+} + w_{i,-} &= \text{Tr } R_a[\rho], \\ w_{i,+} - w_{i,-} &= \text{Tr}(R_a[\rho] \sigma_i), \quad i = 1, 2, 3. \end{aligned} \quad (6.27)$$

The three settings read the real and imaginary parts of local coherence and the difference in position occupations, respectively. Together with the total response share of the unit, they reconstruct $R_a[\rho]$ through repeated readouts of the same preparation. The settings are implemented separately and do not assign simultaneous sharp readings to noncommuting quantities. The four-dimensional geometry in Section 5 comes from the self-adjoint observable space and its determinant; event time and position readings use the calibration of actual instruments. The matrix trace is a unit-response share, not an event time.

For a single stable mode P_ν on shell S_j , Eq. (5.24) gives

$$R_a[P_\nu] = \frac{1}{|S_j|} \begin{pmatrix} 1 & \theta_\nu(-1) \\ \theta_\nu(-1) & 1 \end{pmatrix}, \quad a \in S_j.$$

Substituting the spectral projections of σ_1 yields

$$w_{1,\pm} = \frac{1 \pm \theta_\nu(-1)}{|S_j|}, \quad w_\perp = 1 - \frac{2}{|S_j|}. \quad (6.28)$$

Conditional on a response within the unit, the normalized shares are $(1 \pm \theta_\nu(-1))/2$ and the normalized weighted reading is $\theta_\nu(-1)$. The instrument thus reads the stable mode's negation parity, while the total response share retains its local occupation. Here P_ν is the actual input at the start of readout. If a cycle has already been completed, its final state must be substituted instead; if a previous record r is also retained, use the joint instrument in Eq. (6.25). For a general stable pattern, substituting Eq. (5.25) into Eq. (6.27) gives readings of local occupation and coherence, with distinguishability still limited by the local compression of Section 5.

Coherence readout between same-shell directions Section 5 also examines phase changes between stable directions. For distinct directions ν, μ in the same shell, define

$$\begin{aligned} Z &= |\mu\rangle\langle\nu|, & P_2 &= P_\mu + P_\nu, & Q_2 &= I - P_2, \\ X &= Z + Z^\dagger, & Y &= -i(Z - Z^\dagger). \end{aligned} \quad (6.29)$$

On their common shell, the ratio of direction functions is a phase multiplier, so $Z = M_{\theta_\mu \bar{\theta}_\nu} P_\nu$, with the multiplier zero outside the shell. This operator uses the direction projections and intra-shell phase multipliers already defined on the state space. The two-dimensional direction subspace on which this operator acts generally has support across several position observation units and must be distinguished from the two-dimensional space associated with a single negation orbit.

For readout phase ϑ , set $C_\vartheta = e^{i\vartheta} Z + e^{-i\vartheta} Z^\dagger$. Since $Z^2 = (Z^\dagger)^2 = 0$ and $ZZ^\dagger + Z^\dagger Z = P_2$, we have $C_\vartheta^\dagger = C_\vartheta$, $C_\vartheta^2 = P_2$, and $P_2 C_\vartheta = C_\vartheta$. Thus

$$Q_{\vartheta, \pm} = \frac{1}{2}(P_2 \pm C_\vartheta), \quad Q_{\vartheta, 0} = Q_2 \quad (6.30)$$

is a complete orthogonal projection family. Explicitly,

$$Q_{\vartheta, \pm}^2 = Q_{\vartheta, \pm}, \quad Q_{\vartheta, +} Q_{\vartheta, -} = 0, \quad Q_{\vartheta, +} + Q_{\vartheta, -} + Q_{\vartheta, 0} = I.$$

Let ρ be the state at the start of readout and set $w_{\vartheta, b} = \text{Tr}(Q_{\vartheta, b} \rho)$. The difference of the two response shares satisfies

$$w_{\vartheta, +} - w_{\vartheta, -} = 2\Re(e^{i\vartheta} \text{Tr}(Z\rho)).$$

In particular,

$$\begin{aligned} w_{0, +} - w_{0, -} &= \text{Tr}(X\rho) = 2\Re \text{Tr}(Z\rho), \\ w_{\pi/2, +} - w_{\pi/2, -} &= -\text{Tr}(Y\rho) = -2\Im \text{Tr}(Z\rho). \end{aligned} \quad (6.31)$$

These two settings determine the real and imaginary parts of same-shell direction coherence separately. Every projection commutes with H_K , so the finite reference and pointer construction of Eq. (6.26) implements a readout preserving total energy and total unit-group action. In the interaction picture, the system instrument is $\mathcal{L}_{\vartheta, b}(\rho) = Q_{\vartheta, b} \rho Q_{\vartheta, b}$. The actual recording stage still includes its duration and bare free evolution; complementary outcomes may contain inter-shell components whose free evolution must be retained. This construction reads same-shell direction coherence; readout of coherence between different shell energies is not included in this implementation.

§6.4 Event Probabilities

Joint evolution and record projections determine positive-operator components and shares for all outcomes. Using these shares in event statistics further requires specifying how statistical weights remain consistent when the same record is represented with different auxiliaries or an outcome is refined into mutually exclusive outcomes. Write p_r for the probability of outcome r . For actual readouts and their auxiliary extensions, adopt the following statistical conditions:

- (F0) **Nonnegativity, normalization, and additivity.** Outcome weights are nonnegative and sum to one; the weight of a union of mutually exclusive outcomes is the sum of their weights.
- (F1) **Unitary covariance.** Weights are determined by the joint state and specified orthogonal readout. A unitary transformation on the outcome side, accompanied by transformation of the readout projections, preserves the corresponding weights.
- (F2) **Auxiliary consistency.** Unitaries acting only on unread auxiliary units do not change original outcome weights. Copying orthogonal records, finite orthogonal refinement of joint record subspaces, and addition of uncoupled auxiliary units preserve the corresponding coarse weights. A mixed state and its purification give the same weights when the purifying reference is unread.
- (F3) **Continuity.** On a fixed outcome space and readout partition, weights depend continuously on the normalized joint state.
- (F4) **Zero-component condition.** An outcome has zero weight when the joint state's component in its subspace vanishes.

These are the statistical requirements adopted here. The following proof shows that they uniquely fix event probabilities as the shares of the joint state in record subspaces. Unitary covariance concerns consistency of statistical assignments under transformations of branches and projections; it does not require every auxiliary transformation to be implementable by the conserving interactions of the original cycle. The equal-amplitude exchanges and auxiliary refinements below follow the idea of environment-assisted invariance (envariance) [26], with a proof under the stated statistical conditions.

Weights of equal-amplitude branches First consider a normalized pure joint final state $|\Psi\rangle$, and set $\mathcal{P}_r = I \otimes \Pi_r^{\text{rec}}$. Since the record projections are orthogonal and complete,

$$v_r = \mathcal{P}_r \Psi, \quad a_r = \|v_r\|, \quad \langle v_s, v_r \rangle = 0 \ (s \neq r), \quad \sum_r a_r^2 = 1. \quad (6.32)$$

For $a_r > 0$, define the normalized branch $\sigma_r = v_r/a_r$, giving $\Psi = \sum_{r:a_r>0} a_r \sigma_r$. Refine the outcome subspace $\text{Ran } \mathcal{P}_r$ into the one-dimensional space $\mathbb{C}\sigma_r$ and its orthogonal complement. By (F2), refinement preserves the coarse weight of r . The joint state has zero component in that complement, so (F4) and additivity in (F0) equate the coarse weight of r with the weight of the one-dimensional branch σ_r .

Introduce an auxiliary unit C and copy these orthogonal branch labels as

$$|\sigma_r\rangle|0\rangle_C \mapsto |\sigma_r\rangle|\eta_r\rangle_C, \quad \langle\eta_s|\eta_r\rangle = \delta_{sr}.$$

This map preserves inner products and extends to a unitary by completing orthonormal bases in the finite joint space. The copied state is

$$|\Phi\rangle = \sum_{r:a_r>0} a_r |\sigma_r\rangle|\eta_r\rangle_C. \quad (6.33)$$

By (F2), original outcome weights are unchanged by copying. Suppose $a_r = a_s$. Exchange the one-dimensional branches σ_r, σ_s on the outcome side, acting as the identity on the remaining orthogonal subspaces, and exchange η_r, η_s on the auxiliary side. The two exchanges restore the joint state:

$$(S_{rs} \otimes T_{rs})|\Phi\rangle = |\Phi\rangle.$$

By (F1), the outcome-side exchange swaps the two fine weights; by (F2), the compensating auxiliary exchange leaves them unchanged. Hence $p_r = p_s$. The exchanged branches are one-dimensional after refinement and have equal dimension, even if their original coarse subspaces do not. For N_* nonzero branches of equal amplitude, normalization gives $p_r = 1/N_*$.

Orthogonal refinement of rational shares Suppose the squared norms of nonzero branches are $a_r^2 = m_r/N_*$, with $m_r \in \mathbb{N}$ and $\sum_r m_r = N_*$. If needed, first add an uncoupled auxiliary unit so that C can accommodate N_* orthogonal labels, and take another unit C' of dimension at least N_* . By (F2), these extensions preserve original weights. For each label, implement

$$|\eta_r\rangle_C|0\rangle_{C'} \mapsto \frac{1}{\sqrt{m_r}} \sum_{\ell=1}^{m_r} |r, \ell\rangle_C |r, \ell\rangle_{C'}. \quad (6.34)$$

The vectors $|r, \ell\rangle$ in each auxiliary space are orthonormal. Different input labels have orthogonal unit-norm images, so the map extends to a unitary on $C \otimes C'$. The transformed state is

$$\begin{aligned} |\Phi'\rangle &= \sum_r \sqrt{\frac{m_r}{N_*}} |\sigma_r\rangle \frac{1}{\sqrt{m_r}} \sum_{\ell=1}^{m_r} |r, \ell\rangle_C |r, \ell\rangle_{C'} \\ &= \frac{1}{\sqrt{N_*}} \sum_r \sum_{\ell=1}^{m_r} |\sigma_r\rangle |r, \ell\rangle_{C'} |r, \ell\rangle_C. \end{aligned} \quad (6.35)$$

The last expression uses the natural tensor isomorphism to order factors as original outcome side, C' , and C . The transformation acts only on auxiliary units and leaves the original coarse weight of r unchanged. Combine the original outcome side with C' as the fine-outcome side. Its nonzero branches are $|\sigma_r\rangle|r, \ell\rangle_{C'}$, with orthogonal copies stored in C . Complete their one-dimensional projections to a full readout. Complementary components vanish and have zero weight by (F4). There are now N_* equal-amplitude nonzero fine branches, each of weight $1/N_*$. Adding the m_r terms belonging to the same coarse outcome r , (F0) and (F2) give

$$p_r = \sum_{\ell=1}^{m_r} \frac{1}{N_*} = \frac{m_r}{N_*} = a_r^2.$$

Continuous extension to general pure states For arbitrary shares $(a_r^2)_r$, choose normalized rational shares on their nonzero support, $q_r^{(n)} > 0$ and $\sum_r q_r^{(n)} = 1$, with $q_r^{(n)} \rightarrow a_r^2$. Keep the normalized branches σ_r fixed and define $|\Psi_n\rangle = \sum_r \sqrt{q_r^{(n)}} |\sigma_r\rangle$. Branch orthogonality gives

$$\|\Psi_n - \Psi\|^2 = \sum_r \left(\sqrt{q_r^{(n)}} - a_r \right)^2 \rightarrow 0.$$

For each n , summing the auxiliary fine outcomes in the rational-refinement proof already gives weights in the original outcome space, $p_r(\Psi_n) = q_r^{(n)}$. Continuity is therefore used on one fixed original outcome space and readout partition, without taking limits between auxiliary spaces of different dimensions. By (F3), $p_r(\Psi) = \lim_n p_r(\Psi_n) = a_r^2$; zero components are handled by (F4). Thus, for any normalized pure joint state,

$$\boxed{p_r(\Psi) = \|\mathcal{P}_r \Psi\|^2.} \quad (6.36)$$

Mixed states and cycle-event probabilities Let the actual joint final state be a density operator Ω . Finite dimension permits a finite purification $|\hat{\Omega}\rangle$ satisfying $\text{Tr}_C |\hat{\Omega}\rangle\langle\hat{\Omega}| = \Omega$. Extend the original readout by the identity on the purifying reference C . The pure-state result and (F2) give

$$\begin{aligned} p_r(\Omega) &= \|(\mathcal{P}_r \otimes I_C) \hat{\Omega}\|^2 \\ &= \langle \hat{\Omega} | (\mathcal{P}_r \otimes I_C) | \hat{\Omega} \rangle = \text{Tr}(\mathcal{P}_r \Omega). \end{aligned}$$

After extending purification spaces to a common dimension, different purifications are related by a unitary on the reference; auxiliary consistency makes the result independent of purification. Substitution of the actual cycle final state, Eq. (6.1), gives

$$\boxed{\begin{aligned} p_r(\rho) &= \text{Tr}[(I \otimes \Pi_r^{\text{rec}}) \Omega^{\text{out}}(\rho)] \\ &= \text{Tr} \mathcal{I}_r(\rho) = w_r(\rho). \end{aligned}} \quad (6.37)$$

The last equality uses projection idempotence and cyclicity of the trace. This trace formula is nonnegative, normalized, and additive on exclusive outcomes, and satisfies covariance, auxiliary consistency, continuity, and the zero-component condition. Under (F0)–(F4), record probabilities in Eq. (6.9) therefore uniquely equal the shares determined by joint evolution. For another characterization of trace-form probability assignments using nonnegativity, normalization, and additivity on all effects, see Busch’s representation theorem [27].

The same proof applies to the full joint final state with a position or spectral detector added, so

$$p_{r,x}^{\text{pos}}(\rho) = \text{Tr} \mathcal{I}_{r,x}^{\text{pos}}(\rho), \quad p_{r,\kappa}^A(\rho) = \text{Tr} \mathcal{I}_{r,\kappa}^A(\rho).$$

These probabilities refer to the instruments actually used, including the effect of added detection on the final system state. For an observation unit of Section 5, $p_\kappa^A(\rho) = \text{Tr}(\mathbf{R}_a[\rho] E_\kappa)$, and the probability-weighted readings inside the unit give

$$\sum_\kappa a_\kappa p_\kappa^A(\rho) = \text{Tr}(\mathbf{R}_a[\rho] A).$$

For nonzero unit-response share, the statistical expectation conditional on readout within the unit is

$$\mathbb{E}[A \mid D_a] = \frac{\text{Tr}(\mathbf{R}_a[\rho]A)}{\text{Tr} \mathbf{R}_a[\rho]}.$$

The state–observable pairing established above thereby acquires an interpretation in event statistics.

Record frequencies in repeated preparations Consider independent repeated preparations with the same initial state and instrument each time. Joint preparations and readouts are tensor products, so the trace formula gives

$$\Pr(r_1, \dots, r_L) = \prod_{k=1}^L p_{r_k}.$$

Outcomes of different trials are therefore independent. For fixed r , let $X_k \in \{0, 1\}$ indicate whether trial k yields that outcome, and define its sample frequency $\hat{p}_r^{(L)} = L^{-1} \sum_{k=1}^L X_k$. Using $\mathbb{E}X_k = p_r$, $\text{Var}(X_k) = p_r(1 - p_r)$, and zero covariance between trials gives

$$\begin{aligned} \mathbb{E}\hat{p}_r^{(L)} &= p_r, & \text{Var}(\hat{p}_r^{(L)}) &= \frac{p_r(1 - p_r)}{L}, \\ \Pr(|\hat{p}_r^{(L)} - p_r| \geq \delta) &\leq \frac{p_r(1 - p_r)}{L\delta^2} \longrightarrow 0 & (\delta > 0). \end{aligned} \tag{6.38}$$

The last inequality is Markov’s inequality applied to the nonnegative variable $(\hat{p}_r^{(L)} - p_r)^2$. Thus, in independent repeated preparations, sample record frequencies converge in probability to the Born weights above. In successive cycles of the same subject, one outcome generally changes the next conditional state, and records are generally correlated. Their probabilities require ordered composition of conditional actions, and long-term frequencies require analysis of record correlations.

§6.5 Event Histories

After each cycle completes free evolution, formation exchange, and pattern reconstruction, its final system state becomes the next input, while existing records remain stored under the preservation conditions. The ordered readout outcomes form an **event history**. Since an outcome is correlated with the final system state, the next outcome distribution is determined by that conditional state. Fix the cycle preparation, interaction, and actual instrument. Each cycle uses an independent fresh environment, and old records persist through later cycles.

Joint probability of two records Take two environment copies $\mathcal{E}_1, \mathcal{E}_2$, both prepared in η . The initial joint state is $\rho \otimes \eta_{\mathcal{E}_1} \otimes \eta_{\mathcal{E}_2}$. The first cycle acts on the system and $\mathcal{E}_1$; outcome r corresponds to the joint component

$$X_r = (I \otimes \Pi_r^{\text{rec}})\Omega^{\text{out}}(\rho)(I \otimes \Pi_r^{\text{rec}}), \quad \text{Tr}_{\mathcal{E}_1} X_r = \mathcal{I}_r(\rho).$$

The second cycle acts on the system and fresh environment $\mathcal{E}_2$, preserving the first record projection. Thus, when calculating the joint share of outcomes r, s , the old

environment $\mathcal{E}_1$ may be traced out first. The unnormalized second-cycle input is $\mathcal{I}_r(\rho) \otimes \eta$. The probability formula above gives

$$p(r, s) = \text{Tr } \mathcal{I}_s(\mathcal{I}_r(\rho)). \quad (6.39)$$

For $p_r(\rho) > 0$, the system state conditional on the first outcome r is

$$\rho_r = \frac{\mathcal{I}_r(\rho)}{p_r(\rho)}. \quad (6.40)$$

This operator is positive with unit trace. Hence

$$p(s | r) = \frac{p(r, s)}{p_r(\rho)} = \text{Tr } \mathcal{I}_s(\rho_r).$$

The conditional state follows from partial trace and normalization of the actual joint component and includes the outcome's effect on subsequent cycles. The same joint probability is obtained by reading all stored pointers after several cycles. As long as old record projections are preserved, delayed readout leaves their joint shares unchanged.

Composition of complete cycles Let $\{\mathcal{I}_r\}_{r \in \mathcal{R}}$ be the complete instrument in each cycle, with $\mathcal{R}$ containing every prescribed outcome, including no response, and nonselective action $\mathcal{F} = \sum_{r \in \mathcal{R}} \mathcal{I}_r$. At each cycle's start, the fresh environment is in a tensor product with the joint state of the system, all previous environments, and old records. Existing system–record correlations may remain; subsequent cycles act only on the system and fresh environment. Applying the two-cycle partial-trace relation repeatedly gives composition of records over multiple cycles.

Composition of Complete Cycles

Fix a complete cycle and its actual instrument $\{\mathcal{I}_r\}$. Each fresh environment has the same preparation, independent of the system and all old environments and records jointly; old record projections are preserved during storage. Under the statistical conditions of Section 6.4, a history $\mathbf{r} = (r_1, \dots, r_N)$ satisfies

$$\begin{aligned} \mathcal{I}_{\mathbf{r}} &= \mathcal{I}_{r_N} \circ \dots \circ \mathcal{I}_{r_1}, \\ p(\mathbf{r}) &= \text{Tr } \mathcal{I}_{\mathbf{r}}(\rho_0), \\ \rho_{\mathbf{r}} &= \frac{\mathcal{I}_{\mathbf{r}}(\rho_0)}{p(\mathbf{r})} \quad (p(\mathbf{r}) > 0), \\ \rho_N &= \mathcal{F}^N(\rho_0), \quad \mathcal{F} = \sum_r \mathcal{I}_r. \end{aligned} \quad (6.41)$$

The preparation, cycle action, and actual pointers are prescribed data. For the original cycle, $\mathcal{F} = \Psi_\beta$; for fixed control and background preparation, use the corresponding $\mathcal{F}_{V,g}$.

Here $\mathcal{I}_{\mathbf{r}}(\rho_0)$ is the unnormalized system component retaining the specified history, $p(\mathbf{r})$ its probability, and $\rho_{\mathbf{r}}$ its conditional state. Specifically, for a history $\mathbf{r}_n$ of the first n cycles,

$$\mathcal{I}_{\mathbf{r}_{n+1}}(\rho_0) = \mathcal{I}_{r_{n+1}}(\mathcal{I}_{\mathbf{r}_n}(\rho_0)).$$

When that history has nonzero probability,

$$\begin{aligned} p(r_{n+1} \mid \mathbf{r}_n) &= \text{Tr } \mathcal{I}_{r_{n+1}}(\rho_{\mathbf{r}_n}), \\ p(\mathbf{r}_{n+1}) &= p(\mathbf{r}_n) p(r_{n+1} \mid \mathbf{r}_n). \end{aligned}$$

Conditional probabilities thus vary with the system state determined by the history. If actual records resolve every individual conditional operator, $\mathcal{I}_r(\rho) = K_r \rho K_r^\dagger$, set $K_{\mathbf{r}} = K_{r_N} \cdots K_{r_1}$. Then

$$p(\mathbf{r}) = \text{Tr}(K_{\mathbf{r}} \rho_0 K_{\mathbf{r}}^\dagger).$$

If an actual outcome includes several unresolved conditional operators, composition still uses the full $\mathcal{I}_r$.

Normalization and marginal consistency of histories Summing all length- N histories, linearity of the instrument gives

$$\begin{aligned} \sum_{\mathbf{r}} \mathcal{I}_{\mathbf{r}}(\rho_0) &= \left(\sum_{r_N} \mathcal{I}_{r_N} \right) \circ \cdots \circ \left(\sum_{r_1} \mathcal{I}_{r_1} \right) (\rho_0) \\ &= \mathcal{F}^N(\rho_0). \end{aligned}$$

Without retaining a particular history, the system state is therefore $\rho_N = \mathcal{F}^N(\rho_0)$, in agreement with the recurrence obtained from full joint evolution in Eq. (4.54). Since $\mathcal{F}$ preserves trace,

$$\begin{aligned} \sum_{r_N} p(r_1, \dots, r_N) &= \text{Tr } \mathcal{F}(\mathcal{I}_{r_{N-1}} \circ \cdots \circ \mathcal{I}_{r_1}(\rho_0)) \\ &= p(r_1, \dots, r_{N-1}). \end{aligned} \tag{6.42}$$

Successive summation gives $\sum_{\mathbf{r} \in \mathcal{R}^N} p(\mathbf{r}) = 1$. Every fixed cycle count therefore has a normalized finite history distribution; the union of all finite-length histories is countable. If a cycle retains only a coarse outcome $\mathcal{A} \subseteq \mathcal{R}$, its conditional action is $\mathcal{I}_{\mathcal{A}} = \sum_{r \in \mathcal{A}} \mathcal{I}_r$. Substitution into the history composition gives the corresponding coarse-history probability. This sum combines existing records without changing the implemented joint evolution.

Histories with position or coherence readout With position detection at every cycle end, the actual single-cycle outcome is (r, x) , and one must use $\mathcal{I}_{r,x}^{\text{pos}}(\rho) = P_x \mathcal{I}_r(\rho) P_x$. The history $((r_1, x_1), \dots, (r_N, x_N))$ has probability

$$\text{Tr}[\mathcal{I}_{r_N, x_N}^{\text{pos}} \circ \cdots \circ \mathcal{I}_{r_1, x_1}^{\text{pos}}(\rho_0)].$$

Summing all outcomes gives $\mathcal{F}_{\text{pos}} = \Delta_{\text{pos}} \circ \mathcal{F}$ per cycle, with period $\tau_{\text{tot}} + t_D$. With spectral readout in every cycle, use the actual conditional actions of Eq. (6.25),

$$\mathcal{I}_{r,\kappa}^A(\rho) = e^{-it_D H_K} Q_\kappa \mathcal{I}_r(\rho) Q_\kappa e^{it_D H_K},$$

including the complementary outcome and free evolution during readout. Both history types follow the same composition rule; normalization and marginal consistency follow from trace preservation of their complete instruments. Event times and readings are calibrated by the actual period and selected instrument, respectively.

Non-exit histories Tracking continuation of a formed excitation requires selecting histories in which no exit record has yet occurred. Suppose the actual instrument distinguishes exit from non-exit, and let $\mathcal{E}_{\text{exit}} \subseteq \mathcal{R}$ be the exit outcomes. Define

$$\begin{aligned}\mathcal{Q}(\rho) &= \sum_{r \notin \mathcal{E}_{\text{exit}}} \mathcal{I}_r(\rho), \\ \text{Tr } \mathcal{Q}(\rho) &= \text{Tr } \rho - \sum_{r \in \mathcal{E}_{\text{exit}}} \text{Tr } \mathcal{I}_r(\rho) \quad (\rho \geq 0).\end{aligned}\tag{6.43}$$

This action is completely positive and trace nonincreasing. Retaining non-exit outcomes for N consecutive cycles gives

$$\mathcal{Q}^N(\rho_0) = \sum_{\substack{r_1, \dots, r_N \\ r_j \notin \mathcal{E}_{\text{exit}}}} \mathcal{I}_{r_N} \circ \dots \circ \mathcal{I}_{r_1}(\rho_0).$$

Its trace is the probability of no exit in the first N cycles; when nonzero, the conditional state is $\mathcal{Q}^N(\rho_0) / \text{Tr } \mathcal{Q}^N(\rho_0)$. Non-exit histories may contain other records, so $\mathcal{Q}$ sums all non-exit outcomes and generally has several conditional operators.

The map $\mathcal{Q}$ is defined on the full state space. For general input, its retained outcomes may include formation or background continuation. To track an original excitation, further take $\rho_0 = P\rho_0P$, where P projects onto $\mathcal{M}$. Under the control conditions of Section 4 preserving $\mathcal{C} \oplus \mathcal{M}$, if records retain the exit distinction and no extra probe changes the evolution, then, for $X = PXP$, exit outcomes collect all components with restoration labels r_{ab} , while other outcomes correspond to the auxiliary returning to g . Using $B_0 = P$, $PB_{ab} = 0$, and Eq. (4.53) gives

$$\begin{aligned}\mathcal{Q}(X) &= \sum_{r \notin \mathcal{E}_{\text{exit}}} \mathcal{I}_r(X) = P\mathcal{F}_{V,g}(X)P \\ &= \sum_{\kappa} K_{0\kappa}^{(V)} X K_{0\kappa}^{(V)\dagger} = \mathcal{Q}_V(X).\end{aligned}$$

The output still satisfies $P\mathcal{Q}(X)P = \mathcal{Q}(X)$, so this identification applies at every step of a non-exit history. Maintenance statistics of the original excitation stop at first exit. Formation after exit belongs to the subsequent complete history and remains described by the full instrument. Successive records can thus be used both to collect statistics on events in the complete cycle and to determine the first exit and maintenance time of the original excitation. Figure 5 relates stored cycle records to observational events and shows ordered composition over successive cycles.

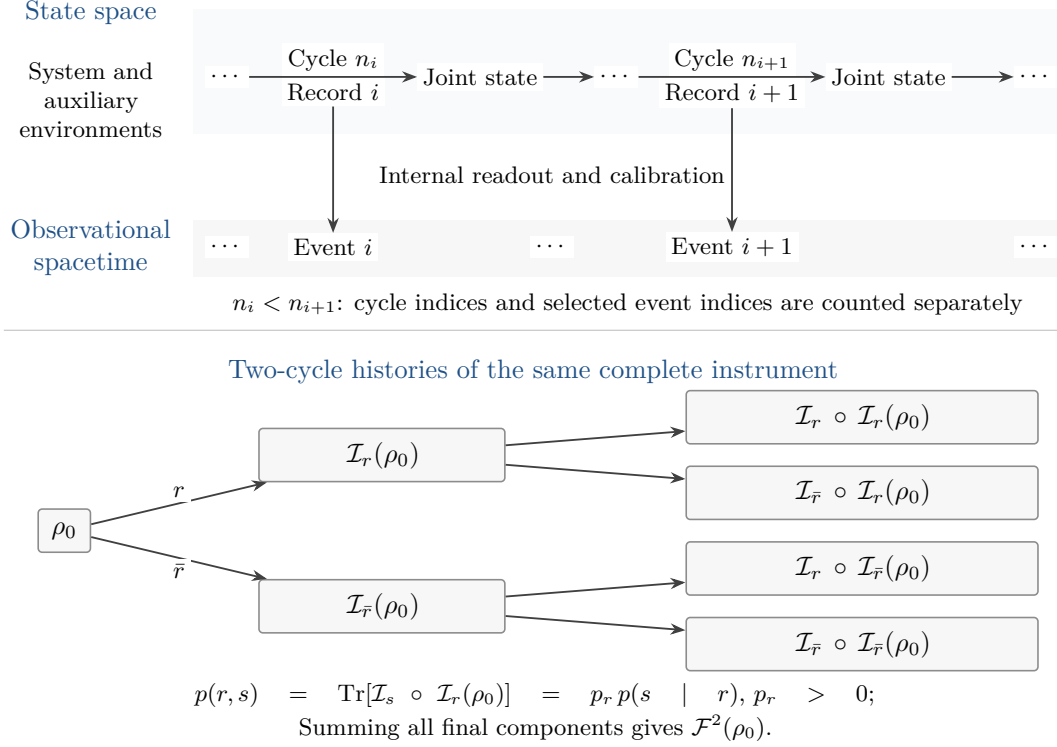


Figure 5: Cycle records, observational events, and event histories. Above, the joint state contains the system and auxiliary environments that have participated; records form within the cycle under the distinction and preservation conditions. The indices n_i, n_{i+1} mark formation cycles of selected records, possibly separated by several cycles. Records may be read out together later; vertical arrows indicate internal readout and calibration of existing records. Below, a selected outcome r and the union $\bar{r}$ of all others form a complete readout. No-response outcomes are included in this complete classification. Each fresh environment is independent of the system and all previous environments and records jointly. Successive outcomes are generally correlated; under (F0)–(F4), ordered composition of actual conditional actions determines their joint probabilities.

§6.6 Rates and Maintenance Times

Event histories specify the order and probabilities of record classes. Together with the time calibration of Section 5, they determine event counts per unit time and changes in maintenance probability, total occupation, and direction coherence with cycle count. Fix the control V , background preparation σ_g^{sh} , and exchange data of Section 4, with actual instrument $\{\mathcal{I}_r\}_r$ and complete cycle $\mathcal{F} = \mathcal{F}_{V,g}$. Each cycle uses an independent identical environment preparation and actual period $\tau_{\text{tot}} > 0$, so

$$\sum_r \mathcal{I}_r = \mathcal{F}, \quad \rho_n = \mathcal{F}^n(\rho_0), \quad t_n = n\tau_{\text{tot}}. \quad (6.44)$$

For $V = I, g = h$, this recovers the original cycle Ψ_β . We consider the original cycles, without additional position detection after each cycle. If position recording is added each time, the complete action and period become $\Delta_{\text{pos}} \circ \mathcal{F}$ and $\tau_{\text{tot}} + t_D$. Counts must use the corresponding instrument, and direction-space preservation and the non-exit map must be determined anew from the actual evolution.

Event-counting rates For a resolved event class $\mathcal{A}$, set $\mathcal{I}_{\mathcal{A}} = \sum_{r \in \mathcal{A}} \mathcal{I}_r$. Let $Z_{\mathcal{A},n+1} \in \{0,1\}$ indicate its occurrence in cycle $n+1$. Its count over the first N cycles is $C_{\mathcal{A}}(N) = \sum_{n=0}^{N-1} Z_{\mathcal{A},n+1}$. Summing all labels of the first n cycles in the history probabilities gives

$$\begin{aligned} \mathbb{E}Z_{\mathcal{A},n+1} &= \sum_{r_1, \dots, r_n} \text{Tr} \mathcal{I}_{\mathcal{A}} \circ \mathcal{I}_{r_n} \circ \dots \circ \mathcal{I}_{r_1}(\rho_0) \\ &= \text{Tr} \mathcal{I}_{\mathcal{A}}(\mathcal{F}^n(\rho_0)) = p_{\mathcal{A}}(\rho_n), \\ \mathbb{E}C_{\mathcal{A}}(N) &= \sum_{n=0}^{N-1} p_{\mathcal{A}}(\rho_n). \end{aligned} \tag{6.45}$$

This follows from linearity of expectation and does not require independence between cycles. Using the actual period, define the **event-counting rate** and its average over a finite observation interval by

$$\begin{aligned} R_{\mathcal{A},n} &= \frac{p_{\mathcal{A}}(\rho_n)}{\tau_{\text{tot}}}, \\ \overline{R}_{\mathcal{A}}^{(N)} &= \frac{\mathbb{E}C_{\mathcal{A}}(N)}{N\tau_{\text{tot}}} = \frac{1}{N} \sum_{n=0}^{N-1} R_{\mathcal{A},n}. \end{aligned} \tag{6.46}$$

The rate $R_{\mathcal{A},n}$ is the expected event count in cycle $n+1$ divided by that cycle's duration. Under the strict-contraction conditions of Section 4, $\rho_n \rightarrow \rho_*$. Since $0 \leq \mathcal{I}_{\mathcal{A}}^*(I) \leq I$,

$$|p_{\mathcal{A}}(\rho_n) - p_{\mathcal{A}}(\rho_*)| \leq \|\rho_n - \rho_*\|_1 \rightarrow 0.$$

Taking arithmetic means of the convergent sequence gives

$$R_{\mathcal{A},n}, \overline{R}_{\mathcal{A}}^{(N)} \rightarrow R_{\mathcal{A},*} = \frac{\text{Tr} \mathcal{I}_{\mathcal{A}}(\rho_*)}{\tau_{\text{tot}}}. \tag{6.47}$$

This is a limit of expected counting rates. Convergence of the actual rate in one long history additionally requires control of record correlations; Section 7 gives a variance bound from contraction of the same cycle.

More generally, for a real reading c_r assigned to outcome r , the expected reading rate in one cycle is

$$R_n = \frac{1}{\tau_{\text{tot}}} \sum_r c_r \text{Tr} \mathcal{I}_r(\rho_n).$$

For the accumulated reading $C(N) = \sum_{n=1}^N c_{r_n}$, define the weighted map

$$\mathcal{F}_{\chi} = \sum_r e^{\chi c_r} \mathcal{I}_r, \quad Z_N(\chi) = \text{Tr} \mathcal{F}_{\chi}^N(\rho_0), \quad \chi \in \mathbb{R}. \tag{6.48}$$

Expanding its N th power and using history probabilities gives

$$Z_N(\chi) = \sum_{\mathbf{r}} e^{\chi \sum_{n=1}^N c_{r_n}} p(\mathbf{r}) = \mathbb{E} e^{\chi C(N)}. \tag{6.49}$$

Thus Z_N is the moment-generating function of the accumulated reading, with $\mathcal{F}_0 = \mathcal{F}$, and satisfies

$$Z_N(0) = 1, \quad Z'_N(0) = \mathbb{E}C(N), \quad \left. \frac{d^2}{d\chi^2} \log Z_N(\chi) \right|_{\chi=0} = \text{Var}(C(N)).$$

Taking c_r as an event-class indicator gives the corresponding count statistics. Higher derivatives of Z_N at the origin give higher moments; those of $\log Z_N$ give cumulants. With several counting parameters, mixed derivatives give cross moments and joint cumulants, respectively.

Maintenance time Maintenance of an original excitation ends at its first exit. Take a normalized initial state $\rho_0 = P\rho_0P$ on the direction space $\mathcal{M}$ and require actual records to distinguish exit and non-exit. By Section 6.5, the non-exit map on direction inputs is

$$\mathcal{Q}_V(X) = \sum_{r \notin \mathcal{E}_{\text{exit}}} \mathcal{I}_r(X), \quad X = PXP. \quad (6.50)$$

First impose the strictly positive exit condition $\epsilon_{\min} := \min_{\nu} \epsilon_{\nu} > 0$ and $a_0 > 0$ of Section 4, where

$$\epsilon_{\nu} = \sum_a a_{a\nu} t_u \sin^2 \alpha_{a\nu}.$$

Each t_u is taken from its exchange channel. Finite inverse temperature, a positive exchange edge for every direction, and the prescribed exchange angles give $\epsilon_{\nu} > 0$ for every direction; finiteness of their number makes the minimum strictly positive.

Let $N_E \geq 1$ be the cycle of first exit. The probability of no exit in the first n cycles is

$$S_n = \Pr(N_E > n) = \text{Tr } \mathcal{Q}_V^n(\rho_0), \quad S_0 = 1, \quad \Pr(N_E = n+1) = S_n - S_{n+1}. \quad (6.51)$$

The second identity is the difference of successive tail probabilities. Only uninterrupted continuation of the original excitation is tracked; formation after exit is excluded from S_n . For a positive operator X on the direction space, set $Y = VXV^\dagger$. The return matrix of Section 4 gives

$$\text{Tr } \mathcal{Q}_V(X) = \text{Tr } X - \sum_{\nu} \epsilon_{\nu} Y_{\nu\nu}.$$

Including the star-return component yields

$$a_0 \text{Tr } X \leq \text{Tr } \mathcal{Q}_V(X) \leq (1 - \epsilon_{\min}) \text{Tr } X.$$

Iteration gives $a_0 S_n \leq S_{n+1} \leq (1 - \epsilon_{\min}) S_n$, and hence

$$a_0^n \leq S_n \leq (1 - \epsilon_{\min})^n.$$

Maintenance probability is therefore positive at every finite count, but $S_n \rightarrow 0$, so first exit occurs with probability one. Conditional on no earlier exit, the exit probability in cycle $n+1$ is

$$\epsilon_n^{\text{cond}} = 1 - \frac{S_{n+1}}{S_n}, \quad S_{n+1} = S_n(1 - \epsilon_n^{\text{cond}}). \quad (6.52)$$

The corresponding conditional exit-counting rate is $\epsilon_n^{\text{cond}}/\tau_{\text{tot}}$, whereas the first-exit counting rate normalized to the initial excitation ensemble is $(S_n - S_{n+1})/\tau_{\text{tot}}$. These rates condition on different ensembles.

To describe changes in maintenance probability between adjacent endpoints, define the **exponential decay rate**; iteration gives

$$\Gamma_n^{\text{exit}} = -\frac{1}{\tau_{\text{tot}}} \log \frac{S_{n+1}}{S_n}, \quad S_N = \exp \left[-\tau_{\text{tot}} \sum_{n=0}^{N-1} \Gamma_n^{\text{exit}} \right]. \quad (6.53)$$

This fixes a decay scale at cycle endpoints, without specifying a continuous exit time within a cycle. Count exit time to the corresponding cycle endpoint and write $T_E = \tau_{\text{tot}} N_E$. Taking expectations in the nonnegative-term identity $N_E = \sum_{n=0}^{\infty} \mathbf{1}_{\{N_E > n\}}$ and using the Neumann inverse of Eq. (4.71) gives

$$\begin{aligned} \mathbb{E}T_E &= \tau_{\text{tot}} \sum_{n=0}^{\infty} S_n \\ &= \tau_{\text{tot}} \text{Tr}[(I - \mathcal{Q}_V)^{-1}(\rho_0)] \leq \frac{\tau_{\text{tot}}}{\epsilon_{\min}}. \end{aligned} \quad (6.54)$$

The strictly positive exit condition ensures convergence of the series; the inverse here acts on $\mathcal{B}(\mathcal{M})$. Here it sums uninterrupted maintenance shares of an original excitation; in the stationary-state formula, Eq. (4.72), it sums direction occupation from continuing formation. For $V = I$ and $\rho_0 = P_\nu$, we have $\mathcal{Q}_I(P_\nu) = (1 - \epsilon_\nu)P_\nu$, so

$$\begin{aligned} S_n &= (1 - \epsilon_\nu)^n, \quad \text{Pr}(N_E = n) = \epsilon_\nu(1 - \epsilon_\nu)^{n-1} \quad (n \geq 1), \\ \mathbb{E}T_E &= \frac{\tau_{\text{tot}}}{\epsilon_\nu}, \quad \Gamma_\nu = -\frac{\log(1 - \epsilon_\nu)}{\tau_{\text{tot}}}. \end{aligned} \quad (6.55)$$

The reciprocal mean maintenance time and exponential decay rate are given by these separate expressions. For weak exit, $\Gamma_\nu = \epsilon_\nu/\tau_{\text{tot}} + O(\epsilon_\nu^2/\tau_{\text{tot}})$. Strictly positive $\epsilon_{\min}$ is a uniform sufficient condition for finite mean maintenance time. If this condition is relaxed, set $N_E = \infty$ for a history that never exits. Since the events $\{N_E > n\}$ decrease with n , $\text{Pr}(N_E = \infty) = \lim_{n \rightarrow \infty} S_n$. The mean maintenance time is still the tail sum $\tau_{\text{tot}} \sum_{n \geq 0} S_n$, possibly infinite. Finiteness depends on the particular $\mathcal{Q}_V$ and initial state; the inverse expression applies only when its Neumann series converges.

Occupation relaxation Total occupation includes continuation and exit of the original excitation, together with re-formation from the background, and is governed by the complete cycle $\mathcal{F}$. Let a nonzero operator B satisfy $\mathcal{F}(B) = \lambda_{\text{rel}} B$, with $0 < |\lambda_{\text{rel}}| < 1$. Trace preservation gives $\text{Tr} B = \text{Tr} \mathcal{F}(B) = \lambda_{\text{rel}} \text{Tr} B$, hence $\text{Tr} B = 0$. Iteration yields $\mathcal{F}^n(B) = \lambda_{\text{rel}}^n B$. At $t_n = n\tau_{\text{tot}}$, this eigencomponent has an exponential amplitude scale with **relaxation rate**

$$\Gamma^{\text{rel}}(\lambda_{\text{rel}}) = -\frac{\log |\lambda_{\text{rel}}|}{\tau_{\text{tot}}}, \quad |\lambda_{\text{rel}}|^n = e^{-t_n \Gamma^{\text{rel}}(\lambda_{\text{rel}})}. \quad (6.56)$$

The eigenvalue argument also produces cyclewise oscillation. A complex eigenoperator and its adjoint enter physical state deviations through self-adjoint combinations. An occupation or reading exhibits this relaxation only if the initial deviation contains the eigencomponent and the readout has nonzero response to it. If the cycle

map is not diagonalizable, write a size- b Jordan block as $J = \lambda_{\text{rel}}I + N$, with $N^b = 0$. Then

$$J^n = \sum_{\ell=0}^{\min(n,b-1)} \binom{n}{\ell} \lambda_{\text{rel}}^{n-\ell} N^\ell.$$

Polynomial factors in n also occur, but the rate above still gives the asymptotic exponential scale for nonzero eigenvalues. A Jordan block with zero eigenvalue vanishes after at most b cycles.

Section 4 gives the contraction decomposition

$$\mathcal{F}(X) = c \text{Tr}(X) \sigma_g^{\text{sh}} + (1-c) \mathcal{G}(X), \quad 0 < c < 1,$$

where $\mathcal{G}$ is completely positive and trace preserving. For the traceless eigenoperator above, $\mathcal{G}(B) = \lambda_{\text{rel}}B/(1-c)$. Positivity and trace preservation make $\mathcal{G}$ contractive in trace norm on self-adjoint operators. Splitting a general operator into real and imaginary parts gives

$$\|\mathcal{G}^n(B)\|_1 \leq \|\Re B\|_1 + \|\Im B\|_1 \leq 2\|B\|_1.$$

Its powers are uniformly bounded, so its eigenvalues have modulus at most one. Therefore

$$|\lambda_{\text{rel}}| \leq 1 - c, \quad \Gamma^{\text{rel}}(\lambda_{\text{rel}}) \geq -\frac{\log(1-c)}{\tau_{\text{tot}}}. \quad (6.57)$$

This gives a uniform decay estimate; specific relaxation rates are still determined by the spectrum of the complete cycle. The stationary limit is a thermal equilibrium state at the given temperature only when the background preparation meets the Gibbs criterion of Section 4.

Coherence decay Take $V = I$ and distinct directions ν, μ . Using the notation of Section 4,

$$d_\nu = \sum_a a_{a\nu} t_u (1 - c_{a\nu}), \quad c_{a\nu} = \cos \alpha_{a\nu}, \quad \gamma_{\nu\mu} = 1 - d_\nu - d_\mu.$$

Restoration removes cross terms between background and directions, while formation adds only direction-diagonal terms. The complete cycle therefore satisfies

$$\begin{aligned} \mathcal{F}_{I,g}(|\nu\rangle\langle\mu|) &= e^{i(\varphi_\nu - \varphi_\mu)} \gamma_{\nu\mu} |\nu\rangle\langle\mu|, \\ \Gamma_{\nu\mu}^{\text{coh}} &= -\frac{\log \gamma_{\nu\mu}}{\tau_{\text{tot}}}. \end{aligned} \quad (6.58)$$

Thus $\rho_{\nu\mu}(n) = \gamma_{\nu\mu}^n e^{in(\varphi_\nu - \varphi_\mu)} \rho_{\nu\mu}(0)$. The prescribed exchange angles and positive star share ensure $a_0 \leq \gamma_{\nu\mu} \leq 1$. The **coherence decay rate** in Eq. (6.58) describes decay of this off-diagonal modulus in the full state; the phase difference $\varphi_\nu - \varphi_\mu$ describes its cyclewise phase change.

The relation between coherence decay and exit follows directly from return amplitudes of the directions. For channel $e = (a, \lambda)$, abbreviate $a_e = a_{a\lambda}$ and $c_e = c_{a\lambda}$. After removing net return phases, define weighted return-amplitude vectors

$$\begin{aligned} (v_\nu)_* &= \sqrt{a_0}, \\ (v_\nu)_{e,u} &= \sqrt{a_e t_u} [1 - (1 - c_e) \delta_{\lambda\nu}], \\ (v_\nu)_{e,v} &= \sqrt{a_e t_v}. \end{aligned} \quad (6.59)$$

Each thermal block satisfies $t_u + t_v = 1$, and $a_0 + \sum_e a_e = 1$. Summing components gives

$$\begin{aligned}\|v_\nu\|^2 &= a_0 + \sum_e a_e [t_u(1 - (1 - c_e)\delta_{\lambda\nu})^2 + t_v] \\ &= 1 - \sum_a a_{a\nu} t_u (1 - c_{a\nu}^2) = 1 - \epsilon_\nu.\end{aligned}$$

For $\nu \neq \mu$, a given channel changes the return amplitude of at most one of the directions, so

$$\langle v_\mu, v_\nu \rangle = 1 - d_\nu - d_\mu = \gamma_{\nu\mu}.$$

The Cauchy–Schwarz inequality gives $\gamma_{\nu\mu} \leq \sqrt{(1 - \epsilon_\nu)(1 - \epsilon_\mu)}$. The vectors have the same nonzero star component. If proportional, their proportionality factor must therefore be one. If a channel connected to ν or μ has $a_e t_u > 0$ and $c_e \neq 1$, the corresponding return-amplitude components differ, making the inequality strict. Finite inverse temperature and positive exchange edges assumed here ensure strictness. Substituting the exit rate $\Gamma_\nu = -\log(1 - \epsilon_\nu)/\tau_{\text{tot}}$ gives

$$\boxed{\begin{aligned}\Gamma_{\nu\mu}^{\text{coh}} &= \frac{\Gamma_\nu + \Gamma_\mu}{2} - \frac{1}{\tau_{\text{tot}}} \log \frac{\gamma_{\nu\mu}}{\sqrt{(1 - \epsilon_\nu)(1 - \epsilon_\mu)}} \\ &> \frac{\Gamma_\nu + \Gamma_\mu}{2}.\end{aligned}} \quad (6.60)$$

The first term describes decay of the two direction-maintenance shares; the second is determined by their different return-amplitude structures. If positivity of exchange is relaxed so that neither direction has effective exit coupling, then $\gamma_{\nu\mu} = 1$ and $\Gamma_{\nu\mu}^{\text{coh}} = 0$. Preservation of coherence modulus and phase change are determined separately by $\gamma_{\nu\mu}$ and $\varphi_\nu - \varphi_\mu$. A state conditional on a non-exit history must also be divided by S_n , so its normalized coherence depends on that history’s maintenance share as well.

Event-counting rates are determined by actual record probabilities, maintenance time by first-exit histories, and occupation and coherence relaxation by the complete cycle.

§7 Observational Manifestations of the Cycle

The preceding section establishes records and events in the cycle, relating changes of excitation states to process distinctions actually retained. We can therefore examine how the position pattern of a formed excitation changes through successive cycles, how its maintenance and decay are identified, and how continuing formation and decay alter readings across the ensemble. These questions require tracking both the post-cycle state and its records.

Use the decomposition $\mathcal{H} = \mathcal{C} \oplus \mathcal{M}$ from Section 4, with projections Π, P and background preparation $\sigma_g^{\text{sh}} = \sum_a g_a P_{e_a}$. Fix a shell-preserving input control V that preserves $\mathcal{C}$ and $\mathcal{M}$ and acts as the identity on $\mathcal{C}$. The same exchange data and independent, identical environment preparations in each cycle determine the complete cycle; Equation (6.41) gives its state iteration

$$\rho_{m+1} = \mathcal{F}_{V,g}(\rho_m), \quad t_m = m\tau_{\text{tot}}. \quad (7.1)$$

Each cycle includes free evolution, formation exchange, and pattern reconstruction; τ_{tot} accounts for the duration of the actual input control and of every interaction. Environment records satisfying the preservation conditions of Section 6 remain after departure from the system. Position readout occurs at selected cycle endpoints, with detection time counted separately. Event-counting rates, exponential decay rates, and mean maintenance times use the definitions of Section 6.6. The propagation range is determined by the criterion in Section 4; Appendix A gives propagation errors, circular-displacement estimates, and compatibility of readouts across depths.

§7.1 Motion and Interference

Applying different numbers of cycles to identical preparations and reading positions at the selected endpoints allows displacement, spreading, and interference of excitation patterns to be compared. Retaining earlier process records also gives position statistics conditional on different histories; summing these histories gives the distribution of the complete cycle.

At the end of cycle m , apply the position instrument $\mathcal{L}_x(\rho) = P_x \rho P_x$ of Section 6. Detection starts at $m\tau_{\text{tot}}$ and recording completes at $m\tau_{\text{tot}} + t_D$, where t_D is the actual detection duration. By Eq. (6.9) and the Born weights of Section 6, the position probability is

$$p_m(x) = \text{Tr}[P_x \mathcal{F}_{V,g}^m(\rho_0)], \quad \sum_x p_m(x) = 1. \quad (7.2)$$

Use the actual instrument corresponding to the same V, g in Eq. (6.44). Separate formation and exit counts require retained thermal-channel and incident and outgoing labels; unresolved labels are summed according to the actual readout. For a record sequence $\mathbf{r} = (r_1, \dots, r_m)$, write

$$\tilde{\rho}_{\mathbf{r}} = \mathcal{I}_{r_m} \circ \dots \circ \mathcal{I}_{r_1}(\rho_0)$$

for the unnormalized conditional state of Eq. (6.41), whose trace is $p(\mathbf{r})$. The joint and conditional probabilities of position and record sequence are

$$p(\mathbf{r}, x) = \text{Tr}(P_x \tilde{\rho}_{\mathbf{r}}), \quad p(x | \mathbf{r}) = \frac{p(\mathbf{r}, x)}{p(\mathbf{r})}. \quad (7.3)$$

The latter applies when $p(\mathbf{r}) > 0$. Equation (6.41) ensures

$$p_m(x) = \sum_{\mathbf{r}} p(\mathbf{r}, x).$$

Position readings give the excitation's position distribution, and process records specify its cycle history under this distinction.

The coherent retrieval component of Section 4 gives an explicitly calculable propagation process. Select histories with the corresponding return label in every cycle; the excitation continues propagating after each pattern reconstruction. As the number of cycles increases, the share of these histories among all outcomes may decrease, while their normalized patterns follow the coherent propagation action. To calculate the history share and its pattern separately, use the control V from the coherent propagation construction of Section 4, with orientation-position space $\mathcal{H}_a \subset \mathcal{M}$, projection Q_a , propagation operator W , and preparation share $a_0 > 0$, satisfying

$$K_{0*}^{(V)} Q_a = \sqrt{a_0} W Q_a.$$

The star thermal block is independent and invariant during exchange, so its final label also fixes the input block. Together with the restoration auxiliary's return to g , it distinguishes this component. Let $*^m$ denote the history in which this outcome occurs in each of m consecutive cycles. For a normalized initial state $\rho_0 = Q_a \rho_0 Q_a$, invariance of $\mathcal{H}_a$ under W gives by composition

$$\tilde{\rho}_{*^m} = a_0^m W^m \rho_0 (W^\dagger)^m.$$

Taking the trace and normalizing gives

$$p(*^m) = a_0^m, \quad \rho_{m|*} = W^m \rho_0 (W^\dagger)^m. \quad (7.4)$$

The four incident/outgoing combinations of ordinary thermal channels remain distinguished by retained labels. Equation (7.4) selects only these coherent histories; all other outcomes remain in Eq. (7.2).

At the same principal-unit position, both orientation components contribute to fine position readout. Changes in their relative phase redistribute probabilities among torsion positions and produce interference patterns. If only the principal-unit position is retained and torsion positions are summed, cross terms may cancel. This distinction depends on the position labels retained by readout; cancellation in a marginal distribution alone does not establish loss of cycle coherence.

Use the finite positions of Section 4,

$$x(s, \ell) = p^j \zeta^s (1 + p)^\ell \pmod{p^K}, \quad 0 \leq j < K,$$

where $q = p - 1$, $s \in \mathbb{Z}_q$, $L = p^{K-j-1}$, and $\ell \in \mathbb{Z}_L$. The orientation conditions are $p \geq 5$ and $2a \not\equiv 0 \pmod{q}$. Let $\mathbf{E} : \mathbb{C}^2 \otimes \mathbb{C}^L \rightarrow \mathcal{H}$ be the isometric embedding already specified, with range $\mathcal{H}_a$ and

$$\mathbf{E}(|\pm\rangle \otimes |\ell\rangle) = q^{-1/2} \sum_s e^{\pm i\theta_s} |x(s, \ell)\rangle, \quad \theta_s = \frac{2\pi a s}{q}, \quad \mathbf{E}\mathbf{E}^\dagger = Q_a.$$

Represent the conditional state by $\varrho_m = \mathbf{E}^\dagger \rho_{m|*} \mathbf{E}$. Compression of actual point projections gives the position effects

$$F_{s,\ell} = \mathbf{E}^\dagger P_{x(s,\ell)} \mathbf{E} = \frac{1}{q} \begin{pmatrix} 1 & e^{-2i\theta_s} \\ e^{2i\theta_s} & 1 \end{pmatrix} \otimes |\ell\rangle\langle\ell|. \quad (7.5)$$

Since $2a \not\equiv 0 \pmod{q}$, the finite geometric sum gives $\sum_s e^{2i\theta_s} = 0$, so

$$\sum_s F_{s,\ell} = I_2 \otimes |\ell\rangle\langle\ell|, \quad \sum_{s,\ell} F_{s,\ell} = I.$$

These effects form a complete position readout on the propagation space. For a normalized pure pattern

$$\psi_m(\ell) = (\psi_{m,+}(\ell), \psi_{m,-}(\ell))^\top,$$

Equation (7.3) becomes

$$\begin{aligned} p(s, \ell \mid *^m) &= \frac{1}{q} |e^{i\theta_s} \psi_{m,+}(\ell) + e^{-i\theta_s} \psi_{m,-}(\ell)|^2, \\ p(\ell \mid *^m) &= |\psi_{m,+}(\ell)|^2 + |\psi_{m,-}(\ell)|^2. \end{aligned} \quad (7.6)$$

The first expression expands as

$$p(s, \ell \mid *^m) = \frac{|\psi_{m,+}(\ell)|^2 + |\psi_{m,-}(\ell)|^2}{q} + \frac{2}{q} \Re \left[e^{2i\theta_s} \psi_{m,+}(\ell) \overline{\psi_{m,-}(\ell)} \right].$$

The second term changes fine position readings through the relative phase; summing over s removes it and gives the second identity in Eq. (7.6). For mixed states, the expression is $\text{Tr}(\varrho_m F_{s,\ell})$, with cross terms determined by off-diagonal entries of the conditional state.

When both phase factors are independent of the principal-unit Fourier-mode index, a pattern at one principal-unit position connects only to adjacent positions in each cycle. For localized preparation, position readings are confined to the range reachable cycle by cycle. For general spectral phases, the actual propagation kernel determines the range. Section 4 gives

$$G_m(v) = \frac{1}{L} \sum_r e^{i\kappa_r v} W_r^m.$$

Substitution of the resulting pattern into Eq. (7.6) gives conditional readouts after each cycle. Under the phase-factor condition of Eq. (4.58), the recurrence of Section 4 reduces to

$$\psi_{m+1}(\ell) = C_0 \begin{pmatrix} \psi_{m,+}(\ell-1) \\ \psi_{m,-}(\ell+1) \end{pmatrix}.$$

The two orientations arrive from neighboring positions and are mixed by the same C_0 . For an initial pattern supported at one position ℓ_0 , induction confines its support after m cycles to graph distance at most m from ℓ_0 on the cycle graph. Position probabilities obey the same support bound. For $L = 1$, only one principal-unit position exists, but orientation and fine position patterns may still change. The step count describes connections between principal-unit positions; interval readings additionally require the actual position calibration.

After applying the interval calibration of Section 5, position records allow comparison of mean position and dispersion. Changes in the mean describe overall displacement in that calibration, while variance describes spreading of readings. Both quantities are computed from actual position probabilities. Let $x_{s,\ell}$ be the fixed integer representative of $x(s, \ell)$ in $1, \dots, d-1$, with $d = p^K$, and choose a nonzero reference representative u_* in the same shell. Write

$$y_{s,\ell} = \log \frac{x_{s,\ell}}{u_*}.$$

The conditional interval distribution is

$$\mu_m = \sum_{s,\ell} \text{Tr}(\varrho_m F_{s,\ell}) \delta_{y_{s,\ell}}, \quad (7.7)$$

where δ_y is the unit point mass at the real number y . Set $Y = \sum_{s,\ell} y_{s,\ell} P_{x(s,\ell)}$, extending it by zero outside the selected shell, and let $A_b = \mathbf{E}^\dagger Y^b \mathbf{E}$, for $b = 1, 2$. Taking moments of the interval distribution gives

$$\mathbb{E}_m(y) = \text{Tr}(\varrho_m A_1), \quad \text{Var}_m(y) = \text{Tr}(\varrho_m A_2) - [\text{Tr}(\varrho_m A_1)]^2. \quad (7.8)$$

The second moment of actual point readings is given by A_2 . Inserting $EE^\dagger = Q_a$ gives

$$\begin{aligned} A_2 - A_1^2 &= E^\dagger Y(I - Q_a)YE \\ &= [(I - Q_a)YE]^\dagger [(I - Q_a)YE] \geq 0. \end{aligned}$$

Continue to write W for its matrix on the abstract propagation space, so $\varrho_{m+1} = W\varrho_m W^\dagger$. Cyclicity of the trace gives

$$\mathbb{E}_{m+1}(y) - \mathbb{E}_m(y) = \text{Tr}[\varrho_m(W^\dagger A_1 W - A_1)]. \quad (7.9)$$

Dividing by τ_{tot} gives the change in mean interval per unit time for identical preparations with adjacent stopping counts. The finite-ring action $x' = [(1+p)x]_d$ may wrap around; the actual interval change is $\log(x'/x)$ using fixed representatives. If the complete cycle outputs the zero element, retain it as a separate outcome without taking its logarithm.

When positions are recorded repeatedly in one history, each post-readout state enters the next cycle. The position sequence then depends on both the original cycle and the position detection already performed. Fine point measurement in every cycle, with outcomes summed, gives the action

$$\Delta_{\text{pos}} \circ \mathcal{F}_{V,g}, \quad \Delta_{\text{pos}}(X) = \sum_x P_x X P_x.$$

The repeated protocol has period $\tau_{\text{tot}} + t_D$. Its histories require composition of the joint instrument in Section 6; the coherent retrieval formula (7.4) applies to the endpoint-only protocol above. Star histories are non-exit histories. Determining maintenance time of the original excitation also requires summing all other non-exit components and using the first exit record as the history's endpoint.

§7.2 Maintenance and Decay

An exit record satisfying the record conditions identifies a decay event of the original excitation when actually read out. For repeated observations of the same initial state, the share with no exit record gives the maintenance probability, and the cycle containing the first exit record determines maintenance time. Here $\rho_0 = P\rho_0 P$ is positive and normalized. If an excitation forms again after exit, its count starts with the new formation record.

First-exit records accumulate over successive cycles, and the share of samples still unexited decreases. Recording exchange channels also distinguishes the direction and background channel through which exit occurs. Use the exit shares $\epsilon_{a\nu}$ from Section 4, retain the thermal channel and incident and outgoing labels, and sum subsequent restoration output labels to obtain

$$\mathcal{I}_{a\nu}^E(X) = \epsilon_{a\nu} \langle \nu | V X V^\dagger | \nu \rangle \sigma_g^{\text{sh}}. \quad (7.10)$$

Here ν labels the direction before exit and a the background exchange channel. Retaining a specific restoration outcome b gives output

$$\epsilon_{a\nu} g_b \langle \nu | V X V^\dagger | \nu \rangle P_{e_b}.$$

Summing over b recovers Eq. (7.10).

For direction-space inputs, combining all non-exit outcomes gives $\mathcal{Q}_V$ of Section 4. The unnormalized state after m successive non-exit cycles is $\mathcal{Q}_V^m(\rho_0)$. Let N_E be the cycle of first exit. Then

$$\begin{aligned} S_m &= \text{Tr } \mathcal{Q}_V^m(\rho_0) = \text{Pr}(N_E > m), \\ \text{Pr}(N_E = m + 1, a, \nu) &= \epsilon_{a\nu} \langle \nu | V \mathcal{Q}_V^m(\rho_0) V^\dagger | \nu \rangle. \end{aligned} \quad (7.11)$$

Summing exit channels, Eq. (6.51) gives

$$\text{Pr}(N_E = m + 1) = S_m - S_{m+1}, \quad \text{Pr}(N_E \leq n) = 1 - S_n.$$

Write $\epsilon_\nu = \sum_a \epsilon_{a\nu}$ and $\epsilon_{\min} = \min_\nu \epsilon_\nu$. For $\epsilon_{\min} > 0$, the bound $S_m \leq (1 - \epsilon_{\min})^m$ in Section 6 gives $S_m \rightarrow 0$, so

$$\sum_{m=0}^{\infty} \text{Pr}(N_E = m + 1) = 1.$$

Strictly positive exit ensures eventual exit of the original excitation with probability one.

At a fixed stopping cycle, select samples with no exit record and collect their position readings. This gives the normalized pattern of the excitations still maintained. Their share of the initial ensemble is S_m . For $S_m > 0$, set

$$\rho_{m|\mathcal{M}} = \frac{\mathcal{Q}_V^m(\rho_0)}{S_m}, \quad p(x | N_E > m) = \text{Tr}(P_x \rho_{m|\mathcal{M}}). \quad (7.12)$$

The joint probability is

$$\text{Pr}(N_E > m, x) = S_m p(x | N_E > m).$$

The conditional states satisfy the recurrence

$$\rho_{m+1|\mathcal{M}} = \frac{\mathcal{Q}_V(\rho_{m|\mathcal{M}})}{\text{Tr } \mathcal{Q}_V(\rho_{m|\mathcal{M}})}.$$

The denominator generally depends on the input, so the conditional pattern also reflects direction-dependent maintenance probabilities. Non-exit conditioning restricts process destinations; preservation of pattern must be assessed from changes in the conditional state.

Without input control, $V = I$, exit probabilities may differ between directions. After many cycles, more slowly exiting directions occupy a larger proportion of non-exit samples and may change the remaining position pattern. Write the initial direction occupations as $x_\nu = \langle \nu | \rho_0 | \nu \rangle$, with $x_\nu \geq 0$ and $\sum_\nu x_\nu = 1$. Multiplying the diagonal entries of Section 4 over successive cycles gives

$$S_m = \sum_\nu x_\nu (1 - \epsilon_\nu)^m, \quad \langle \nu | \rho_{m|\mathcal{M}} | \nu \rangle = \frac{x_\nu (1 - \epsilon_\nu)^m}{S_m}. \quad (7.13)$$

For $x_\nu, x_\mu > 0$, their conditional occupation ratio is

$$\frac{\langle \nu | \rho_{m|\mathcal{M}} | \nu \rangle}{\langle \mu | \rho_{m|\mathcal{M}} | \mu \rangle} = \frac{x_\nu}{x_\mu} \left(\frac{1 - \epsilon_\nu}{1 - \epsilon_\mu} \right)^m.$$

Even if each individual direction preserves its spatial pattern, a multidirection conditional pattern may change as relative maintenance shares change.

Interference in non-exit samples also depends on coherence between directions. Both return overlaps and conditional normalization affect it; decay of unnormalized coherence does not ensure a decrease of normalized coherence in every cycle. Using the return-overlap coefficient $\gamma_{\nu\mu} = 1 - d_\nu - d_\mu$ of Section 6 gives

$$[\mathcal{Q}_I^m(\rho_0)]_{\nu\mu} = e^{im(\varphi_\nu - \varphi_\mu)} \gamma_{\nu\mu}^m \rho_{0,\nu\mu}, \quad \nu \neq \mu. \quad (7.14)$$

Under the finite inverse temperature β , positive star share, and positive exchange-edge conditions used in Eq. (6.60), the strict Cauchy–Schwarz inequality for return amplitudes gives

$$0 < \gamma_{\nu\mu} < \sqrt{(1 - \epsilon_\nu)(1 - \epsilon_\mu)}. \quad (7.15)$$

Dividing Eq. (7.14) by S_m gives the off-diagonal component of the conditional pattern. For a component initially nonzero, its modulus satisfies cyclewise

$$\frac{|[\rho_{m+1|\mathcal{M}}]_{\nu\mu}|}{|[\rho_{m|\mathcal{M}}]_{\nu\mu}|} = \gamma_{\nu\mu} \frac{S_m}{S_{m+1}}.$$

Taking $-\tau_{\text{tot}}^{-1} \log$ and using Eqs. (6.53) and (6.58) gives the conditional logarithmic decay rate $\Gamma_{\nu\mu}^{\text{coh}} - \Gamma_m^{\text{exit}}$. This difference may vary with cycle count and may temporarily be negative, indicating a temporary increase in normalized coherence.

Under these strict conditions and $V = I$, long-lived non-exit samples concentrate on the slowest-exiting directions initially occupied, retaining their initial relative occupations while interdirection coherence tends to zero. Set

$$\epsilon_* = \min_{\nu: x_\nu > 0} \epsilon_\nu, \quad I_* = \{\nu : x_\nu > 0, \epsilon_\nu = \epsilon_*\}.$$

Then

$$\lim_{m \rightarrow \infty} \rho_{m|\mathcal{M}} = \frac{\sum_{\nu \in I_*} x_\nu P_\nu}{\sum_{\nu \in I_*} x_\nu} \quad (V = I). \quad (7.16)$$

The proof is entrywise. First,

$$\frac{S_m}{(1 - \epsilon_*)^m} = \sum_{\nu} x_\nu \left(\frac{1 - \epsilon_\nu}{1 - \epsilon_*} \right)^m \rightarrow \sum_{\nu \in I_*} x_\nu > 0.$$

Conditional occupation of faster-exiting directions therefore vanishes, while the slowest retain their initial occupation ratios. Second, positivity of the initial state gives $|\rho_{0,\nu\mu}|^2 \leq x_\nu x_\mu$, so every row and column with $x_\nu = 0$ vanishes. For other off-diagonal entries,

$$\gamma_{\nu\mu} < \sqrt{(1 - \epsilon_\nu)(1 - \epsilon_\mu)} \leq 1 - \epsilon_*.$$

After division by the same leading factor, these terms also tend to zero. In finite dimension, entrywise convergence proves Eq. (7.16). The position readings converge to

$$\lim_{m \rightarrow \infty} p(x \mid N_E > m) = \frac{\sum_{\nu \in I_*} x_\nu \text{Tr}(P_x P_\nu)}{\sum_{\nu \in I_*} x_\nu}.$$

This limit describes the pattern of long-lived non-exit samples, while their share S_m of the initial ensemble tends to zero.

For general shell-preserving input control V , an excitation initially in one shell remains there at non-exit cycle endpoints, so this conditional history does not produce thermal occupation across shells. Set $P_j^{\mathcal{M}} = \sum_{\nu: j_\nu=j} P_\nu$. It commutes with V, U_φ , and

$$\mathcal{Q}_V(P_j^{\mathcal{M}} X P_j^{\mathcal{M}}) = P_j^{\mathcal{M}} \mathcal{Q}_V(X) P_j^{\mathcal{M}}.$$

Thus a single-shell initial state's non-exit final state always remains supported on that shell. For an initial state occupying several shells, conditional shell shares may still change through different exit probabilities. This preservation concerns non-exit states at cycle endpoints; energy exchange during the round trip through the zero element within formation exchange remains determined by the joint process of Section 4.

A first-exit record assigns each maintenance history a duration counted in cycles. Count exit time to its cycle endpoint, $T_E = \tau_{\text{tot}} N_E$. For $\epsilon_{\min} > 0$, Eq. (6.54) gives

$$\mathbb{E}T_E = \tau_{\text{tot}} \sum_{m=0}^{\infty} S_m = \tau_{\text{tot}} \text{Tr}[\mathcal{R}_V(\rho_0)], \quad \mathcal{R}_V = (I - \mathcal{Q}_V)^{-1}.$$

This resolves time at cycle endpoints, not the continuous exit instant within formation exchange. For $V = I$ and initial state P_ν ,

$$S_m = (1 - \epsilon_\nu)^m, \quad \rho_{m|\mathcal{M}} = P_\nu.$$

The direction retains its pattern until exit and has a finite mean maintenance time. Equation (6.55) gives

$$\mathbb{E}T_E = \frac{\tau_{\text{tot}}}{\epsilon_\nu}, \quad \Gamma_\nu = -\frac{\log(1 - \epsilon_\nu)}{\tau_{\text{tot}}}.$$

The conditional exit-counting rate for the next cycle among non-exit samples is $\epsilon_\nu/\tau_{\text{tot}}$; the rate normalized to the initial ensemble is $S_m \epsilon_\nu/\tau_{\text{tot}}$. For general control, use Eqs. (7.11) and (6.54).

If exit records also resolve the thermal channel, each exit has a cycle label and a thermal-environment energy reading. The thermal-environment energy increment in channel (a, ν) is $\Delta E_{a\nu} = -\Delta_{a\nu} = E_\nu - E_a$, fixed by energy matching in Section 4. Assigning channel probabilities to these readings gives

$$\begin{aligned} \Pr(N_E = m + 1, \Delta E \in A) \\ = \sum_{a, \nu: E_\nu - E_a \in A} \epsilon_{a\nu} \langle \nu | V \mathcal{Q}_V^m(\rho_0) V^\dagger | \nu \rangle. \end{aligned} \quad (7.17)$$

Energy-lowering exit gives a positive thermal-environment energy increment. The matrix elements in Section 4 show that such channels with nonzero coupling can remain active in the zero-temperature limit. Background restoration has its own energy readings, which must be included in the full-cycle balance. For decay of a specific particle, the parent state and products must additionally be related to these channel records. First exit ends maintenance of the original excitation, while the resulting background continues cycling; a new formation record marks the beginning of another excitation history.

§7.3 Formation

Participation of the background in formation exchange can produce new directional excitations. An actual formation record labels the cycle in which this occurs and, if the channel is resolved, the direction formed. Position readout then gives the new excitation's position distribution. Repeated observations of the same background preparation provide statistics on waiting times before formation, the distribution of formation directions, and continuation after formation.

Different formation channels may produce different directions. With channel labels retained, position readings give the corresponding direction's distribution; with direction labels summed, they are weighted by actual formation shares. Use $f_{\nu a}$ from Section 4. Since V acts as the identity on $\mathcal{C}$, retaining formation channels and summing the associated auxiliary output labels gives the instrument

$$\mathcal{I}_{\nu a}^F(X) = f_{\nu a} \langle e_a | X | e_a \rangle P_\nu. \quad (7.18)$$

Here a labels the background exchange channel and ν the direction formed. For background preparation σ_g^{sh} , formation through (ν, a) in one cycle has probability $g_a f_{\nu a}$. Write

$$f_\nu(g) = \sum_a g_a f_{\nu a}, \quad f_\Sigma = \sum_\nu f_\nu(g), \quad B_g = \sum_\nu f_\nu(g) P_\nu.$$

The total formation probability per cycle is f_Σ . Assume $f_\Sigma > 0$; finite inverse temperature, positive exchange edges, and positive background shares in Section 4 ensure this, while a positive star share also gives $f_\Sigma < 1$. Conditional on formation, the direction distribution and combined state are

$$\Pr(\nu | F) = \frac{f_\nu(g)}{f_\Sigma}, \quad \rho_F = \frac{B_g}{f_\Sigma}.$$

If position readout is performed at the end of the formation cycle, then

$$p(x | F) = \text{Tr}(P_x \rho_F) = \frac{1}{f_\Sigma} \sum_\nu f_\nu(g) \text{Tr}(P_x P_\nu). \quad (7.19)$$

When a record resolves a direction ν with positive formation probability, its position reading has distribution $p(x | F, \nu) = \text{Tr}(P_x P_\nu)$. The formation record fixes the direction entered, and position readings give that conditional state's distribution. Different directions in the same shell may have identical position distributions; their distinction remains in the formation labels.

Starting from a fixed background preparation, first formation may occur in the first cycle or later. After summing restoration outputs in a nonformation cycle, the normalized background returns to the same preparation, giving the same formation probability in the next cycle. Denote the summed nonformation action by $\mathcal{I}^{\text{NF}}$. Restoration in Section 4 gives

$$\mathcal{I}^{\text{NF}}(\sigma_g^{\text{sh}}) = (1 - f_\Sigma) \sigma_g^{\text{sh}}.$$

Let $L_F \geq 1$ be the number of cycles until first formation. After $\ell - 1$ nonformation cycles followed by formation through (ν, a) ,

$$\Pr(L_F = \ell, \nu, a) = (1 - f_\Sigma)^{\ell-1} g_a f_{\nu a}.$$

Summing the relevant labels gives

$$\Pr(L_F = \ell, \nu) = (1 - f_\Sigma)^{\ell-1} f_\nu(g), \quad \Pr(L_F = \ell) = (1 - f_\Sigma)^{\ell-1} f_\Sigma. \quad (7.20)$$

The geometric series yields $\sum_{\ell \geq 1} \Pr(L_F = \ell) = 1$. The tail is $\Pr(L_F > m) = (1 - f_\Sigma)^m$, so

$$\mathbb{E}L_F = \sum_{m=0}^{\infty} \Pr(L_F > m) = \frac{1}{f_\Sigma}.$$

Summing over formation times gives $\Pr(\nu \mid \text{first formation}) = f_\nu(g)/f_\Sigma$, and hence

$$\Pr(L_F = \ell, \nu) = \Pr(L_F = \ell) \frac{f_\nu(g)}{f_\Sigma}.$$

For fixed background preparation, the joint waiting-time and formation-direction distribution therefore factorizes.

First formation starts a new maintenance history. A position readout at the end of the n th subsequent cycle examines its maintenance probability and conditional pattern. This protocol inserts no position probe in the formation cycle, so after summing formation labels the initial state remains $\rho_F = B_g/f_\Sigma$. Using the non-exit map above gives

$$\begin{aligned} S_n^F &= \text{Tr } \mathcal{Q}_V^n(\rho_F), \\ p(x \mid F, N_E > n) &= \frac{\text{Tr}[P_x \mathcal{Q}_V^n(\rho_F)]}{\text{Tr } \mathcal{Q}_V^n(\rho_F)}. \end{aligned} \quad (7.21)$$

The latter applies when the denominator is positive. Formation channels determine the initial direction distribution; subsequent cycles determine relative maintenance shares and position patterns. If position detection has already occurred at the end of the formation cycle, its conditional action must be included in subsequent evolution according to the history-composition rule of Section 6.

Restoration records after an exit may retain different amounts of information. With the coarse exit record in Eq. (7.10), summing restoration output labels gives normalized state σ_g^{sh} , so the next formation obeys Eq. (7.20). Retaining a specific restoration outcome b gives conditional initial state P_{e_b} , whose first-cycle formation distribution may differ. Write $f_b^C = \sum_\nu f_{\nu b}$. If the first cycle does not form an excitation and restoration outputs are summed in that and subsequent cycles, the background returns to the common preparation, so

$$\begin{aligned} \Pr(L_F = 1, \nu \mid b) &= f_{\nu b}, \\ \Pr(L_F = \ell, \nu \mid b) &= (1 - f_b^C)(1 - f_\Sigma)^{\ell-2} f_\nu(g), \quad \ell \geq 2. \end{aligned} \quad (7.22)$$

Summing over b with post-exit weights g_b and using $\sum_b g_b f_b^C = f_\Sigma$ recovers $f_\nu(g)$ in the first cycle. For $\ell \geq 2$,

$$\sum_b g_b \Pr(L_F = \ell, \nu \mid b) = (1 - f_\Sigma)^{\ell-1} f_\nu(g).$$

Statistics with resolved restoration outputs are thus consistent with those obtained by summing them.

The interval between successive exit records contains waiting for formation in the background and maintenance after a new excitation forms. The first part determines

when an excitation reappears; the second determines how long it continues. For $f_\Sigma > 0$ and $\epsilon_{\min} > 0$, both mean durations are finite. Using $\mathcal{R}_V = (I - \mathcal{Q}_V)^{-1}$, the formation waiting law and Eq. (6.54) give

$$\bar{T}_c = \frac{\tau_{\text{tot}}}{f_\Sigma}, \quad \bar{T}_M = \frac{\tau_{\text{tot}}}{f_\Sigma} \text{Tr } \mathcal{R}_V(B_g). \quad (7.23)$$

Here $\bar{T}_M$ uses the actual formation state B_g/f_Σ as its initial state. Adding the two durations gives

$$\mathbb{E}(T_{\text{EE}}) = \bar{T}_c + \bar{T}_M.$$

Formation waiting is counted to the end of its cycle; direction maintenance runs from that endpoint to the endpoint of the next exit cycle. Summing restoration labels always returns the same background, and later environments are independent fresh preparations, so successive exit intervals have the same statistical law above.

In histories whose initial background or direction block is known or recorded, formation enters the direction space and exit returns to the background. The control preserves both blocks, so the difference between formation and exit counts depends only on initial and final block membership. Let $I_M(m) \in \{0, 1\}$ indicate direction occupation at cycle m 's end, and let $C_F(m), C_E(m)$ be formation and exit counts over the first m cycles. Formation increases I_M by one, exit decreases it by one, and other outcomes leave it unchanged. Summing cycles gives

$$C_F(m) - C_E(m) = I_M(m) - I_M(0). \quad (7.24)$$

The right side has absolute value at most one, so

$$\left| \frac{C_F(m)}{m} - \frac{C_E(m)}{m} \right| \leq \frac{1}{m}.$$

Summing restoration outputs does not change formation/exit classification, and this relation remains valid. For general initial states, the mean relation follows from linearity of the instrument and the occupation recurrence of Section 4.

§7.4 Occupation and Thermalization

As formation and exit continue, individual maintenance histories repeatedly begin and end, changing total direction occupation, shell occupation, and position distributions. Under appropriate conditions these readings across the ensemble approach stationarity while formation and exit records continue. Changes in total occupation follow formation/exit balance, and stationary reading limits follow the complete cycle. When the background preparation meets the thermal equilibrium condition, the readings exhibit thermalization.

Repeated observation of identical preparations gives direction occupations and formation and exit counts for each cycle. Direction occupation increases when formation exceeds exit and decreases in the opposite case. This occupation includes excitations formed again later. For general ρ_0 , Section 4 shows that, after the first cycle, the state has the following form, which is preserved by subsequent cycles:

$$\rho_m = X_m + y_m \sigma_g^{\text{sh}}, \quad m \geq 1,$$

where $X_m = PX_mP \geq 0$, $y_m \geq 0$, and $\text{Tr } X_m + y_m = 1$. Equation (4.68) gives $X_{m+1} = \mathcal{Q}_V(X_m) + y_mB_g$. The mean formation and exit counts in cycle $m+1$ are

$$r_{F,m} = f_\Sigma y_m, \quad r_{E,m} = \text{Tr}[X_m - \mathcal{Q}_V(X_m)]. \quad (7.25)$$

Taking the trace of the direction recurrence gives

$$\text{Tr } X_{m+1} - \text{Tr } X_m = r_{F,m} - r_{E,m}.$$

With the time calibration of Section 6, the counting rates are

$$R_{F,m} = \frac{r_{F,m}}{\tau_{\text{tot}}}, \quad R_{E,m} = \frac{r_{E,m}}{\tau_{\text{tot}}}.$$

Here r is a mean count per cycle, and R a mean count per unit time. If the initial state lacks the closed form, compute the first cycle from the actual instrument. Let $\mathcal{I}_F$ and $\mathcal{I}_E$ sum formation and exit actions, respectively, and take

$$r_{F,0} = \text{Tr } \mathcal{I}_F(\rho_0), \quad r_{E,0} = \text{Tr } \mathcal{I}_E(\rho_0).$$

Expected cumulative counts are

$$\mathbb{E}C_F(N) = \sum_{m=0}^{N-1} r_{F,m}, \quad \mathbb{E}C_E(N) = \sum_{m=0}^{N-1} r_{E,m}.$$

If the initial state already has form $X_0 + y_0\sigma_g^{\text{sh}}$, the recurrence and count formulas apply from $m = 0$. For N_{samp} independent systems with the same preparation, the expected number occupying the direction space is $N_{\text{samp}} \text{Tr } X_m$; maintenance probability of the original excitation remains S_m .

Within the parameter range of Section 4, take finite inverse temperature β , positive restoration shares g_a , and at least one positive exchange edge for every direction. The complete cycle then has a unique stationary state approached from every initial state. At stationarity, total direction occupation is constant and mean formation and exit counts agree; formation continually replenishes direction occupation after original excitations exit.

Write the stationary state as $\rho_* = X_* + y_*\sigma_g^{\text{sh}}$. Equation (4.72) gives

$$X_* = y_*\mathcal{R}_V(B_g), \quad y_* = [1 + \text{Tr } \mathcal{R}_V(B_g)]^{-1}.$$

Taking the trace of the stationary direction equation gives

$$r_{F,*} = r_{E,*} = f_\Sigma y_*, \quad R_{F,*} = R_{E,*} = \frac{f_\Sigma y_*}{\tau_{\text{tot}}}. \quad (7.26)$$

Stationary direction occupation is directly related to mean maintenance time following formation. For a formed excitation, $\mathcal{R}_V$ sums maintenance shares; for formation components supplied each cycle, it sums stationary occupation. Multiplying $\bar{T}_{\mathcal{M}}$ in Eq. (7.23) by the stationary formation rate gives

$$R_{F,*}\bar{T}_{\mathcal{M}} = y_* \text{Tr } \mathcal{R}_V(B_g) = \text{Tr } X_*.$$

Therefore

$$\boxed{\text{Tr } X_* = R_{F,*}\bar{T}_{\mathcal{M}}}. \quad (7.27)$$

This relation uses mean maintenance time for the actual formation state B_g/f_Σ and does not require equal exit shares for all directions. Substituting the background waiting time further gives

$$\mathrm{Tr} X_* = \frac{\bar{T}_M}{\bar{T}_C + \bar{T}_M}, \quad R_{F,*} = R_{E,*} = \frac{1}{\mathbb{E}(T_{EE})}. \quad (7.28)$$

Stationary direction occupation equals mean maintenance duration divided by mean exit interval. This mean relation does not specify time fractions in every infinite history. Even as original-excitation maintenance probability tends to zero, later formation can sustain nonzero stationary total occupation.

Convergence of the full system state to stationarity means that, for a fixed read-out scheme, the distributions obtained from different initial preparations approach the same limit. This provides estimates of convergence to stationarity for position distributions, shell distributions, and other fixed readings. Use the observation units and actual compressions of Section 5. In particular, Eq. (5.23) gives the compression of a general restoration preparation on the nonzero observation unit associated with the doublet $D_u = \{u, -u\}$ as

$$\mathbf{R}_u[\sigma_g^{\mathrm{sh}}] = \frac{g_{j(u)}}{|S_{j(u)}|} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}. \quad (7.29)$$

For $g = h$, the coefficient is $1/d$; actual weights depend on the chosen preparation. For any complete endpoint readout $\{E_r\}_r$, with $E_r \geq 0$ and $\sum_r E_r = I$, write

$$p_m(r) = \mathrm{Tr}(E_r \rho_m), \quad p_*(r) = \mathrm{Tr}(E_r \rho_*).$$

Set $\Delta_m = \rho_m - \rho_*$ and decompose $\Delta_m = \Delta_m^+ - \Delta_m^-$. Then

$$\sum_r |\mathrm{Tr}(E_r \Delta_m)| \leq \sum_r \mathrm{Tr}[E_r (\Delta_m^+ + \Delta_m^-)] = \|\Delta_m\|_1.$$

Combining with the contraction estimate (4.70) gives

$$\begin{aligned} \frac{1}{2} \sum_r |p_m(r) - p_*(r)| &\leq \frac{1}{2} \|\rho_m - \rho_*\|_1 \\ &\leq \frac{1}{2} (1 - c)^m \|\rho_0 - \rho_*\|_1 \leq (1 - c)^m. \end{aligned} \quad (7.30)$$

Here $c > 0$ provides a uniform convergence bound; the relaxation rates of particular deviation components are determined by the participating eigenvalues of the complete-cycle map, as in Section 6. This result sums prior process records; readings conditional on a specified history use its conditional state.

Thermalization requires stationary readings to match the prescribed energy and temperature. Under the thermal-background preparation condition of Section 4, positions within a shell receive equal weights, and total occupation of each shell depends on both its energy and number of positions. Specifically, before iteration choose

$$g_a = \pi_a = \frac{e^{-\beta E_a}}{Z_C}, \quad Z_C = \sum_a e^{-\beta E_a}.$$

Section 4 proves that the stationary state is then

$$\rho_\beta = \frac{e^{-\beta H_K}}{Z_{\text{full}}}, \quad Z_{\text{full}} = \sum_j |S_j| e^{-\beta E_j}.$$

Since H_K has value E_j on S_j , position and shell probabilities are

$$p_*(x) = \frac{e^{-\beta E_{j(x)}}}{Z_{\text{full}}}, \quad \Pr_*(S_j) = \frac{|S_j| e^{-\beta E_j}}{Z_{\text{full}}}. \quad (7.31)$$

For a nonzero observation unit associated with the doublet D_u , both points lie in one shell, giving stationary compression

$$R_u[\rho_\beta] = \frac{e^{-\beta E_{j(u)}}}{Z_{\text{full}}} I_2. \quad (7.32)$$

Its off-diagonal coherence readings vanish, and its trace gives the total weight. Together with Eq. (7.30), the stated conditions make all initial states approach the same thermal equilibrium readings. A stationary distribution for a general restoration preparation is Gibbs only if the background criterion of Section 4 holds. The shell-dephased preparation $g = h$ is subject to the same criterion; stationarity alone does not ensure thermal equilibrium.

If readout retains only histories that never exited, maintenance shares reweight direction probabilities. Even when the complete cycle has thermal equilibrium, these conditional samples need not have the same thermal distribution. Define the conditional Gibbs state on the direction space by

$$\sigma_{\beta, \mathcal{M}} = \frac{P e^{-\beta H_K} P}{Z_{\mathcal{M}}}, \quad Z_{\mathcal{M}} = \text{Tr}(P e^{-\beta H_K}).$$

At finite β , every direction weight is positive. Shell-preserving V leaves this state invariant, and the non-exit map gives

$$\mathcal{Q}_V(\sigma_{\beta, \mathcal{M}}) = \sum_\nu (1 - \epsilon_\nu) \frac{e^{-\beta E_\nu}}{Z_{\mathcal{M}}} P_\nu. \quad (7.33)$$

The conditional direction weights are obtained by multiplying the Gibbs weights by their maintenance shares and then normalizing. Since all original weights are nonzero, the result equals $\sigma_{\beta, \mathcal{M}}$ if and only if every ϵ_ν is equal. Even then, the non-exit map preserves shells. Whenever the direction space contains several energy shells, a conditional state starting in one shell cannot approach a Gibbs state positive on all these shells. Thermalization readings here therefore use the complete cycle, including exit, background restoration, and formation again.

With the system initially in its stationary state, recording formation, exit, or another specified event over successive cycles estimates its stationary counting rate. Records in adjacent cycles are generally correlated; approach of long-history sample frequencies to stationary probabilities depends on accumulated correlations. Let $\mathcal{J}$ be the instrument action for a fixed record class and $r_* = \text{Tr } \mathcal{J}(\rho_*)$. Let $Z_m \in \{0, 1\}$ indicate its occurrence in cycle m . With the system initially in its stationary state, $\mathbb{E} Z_m = r_*$. Unspecified outcomes between two records are summed by the complete cycle, so for $k \geq 1$,

$$\mathbb{E}(Z_m Z_{m+k}) = \text{Tr}[\mathcal{J} \mathcal{F}_{V,g}^{k-1} \mathcal{J}(\rho_*)]. \quad (7.34)$$

If $r_* > 0$, set

$$\rho_{\mathcal{J}} = \frac{\mathcal{J}(\rho_*)}{r_*}, \quad E_{\mathcal{J}} = \mathcal{J}^*(I), \quad 0 \leq E_{\mathcal{J}} \leq I.$$

Then

$$\text{Cov}(Z_m, Z_{m+k}) = r_* \text{Tr}[E_{\mathcal{J}}(\mathcal{F}_{V,g}^{k-1}(\rho_{\mathcal{J}}) - \rho_*)].$$

For a traceless self-adjoint Δ , positive and negative decomposition gives

$$|\text{Tr}(E_{\mathcal{J}}\Delta)| \leq \frac{1}{2}\|\Delta\|_1.$$

Combining with Eq. (4.70) yields

$$|\text{Cov}(Z_m, Z_{m+k})| \leq r_*(1-c)^{k-1}. \quad (7.35)$$

For $r_* = 0$, the record does not occur at stationary preparation, and the bound still holds. Define the sample frequency over the first M cycles by $\bar{Z}_M = M^{-1} \sum_{m=1}^M Z_m$. Expanding its variance and using the geometric correlation bound gives

$$\begin{aligned} \text{Var}(\bar{Z}_M) &\leq \frac{r_*(1-r_*)}{M} + \frac{2r_*}{M^2} \sum_{k=1}^{M-1} (M-k)(1-c)^{k-1} \\ &\leq \frac{r_*(1-r_*) + 2r_*/c}{M}. \end{aligned}$$

Hence

$$\text{Var}(\bar{Z}_M) \leq \frac{1+2/c}{M} \rightarrow 0. \quad (7.36)$$

For any $\delta > 0$, Chebyshev's inequality gives

$$\Pr(|\bar{Z}_M - r_*| \geq \delta) \leq \frac{1+2/c}{M\delta^2} \rightarrow 0.$$

The sample frequency converges in probability to r_* , and the counting rate to $R_* = r_*/\tau_{\text{tot}}$. Taking $\mathcal{J}$ as the formation or exit instrument gives record-based estimates of both stationary rates in Eq. (7.26). The corresponding formation and exit counts also satisfy Eq. (7.24) on every history with definite block membership.

§8 Conservation and Irreversibility

The system and auxiliary units jointly account for energy and charge balances in formation, exit, and restoration. Post-exchange correlations and the environment preparation supplied each cycle also enter the entropy balance of repeated cycles.

Use $\mathcal{H} = \mathcal{C} \oplus \mathcal{M}$ from Section 4, with projections Π, P and shell energy

$$H_K = \sum_{j=0}^K E_j \Pi_j, \quad E_j = j \log p.$$

Shell-preserving input control V , exchange data, restoration preparation, and actual period are fixed across cycles. Each fresh environment is independent of the system and all existing environments and records jointly. The complete cycle is $\mathcal{F}_{V,g}$, with actual period τ_{tot} . The thermal preparation remains η_T , and the retrieval/restoration auxiliary preparation is P_g . The control auxiliary and finite group reference use the preparations of Section 4 and return to their initial states at the end of control. We first calculate balances before record readout; any label copier added is included in the joint system.

§8.1 Exchange and Conservation

The complete cycle preserves the joint total energy and the joint unit-group charge of the system and auxiliary units. Formation, exit, and restoration redistribute these quantities, subject to the same joint conservation relations. The shares of all conditional outcomes continue to sum to one. Energy-resolving records express the joint relations as branchwise exchange differences; summing all outcomes gives the mean balance for one cycle.

Let $\mathcal{E}$ contain every auxiliary unit actually participating in a cycle, with total energy $H_{\mathcal{E}}$. At the input and output endpoints, total energy is

$$H_{\text{tot}} = H_K \otimes I_{\mathcal{E}} + I_{\mathcal{H}} \otimes H_{\mathcal{E}}.$$

The auxiliaries include the exchange environment, restoration auxiliary, and references required by actual control. Free evolution preserves shell energy, and every joint rotation in formation exchange connects states with equal total energy and joint group charge. By Eq. (4.13), symmetries of the exchange matrix elements pass to the rotations. The finite-reference implementation in Eq. (4.43) also preserves total energy and total unit-group action, so the product of all stages preserves the same total quantities. The complete joint cycle in Eq. (4.46) satisfies

$$[\mathbb{U}_{\text{tot}}, H_{\text{tot}}] = 0, \quad [\mathbb{U}_{\text{tot}}, R_u^{\text{tot}}] = 0 \quad (u \in G), \quad G = (\mathbb{Z}/p^K\mathbb{Z})^\times. \quad (8.1)$$

Here R_u^{tot} is the joint unit-group action on the system, exchange auxiliaries, and group reference. Pattern reconstruction gives the position representation of the joint final state and adds no energy transfer. If the recording operations of Section 6 follow, their detectors and references must also enter the joint system, with the conservation relations verified for those recording implementations.

Let the full auxiliary preparation be $\eta_{\mathcal{E}} = \eta_{\text{full}}$, the joint input $\rho \otimes \eta_{\mathcal{E}}$, and the joint output

$$\Omega' = \mathbb{U}_{\text{tot}}(\rho \otimes \eta_{\mathcal{E}})\mathbb{U}_{\text{tot}}^\dagger.$$

Denote its system and environment marginals by ρ' and $\eta'_{\mathcal{E}}$. Commutation with total energy and cyclicity of the trace give

$$\begin{aligned} \text{Tr}(H_{\text{tot}}\Omega') &= \text{Tr}[H_{\text{tot}}(\rho \otimes \eta_{\mathcal{E}})], \\ \text{Tr}[H_K(\rho' - \rho)] &= -\text{Tr}[H_{\mathcal{E}}(\eta'_{\mathcal{E}} - \eta_{\mathcal{E}})]. \end{aligned} \quad (8.2)$$

The system's mean energy gain equals the combined mean energy loss of all environments. Energy-resolving records refine this relation into individual exchange outcomes.

In this model, the auxiliary preparation commutes with auxiliary energy, so choose a common eigenbasis with

$$\eta_{\mathcal{E}} = \sum_i \pi_i |i\rangle\langle i|, \quad H_{\mathcal{E}}|i\rangle = \mathcal{E}_i|i\rangle.$$

Write an outgoing energy eigenstate as $|o\rangle$, of energy $\mathcal{E}_o$. The actual energy-label protocol of Section 6 gives conditional operators

$$J_{oi} = \sqrt{\pi_i} \langle o | \mathbb{U}_{\text{tot}} | i \rangle.$$

Taking environment matrix elements of joint energy commutation yields

$$H_K J_{oi} - J_{oi} H_K = (\mathcal{E}_i - \mathcal{E}_o) J_{oi}. \quad (8.3)$$

Thus a nonzero matrix element $\langle b | J_{oi} | a \rangle$ between system energy eigenstates requires

$$E_b + \mathcal{E}_o = E_a + \mathcal{E}_i.$$

System and auxiliary energy changes balance term by term. The group reference has zero energy and its coherent preparation lies within one energy eigenspace, leaving the decomposition valid. If an actual record combines different energy differences, its conditional action sums the corresponding components and generally no longer has a unique energy difference.

Joint unitarity gives $\sum_{o,i} J_{oi}^\dagger J_{oi} = I$. Multiplying Eq. (8.3) on the left by $J_{oi}^\dagger$ and summing gives

$$\mathcal{F}_{V,g}^*(H_K) - H_K = \sum_{o,i} (\mathcal{E}_i - \mathcal{E}_o) J_{oi}^\dagger J_{oi}, \quad \mathcal{F}_{V,g}^*(A) = \sum_{o,i} J_{oi}^\dagger A J_{oi}. \quad (8.4)$$

Taking the input-state expectation makes the right side the outcome-share-weighted sum of energy differences. Under the statistical conditions of Section 6, it corresponds to weighted counting of energy records. At the same time,

$$\mathcal{F}_{V,g}^*(I) = I, \quad \sum_r p_r = \sum_r \text{Tr } \mathcal{I}_r(\rho) = \text{Tr } \rho = 1, \quad (8.5)$$

ensures normalization of the total share of all exchange outcomes. This normalization holds together with occupation transfer between directions and background.

For a specified record, energy comparisons must use incident and outgoing states under the same condition. Suppose outcome r resolves energy difference q_r , with conditional operators satisfying

$$[H_K, J_{r\ell}] = q_r J_{r\ell}.$$

Set $E_r = \sum_\ell J_{r\ell}^\dagger J_{r\ell}$ and assume $p_r = \text{Tr}(E_r \rho) > 0$. The commutation relation and its adjoint give $[H_K, E_r] = 0$, and hence

$$\frac{\text{Tr}[H_K \mathcal{I}_r(\rho)]}{p_r} = \frac{\text{Tr}(H_K E_r \rho)}{p_r} + q_r. \quad (8.6)$$

The first term on the right is the incident energy mean weighted by this outcome. It differs from the conditional outgoing mean by q_r and generally differs from the unconditional incident mean.

Unit-group charges obey the corresponding character multiplication relation. For joint input and output states of definite character, conservation of the total unit-group action implies

$$\langle b, o | \mathbb{U}_{\text{tot}} | a, i \rangle \neq 0 \implies \chi_b(u) \chi_o(u) = \chi_a(u) \chi_i(u) \quad \text{for all } u \in G. \quad (8.7)$$

This follows by taking matrix elements of the commutation relation between joint character eigenstates. For charge superpositions, the projections onto joint charge sectors also commute with $\mathbb{U}_{\text{tot}}$, so their total shares are preserved. The group

reference $|1_G\rangle$ used for input control has no definite character. Joint conservation then includes its coherent components, and a conditional system outcome need not have a single character ratio. Energy and group charge balance additively and by character multiplication, respectively.

To apply these relations to one cycle, set

$$\bar{\rho} = V\rho V^\dagger, \quad x_\nu = \langle \nu | \bar{\rho} | \nu \rangle, \quad z_a = \langle e_a | \bar{\rho} | e_a \rangle.$$

Here x_ν and z_a are direction and shell-average occupations. In the interaction picture, the energy-dependent phases from free evolution do not change these occupations. Thermal exchange in Section 4 gives the net exit share

$$J_{a\nu} = \epsilon_{a\nu}x_\nu - f_{\nu a}z_a. \quad (8.8)$$

The first term is transfer from a directional excitation to the background, and the second formation from the background. The post-exchange diagonal entries are therefore

$$\tilde{x}_\nu = x_\nu - \sum_a J_{a\nu}, \quad \tilde{z}_a = z_a + \sum_\nu J_{a\nu}.$$

After thermal exchange, the total background share is $\tilde{y} = \sum_a \tilde{z}_a$. Restoration redistributes it as $\tilde{y}\sigma_g^{\text{sh}}$. Write $\bar{E}_g = \sum_a g_a E_a$. Taking energy gained by the system as positive, the two stages give

$$Q_T = \sum_{a,\nu} (E_a - E_\nu) J_{a\nu}, \quad Q_R = \tilde{y} \bar{E}_g - \sum_a E_a \tilde{z}_a. \quad (8.9)$$

The first identity follows from occupation changes during thermal exchange; the second compares background energies before and after restoration. For a restoration component $e_b \rightarrow e_a$, system energy changes by $E_a - E_b$, while the restoration auxiliary's energy changes from E_K to $E_K + E_b - E_a$, maintaining branchwise balance.

The coherent round-trip auxiliary returns to its input state and has zero net energy change over that stage; shell-preserving input control satisfies

$$\text{Tr}[H_K(V\rho V^\dagger - \rho)] = 0.$$

The one-cycle system energy balance is therefore

$$\boxed{\Delta E_S = \text{Tr}[H_K(\mathcal{F}_{V,g}(\rho) - \rho)] = Q_T + Q_R.} \quad (8.10)$$

This also holds for coherent inputs, since the expectation of H_K depends on these energy-basis diagonal entries.

Summing balances over N consecutive cycles gives

$$\begin{aligned} \sum_{n=0}^{N-1} (Q_{T,n} + Q_{R,n}) &= \text{Tr}[H_K(\rho_N - \rho_0)], \\ \left| \frac{1}{N\tau_{\text{tot}}} \sum_{n=0}^{N-1} (Q_{T,n} + Q_{R,n}) \right| &\leq \frac{K \log p}{N\tau_{\text{tot}}}. \end{aligned} \quad (8.11)$$

The second bound uses $0 \leq H_K \leq K \log p I$, showing that the finite energy spectrum bounds the system's accumulated net energy change. At stationarity, $Q_{T,*} + Q_{R,*} =$

0. The two terms come from thermal exchange and background restoration and need not vanish separately. Thus, even with stationary system energy, energy may continue flowing between environments through the system. Dividing each term by τ_{tot} gives its mean energy-transfer rate.

Joint conservation does not require preservation of every system occupation. The propagation operator $W = \bigoplus_r W_r$ of Section 4 preserves the principal-unit Fourier-mode projections P_r . Within the selected propagation space, the specified coherent outcome $K_{0*}^{(V)} = \sqrt{a_0}W$ satisfies

$$K_{0*}^{(V)\dagger} P_r K_{0*}^{(V)} = a_0 P_r.$$

Its total share is a_0 , so mode occupations are preserved after conditional normalization. Summing every exchange outcome still gives occupation dynamics of the complete cycle.

Now impose the strict-contraction conditions of Section 4: finite inverse temperature β , at least one positive exchange edge for every direction, all $g_a > 0$, and fixed cycle data across iterations. All statements below about strict contraction, a unique stationary state, and inverse system actions use these conditions. By Eq. (4.70), every initial state approaches the same full-rank stationary state ρ_* . If a self-adjoint system quantity A satisfies $\mathcal{F}_{V,g}^*(A) = A$, then, for every initial ρ ,

$$\text{Tr}(A\rho) = \text{Tr}[A\mathcal{F}_{V,g}^n(\rho)] \longrightarrow \text{Tr}(A\rho_*).$$

Choosing the eigenstates of A as inputs shows that all eigenvalues coincide. Therefore

$$\boxed{\mathcal{F}_{V,g}^*(A) = A \iff A = \text{Tr}(A\rho_*)I \quad (A = A^\dagger).} \quad (8.12)$$

The only fixed system projections are 0 and I . Nor does the complete cycle have a nonzero proper invariant support subspace: all iterates from that subspace and their limit would remain supported there, contradicting full rank of ρ_* . These conclusions apply to the cycle summed over all outcomes and differ from conditional preservation on a specified branch. Unchanging occupations at stationarity are properties of a particular input; preservation of single-shell support by the non-exit map also does not imply preservation of relative occupations for every multishell input after normalization.

Energy fixed by exchange conservation must also be related to the cycle phase read out in propagation. First, for the same joint cycle and environment energy, system energy is determined up to an additive constant. If another self-adjoint system quantity $\tilde{H}$ satisfies

$$[\mathbb{U}_{\text{tot}}, \tilde{H} \otimes I + I \otimes H_{\mathcal{E}}] = 0,$$

set $D = \tilde{H} - H_K$ and subtract to obtain $[\mathbb{U}_{\text{tot}}, D \otimes I] = 0$. Taking environment matrix elements gives $[D, J_{oi}] = 0$, hence $\mathcal{F}_{V,g}^*(D) = D$. By Eq. (8.12),

$$\tilde{H} = H_K + \kappa I, \quad \kappa \in \mathbb{R}. \quad (8.13)$$

On the other hand, the full phase of a returning direction is

$$e^{i\varphi_\nu} = e^{-i\tau_{\text{tot}} E_\nu} w_\nu.$$

Using the free-evolution convention e^{-itH} , choose an integer m_ν on each common eigendirection. The corresponding self-adjoint cycle generator has eigenvalues $(-\varphi_\nu + 2\pi m_\nu)/\tau_{\text{tot}}$. There exists a choice of integers for which this generator equals $H_K + \kappa I$ on the direction space if and only if

$$\begin{aligned} e^{i\varphi_\nu} &= e^{-i\tau_{\text{tot}}(E_\nu + \kappa)} \quad \text{for all } \nu, \\ \text{equivalently, } w_\nu &= e^{-i\tau_{\text{tot}}\kappa} \quad \text{independent of direction.} \end{aligned} \tag{8.14}$$

For the directionwise phase readout with $V = I$ in Section 5.4, $v = 1/\tau_{\text{tot}}$ and $\omega_\nu = v\varphi_\nu$. Under Eq. (8.14), their relation to exchange energy is

$$\omega_\nu = -(E_\nu + \kappa) + 2\pi v m_\nu, \quad m_\nu \in \mathbb{Z}.$$

The integer term labels phase branches indistinguishable at cycle endpoints. Non-scalar propagation within a shell likewise cannot be generated solely by $H_K + \kappa I$, which is scalar on that shell. Cycle phase combines contributions from free evolution and energy-preserving interactions; identifying it with a phase generated by exchange energy requires the compatibility condition above.

§8.2 Formation of Correlations

After exchange, distinctions between inputs may reside in the system state, the environment state, or their correlations.

A complete joint action preserves the joint von Neumann entropy. Individual system and environment entropies may change, with their changes related to formation of correlations. Use von Neumann entropy and quantum relative entropy [28],

$$S(\rho) = -\text{Tr}(\rho \log \rho), \quad D(\rho \parallel \sigma) = \text{Tr}[\rho(\log \rho - \log \sigma)]. \tag{8.15}$$

We use the convention $0 \log 0 = 0$ and set the relative entropy to $+\infty$ when $\text{supp } \rho \not\subseteq \text{supp } \sigma$.

Suppose the system is initially independent of all auxiliary units taken jointly, with input $\Omega_{\text{in}} = \rho \otimes \eta_{\mathcal{E}}$. Joint unitary evolution preserves its spectrum, and hence

$$S(\Omega_{\text{out}}) = S(\Omega_{\text{in}}) = S(\rho) + S(\eta_{\mathcal{E}}).$$

Write the output marginals as ρ' and $\eta'_{\mathcal{E}}$. Define their output **mutual information** by

$$\begin{aligned} I(S : \mathcal{E})_{\text{out}} &= S(\rho') + S(\eta'_{\mathcal{E}}) - S(\Omega_{\text{out}}) \\ &= D(\Omega_{\text{out}} \parallel \rho' \otimes \eta'_{\mathcal{E}}) \geq 0. \end{aligned}$$

Substituting joint entropy preservation gives

$$\boxed{\Delta S_S + \Delta S_{\mathcal{E}} = I(S : \mathcal{E})_{\text{out}}}. \tag{8.16}$$

Joint entropy is therefore unchanged, while the increase in the sum of marginal entropies exactly equals the correlations formed. System entropy alone may increase or decrease, as determined jointly by the environment entropy change and their correlations. The initial environment may have internal correlations; the formula treats it as the single system $\mathcal{E}$.

Mutual information measures departure of the joint state from the product of its marginals. Whether process distinctions in correlations constitute records additionally requires the distinction and preservation conditions of Section 6. Nonzero mutual information alone does not establish a persistently readable record.

For thermal exchange, use the entropy-balance decomposition into mutual information and relative entropy [29, 30], with its form fixed by the actual environment preparation. Take the controlled input $\bar{\rho}$ above and thermal environment T prepared in η_T . The joint output is

$$\tilde{\Omega}_{ST} = U_{\text{th}}(\bar{\rho} \otimes \eta_T)U_{\text{th}}^\dagger,$$

with marginals $\tilde{\rho}$ and η'_T . Writing the mutual information of this stage as I_T gives

$$I_T = \Delta S_{\text{th}}^{\text{sys}} + \Delta S_T, \quad \Delta S_T = S(\eta'_T) - S(\eta_T),$$

where $\Delta S_{\text{th}}^{\text{sys}} = S(\tilde{\rho}) - S(\bar{\rho})$.

The actual thermal preparation consists of the direct-sum blocks in Section 4. Let P_e^T project onto each block, including the nonexchanging star block. Retained blocks have weights $a_e > 0$, with $\sum_e a_e = 1$. At finite inverse temperature β ,

$$\eta_T = \bigoplus_e a_e \frac{e^{-\beta H_T|_e}}{Z_e}, \quad Z_e = \text{Tr}_{P_e^T} e^{-\beta H_T}.$$

Thus η_T has full rank on the auxiliary space used, and

$$\log \eta_T = -\beta H_T + \sum_e (\log a_e - \log Z_e) P_e^T. \quad (8.17)$$

All blocks have the same internal temperature, while their weights are fixed by the actual preparation. Thermal exchange preserves each direct-sum block, so

$$\text{Tr}[P_e^T(\eta'_T - \eta_T)] = 0.$$

The constant block terms therefore cancel in expansion of relative entropy. Using the energy balance $\text{Tr}[H_T(\eta'_T - \eta_T)] = -Q_T$ gives

$$\begin{aligned} D(\eta'_T \| \eta_T) &= -\Delta S_T - \text{Tr}[(\eta'_T - \eta_T) \log \eta_T] \\ &= -\Delta S_T - \beta Q_T. \end{aligned}$$

Combining with the mutual-information relation yields

$$\boxed{\Delta S_{\text{th}}^{\text{sys}} - \beta Q_T = I_T + D(\eta'_T \| \eta_T) \geq 0.} \quad (8.18)$$

The two terms describe system–thermal–environment correlations formed in exchange and the change of the final thermal state relative to its preparation. The identity uses the actual block preparation and preservation of its weights; it requires no global Gibbs assumption for the entire thermal environment. If incident energy labels are retained, this formula still concerns the physical thermal–environment marginal. Label copies correlated with incident energy levels must enter the extended joint system separately, rather than being inserted directly into this full-rank comparison state.

Background restoration uses the pure auxiliary preparation $P_g = |g\rangle\langle g|$. After the coherent round trip, this auxiliary returns to g and the system acquires the unitary phase U_w of Section 4. Even if the system is then correlated with the thermal environment, its marginal input with the restoration auxiliary remains

$$(U_w \tilde{\rho} U_w^\dagger) \otimes P_g.$$

Here U_w is the identity on the background subspace and applies unitary phases on the direction subspace, preserving system entropy and both subspace occupations. Write

$$\tilde{\rho}_C = \Pi \tilde{\rho} \Pi, \quad \tilde{\rho}_M = P \tilde{\rho} P, \quad y = \text{Tr } \tilde{\rho}_C.$$

For $y > 0$, define the conditional background state $\xi_C = \tilde{\rho}_C / y$. Restoration in Eq. (4.21) satisfies

$$|e_b, g\rangle \mapsto \sum_a \sqrt{g_a} |e_a, r_{ab}\rangle,$$

and preserves the product of direction components with auxiliary state g . Tracing the post-restoration joint state over the system removes cross terms by orthogonality of direction and background subspaces. Within a fixed- a auxiliary subspace, identify matrices through $b \mapsto r_{ab}$ to obtain

$$\eta'_R = (1 - y)P_g \oplus \bigoplus_a g_a \tilde{\rho}_C. \quad (8.19)$$

Other unoccupied auxiliary subspaces are filled with zeros. Thus the matrix structure of the background input is retained in the corresponding auxiliary blocks.

Set $H(g) = -\sum_a g_a \log g_a$ and $h_2(y) = -y \log y - (1 - y) \log(1 - y)$. If ξ_C has eigenvalues λ_b , the nonzero auxiliary spectrum consists of $1 - y$ and $y g_a \lambda_b$. Hence

$$S(\eta'_R) = h_2(y) + y[H(g) + S(\xi_C)]. \quad (8.20)$$

The system entropy change also contains a contribution from background–direction coherence. Define

$$C_{\text{blk}} = S(\tilde{\rho}_M + \tilde{\rho}_C) - S(\tilde{\rho}).$$

This block decomposition satisfies

$$\tilde{\rho}_M + \tilde{\rho}_C = \frac{1}{2}[\tilde{\rho} + (P - \Pi)\tilde{\rho}(P - \Pi)].$$

Since $P - \Pi$ is unitary, concavity and unitary invariance of entropy give $C_{\text{blk}} \geq 0$. The restored system has two blocks: a unitary conjugate of $\tilde{\rho}_M$ and background block $y\sigma_g^{\text{sh}}$. Relative to the block-dephased input, the direction-block spectrum and the two total shares are unchanged, while the conditional background state changes from ξ_C to σ_g^{sh} . Expanding direct-sum entropies and subtracting gives

$$\Delta S_R^{\text{sys}} = y[H(g) - S(\xi_C)] + C_{\text{blk}}. \quad (8.21)$$

The first term comes from the conditional background change; the second from removing background–direction coherence in the system marginal. For a general coherent input, the second term must be retained.

The restoration auxiliary initially has zero entropy and is marginally independent of the system. Joint entropy preservation during this stage gives $I_R = \Delta S_R^{\text{sys}} + S(\eta'_R)$. Substituting the preceding formulas cancels the conditional background entropy, yielding

$$I_R = h_2(y) + 2yH(g) + C_{\text{blk}}. \quad (8.22)$$

At $y = 0$ or $y = 1$, zero-weight entropy terms are treated by continuity. Restoration both redistributes background occupation and retains input distinctions in the auxiliary marginal and joint correlations.

The mutual informations I_T at thermal-exchange end and I_R at restoration end refer to different stages. Their sum must be compared with correlations in the same final joint state. Later restoration does not alter the thermal marginal, and the round-trip unitary phase does not alter system entropy. Hence

$$I_T + I_R = S(\rho_{\text{fin}}) - S(\bar{\rho}) + S(\eta'_T) - S(\eta_T) + S(\eta'_R).$$

The initial joint state of system, restoration auxiliary, and thermal environment is $\bar{\rho} \otimes P_g \otimes \eta_T$, whose entropy is preserved by the complete joint action. Thus

$$S(\Omega_{SRT, \text{fin}}) = S(\bar{\rho}) + S(\eta_T).$$

Combining these identities and expanding final mutual information gives

$$I_T + I_R = I(S : RT)_{\text{fin}} + I(T : R)_{\text{fin}}. \quad (8.23)$$

The second term on the right is the final correlation between the two auxiliary environments. Thus the sum of correlations formed stage by stage generally differs from final mutual information between the system and the entire environment.

These calculations concern the actual joint process before a label copier is added. After record copying, the copier also belongs to the joint system, and entropies and correlations must be computed from the same extended final state.

§8.3 Environment Preparation and Irreversibility

After a cycle, auxiliary units participating in exchange have generally changed and become correlated with the system or other auxiliaries. Repeating the same cycle requires fresh auxiliary units in the prescribed states, independent of the system and all existing environments jointly. Earlier correlations remain in the joint state containing old environments; the marginal input of the system and fresh environment again has product form.

The restoration auxiliary has actual input P_g and output η'_R . To compare them, fix a full-rank state

$$\omega_R = \frac{e^{-\beta_R H_R}}{Z_R}, \quad Z_R = \text{Tr } e^{-\beta_R H_R},$$

with finite β_R . Define

$$C_R = D(P_g \| \omega_R) - D(\eta'_R \| \omega_R). \quad (8.24)$$

Here ω_R is a comparison state; the actual restoration preparation remains P_g . The quantity C_R is the difference of relative entropies to the same comparison state before and after the action; no sign is assumed for general inputs. Using

$$D(\eta \| \omega_R) = -S(\eta) + \beta_R \text{Tr}(H_R \eta) + \log Z_R,$$

and $S(P_g) = 0$ gives

$$C_R = S(\eta'_R) - \beta_R \text{Tr}[H_R(\eta'_R - P_g)].$$

Energy balance above gives $\text{Tr}[H_R(\eta'_R - P_g)] = -Q_R$, so

$$C_R = S(\eta'_R) + \beta_R Q_R. \quad (8.25)$$

This expresses the change in restoration preparation through auxiliary entropy and exchanged energy. Since an actual reparation operation has not been specified, C_R cannot be directly identified with its work cost.

Control and coherent round trips act unitarily on the system at their endpoints and do not change system entropy. The full-cycle system entropy change is therefore

$$\Delta S_S = \Delta S_{\text{th}}^{\text{sys}} + \Delta S_R^{\text{sys}}.$$

Equations (8.18) and $\Delta S_R^{\text{sys}} = I_R - S(\eta'_R)$, combined with Eq. (8.25) give

$$\boxed{\Delta S_S - \beta Q_T - \beta_R Q_R + C_R = I_T + D(\eta'_T \| \eta_T) + I_R \geq 0.} \quad (8.26)$$

All terms arise from the same actual joint cycle. The left side relates system entropy change, two energy exchanges, and the change in restoration preparation; the right side contains mutual information from the two stages and relative entropy of the final thermal environment to its original preparation.

For the comparison $\beta_R = \beta$, a stationary system has $\Delta S_S = 0$ and $Q_T + Q_R = 0$. Then

$$C_R = I_T + D(\eta'_T \| \eta_T) + I_R \geq 0. \quad (8.27)$$

Stationary system readings do not require the auxiliary preparations to remain unchanged. Every cycle may still create correlations and change the departing auxiliary units.

Under the same-temperature Gibbs criterion, this distinction can be computed explicitly. Take

$$g_a = \pi_a = \frac{e^{-\beta E_a}}{Z_C}, \quad Z_C = \sum_a e^{-\beta E_a}, \quad \rho_\beta = \frac{e^{-\beta H_K}}{Z_{\text{full}}}.$$

Within every thermal-exchange block, states coupled by the joint rotation have equal total energy and equal weights in $\rho_\beta \otimes \eta_T$. The rotation therefore preserves the joint thermal-exchange input, giving $I_T = D(\eta'_T \| \eta_T) = Q_T = 0$. The system background block is $y_* \sigma_g^{\text{sh}}$, with $y_* = Z_C / Z_{\text{full}}$. Restoration preserves the system state, so $Q_R = 0$ and $C_{\text{blk}} = 0$. Nevertheless, Eq. (8.22) gives

$$C_R = I_R = h_2(y_*) + 2y_* H(g) > 0. \quad (8.28)$$

For an odd prime p and $K \geq 1$, the direction-space dimension is $d - K - 1 = p^K - K - 1 > 0$. All Gibbs weights are positive at finite β , so $0 < Z_C < Z_{\text{full}}$ and $0 < y_* < 1$. Hence $h_2(y_*) > 0$, establishing the strict inequality above. Even in system thermal equilibrium, the restoration auxiliary changes from its prescribed pure preparation to a correlated final state.

Actual record storage requires the label copier to enter the joint balance as well. Let B copy the prescribed input and output labels of Section 6, with a pure initial state and zero energy. On orthogonal labels that can be copied, the action is

$$|r, 0_B\rangle \mapsto |r, r_B\rangle.$$

The copied objects are specified orthogonal labels, including retained thermal labels and restoration labels g, r_{ab} . The isometry preserves the full joint spectrum; tracing out the copier removes cross terms between the corresponding labels. Copying these energy-level and restoration labels preserves auxiliary energy and thermal-block projections.

Start from the independent input before copying and denote the enlarged joint final state by Ω'_{SRTB} , with marginals ρ'_S , η'_R , η'_T , and η'_B . All entropies, energy exchanges, and C_R below are recalculated from the actual final state of this extended implementation. Define the total correlation of the four parts by

$$I_{\text{tot}} = S(\rho'_S) + S(\eta'_T) + S(\eta'_R) + S(\eta'_B) - S(\Omega'_{SRTB}). \quad (8.29)$$

Equivalently, it is the relative entropy of the joint state to the product of its marginals,

$$I_{\text{tot}} = D(\Omega'_{SRTB} \| \rho'_S \otimes \eta'_R \otimes \eta'_T \otimes \eta'_B) \geq 0.$$

The restoration auxiliary and label copier both start pure. Joint entropy preservation therefore gives

$$I_{\text{tot}} = \Delta S_S + \Delta S_T + S(\eta'_R) + S(\eta'_B).$$

Label copying preserves thermal-block weights and auxiliary energy, so $D(\eta'_T \| \eta_T) = -\Delta S_T - \beta Q_T$. Again taking $\beta_R = \beta$ and substituting Eq. (8.25) yields

$$\Delta S_S - \beta(Q_T + Q_R) + C_R + S(\eta'_B) = I_{\text{tot}} + D(\eta'_T \| \eta_T). \quad (8.30)$$

The copier's entropy and its correlations thereby enter the same joint balance. Environment entropy formulas obtained before copying cannot be applied to post-copying marginals without verification. Control auxiliaries and group references returning to their original pure states contribute no final correlations and may be omitted here; other detectors and references actually used in readout must be included according to their final states.

Successive system states are obtained by iterating the same reduced map:

$$\rho_{n+1} = \mathcal{F}_{V,g}(\rho_n).$$

This iteration uses fixed control, exchange data, and restoration preparation, with each fresh environment independent of the system and all previous environments. Under the strict-contraction conditions in Section 8.1, Section 4 proves a unique full-rank stationary state ρ_* and convergence $\rho_n \rightarrow \rho_*$.

Departure from stationarity can be measured by relative entropy. The data-processing inequality for relative entropy under completely positive trace-preserving maps [31, 32], together with $\mathcal{F}_{V,g}(\rho_*) = \rho_*$, gives

$$\begin{aligned} D(\mathcal{F}_{V,g}(\rho) \| \rho_*) &= D(\mathcal{F}_{V,g}(\rho) \| \mathcal{F}_{V,g}(\rho_*)) \\ &\leq D(\rho \| \rho_*). \end{aligned} \quad (8.31)$$

Thus $D(\rho_{n+1}||\rho_*) \leq D(\rho_n||\rho_*)$. Full rank makes $\log \rho_*$ bounded. Using the proved state convergence and continuity of entropy in finite dimension [33],

$$\begin{aligned} D(\rho_n||\rho_*) &= -S(\rho_n) - \text{Tr}(\rho_n \log \rho_*) \\ &\longrightarrow -S(\rho_*) - \text{Tr}(\rho_* \log \rho_*) = 0. \end{aligned}$$

Relative entropy to ρ_* is nonincreasing cycle by cycle; system von Neumann entropy need not increase monotonically. Whether the stationary state is Gibbs at the same temperature remains determined by the restoration-share criterion of Section 4.

Strict contraction also limits inverse processes implementable on the system alone. With contraction constant $c > 0$ from Section 4, for all system states ρ, σ ,

$$\|\mathcal{F}_{V,g}(\rho) - \mathcal{F}_{V,g}(\sigma)\|_1 \leq (1 - c)\|\rho - \sigma\|_1.$$

Suppose a completely positive trace-preserving map $\mathcal{R}$ acting only on the system restores every initial state, so $\mathcal{R} \circ \mathcal{F}_{V,g} = \text{id}$. Trace-distance contraction under $\mathcal{R}$ [34] gives

$$\begin{aligned} \|\rho - \sigma\|_1 &= \|\mathcal{R}(\mathcal{F}_{V,g}(\rho)) - \mathcal{R}(\mathcal{F}_{V,g}(\sigma))\|_1 \\ &\leq \|\mathcal{F}_{V,g}(\rho) - \mathcal{F}_{V,g}(\sigma)\|_1 \\ &\leq (1 - c)\|\rho - \sigma\|_1. \end{aligned} \tag{8.32}$$

Taking $\rho \neq \sigma$ yields a contradiction. Under these conditions, no physical inverse acting only on the system can restore every initial state.

The complete cycle has a joint inverse because it is implemented by a joint system–auxiliary unitary. Restoring the input requires retaining and using the corresponding auxiliary outputs and their correlations with the system. On the system alone, restoration maps different background inputs of the same share to the same prescribed background; its input cannot be determined from its system output alone. For the cycle summed over all outcomes, under this section’s strict-contraction conditions, Equation (8.32) likewise excludes a system-only physical inverse recovering every initial state.

Successive cycles use independent fresh environment preparations; earlier auxiliary outputs and correlations no longer participate in the next action. The system can therefore relax cycle by cycle while the full joint process remains reversible. Figure 6 relates joint conservation to reduced irreversibility.

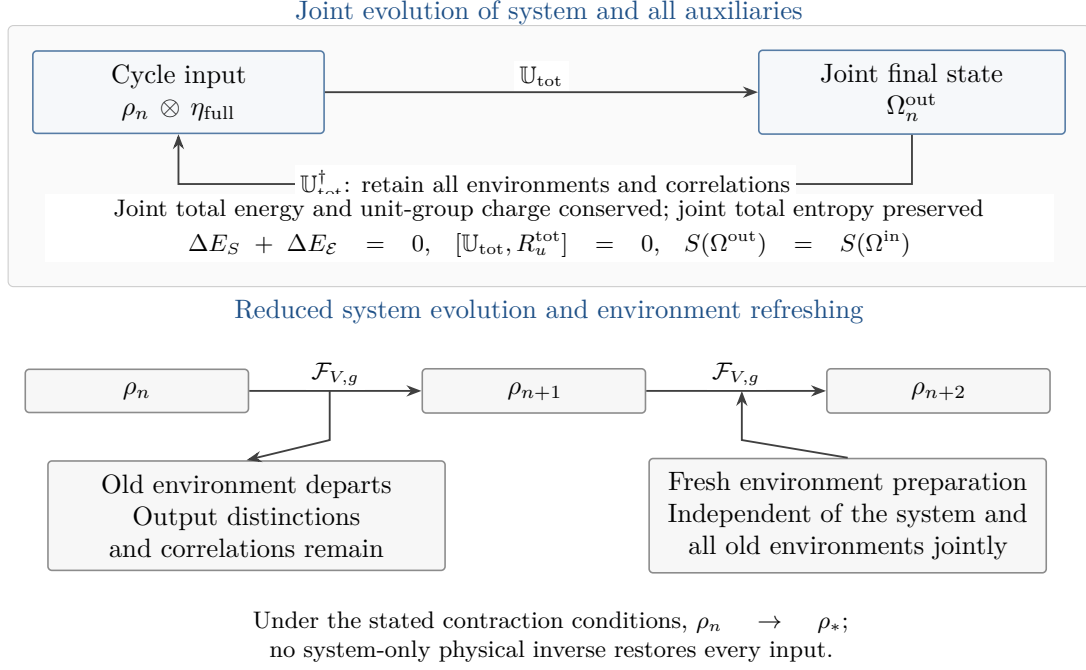


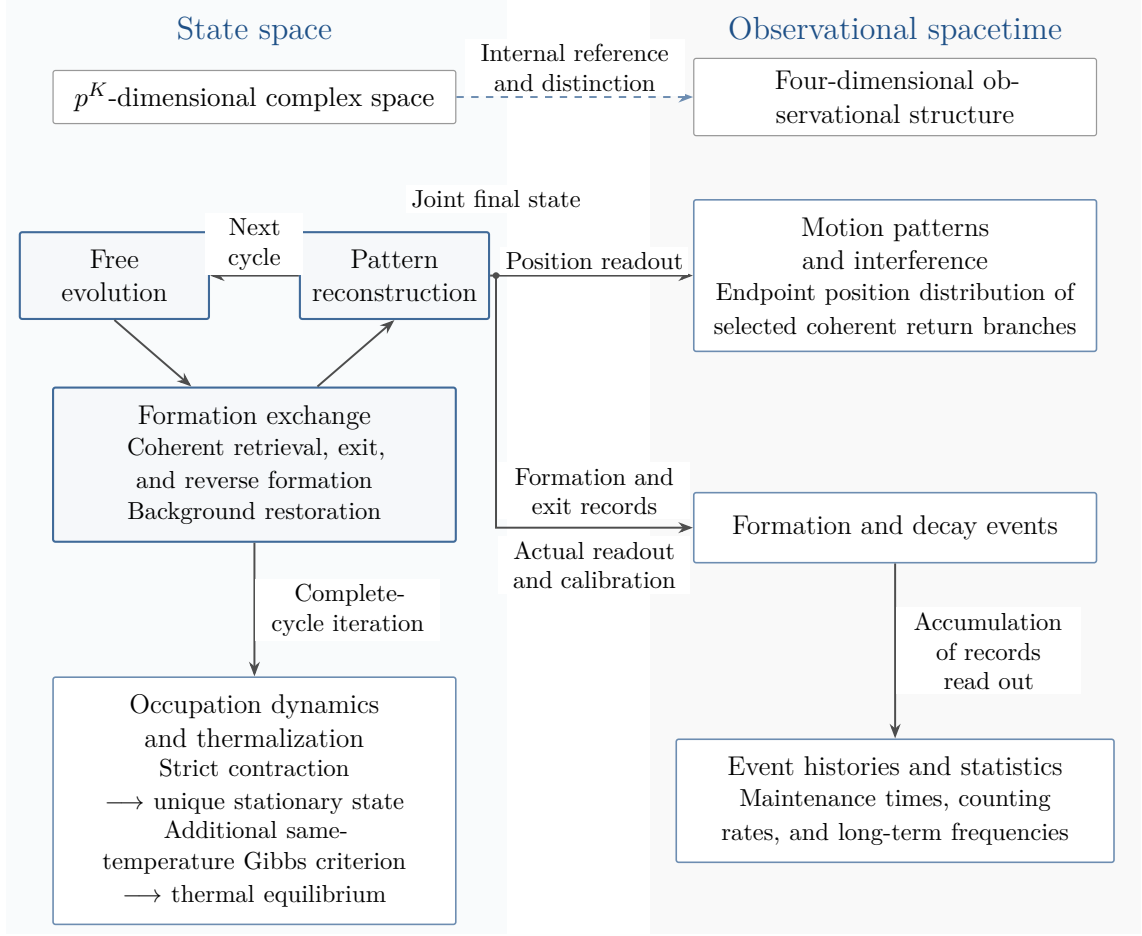
Figure 6: Joint conservation and reduced irreversibility. Above, all participants in a cycle are retained, including any label copier added. The joint unitary evolution constructed here preserves total energy, unit-group charge, and total entropy; reversing it requires the corresponding complete joint final state. Here $\Delta E_{\mathcal{E}} = \text{Tr}[H_{\mathcal{E}}(\eta'_{\mathcal{E}} - \eta_{\mathcal{E}})]$ is the mean energy change of all auxiliaries, consistent with Eq. (8.2). Below, successive cycles use independent fresh environment preparations. Under the stated contraction conditions, the system approaches stationarity cycle by cycle and no system-only action recovers every input. Preparation and record entropy balances are given in Eqs. (8.26) and (8.30).

§9 Conclusion

On the state space obtained by finite p -adic reduction of the BC system, we unify formation, motion, decay, and thermalization in a complete cycle of exchange between stable states and the background. Formation, motion, and decay of excitations, together with relaxation of the ensemble occupation distribution, are determined by the same joint evolution and its iteration (Fig. 7).

The structure of observational spacetime is determined by references and the observable algebra internal to the state space. Records left by the cycle appear as events through internal readout and calibration, relating the evolution of the dynamical subject to its observed outcomes. Dynamical processes and their observational manifestations are thereby described within the same construction.

These dynamics are based on the structure of the state space itself. Subject and background, evolution and exchange, and records and observation are described within one joint construction of system and auxiliary spaces, without an additional external observer or an observation space independent of the state space. The evolution of the dynamical subject and its observable manifestations thus share a common internal basis, allowing its physical properties to be studied within a self-consistent dynamical framework. Future work will develop the quantitative evaluation of dynamical subjects, focusing on mass and interaction forces.



Cycle counts and counts of selected event records are distinct

Figure 7: Cycles, records, and their observational manifestations. The left side includes the joint state space of the system and auxiliary environments. Records labeled on arrows are supported by these environments or auxiliary units and satisfy the stated distinction and preservation conditions. Cross-column arrows relate records to events by internal readout and calibration; decay is identified by the record of the original excitation's first exit. The dashed arrow indicates the four-dimensional observational structure determined by internal reference and distinction. Motion patterns use independent repeated preparations of selected coherent return branches, with endpoint position readout at different stopping times. Several cycles may separate selected records. Iteration uses fixed independent environment preparations; stationarity and thermal equilibrium use the stated strict-contraction conditions and same-temperature Gibbs criterion. Event probabilities and counts use the readout conditions of Section 6; long-term frequencies additionally use stationary preparation and contraction from Section 7.

§A Estimates and Compatibility of Position Propagation

Section 4 establishes coherent propagation and its criterion for a nearest-neighbor recurrence; Section 7 gives the corresponding position readout. This appendix examines the spectral range required for position preparation, displacement estimates for general spectral phases, and propagation/readout relations across truncation depths.

Use the shell and orientation space of Eq. (4.32), with $p \geq 5$, $0 \leq j < K$, and $2a \not\equiv 0 \pmod{p-1}$, and write $n = L = p^{K-j-1}$. Calculations use the abstract space $\mathbb{C}^2 \otimes \mathbb{C}^n$; actual states enter $\mathcal{H}_a$ through the isometric embedding in that equation. The propagation operator remains $W_r = C_r e^{-i\kappa_r \sigma_3}$, with $\kappa_r = 2\pi r/n$ and $z_{\pm,r} = e^{i\varphi_{\pm,r}}$. Below t labels principal-unit positions and m counts cycles, with elapsed cycle time $m\tau_{\text{tot}}$. Propagation statistics are conditional on the actual star-return history in Eq. (7.4), with position read at the selected endpoint; the complete cycle still iterates Eq. (4.53).

§A.1 Position Preparation and Propagation Error

Spectral-mode selection also limits localization of the initial pattern. Let a nonempty set $\mathcal{A} \subseteq \mathbb{Z}_n$ label selected full two-dimensional spectral blocks, $P_{\mathcal{A}} = I_2 \otimes \sum_{r \in \mathcal{A}} P_r$. For a normalized pure state supported on the selected blocks, the Fourier coefficients satisfy $\sum_{r \in \mathcal{A}} \|\tilde{\psi}(r)\|^2 = 1$. From $\psi(t) = n^{-1/2} \sum_{r \in \mathcal{A}} e^{i\kappa_r t} \tilde{\psi}(r)$, vector-valued Cauchy–Schwarz gives $P(t) = \|\psi(t)\|^2 \leq |\mathcal{A}|/n$; convex decomposition gives the same bound for mixed states. Preparing $P(t_0) \geq 1 - \eta$ therefore requires $|\mathcal{A}| \geq n(1 - \eta)$, and point preparation requires every position mode. For $n > 1$, retaining only $r = 0$ gives the uniform distribution $P(t) = 1/n$; narrow-band localized preparations must retain their position tails. For $n = 1$, there is only one position and uniform preparation is point preparation.

Subspaces preserved by local preparation and propagation can also be classified directly. Set $Q_t = I_2 \otimes P_t$. For $n \geq 3$, subspaces $\mathcal{L}$ satisfying both $Q_t \mathcal{L} \subseteq \mathcal{L}$ for all t and $T\mathcal{L} \subseteq \mathcal{L}$ have only four forms:

$$0, \quad |+\rangle \otimes \mathbb{C}^n, \quad |-\rangle \otimes \mathbb{C}^n, \quad \mathcal{H}_a. \quad (\text{A.1})$$

To prove this, extract a nonzero position component with Q_t . Since $n \geq 3$, $t+1 \neq t-1$. If that component has a nonzero $+$ orientation, $Q_{t+1}TQ_t$ extracts $|+, t+1\rangle$ alone, and repeated shifts generate every $+$ position state. The same holds for $-$. If both orientations occur, the subspace contains all of $\mathcal{H}_a$, proving Eq. (A.1). Thus, retaining all local preparations and both propagation orientations prevents turning a nonconstant spectrum into a constant one by deleting position modes. Here Q_t is a mathematical projection within the propagation space. Even after coarse graining, the actual fine-point instrument in Eq. (6.21) may produce states outside $\mathcal{H}_a$; the classification therefore does not assert subspace preservation under actual point measurement every cycle.

Approximation error for nearest-neighbor propagation When the original spectrum varies with the position-mode index, the resulting propagation can be

compared with the nearest-neighbor propagation of Section 4. Set $C_* = C_0$ and define

$$\varepsilon_{\mathcal{A}} = \max_{r \in \mathcal{A}, \sigma = \pm} |z_{\sigma, r} - z_{\sigma, 0}|, \quad W_* = C_* T. \quad (\text{A.2})$$

Use the same normalized initial state $\rho_0 = P_{\mathcal{A}} \rho_0 P_{\mathcal{A}}$ for both propagators. Both W and W_* preserve the selected band, and the same B diagonalizes every C_r , so $\|(W - W_*)P_{\mathcal{A}}\| = \varepsilon_{\mathcal{A}}$. Using

$$W^m - W_*^m = \sum_{b=0}^{m-1} W^{m-1-b} (W - W_*) W_*^b$$

and unitarity of every factor gives

$$\|(W^m - W_*^m)P_{\mathcal{A}}\| \leq \min(2, m\varepsilon_{\mathcal{A}}), \quad \text{TV}(P_m, P_m^*) \leq \min(1, m\varepsilon_{\mathcal{A}}). \quad (\text{A.3})$$

Here P_m, P_m^* denote the principal-unit position distributions after W^m, W_*^m . Their total variation distance is $\text{TV}(P, Q) = \frac{1}{2} \sum_t |P(t) - Q(t)|$. For the second bound, apply the first to a purification of the common initial state and then use trace-distance contraction under partial trace and endpoint measurement.

With representatives $0, \dots, n-1$, define the cycle-graph distance $d_n(t, u) = \min(|t - u|, n - |t - u|)$. For a point initial state, use the full-mode error $\varepsilon = \varepsilon_{\mathbb{Z}_n}$. The reference propagation W_*^m vanishes beyond graph distance m . Projecting onto that region and applying Eq. (A.3) gives

$$P_m \{d_n(t, t_0) > m\} \leq \min(1, m^2 \varepsilon^2). \quad (\text{A.4})$$

Narrow-band preparations generally have nonzero position tails. For $X \subseteq \mathbb{Z}_n$, write $P_X = \sum_{t \in X} Q_t$ and $P_{X^c} = I - P_X$. If a normalized ψ is restricted to $\mathcal{A}$ and initially has probability at most η outside X , let $X^{(m)}$ be its m -step neighborhood and Q_{out} the projection outside that neighborhood. Equation (A.3) and the reference support restriction give

$$\begin{aligned} \|Q_{\text{out}} W^m \psi\| &\leq \|(W^m - W_*^m)P_{\mathcal{A}} \psi\| + \|Q_{\text{out}} W_*^m P_{X^c} \psi\| \\ &\leq m\varepsilon_{\mathcal{A}} + \sqrt{\eta}. \end{aligned}$$

Position decomposition is used only for the reference propagation, so $P_X \psi$ need not remain in the same band. Squaring gives

$$P_m \{t \notin X^{(m)}\} \leq \min(1, (\sqrt{\eta} + m\varepsilon_{\mathcal{A}})^2). \quad (\text{A.5})$$

Point and narrow-band preparations use Eqs. (A.4) and (A.5), respectively; the full-spectrum error in the former cannot simply be replaced by a narrow-band error. The utility of the bound depends on the actual $\varepsilon_{\mathcal{A}}$ and initial tail. The compared label shares are both a_0^m , so unnormalized joint label–position probabilities have the bound multiplied by a_0^m . These results concern only the selected coherent history; they do not establish a strict finite propagation range for the complete channel with formation and restoration.

§A.2 Circular-Displacement Estimates

Principal-unit positions are arranged modulo n , so endpoint position probabilities determine displacement statistics on a circle. Take $n > 1$, hence $n \geq 5$; for $n = 1$ the position is constant, and expressions with denominator $\sin(\pi/n)$ are not used. Define the unitary phase operator on principal-unit positions by

$$M_\ell|t\rangle = e^{i\kappa_\ell t}|t\rangle, \quad M_\ell|r\rangle = |r + \ell\rangle. \quad (\text{A.6})$$

The real and imaginary parts of its expectation are obtained by averaging $\cos(\kappa_\ell t)$ and $\sin(\kappa_\ell t)$, respectively, over the recorded positions. Write $\mathcal{J}_\ell = W^\dagger M_\ell W - M_\ell$. Modewise calculation in the Fourier basis gives

$$\begin{aligned} W^\dagger M_\ell W &= \sum_r (W_{r+\ell}^\dagger W_r) \otimes |r + \ell\rangle\langle r|, \\ \|\mathcal{J}_\ell\| &= \max_r \|W_{r+\ell}^\dagger W_r - I\| = \max_r \|W_{r+\ell} - W_r\| =: \eta_\ell. \end{aligned} \quad (\text{A.7})$$

Since $\langle M_\ell \rangle_{b+1} - \langle M_\ell \rangle_b = \text{Tr}(\rho_b \mathcal{J}_\ell)$, the change per cycle satisfies $|\langle M_\ell \rangle_{b+1} - \langle M_\ell \rangle_b| \leq \eta_\ell$. For initial position t_0 and orientation density matrix ρ_{or} , substituting the finite Fourier expansion of W^m into position probabilities and summing positions gives the displacement characteristic function

$$\mathbb{E}_m(e^{-i\kappa_\ell(t-t_0)}) = \frac{1}{n} \sum_r \text{Tr} [\rho_{\text{or}} (W_r^\dagger)^m W_{r+\ell}^m]. \quad (\text{A.8})$$

The negative-sign reading here corresponds to $M_{-\ell}$, so matrix order differs from the positive-sign reading in Eq. (A.7).

Displacement bounds Take $\ell = 1$ and write $\eta_1 = \eta_{\text{pos}}$. A point initial vector satisfies $M_1\psi = e^{i\kappa_1 t_0}\psi$. Expanding the commutator term by term gives

$$\|(M_1 - e^{i\kappa_1 t_0})W^m\psi\| = \|[M_1, W^m]\psi\| \leq m\eta_{\text{pos}}. \quad (\text{A.9})$$

The squared left side is the mean squared chord length on the unit circle, so

$$\mathbb{E}_m \left[4 \sin^2 \frac{\pi(t-t_0)}{n} \right] \leq \min(4, m^2 \eta_{\text{pos}}^2). \quad (\text{A.10})$$

Normalize by the chord length between adjacent positions and set $d_{\text{circ}}(t, t_0) = |\sin(\pi(t-t_0)/n)| / \sin(\pi/n)$. Then

$$\sqrt{\mathbb{E}_m(d_{\text{circ}}^2)} \leq \min \left\{ \frac{\sin(\pi \lfloor n/2 \rfloor / n)}{\sin(\pi/n)}, \frac{m\eta_{\text{pos}}}{2 \sin(\pi/n)} \right\}. \quad (\text{A.11})$$

Use the cycle-graph distance d_n above. Where $d_n(t, t_0) \geq R$, the squared chord length is at least $4 \sin^2(\pi R/n)$; for $1 \leq R \leq \lfloor n/2 \rfloor$,

$$P_m\{d_n(t, t_0) \geq R\} \leq \min \left(1, \frac{m^2 \eta_{\text{pos}}^2}{4 \sin^2(\pi R/n)} \right). \quad (\text{A.12})$$

The one-cycle spectral difference can be bounded further by the original phases:

$$\eta_{\text{pos}} \leq \min \left\{ 2, 2 \sin(\pi/n) + \max_{\sigma, r} |z_{\sigma, r+1} - z_{\sigma, r}| \right\}. \quad (\text{A.13})$$

Set $T_r = e^{-i\kappa_r\sigma_3}$, add and subtract $C_r T_{r+1}$ in $W_{r+1} - W_r$, and use $\|T_{r+1} - T_r\| = 2\sin(\pi/n)$ and common diagonalization of C_r to obtain this bound. For constant C_r , $\eta_{\text{pos}} = 2\sin(\pi/n)$, and Eq. (A.11) bounds root-mean-square circular displacement by m . Differences between modes of the original spectrum remain in Eq. (A.13); for large η_{pos} , the bound reaches the circle's diameter rapidly. These estimates concern circular readings under conditional propagation. Physical speeds additionally require the actual period and spacetime-coordinate calibration; the bound alone gives no parameter-independent speed limit.

§A.3 Propagation and Position Readout across Depths

Comparison across truncation depths depends on the embedding used. Increasing depth and shell index together preserves the effective unit group; increasing depth alone increases the number of principal-unit positions. These compare shell scaling and increased position resolution, respectively.

Use the position embedding of Section 7, abbreviated as E . Take fixed integer representatives $x_{s,t} = p^j[\zeta^s(1+p)^t]_{p^{K-j}}$, a nonzero same-shell reference u_* , and $Y = \sum_{s,t} \log(x_{s,t}/u_*) P_{x_{s,t}}$, zero outside the shell. Position intervals and moments follow from Eqs. (7.7) and (7.8). Write the abstract normalized conditional state as $\rho_m = W^m \rho_0 (W^\dagger)^m$, with actual initial state $E \rho_0 E^\dagger$.

Shell scaling Define a point isometry between successive finite spaces by

$$\mathcal{I}_K |x \bmod p^K\rangle = |px \bmod p^{K+1}\rangle. \quad (\text{A.14})$$

It sends shell j to $j+1$, preserving the effective modulus p^{K-j} and $n = p^{K-j-1}$. The representative formula gives $x'_{s,t} = px_{s,t}$ pointwise. Sending the reference to $u'_* = pu_*$ as well gives

$$\log \frac{x'_{s,t}}{u'_*} = \log \frac{x_{s,t}}{u_*}, \quad Y' \mathcal{I}_K E = \mathcal{I}_K Y E. \quad (\text{A.15})$$

For absolute coordinates $\log x$, an additional $\log p$ times the identity appears. Bare shell energy also satisfies $H_{K+1} \mathcal{I}_K = \mathcal{I}_K (H_K + \log p I)$, including the finite zero element.

The two shells have the same effective modulus p^{K-j} . Their principal-unit phases, full characters, primitive conductors, and resonance ratios coincide componentwise. With matching orientations and zero-response extensions, Eq. (4.36) gives $W_{K+1,j+1} = W_{K,j}$ on the two abstract propagation spaces. Set $E' = \mathcal{I}_K E$ and $\widehat{W}' = E' W_{K+1,j+1} E'^\dagger$ on its range. Then

$$\widehat{W}' \mathcal{I}_K E = \mathcal{I}_K E W_{K,j}. \quad (\text{A.16})$$

Induction gives the same intertwining for any m cycles. With Eq. (A.15), all end-point relative-interval probabilities, characteristic functions, and moments coincide. Shell scaling thus preserves both the selected conditional propagation and relative-interval readout. Joint label-position probabilities also include the respective factors $a_{0,K}^m$; their unnormalized distributions coincide only when these label shares agree. Conversion to time likewise uses the actual period at each depth.

This relation does not extend to the complete thermal cycle with background restoration. Set $P_{\mathcal{I}} = \mathcal{I}_K \mathcal{I}_K^\dagger$ and take the old zero-element state $\rho = P_z$. The finer zero-element state is $z' = e'_{K+1}$, with $\mathcal{I}_K z = z'$ by definition. Input control is the identity on the shell-average subspace, so Eq. (4.61) gives for the finer complete cycle

$$\mathcal{F}'(P_{z'}) = \sum_{\nu} f'_{\nu, K+1} P_{\nu'} + k'_{K+1} (\sigma_g^{\text{sh}})', \quad k'_{K+1} \geq a'_0 > 0.$$

The new unit shell S'_0 lies entirely in the orthogonal complement of $P_{\mathcal{I}}$, and the restoration share satisfies $g'_0 > 0$, so

$$\text{Tr}[(I - P_{\mathcal{I}}) \mathcal{F}'(P_{z'})] \geq a'_0 g'_0 > 0. \quad (\text{A.17})$$

But $\mathcal{I}_K \mathcal{F}(P_z) \mathcal{I}_K^\dagger$ remains supported on $P_{\mathcal{I}}$, and the two differ. The complete thermal update therefore fails to satisfy the same intertwining identity for this embedding. The original Haar preparation has $g'_0 = (p-1)/p$.

Increasing position resolution Keeping j fixed and increasing K by one changes the principal-unit position count to pn . Choose compatible Teichmüller generators so the finer ζ' reduces to ζ . Reduction sends (s, t') to $(s, t' \bmod n)$. The uniform coherent lift on the propagation space is

$$\mathcal{L}|\pm, t\rangle = p^{-1/2} \sum_{b=0}^{p-1} |\pm, t + bn\rangle', \quad \mathcal{L}^\dagger \mathcal{L} = I. \quad (\text{A.18})$$

Finite Fourier summation gives $\mathcal{L}|\pm; r\rangle = |\pm; pr\rangle'$. It intertwines principal-unit translation in both orientations, since cyclic wraparound merely permutes b .

The finer characters $(\pm a, pr)$ are precisely lifts of the old characters. Extending both to integer characters by zero on multiples of p gives equal values at every integer; their Dirichlet series therefore agree term by term for $\Re s > 1$ and remain equal under analytic continuation. Primitive conductors and resonance ratios also coincide. With identical phase orientations and zero-response conventions,

$$z'_{\pm, pr} = z_{\pm, r}, \quad W'_{pr} = W_r, \quad W' \mathcal{L} = \mathcal{L} W. \quad (\text{A.19})$$

For any initial ρ , it follows that $\rho'_m = \mathcal{L} \rho_m \mathcal{L}^\dagger$. Equation (7.5) gives the fine endpoint distribution pointwise:

$$\Pr'(s, t + bn \mid *^m) = \frac{1}{p} \Pr(s, t \mid *^m). \quad (\text{A.20})$$

Summing the p outcomes reducing to the same position recovers the original distribution. This uses the coherent preparation of Eq. (A.18) and endpoint measurement; fine point measurements every cycle need not leave states in the lifted space.

The lift in Eq. (A.18) does not preserve actual logarithmic coordinates at the finer depth. An old representative x has p lifted representatives $x + bp^K$, possibly in a permuted order. Using the same coarse reference u_* at the finer depth and conditioning on a positive-probability coarse outcome x gives

$$\mathbb{E}' \left(\log \frac{x'}{u_*} \middle| x \right) - \log \frac{x}{u_*} = \frac{1}{p} \sum_{b=0}^{p-1} \log \left(1 + \frac{bp^K}{x} \right) > 0. \quad (\text{A.21})$$

Thus coherent lifting preserves spectral propagation and the reduced position distribution but not fine logarithmic position intervals. Reducing position labels before logarithmic calibration recovers the old readings exactly; direct fine-coordinate readout changes the conditional mean according to Eq. (A.21). The distinction depends on actual readout. Density of the scalar interval set in Section 5 cannot replace these propagation and readout relations.

Data Availability

The main results are obtained by analytical derivation; no experimental or observational data were used. Related supplementary derivations and numerical checks are provided in the accompanying supplementary materials (in Chinese).

Conflict of Interest

The author declares no conflict of interest.

References

- [1] J.-B. Bost and A. Connes, *Hecke algebras, type III factors and phase transitions with spontaneous symmetry breaking in number theory*, *Selecta Mathematica* (N.S.) **1** (1995), 411–457. [doi:10.1007/BF01589495](https://doi.org/10.1007/BF01589495).
- [2] M. Laca and I. Raeburn, *A semigroup crossed product arising in number theory*, *Journal of the London Mathematical Society* (2) **59** (1999), 330–344. [doi:10.1112/S0024610798006620](https://doi.org/10.1112/S0024610798006620).
- [3] P. Erdős, *On the distribution function of additive functions*, *Annals of Mathematics* (2) **47** (1946), 1–20, Theorem XIII. [doi:10.2307/1969031](https://doi.org/10.2307/1969031); [Scan of the original article](#).
- [4] G. B. Folland, *A Course in Abstract Harmonic Analysis*, 2nd ed., CRC Press, Boca Raton, 2015. [doi:10.1201/b19172](https://doi.org/10.1201/b19172).
- [5] F. Q. Gouvêa, *p-adic Numbers: An Introduction*, 3rd ed., Universitext, Springer, Cham, 2020. [doi:10.1007/978-3-030-47295-5](https://doi.org/10.1007/978-3-030-47295-5).
- [6] T. M. Apostol, *Introduction to Analytic Number Theory*, Undergraduate Texts in Mathematics, Springer-Verlag, New York, 1976. [doi:10.1007/978-1-4757-5579-4](https://doi.org/10.1007/978-1-4757-5579-4).
- [7] J.-P. Serre, *Local Fields*, Graduate Texts in Mathematics, Vol. 67, Springer-Verlag, New York, 1979. [doi:10.1007/978-1-4757-5673-9](https://doi.org/10.1007/978-1-4757-5673-9).
- [8] A. Terras, *Fourier Analysis on Finite Groups and Applications*, London Mathematical Society Student Texts, Vol. 43, Cambridge University Press, Cambridge, 1999. [doi:10.1017/CBO9780511626265](https://doi.org/10.1017/CBO9780511626265).

-
- [9] J.-P. Serre, *Linear Representations of Finite Groups*, Graduate Texts in Mathematics, Vol. 42, Springer-Verlag, New York, 1977. doi:10.1007/978-1-4684-9458-7.
- [10] F. W. J. Olver, A. B. Olde Daalhuis, D. W. Lozier, B. I. Schneider, R. F. Boisvert, C. W. Clark, B. R. Miller, B. V. Saunders, H. S. Cohl and M. A. McClain (eds.), *NIST Digital Library of Mathematical Functions*, Release 1.2.8, 2026-09-15, Chapter 25, §25.15(i). <https://dlmf.nist.gov/25.15>.
- [11] H. L. Montgomery and R. C. Vaughan, *Multiplicative Number Theory I: Classical Theory*, Cambridge Studies in Advanced Mathematics, Vol. 97, Cambridge University Press, Cambridge, 2007. doi:10.1017/CBO9780511618314.
- [12] G. H. Hardy and M. Riesz, *The General Theory of Dirichlet's Series*, Cambridge Tracts in Mathematics and Mathematical Physics, No. 18, Cambridge University Press, Cambridge, 1915.
- [13] L. Bruneau, A. Joye and M. Merkli, *Asymptotics of repeated interaction quantum systems*, Journal of Functional Analysis **239** (2006), 310–344. doi:10.1016/j.jfa.2006.02.006.
- [14] W. F. Stinespring, *Positive functions on C^* -algebras*, Proceedings of the American Mathematical Society **6** (1955), 211–216. doi:10.1090/S0002-9939-1955-0069403-4.
- [15] K. Kraus, *General state changes in quantum theory*, Annals of Physics **64** (1971), 311–335. doi:10.1016/0003-4916(71)90108-4.
- [16] M.-D. Choi, *Completely positive linear maps on complex matrices*, Linear Algebra and its Applications **10** (1975), 285–290. doi:10.1016/0024-3795(75)90075-0.
- [17] Y. Aharonov, L. Davidovich and N. Zagury, *Quantum random walks*, Physical Review A **48** (1993), 1687–1690. doi:10.1103/PhysRevA.48.1687.
- [18] M. Raginsky, *Strictly contractive quantum channels and physically realizable quantum computers*, Physical Review A **65** (2002), 032306. doi:10.1103/PhysRevA.65.032306.
- [19] J. Faraut and A. Korányi, *Analysis on Symmetric Cones*, Oxford University Press, Oxford, 1994. doi:10.1093/oso/9780198534778.001.0001.
- [20] R. Penrose and W. Rindler, *Spinors and Space-Time, Vol. 1: Two-Spinor Calculus and Relativistic Fields*, Cambridge University Press, Cambridge, 1984. doi:10.1017/CBO9780511564048.
- [21] G. Kimura, *The Bloch vector for N -level systems*, Physics Letters A **314** (2003), 339–349. doi:10.1016/S0375-9601(03)00941-1.
- [22] R. P. Boas, Jr., *Entire Functions*, Pure and Applied Mathematics, Vol. 5, Academic Press, New York, 1954, Chapter 9, §9.2. doi:10.1016/S0079-8169(08)60152-X.

-
- [23] B.-G. Englert, *Fringe visibility and which-way information: An inequality*, Physical Review Letters **77** (1996), 2154–2157. doi:[10.1103/PhysRevLett.77.2154](https://doi.org/10.1103/PhysRevLett.77.2154).
- [24] E. B. Davies and J. T. Lewis, *An operational approach to quantum probability*, Communications in Mathematical Physics **17** (1970), 239–260. doi:[10.1007/BF01647093](https://doi.org/10.1007/BF01647093).
- [25] M. Ozawa, *Quantum measuring processes of continuous observables*, Journal of Mathematical Physics **25** (1984), 79–87. doi:[10.1063/1.526000](https://doi.org/10.1063/1.526000).
- [26] W. H. Zurek, *Probabilities from entanglement, Born’s rule $p_k = |\psi_k|^2$ from enviance*, Physical Review A **71** (2005), 052105. doi:[10.1103/PhysRevA.71.052105](https://doi.org/10.1103/PhysRevA.71.052105).
- [27] P. Busch, *Quantum states and generalized observables: A simple proof of Gleason’s theorem*, Physical Review Letters **91** (2003), 120403. doi:[10.1103/PhysRevLett.91.120403](https://doi.org/10.1103/PhysRevLett.91.120403).
- [28] H. Umegaki, *Conditional expectation in an operator algebra, IV (Entropy and information)*, Kodai Mathematical Seminar Reports **14** (1962), 59–85. doi:[10.2996/kmj/1138844604](https://doi.org/10.2996/kmj/1138844604).
- [29] M. Esposito, K. Lindenberg and C. Van den Broeck, *Entropy production as correlation between system and reservoir*, New Journal of Physics **12** (2010), 013013. doi:[10.1088/1367-2630/12/1/013013](https://doi.org/10.1088/1367-2630/12/1/013013).
- [30] D. Reeb and M. M. Wolf, *An improved Landauer principle with finite-size corrections*, New Journal of Physics **16** (2014), 103011. doi:[10.1088/1367-2630/16/10/103011](https://doi.org/10.1088/1367-2630/16/10/103011).
- [31] G. Lindblad, *Completely positive maps and entropy inequalities*, Communications in Mathematical Physics **40** (1975), 147–151. doi:[10.1007/BF01609396](https://doi.org/10.1007/BF01609396).
- [32] A. Uhlmann, *Relative entropy and the Wigner–Yanase–Dyson–Lieb concavity in an interpolation theory*, Communications in Mathematical Physics **54** (1977), 21–32. doi:[10.1007/BF01609834](https://doi.org/10.1007/BF01609834).
- [33] K. M. R. Audenaert, *A sharp continuity estimate for the von Neumann entropy*, Journal of Physics A: Mathematical and Theoretical **40** (2007), 8127–8136. doi:[10.1088/1751-8113/40/28/S18](https://doi.org/10.1088/1751-8113/40/28/S18).
- [34] J. Watrous, *The Theory of Quantum Information*, Cambridge University Press, Cambridge, 2018. doi:[10.1017/9781316848142](https://doi.org/10.1017/9781316848142).